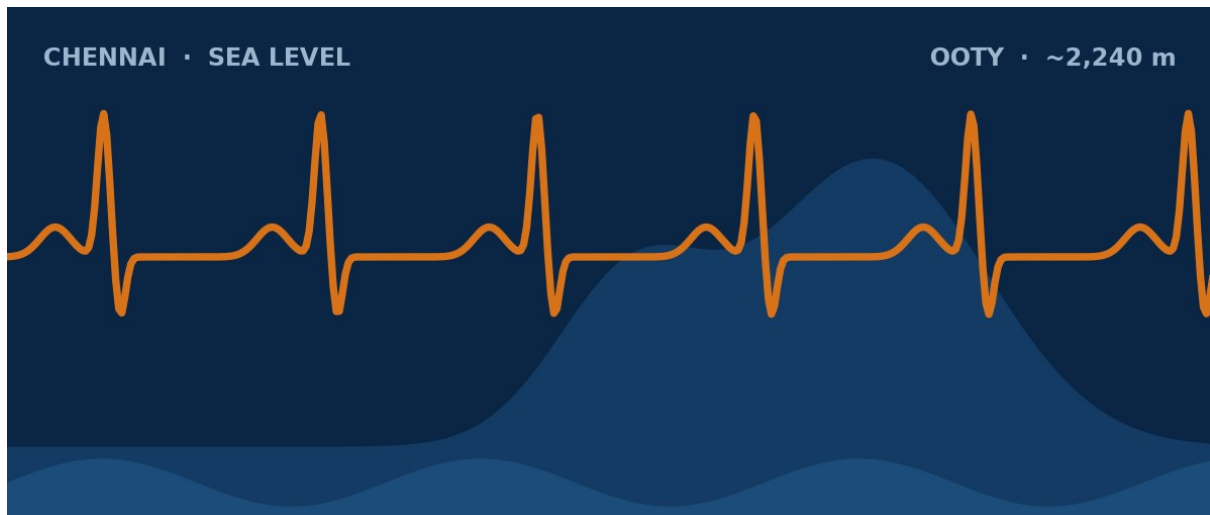




AKURA

WORLD ATHLETE RESEARCH BOOK

The AKURA Endurance Index™ (AEI)

From Myth to Measurement

MEASURE THE ATHLETE • UNDERSTAND THE BODY • BUILD THE
PERFORMANCE

Complete English Edition • Indian Sports Context

KURA

AKURA Endurance Research System

Prototype Edition 0.1 • 2026

AKURA World Athlete Research Book — The AKURA Endurance Index™ (AEI): From Myth to Measurement

Complete English Edition. Prototype Edition 0.1, 2026.

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Important notice. This book is an educational and research document. It is not medical advice. The AKURA Endurance Index is a proposed research framework that has not yet been validated; its equations and categories must not be used for high-stakes selection, medical or safety decisions. Blood testing, cardiac screening, hypoxic exposure and heat-acclimation interventions should be conducted only with appropriate medical oversight and, for research, with informed consent and ethics committee approval.

Chennai, Tamil Nadu, India.

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Author's Note

This book began with a practical question: when endurance athletes perform at a world-class level, which parts of that performance are inherited, which are trained, which are environment-dependent, and which can be measured and developed in an Indian athlete? It is a simple question to ask on a track at six in the morning. It is a much harder question to answer honestly.

For years I have watched runners in Chennai finish a humid long run soaked to the skin, heart rate climbing while the pace slowly falls, and then ask me what they are missing. Some have been told they need altitude. Some have been told their lungs are weak, or that they were simply not born in the right place. Some have been sold a supplement, a watch metric or a training plan that promised a shortcut. Very few have been given the one thing that actually helps: a clear way to measure what is happening in their own body, and a method for turning that measurement into a better training decision.

The purpose of this book is not to copy another country. Kenya and Ethiopia have given the world extraordinary distance runners, and there is much to learn from how those athletes live, train and progress. But learning is not copying. An Indian athlete cannot import a Rift Valley childhood, and should not have to. What an Indian athlete can do is understand the human endurance system well enough to build an evidence-led pathway for training in our own climate, on our own roads, tracks and hills, and in our own cities.

We don't have to copy their mountain. We have to understand the science behind their success.

— Kura

That sentence is the spirit of the whole project. The mountain is not the magic. The mechanisms are what matter: how oxygen is delivered and used, how heat and hypoxia challenge the body, how an athlete absorbs training load, and how performance holds up in the last ten kilometres of a marathon. Mechanisms can be studied anywhere. Many of them can be measured with tools already in a coach's hands, and others with the help of a sports-science laboratory or a sports-medicine team.

The AKURA Endurance Index (AEI) is my attempt to organise that measurement into a single, transparent research framework. I want to be clear from the first page about what it is and what it is not. It is a structured way of collecting and interpreting an athlete's performance, physiology, environment and recovery. It is not yet a validated score. It is not a replacement for established tools such as VDOT, laboratory testing or medical assessment. It is a Prototype Edition, numbered 0.1, because I expect it to be tested, corrected and improved by the athletes and coaches who use it.

I have tried to write with humility. Where the evidence is strong, I say so. Where a mechanism is plausible but unproven, I say that too. Where I am sharing coaching experience, or proposing an AKURA hypothesis, the text is labelled accordingly. If some of those hypotheses turn out to be wrong, the book will have done its job, because a wrong hypothesis that is tested properly still moves the field forward.

This book is dedicated to every Indian runner who has been told what they cannot become, and to every coach who would rather measure than guess.

How to Use This Book

This book is deliberately written in layers, so that different readers can enter at the level that suits them and then go deeper when they are ready.

The first layer explains physiology in plain language. Parts I to VI describe the question behind the book, the human endurance engine, heat, altitude, heat versus altitude, and the marathon as a systems test. You do not need a degree in physiology to follow these chapters, although sports-science students will find the concepts consistent with their textbooks.

The second layer converts physiology into measurable variables. Throughout the book, and especially in the Testing Lab and Environment sections, each idea is translated into something that can be recorded: a pace at a fixed heart rate, a haemoglobin-mass value, a sweat-loss estimate, a durability measure from a long run.

The third layer proposes the AEI research framework itself: its domains, its separation of Fitness from Readiness, its approach to normalisation, missing data and confidence, and its validation plan.

The fourth layer connects AEI to day-to-day readiness and training decisions: how a coach might use the profile to modify a session, plan a heat or altitude block, or interpret a disappointing race.

A research prototype is not presented as a proven clinical or sporting standard. Its equations and domains must be validated against athlete data before being used as a high-stakes decision system, such as team selection, return-to-play decisions or medical judgements.

Boxes you will meet

Five types of box appear throughout the chapters. Each has a specific job.

Box	Purpose
Key idea	The central point of a section in one to three sentences
Coach's corner	How to apply the idea in real training
Field protocol	A concrete, step-by-step test or session you can run
Myth vs evidence	A common belief set beside what the evidence supports
Checkpoint	Reflection questions to close each chapter

The Checkpoint questions are not a test. They are designed to be discussed between coach and athlete, or used in a classroom or team meeting.

Who this book is for

- Coaches, from school and college coaches to club and elite endurance coaches, who want to move from rules of thumb to repeatable, measured decisions.
- Athletes, especially club runners and aspiring competitive runners, who want to understand why their training works or fails and what their data is really saying.
- Sports-science students, who need a bridge between laboratory physiology and the messy reality of training in Indian heat, humidity and traffic.
- Sports-medicine staff, including physicians, physiotherapists and dietitians, who work alongside coaches and need a shared vocabulary for testing, monitoring and safety.

You can read the book from start to finish, or begin with the part most relevant to your current problem. If you are a coach preparing a Chennai marathon group, Part III on heat may be your starting point. If you are planning a camp in Ooty or Kodaikanal, begin with Part IV. Wherever you begin, return to Part I, because it explains the attitude the rest of the book depends on.

Important Scientific Boundary

Elite East African endurance performance is multifactorial. This book does not claim that nationality, ethnicity, altitude or genetics alone explains marathon success. No serious body of research supports such a single-cause explanation, and repeating it harms athletes on every side of the argument: it reduces East African champions to their birthplace and ignores the years of work behind their performances, and it tells athletes elsewhere that their ceiling has been set before they begin.

This book also does not claim that heat training is a universal substitute for altitude training. Heat and altitude are different environmental stimuli. They can overlap in some cardiovascular and performance outcomes, but their primary mechanisms differ.

Where later chapters compare them, the comparison is made carefully and with explicit uncertainty.

The purpose throughout is to separate four kinds of statement that are often mixed together in coaching conversations: evidence, plausible mechanisms, coach experience and hypotheses. A statement can be useful without being proven, and a mechanism can be plausible without being the explanation for a given athlete. Keeping these categories apart is the single most important habit this book hopes to build.

Nothing in this book is medical advice. Blood tests, cardiac screening, hypoxic exposure and heat exposure all carry risks and require appropriate supervision. Any abnormal finding, symptom or concern should be referred to a qualified clinician. A coach or an algorithm should never diagnose from a watch reading, an ECG screenshot or a single blood value.

Finally, the AKURA Endurance Index is presented as an original research framework in its prototype form. It must be validated prospectively and independently before being used to make high-stakes decisions about any athlete.

Evidence Levels Used in This Book

To keep the boundary described above visible, claims in this book are written with one of five levels of confidence in mind. When the language says "research shows", it refers to the first two levels. When it says "may", "could" or "AKURA proposes", it refers to the later levels.

Level	Meaning	Example
Established evidence	Consistent findings across many studies and reviews	Heat acclimation lowers heart rate at a fixed workload
Emerging evidence	Supported by some studies; results vary or are still developing	Heat blocks improving sea-level performance in cool conditions
Plausible mechanism	Physiologically reasonable but not confirmed as the cause	Plasma expansion supporting stroke volume late in a marathon
Coach experience	Repeated practical observation, not formally tested	Runners in humid Chennai summers adapting over four to six weeks
AKURA hypothesis	A proposal of this book, awaiting validation	AEI domains improving training decisions beyond VDOT alone

Readers should expect statements to move between these levels as new research appears. That movement is a sign of a healthy science, not a weakness.

PART I — THE QUESTION

Every serious research programme begins with a well-framed question. Part I asks why some athletes reach world-class endurance, what we can honestly say about East African success, how birthplace, biology and genetics fit together, and whether Indian athletes can build comparable endurance in their own environments. It rejects both easy answers: that champions are simply born, and that a few weeks in the heat or the hills can reproduce their physiology. Instead, it sets out the AKURA method of turning coaching beliefs into measurable hypotheses, testing them in individual athletes, and accepting the result even when it disagrees with our expectations.

IN THIS PART

- 1 Why This Book Exists
- 2 The East African Marathon Question
- 3 Birthplace Versus Biology
- 4 Genetics: A Contribution, Not a Shortcut
- 5 Can Indian Athletes Build World-Class Endurance?
- 6 From Belief to Measurement

PART I • THE QUESTION

From Belief to Measurable Hypothesis

Why do some athletes reach world-class endurance, and can Indian athletes build it in their own environments?

2:11:37

Sawan Barwal's national marathon record, Asian Games, Nagoya 2026

 **48 yrs**

The record that waited

Shivnath Singh's 2:12:00 from 1978 stood for 48 years before Sawan Barwal broke it in 2026.



Many systems, not one

East African success comes from training, economy, altitude, culture and motivation interacting. No single cause explains it.



Birthplace is not destiny

Birthplace shapes development, but training history, active childhood, nutrition and coaching can be built in many places.

 **Not scored**

Genetics: no shortcut

Genes contribute, but cannot be read from nationality. AEI does not score ancestry or consumer genetic tests.

 **4 settings**

India as a laboratory

Chennai heat, Bengaluru plateau, Ooty/Kodaikanal hills and the Himalayas give different stimuli, not interchangeable ones.



Write a hypothesis

Define the intervention, the measurement and the expected change: a 10–14 day heat block lowers HR at fixed pace.

THE PROCESS



A single number rarely explains performance: turn beliefs into hypotheses and accept what the data show.

THE STORY

The Record That Waited Forty-Eight Years

PART I — THE QUESTION

The number is 2:12:00. Two hours, twelve minutes, zero seconds. For most of the history of Indian distance running, it sat at the end of every serious marathon conversation like a milestone nobody could pass.

It was set in 1978, at the National Inter-State Championship in Jalandhar, by an Army runner named Shivnath Singh. Biographical accounts say he was born in 1946 in Buxar district in Bihar, served in the Bihar Regiment and ran barefoot for his whole career. Two years before Jalandhar he had raced the marathon at the Montreal Olympics, where he is reported to have finished 11th, and he had reportedly won Asian Championship silver medals on the track at 5000 m and 10,000 m. On that day in Punjab he ran a time that no Indian would better for forty-eight years.

Forty-eight years. Think about what fits inside that. Runners who were not yet born in 1978 grew up, joined the Army or the Railways, raced, retired and began coaching, and the record was still there. Other ancient marks fell one by one. Hari Chand's barefoot 10,000 m record from Montreal lasted 32 years. Gopal Saini's steeplechase record from 1981 lasted 37. Bahadur Prasad's 5000 m record from 1992 lasted about 30. The marathon outlived them all. By Olympics.com's reckoning it became the longest-standing record in Indian athletics.

The near misses

People came close, and the closeness was part of the torment. At the 2016 Mumbai Marathon, three Army runners, Nitendra Singh Rawat, Thonakal Gopi and Kheta Ram, qualified for the Rio Olympics. In Rio, Gopi finished 25th in 2:15:25, a personal best, and Kheta Ram was one second behind him. Good runs. Still more than three minutes short.

Then, on 7 December 2025, in Valencia, Gopi ran 2:12:23 after four months of altitude training in Ooty. Twenty-three seconds. That was all that stood between him and Shivnath Singh. In the same race Man Singh ran 2:13:25 and A.B. Belliappa 2:14:07. For the first time in a generation, the old number was not a distant rumour. It was within touching distance, and several men were reaching for it.

Two seconds in Rotterdam

Sawan Barwal comes from the Joginder Nagar area of Mandi district in Himachal Pradesh. He grew up in a working family: his father worked as a driver in New Delhi. He started with the 1500 m at fourteen and joined the Army in 2019. In 2024 he won the Indian elite men's race at the Delhi Half Marathon in 1:02:46. In early 2026 he finished 60th at the World Cross Country Championships in Tallahassee.

He was also part of something new: a Reliance Foundation programme coached by Ajith Markose and named, with no false modesty, Project 2:09. The name was a target,

and the target was three minutes faster than any Indian had ever run. According to Markose, the programme normally spends about four months, from November to February, around Ooty and Wellington in the Nilgiris, then moves to Bengaluru for pre-competition work, and Sawan spent most of his time at altitude. The Tribune reported a training load of about 200 kilometres a week, with long runs of 40 to 42 kilometres on alternate weeks.

On 12 April 2026, Sawan ran his first marathon, in Rotterdam. He finished 20th, in 2:11:58. After forty-eight years, the record fell by two seconds. Gopi, still chasing, finished 23rd in the same race in 2:13:16, inside the Asian Games qualifying standard.

Two seconds is a strange margin for a record that waited so long. It is less than the time it takes to tie a shoelace. And if the moment invites celebration, the wider picture invites humility. Two weeks later, in London, Kenya's Sabastian Sawe ran 1:59:30, reported as the first sub-two-hour marathon in a record-eligible race. When Sawe crossed the finish line, a runner holding Sawan's Rotterdam pace would still have had nearly four kilometres to go.

Nagoya

Sawan did not stop at two seconds. On 26 September 2026, at the Asian Games in Aichi-Nagoya, he ran 2:11:37 and won the silver medal, India's first men's marathon medal at the Asian Games since 1982. He had lowered his own national record by another 21 seconds, this time in a championship race where medals mattered more than the clock.

There was only one thing in my mind: I have to go till the finishing line and how I can maintain the pace.

— Sawan Barwal, to The Tribune

Ajith Markose, his Project 2:09 coach, had said it plainly months earlier, speaking to The Bridge: "When we started this Project 2:09, we knew that he is the one who is able to do 2:09." Even after Nagoya, that target is still more than two and a half minutes away. No Indian has ever run a marathon under 2:10. In 2019 alone, the world produced 293 performances under 2:10, and Kenyan men accounted for 123 of them.

Myth or measurement?

So what finally moved the number? The easy answers arrive quickly, and they arrive in familiar forms.

He is a hill boy, some will say, born for thin air. But Shivnath Singh, who set the old record, was reportedly born on the plains of Bihar. Gopi comes from Wayanad, about 1,000 metres up. Kheta Ram is reported to have grown up in the Barmer desert. If birthplace were destiny, these men would not share the same page of the record book.

It was the altitude, others will say. Perhaps altitude played a part. Sawan's coach says he spent most of his time at altitude, and Gopi credits Ooty for his Valencia run. But those altitude blocks came bundled with other things: 200-kilometre weeks, 40-kilometre long runs, an Army job that allowed full-time training, a privately funded programme with a stated target, and fast, well-paced races in Europe. Almost every Indian distance record since 2022 has been set abroad, in races with the pacing and competition that home meets rarely offer. Avinash Sable has made the point, as reported by ESPN, that nobody in India could set the pace a record required. Altitude was one ingredient in a crowded recipe, and nobody has shown how much of the two seconds, or the 21 seconds that followed, it was responsible for.

Here is what the public record contains: splits, finishing times, weekly kilometres, training bases. Here is what it does not contain: Sawan's haemoglobin mass before and after the Nilgiris, his running economy at marathon pace, his lactate curve, how his heart rate drifted in the heat, how much carbohydrate he took in and when. Those numbers may exist in a notebook or a laboratory somewhere. They are not part of the story India tells itself about its runners. Instead, the story is filled with birthplace, grit and mountain air.

There is a systemic side too. The Army Sports Institute in Pune, set up in 2001, and the sports-quota jobs of the services and Railways have carried Indian distance running for decades. India now has high-altitude centres at Ooty and Shilaroo and a planned one in Shillong, yet a ministry official told the Hindustan Times in 2026 that the country is "short of roughly 750 coaches". A national sports-science centre opened in New Delhi only in April 2025. And in April 2026 the Athletics Integrity Unit placed India in its highest doping-risk category, which means every fast Indian time must now also be a well-tested one before the world fully trusts it.

This is the question the book is built on. Not whether Indians can run fast, because they plainly can, but what it actually takes, physiologically and systemically, for an Indian runner to go faster. Which parts of the engine limit performance? Which of them can be measured, which can be trained, and how large is the response in one particular athlete? What does heat do, what does altitude do, and what can a coach in Chennai or Pudukkottai do with that knowledge on an ordinary Tuesday morning?

For forty-eight years the answer was mostly belief. The next record will be easier to understand, and perhaps easier to reach, if it comes with measurement. To see why, we need to start by asking the question properly.

Sources: <https://www.olympics.com/en/news/sawan-barwal-breaks-marathon-national-record-india> ; <https://www.tribuneindia.com/news/sports/who-is-sawan-barwal-himachal-runner-who-challenged-marathon-myth-to-win-asian-games-silver/> ; <https://thebridge.in/athletics/ajith-markose-decode-sawan-barwal-48-year-old-record-breaking-run-56362> ; <https://worldathletics.org/athletes/india/sawan-barwal-14734651> ; <https://www.olympics.com/en/news/athletics-list-of-indian-national-record-holders> ; https://en.wikipedia.org/wiki/Shivnath_Singh ; <https://thecronica.substack.com/p/a-barefoot-record-of-an-indian-marathoner> ; <https://shyamgopan.com/2025/12/11/gopi-23-secs-short-of-national-record-sets-new-pb-at-valencia-marathon/> ; <https://www.indiasportshub.com/articles/gopi-thonakals-valencia-marathon-run-rekindles-indian-marathon-hope> ; <https://www.firstpost.com/sports/rio-olympics-2016-thanackal-gopi-and-kheta-ram-clock-personal->

best-to-finish-25th-26th-in-marathon-2969066.html ; <https://www.npr.org/2026/04/27/nx-s1-5800113/kenya-sabastian-sawe-east-africa-london-marathon-2-hour-barrier> ; <https://tonireavis.com/2021/12/21/sub-210-marathoning-2021/> ; <https://www.hindustantimes.com/sports/others/high-altitude-training-centre-to-come-up-in-shillong-101780588041378.html> ; <https://www.aljazeera.com/sports/2026/4/20/india-in-extremely-high-doping-risk-bracket-athletics-integrity-unit>

CHAPTER 1

1. Why This Book Exists

PART I — THE QUESTION

It is 5:40 a.m. on the Marina, and the sea is the colour of weak tea. A club runner slows to a walk near the lighthouse, hands on knees, shirt dark with sweat after thirty kilometres in air that already feels like a steam room. His watch buzzes. Training load: high. Recovery: 42 hours. VO_2 max estimate: down one point. Later that morning, over filter coffee, three people will explain his slow finish to him with complete confidence. A friend says he needs a month in Ooty. An uncle says some people simply have weak lungs. A stranger in an online running group says he was not born in the right place. Nobody asks what the temperature was, what he ate, how he slept or what his heart rate did at the same pace a month ago. Each explanation is a single sentence. His body, like every runner's body, is not. The distance between those two things, the one-line verdict and the many-sided athlete, is where this book begins.

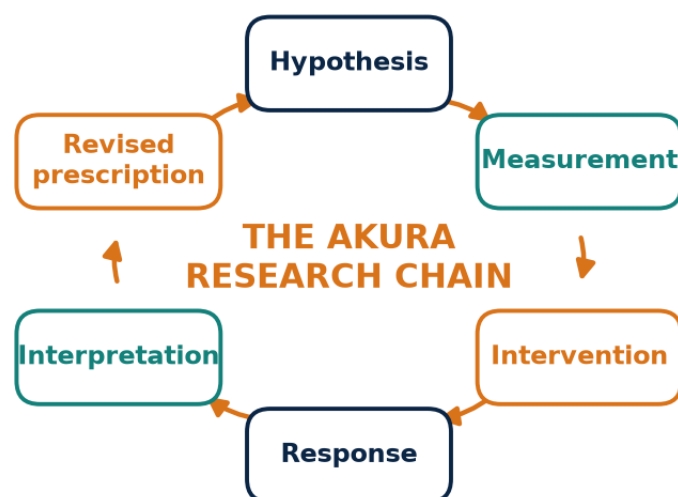


Figure 1. The AKURA research chain. Every coaching belief becomes a testable hypothesis; the loop repeats for the athlete's entire career.

Endurance coaching is full of numbers: pace, heart rate, VO_2max , lactate, kilometres, elevation, sleep score and training load. A club runner in Bengaluru can finish a Sunday long run and receive, within a minute, a dozen metrics from a watch and an app. Yet athletes often receive a single conclusion from a complex body: "you need altitude", "your lungs are weak", "your haemoglobin is high" or "your VDOT is low". Real physiology is rarely that simple. This chapter explains why AKURA exists, and why it asks a different kind of question.

Too many numbers, too few answers

The problem is not a shortage of data. Modern athletes are surrounded by it. The problem is that the data is rarely organised around a clear question, rarely collected under consistent conditions, and rarely interpreted with its limitations in view. A heart-rate value from a humid evening run in Chennai is compared directly with one from a cool morning in Ooty. A single blood test is treated as a verdict. A watch estimate of VO_2max is quoted as if it were a laboratory result.

When numbers are used this way, they do not replace the old coaching myths. They simply dress them up. "Your lungs are weak" becomes "your VO_2max is low", and the athlete is still left with a single explanation for a performance that almost certainly has several causes.

Physiologists have long described endurance performance as the product of several interacting determinants. Joyner and Coyle's 2008 review of the physiology of endurance champions, for example, framed performance around maximal oxygen uptake, the fraction of that capacity an athlete can sustain, and running economy, each shaped by further underlying factors. No single number in that model tells the whole story, and this book takes the same view.

KEY IDEA

A single number rarely explains an endurance performance. AKURA's purpose is to organise many measurements around clear questions, so that each number is interpreted in context rather than treated as a verdict.

The central AKURA question

The central AKURA question is therefore not whether one environment is magical. It is not "Is altitude better than heat?" or "Are some people simply born to run?" It is a set of four practical questions:

1. Which performance determinants can be measured?
2. Which of those determinants can be trained?
3. How large is the individual response to training or environmental exposure?
4. How should that information change the athlete's training?

These questions turn coaching from a collection of beliefs into a repeatable research process. They also protect athletes. A college runner who is told "you need altitude" may spend money and weeks away from home on a camp that does not address the actual limitation. A runner whose real problem is late-race durability or fuelling will not be helped by chasing a higher haemoglobin value.

MYTH VS EVIDENCE

Myth: My watch says my VO_2max is low, so that is why my marathon is slow.

Evidence: A watch estimate is an indirect calculation, and even a laboratory VO_2max is only one determinant. Marathon performance also depends on economy, sustainable intensity, fuelling, heat and durability.

The AKURA research chain

The book uses a simple chain that appears again and again in later chapters:

hypothesis → measurement → intervention → response → interpretation → revised prescription

A good coach may start with intuition. Experienced coaches often see patterns long before they can explain them, and that intuition is valuable. But intuition is where the process starts, not where it ends. A good research system must finish with data. The coach's hunch becomes a hypothesis; the hypothesis is tested with a defined measurement; an intervention is applied; the athlete's response is recorded; the result is interpreted honestly; and the training prescription is revised.

Step	Question it answers	Example
Hypothesis	What do we think is happening?	Heat is limiting this runner's long runs
Measurement	What will we record?	Pace, HR, body-mass loss, WBGT
Intervention	What will we change?	A 10–14 day heat block
Response	What happened?	HR at fixed pace fell by a measurable amount
Interpretation	What does it mean?	A heat response, not proof of elite status
Revised prescription	What do we do next?	Adjust summer long-run pacing

The chain is repeated for the athlete's entire career. Each cycle makes the athlete more understandable to themselves and their coach, which is ultimately what the AKURA Endurance Index is for.

COACH'S CORNER

Before you give an athlete an explanation, write it down as a hypothesis and name one measurement that could prove it wrong. If you cannot name such a measurement, you are holding a belief, not a coaching conclusion. Share the hypothesis with the athlete so they understand why they are being tested.

Why an Indian research system

Most of the physiology behind endurance performance was first described in laboratories in Europe and North America, and later refined through studies of East African runners. Those findings are valuable, but Indian athletes train in conditions that are rarely studied: monsoon humidity, coastal heat, dense urban traffic, school and college schedules, and hill stations that offer moderate rather than high altitude. A research system built in India can collect local, longitudinal data rather than assuming that findings from elsewhere will transfer unchanged.

That is why this book exists: not to announce a new truth, but to provide a structured way of finding out what is true for each athlete.

CHECKPOINT

Think of the last explanation you gave, or were given, for a poor race. Was it a single cause? What measurement could have tested it, and what else might have contributed?

CHAPTER 2**2. The East African Marathon Question****PART I — THE QUESTION**

On 26 April 2026, on the streets of London, Sabastian Sawe of Kenya ran the marathon in 1:59:30, reported as the first sub-two-hour marathon in a record-eligible race. Ethiopia's Yomif Kejelcha finished eleven seconds behind him. Two men under two hours in one race, from two neighbouring countries. The numbers behind that morning are just as startling. In 2019 alone, runners broke 2:10 for the marathon on 293 occasions worldwide, and Kenyans accounted for 123 of them. An Indian man has never done it once. It is the kind of statistic that tempts people to reach for a single sentence, and the sentence is almost always the same: they are born for running. The town of Iten, at about 2,400 metres in Kenya's highlands, even calls itself the Home of Champions, which seems to settle the argument. It

does not. The more closely scientists have looked, the less that one sentence has held up.

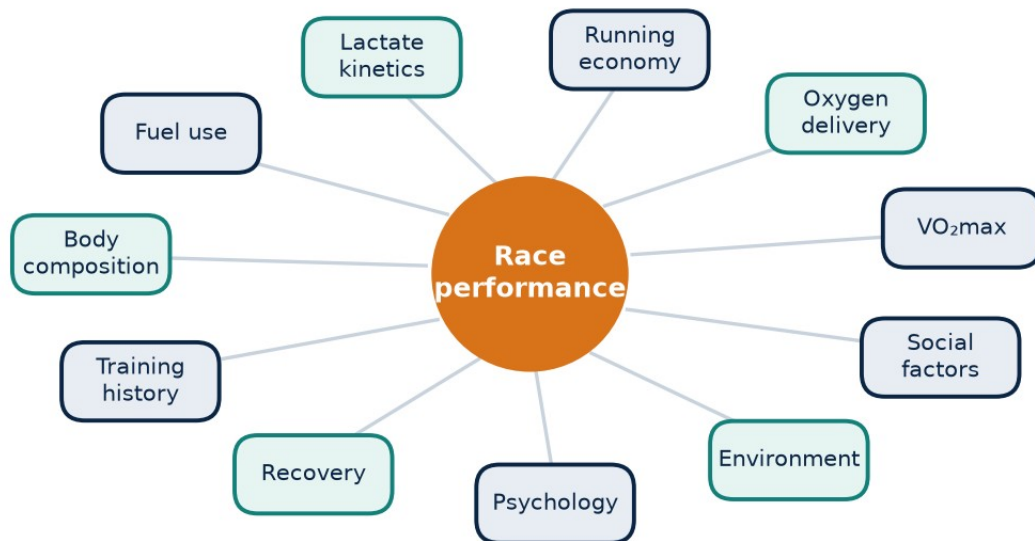


Figure 2. Elite endurance performance emerges from many interacting systems. No single factor, including birthplace or genetics, explains it alone.

Kenyan and Ethiopian runners have produced a remarkable body of elite distance-running performances, from Olympic medals on the track to world records on the roads. It is tempting to explain that success with one sentence: "they are born for running". That sentence is too small for the problem. This chapter explains why, and how AKURA uses the East African example as a source of questions rather than a stereotype.

A performance built from many systems

Performance is the result of interacting systems: maximal oxygen uptake, oxygen delivery, running economy, lactate kinetics, substrate use, body composition, training history, recovery, psychology, environment and social factors. Any explanation that picks one of these and ignores the rest will be incomplete.

Researchers have studied this question for decades. Saltin and colleagues' studies of Kenyan and Scandinavian runners in the 1990s compared aerobic capacity and muscle characteristics and did not find that a single variable such as maximal oxygen uptake explained the performance gap. Later reviews, including Wilber and Pitsiladis's 2012 paper on why Kenyan and Ethiopian runners are so successful, concluded that the answer is multifactorial, involving physiological and biomechanical traits, the training environment, a culture of running, and strong socio-economic motivation, with no convincing evidence that a specific genetic profile explains the dominance.

KEY IDEA

East African distance-running success is best understood as the interaction of many physiological, environmental, training and social factors. No single cause, including birthplace, altitude or genetics, is sufficient to explain it.

Altitude is part of the story, not the whole story

Many successful East African runners live and train at moderate altitude, and altitude exposure may contribute for some athletes. But living at altitude is not the same thing as possessing a permanent physiological advantage. Adaptations to altitude, such as changes in haemoglobin mass, depend on the dose of exposure, iron availability and the individual, and some of them fade after return to lower elevation.

It is also worth noting what altitude comes bundled with. Training camps in the Kenyan and Ethiopian highlands often combine high training volume, group training with other strong runners, soft trails, simple routines and a strong focus on running as a career. Separating the effect of the altitude from the effect of everything else is extremely difficult.

Factor	What it may contribute	How confident are we?
Training volume and history	Aerobic development over many years	Established as important
Running economy	Lower oxygen cost at race pace	Emerging, varies by study
Altitude exposure	Possible oxygen-transport adaptations	Plausible, individual
Group and camp culture	Consistency, motivation, pacing partners	Coach experience
Socio-economic motivation	Commitment and depth of talent pool	Discussed in reviews
Specific genes	Uncertain contribution	Not demonstrated as decisive

MYTH VS EVIDENCE

Myth: East Africans win because they are born at altitude.

Evidence: Altitude may help some athletes, but many runners born at altitude elsewhere in the world do not become elite, and the evidence points to a combination of training, environment, economy, culture and individual biology.

A source of questions, not a stereotype

AKURA treats the East African model as a source of questions, not a stereotype. Instead of asking "What are they?", it asks "What do they do, and what might it change?" For example:

- How much easy running do successful East African athletes accumulate over years, and at what intensity?

- How does running on soft, hilly terrain influence economy and resilience?
- How do group training and simple daily routines support consistency and recovery?
- What changes in oxygen transport, if any, occur with the altitude doses they experience?

Each of these can become an AKURA research question for Indian athletes. If an adaptation can be measured, it should be measured. If it cannot yet be separated from confounding factors, the correct scientific answer is uncertainty. That answer may feel unsatisfying, but it is more honest, and more useful, than a confident myth.

COACH'S CORNER

When an athlete brings up the "born to run" idea, redirect the conversation to what can be copied: consistent easy volume, group training, patient progression and good recovery. These are available in Chennai, Bengaluru or a small-town college ground, and they are among the most transferable lessons from successful endurance cultures.

CHECKPOINT

List three habits from successful endurance cultures that your athletes could adopt this season. Which of them could you measure, and which would you have to observe more informally?

CHAPTER 3

3. Birthplace Versus Biology

PART I — THE QUESTION

In Mohi village in Satara district, in a drought-prone corner of Maharashtra, a girl from a farming family ran 4 kilometres each way to school and helped fetch water for a household of seventeen people. Her name was Lalita Babar, and in 2016 she reached the Olympic steeplechase final in Rio, finishing tenth. Further north, near Nashik, Kavita Raut, reportedly from a tribal family in Sawarpada village, went on to win 10,000 m silver at the 2010 Asian Games. Her Nashik coach, Vijender Singh, once told the Indian Express: "I am telling you Nashik is the Kenya of India." Stories like these are true, and they are moving. They are also easy to misread. Was it the village that made these runners, the water pots and the long road to school? Or was it the years of running, the coaches who found

them and the jobs that let them keep training? The answer matters for every child who was born somewhere else.

A birthplace can influence an athlete's developmental environment without being a biological destiny. Where a child grows up shapes how much they move, what they eat, how far they walk or run to school, what sport they see around them and what opportunities they receive. These are real influences. But they are not the same as a fixed biological ceiling, and confusing the two leads to poor coaching decisions.

Environment during development

A person raised at altitude may experience repeated hypoxic exposure over many years, alongside whatever physical activity their daily life involves. Another person may develop exceptional endurance through years of running, an active childhood, good nutrition, sound coaching and selection into strong training groups. Both routes can produce a highly capable endurance athlete. Only one of them depends on geography.

It is also important to remember selection. Every elite athlete who emerges from any region is the survivor of a large pool of young people who tried the sport. When we look only at the winners, it is easy to credit the place rather than the process that filtered and developed them.

KEY IDEA

Birthplace shapes the developmental environment, but it is not a biological destiny. Most of the factors that build endurance, such as training history, active childhood, nutrition and coaching, can be developed in many places.

What this means in India

This distinction matters in India. Athletes in Chennai, Bengaluru, Delhi or coastal towns should not be told that world-class endurance is impossible because they were not born in a mountain region. That message is both scientifically weak and practically damaging, because it discourages young athletes before they have had a chance to develop.

Conversely, they should not be promised that a few weeks of heat or altitude will reproduce the complete physiology of an elite East African athlete. A two-week hill camp in Kodaikanal, or a summer of humid running in Chennai, may produce useful and measurable adaptations. It will not recreate years of development, and it should not be sold as if it could.

Statement	Why it is a problem	A better framing
You were not born at altitude, so forget the marathon	Treats birthplace as a ceiling	Let us measure what you can develop
Two weeks in Ooty will make you like a Kenyan	Overpromises a short exposure	Let us test your response to a defined dose
Chennai heat is the same as altitude	Confuses different stimuli	Heat and altitude produce different adaptations
Your background decides your potential	Ignores training history	Your trend over time will tell us more

MYTH VS EVIDENCE

Myth: Athletes from the plains cannot reach high endurance levels.

Evidence: Endurance is built through years of training, nutrition, recovery and coaching. Geography can shape opportunities, but it does not set a fixed biological limit for any individual.

A narrower, more useful objective

The research objective is narrower and more useful: identify modifiable determinants, quantify them, and build the training environment around the individual response. Modifiable determinants include aerobic fitness, running economy, threshold, durability, heat tolerance, iron status (under clinical supervision), body composition, strength and recovery habits. Each can be measured to some degree, and each can respond to training or environmental change.

FIELD PROTOCOL

1. Development history interview for a new athlete
2. Record where the athlete grew up and the approximate elevation
3. Ask how active their childhood was, including walking or cycling to school and other sports
4. Record years of structured running and typical weekly volume in each year
5. Note current training environment, temperature and humidity range
6. Store these answers in the athlete's baseline file as context, not as a score

COACH'S CORNER

Use background information to understand the athlete, never to label them. A school runner from a coastal town with little childhood sport simply has a shorter training history, which usually means more room for development, not less potential.

CHECKPOINT

What assumptions have you made about an athlete based on where they come from?
Which modifiable determinants could you measure instead?

CHAPTER 4**4. Genetics: A Contribution, Not a Shortcut**

PART I — THE QUESTION

Harmilan Bains grew up in a family of runners. Her mother, Madhuri Singh, won 800 m silver at the 2002 Asian Games. Her father, Amandeep Bains, is a South Asian Games medallist and a former national 1500 m champion. In September 2021, at the National Open in Warangal, Harmilan ran the 1500 m in 4:05.39, breaking a national record that had stood since 2002. Two years later she won silver in both the 800 m and the 1500 m at the Asian Games in Hangzhou. It is almost too neat a story: talent in the blood, passed from parents to daughter. But look again at what else passes through a family like that. Knowledge of training. Early access to a track. Parents who know what a good coach looks like, what an overloaded week feels like and why an injury must be respected. Genes and upbringing arrive in the same household at the same time, and no stopwatch can tell them apart.

Genetics can influence endurance-related traits. Differences between people in heart size, muscle-fibre composition, trainability and many other characteristics have some inherited component. But a simple national or ethnic genetic explanation is not a sufficient model of elite performance. This chapter explains why genetic arguments are harder than they appear, and why AEI deliberately leaves genetics out of the score.

Why genetic explanations are difficult

Genetic studies of endurance performance face four major problems.

- Performance is polygenic. Endurance depends on many traits, each influenced by many genes with small individual effects. There is no single "endurance gene".
- Populations are heterogeneous. A country or broad ethnic group contains enormous genetic diversity, often more within the group than between groups.
- Environmental exposure is entangled with ancestry. People who share ancestry often share environments, diets, activities and cultures, which makes it very difficult to separate genetic effects from environmental ones.

- Elite selection is strong. Elite athletes are an extremely filtered group, so comparisons between champions and the general population can exaggerate or distort genetic signals.

Family and training studies show that genetics matters in a more individual way. The HERITAGE Family Study led by Bouchard and colleagues found that people given the same standardised endurance-training programme showed very different improvements in VO_2max , and that the size of the response clustered within families. In other words, inherited factors may influence how much an individual responds to training, but the only way to know an individual's response is to train and measure them.

KEY IDEA

Genetics contributes to endurance traits and to how strongly individuals respond to training, but it cannot be read from nationality or ethnicity. An athlete's ceiling must be observed through testing, not assumed from background.

The practical coaching rule

For coaching, the practical rule is straightforward: do not infer an athlete's physiological ceiling from nationality. Test the athlete. Measure performance and physiology longitudinally, over months and seasons rather than a single test day.

When a trait is strongly inherited or poorly trainable, that is still useful information. For example, if repeated testing over two years shows that an athlete's VO_2max has changed very little while their economy and threshold have improved considerably, the coach can shift attention towards the qualities that are still moving. But this conclusion must be demonstrated, not assumed. It comes from the athlete's own data, not from a stereotype about a group.

MYTH VS EVIDENCE

Myth: A genetic test can tell you whether you should be a marathon runner.

Evidence: Current consumer genetic tests cannot reliably predict endurance performance or talent. Performance depends on many genes and on years of training and environment; longitudinal testing of the athlete is far more informative.

COACH'S CORNER

If an athlete asks whether they have the right genes, answer honestly: nobody can tell from a spit test or a surname. Then show them their trend data, such as easy pace at a fixed heart rate over the last six months. Trainability is revealed by training, and the trend is the athlete's own evidence.

Why AEI excludes genetics from the score

AEI therefore treats genetic explanations as background context rather than a score component. A score should reward what can be observed, measured and validated. Including a genetic or ancestry component would introduce three problems: it would reward something the athlete cannot change, it would rest on weak predictive evidence, and it could reinforce the very stereotypes this book rejects.

Type of information	Role in AEI	Reason
Race performance	Primary anchor	Directly observed and meaningful
Physiological tests	Supporting domains	Measurable and repeatable
Training response over time	Trend and confidence	Shows trainability in the individual
Family or ancestry information	Not scored	Weak prediction, risk of stereotyping
Consumer genetic tests	Not scored	Not validated for performance prediction

This does not mean genetics is unimportant for science. Research into how genes influence training response is valuable and will continue. It means that for coaching an individual athlete, measured response is more useful than inferred potential.

CHECKPOINT

Which of your athletes has improved most over the last year? Would you have predicted that from their background? What does their trend tell you that a prediction could not?

CHAPTER 5

5. Can Indian Athletes Build World-Class Endurance?

PART I — THE QUESTION

Thonakal Gopi grew up in Sulthan Bathery in Wayanad, about 1,000 metres above the sea, the son of farmers who grew rice and ginger. He joined the Army through the sports quota. Before the Rio Olympics he spent ten months training at a high-altitude facility in Ooty, and in Rio he finished 25th in what was then a personal best. Nine years later, in 2025, he went back up the Nilgiri road for four months before the Valencia Marathon, and in December he ran 2:12:23, just 23 seconds outside the national record of the time. "I think the high altitude at Ooty helped my performance," he said afterwards. For the previous six years his base had been Bengaluru, a city

on the Deccan plateau with a milder climate. Wayanad, Ooty, Bengaluru, Valencia: four environments in one career, three of them within a day's travel of each other. That variety is not a curiosity. Used carefully, it is one of the most valuable research tools a coach in India has.

Yes, athletes can develop very high endurance performance in India, but the route is individual. There is no single Indian formula, just as there is no single East African one. What India offers, perhaps more than most countries, is variety: a wide range of climates and elevations within reach of most athletes. This chapter describes that variety and reframes the national question as an individual one.

India as a natural laboratory

India contains hot and humid coastal environments, warm inland climates, cool highlands and substantial altitude zones. This creates a natural research laboratory. An athlete can, within a day's travel, move from sea-level humidity to a cooler hill station, or from a dry inland city to a coastal one.

Chennai can provide repeated heat and humidity exposure for much of the year. Bengaluru, at roughly 900 metres above sea level, provides a different climate and elevation profile, with generally milder temperatures. Ooty and Kodaikanal in the Western Ghats sit at roughly 2,000 metres or more and provide access to higher elevation and cooler conditions. The Himalayan regions offer considerably higher altitude, with correspondingly greater physiological and safety demands.

Environment	Typical stimulus	Research use	Main caution
Chennai (coastal)	Heat and high humidity	Heat and humidity adaptation	Heat illness, dehydration
Bengaluru (plateau)	Milder climate, modest elevation	Quality training, comparison	Not a strong hypoxic dose
Ooty / Kodaikanal (hills)	Moderate altitude, cooler air	Altitude response studies	Dose depends on sleep altitude
Himalayan regions	High altitude	Specialist hypoxia research	Requires medical oversight

KEY IDEA

India's range of climates and elevations is a genuine research asset. But these environments are not interchangeable, and each produces a different stimulus that must be defined and measured.

Environments are not interchangeable

These environments are not interchangeable, and the useful stimulus depends on the athlete, the season, the dose and recovery. A Chennai summer and a Chennai December are very different heat exposures. A week in Kodaikanal during the monsoon brings rain and cool temperatures along with the elevation. An athlete who travels overnight by bus to a hill station starts the camp with poor sleep, which can blunt the benefit of the first days.

The dose also matters. Spending a weekend in Ooty is not the same as sleeping there for three weeks. Running one humid long run is not the same as a structured, progressive heat-acclimation block. Later chapters on heat and altitude describe how to record the dose properly, but the principle begins here: name the stimulus, measure the dose, and record the response.

FIELD PROTOCOL

1. Environment log for any training block
2. Record location and approximate elevation of sleeping and training sites
3. Log temperature and relative humidity for key sessions, or WBGT if available
4. Note total days and nights spent in the environment
5. Record sleep duration and quality daily
6. Note travel, illness and any change in diet or routine

MYTH VS EVIDENCE

Myth: Training in Chennai heat is the same as training at altitude.

Evidence: Heat and altitude challenge the body in different ways. Heat mainly stresses thermoregulation and fluid balance; altitude reduces the pressure of inspired oxygen. Some outcomes may overlap, but the primary adaptations differ.

From a national question to an individual one

The research question is therefore not "Can India produce elite runners?" It is "Which combination of training, environment, recovery and athlete characteristics produces the best response for this athlete?"

This framing matters because national questions invite national answers, and national answers invite stereotypes. Individual questions invite measurement. A college athlete in Coimbatore, a club runner in Bengaluru and a school runner in a coastal town will each respond differently to the same plan. The goal of AEI is to learn those differences and use them.

COACH'S CORNER

Before planning a hill camp or a summer heat block, write down the specific adaptation you want, the measurement you will use before and after, and how you will control for travel, sleep and training load. If you cannot write those three things, the camp is a holiday with running, not an intervention.

CHECKPOINT

What environments can your athletes realistically access within a season? For each one, what adaptation would you target, and how would you know if it happened?

CHAPTER 6**6. From Belief to Measurement**

PART I — THE QUESTION

Every coach has a sentence they trust. On a school ground in Pudukkottai, a coach tells his runners that afternoon runs on hot tar roads will make them tough. In a college hostel in Coimbatore, another insists that a week in Kodaikanal before selection trials is worth a month of training anywhere else. Both may be right. Both may be wrong. Neither has written the sentence down in a form that could be proved wrong. Numbers alone do not solve the problem either. When Sawan Barwal broke the 48-year-old national marathon record in Rotterdam in April 2026, one published report gave his margin as "0.42 seconds". The real margin was two seconds: 2:11:58 against 2:12:00. It was a formatting slip, easily caught by anyone who checked. But it is a useful reminder. A belief becomes knowledge only when it is stated precisely, measured carefully and checked against something that could contradict it.

Every coach holds beliefs about training. Many of them are valuable, built from years of watching athletes succeed and fail. But a belief only becomes useful knowledge when it can be tested. This chapter explains how a coaching belief becomes a research hypothesis, what every AKURA intervention must define, and why accepting an unwelcome result is the most important discipline in the whole system.

Turning a belief into a hypothesis

A coach's belief becomes a research hypothesis when it is stated in measurable terms. "Heat makes you stronger" is vague. It does not say what "stronger" means, how much heat, for how long, or how the change would be detected. "A 10–14 day controlled heat-

acclimation block increases plasma volume and lowers heart rate at a fixed submaximal workload" is testable. It names the dose, the expected adaptations and the measurements that could confirm or refute them. It is also consistent with the heat-acclimation literature, which, as reviewed by Périard, Racinais and Sawka in 2015, describes lower exercise heart rate and plasma-volume expansion among the adaptations to repeated heat exposure, while noting that the size of response varies.

Vague belief	Testable hypothesis
Heat makes you stronger	A 10–14 day heat block lowers HR at a fixed pace by a measurable amount
Hill camps build lungs	Two weeks sleeping in Ooty changes Hb mass, measured before and after
Long runs build endurance	Eight weeks of progressive long runs reduce late-run pace decline
Strength work makes you faster	Twice-weekly strength training improves economy at marathon pace

KEY IDEA

A belief becomes useful when it is written as a testable hypothesis: a defined intervention, a named measurement and an expected direction of change.

What every AKURA intervention must define

Every AKURA intervention should define six elements before it begins:

1. Baseline: the athlete's current performance and relevant physiology, measured under recorded conditions.
2. Exposure dose: what the intervention is, how long it lasts, and how intense it is, such as temperature, humidity, altitude and hours of exposure.
3. Performance test: a standardised test that will be repeated, such as a fixed-pace run or a time trial.
4. Physiological markers: the variables expected to change, such as heart rate at a fixed pace, body-mass loss or haemoglobin mass.
5. Timing of the post-test: when the repeat test will occur, since some adaptations appear quickly and others take weeks.
6. Possible confounders: other factors that could explain any change.

Confounders are where many coaching conclusions quietly fail. If the athlete sleeps less, loses weight, changes caffeine intake or suddenly adds mileage during the intervention, the interpretation becomes weaker. A runner who improves after a hill camp may have improved because of the altitude, or because they slept ten hours a night away from college exams, or because the camp included more total running. Recording these factors does not remove them, but it keeps the interpretation honest.

Measurement error must also be respected. Every test has some natural variation from day to day. Hopkins's widely used work on reliability in sports science emphasised that a change that is not clearly larger than a test's typical error should not be interpreted as a real adaptation. A two-beat drop in heart rate on a single humid morning may be noise.

FIELD PROTOCOL

1. Standard fixed-pace heart-rate test
2. Choose a flat, measured route or track and a pace the athlete finds comfortable
3. Test at the same time of day, after a similar previous day and similar breakfast
4. Warm up for 10 minutes, then run 20 minutes at the fixed pace
5. Record average HR for the final 10 minutes, plus temperature, humidity and body mass before and after
6. Repeat on two separate days at baseline to see normal variation before comparing with post-intervention results

Accepting results that disagree with you

The most important discipline is to accept a result that disagrees with the coach's expectation. It is natural to want the heat block to work, the hill camp to pay off or the new plan to succeed. But if the heart rate at a fixed pace does not change, or haemoglobin mass stays the same, that is information. It may mean the dose was too small, the measurement too noisy, or the athlete not a strong responder to that particular stimulus. Each of those is worth knowing.

Data that proves a hypothesis wrong is still valuable data. It saves the athlete from repeating an intervention that does not help them, and it pushes the coach towards a better question. Over time, a record of honest results, positive and negative, becomes the athlete's most valuable coaching asset.

MYTH VS EVIDENCE

Myth: If the test did not show improvement, the test was wrong.

Evidence: Sometimes a test is poorly controlled, which is why protocols matter. But often a null result is real. Accepting it, and asking why, is how coaching knowledge improves.

COACH'S CORNER

Keep a simple intervention log for each athlete: the hypothesis, the dose, the before-and-after results and your honest interpretation. Review it at the end of each season with the athlete. You will quickly see which interventions consistently help that individual and which were based on hope.

CHECKPOINT

Choose one belief you hold about training. Rewrite it as a testable hypothesis with a dose, a measurement and possible confounders. What result would convince you that you were wrong?

PART I AT A GLANCE

The key idea from each chapter, for quick revision.

- 1 Why This Book Exists**
 A single number rarely explains an endurance performance. AKURA's purpose is to organise many measurements around clear questions, so that each number is interpreted in context rather than treated as a verdict.

- 2 The East African Marathon Question**
 East African distance-running success is best understood as the interaction of many physiological, environmental, training and social factors. No single cause, including birthplace, altitude or genetics, is sufficient to explain it.

- 3 Birthplace Versus Biology**
 Birthplace shapes the developmental environment, but it is not a biological destiny. Most of the factors that build endurance, such as training history, active childhood, nutrition and coaching, can be developed in many places.

- 4 Genetics: A Contribution, Not a Shortcut**
 Genetics contributes to endurance traits and to how strongly individuals respond to training, but it cannot be read from nationality or ethnicity. An athlete's ceiling must be observed through testing, not assumed from background.

- 5 Can Indian Athletes Build World-Class Endurance?**
 India's range of climates and elevations is a genuine research asset. But these environments are not interchangeable, and each produces a different stimulus that must be defined and measured.

- 6 From Belief to Measurement**
 A belief becomes useful when it is written as a testable hypothesis: a defined intervention, a named measurement and an expected direction of change.

PART II — THE HUMAN ENDURANCE ENGINE

Before we can measure an endurance athlete, we have to understand what we are measuring. Part II opens the engine. It follows oxygen from the air of a Chennai beach road or an Ooty hillside into the lungs, across into blood, around the body on haemoglobin, through the heart and into the working muscle, where mitochondria turn fuel into movement. Each chapter takes one link of that chain, explains its physiology in plain language, separates evidence from myth, and shows how it becomes a variable inside the AKURA Endurance Index. The lesson throughout is the same: no single number is the athlete.

IN THIS PART

- 7 The Endurance System as a Chain
- 8 The Heart: Pump, Not Just Heart Rate
- 9 The Lungs and the Oxygen Myth
- 10 Haemoglobin Mass
- 11 Plasma Volume
- 12 Red Blood Cells and Erythropoiesis
- 13 VO_2max
- 14 Running Economy
- 15 Mitochondria and Muscle Oxygen Use
- 16 Lactate Is a Messenger
- 17 Lactate Threshold
- 18 Fuel, Glycogen and Fat Oxidation

PART II • THE ENDURANCE ENGINE

Following Oxygen from Air to Motion

Endurance is a chain of linked systems — lungs, blood, heart, muscle and mechanics — and no single number is the athlete.

6 km Avinash Sable's daily run to school from Mandwa village

♥ HR × SV

The heart is a pump

Cardiac output = heart rate × stroke volume; oxygen uptake = cardiac output × oxygen extraction.

🌡️ 20.9%

Pressure, not percentage

At altitude air is still 20.9% oxygen, but lower pressure reduces the driving force for gas exchange.

💧 Hb mass

Mass beats concentration

Hb concentration shifts with plasma volume; total Hb mass better marks oxygen-carrying capacity.

📊 70 = 70

Same VO₂max, different races

Two runners at VO₂max 70: threshold at 85% and lower oxygen cost likely races faster.

🏃 Economy

Oxygen cost of speed

A more economical runner has more reserve at race pace, even with the same VO₂max.

💧 Lactate

A messenger, not waste

Blood lactate reflects production versus clearance. Store every threshold value with its protocol.

THE PROCESS



Strengthening one link helps only if it is the link that limits the athlete.

THE STORY

The Boy from Mandwa

PART II — THE HUMAN ENDURANCE ENGINE

Mandwa is a village near Ashti in Beed district, in the Marathwada region of Maharashtra, a part of the Deccan where the land is dry for much of the year and the monsoon is watched anxiously. Avinash Sable was born there on 13 September 1994, the son of farmers. The detail that every profile of him repeats is the road to school. It was 6 kilometres away, and he ran.

It is tempting to stop the story there, at the image of a boy on a dusty road, and declare that the road explains everything. It does not. The boy from Mandwa did not become a runner on that road. He became a soldier first.

The soldier

After Class 12, Sable joined the Army and was posted to 5 Mahar Regiment. His postings took him to the Siachen region, to Rajasthan and to Sikkim. In one career he met some of the harshest environments on the subcontinent: the cold of Siachen and the desert heat of Rajasthan, where temperatures can reach 50 °C. None of it was a training plan. It was service.

Running entered his life through Army cross country in 2015. Two years later, an Army coach named Amrish Kumar, himself a former South Asian Games steeplechase champion, spotted him and moved him into the 3000 m steeplechase, the event that asks a distance runner to clear 28 barriers and 7 water jumps while racing at close to full aerobic capacity.

The record that fell ten times

The Indian steeplechase record then belonged to Gopal Saini: 8:30.88, set at the Asian Championships in Tokyo in 1981. It had stood for 37 years. In September 2018, at the Open National Championships in Bhubaneswar, Sable broke it with 8:29.80.

What followed was one of the most sustained assaults on a national record in Indian athletics. In March 2019 he lowered it to 8:28.94 at the Federation Cup in Patiala. At the 2019 World Championships in Doha he ran 8:25.23 in the heats and then, after an appeal by the Athletics Federation of India got him into the final, 8:21.37 there, finishing 13th. At the Tokyo Olympics in 2021 he ran 8:18.12 in his heat, a national record that still was not enough for a place in the final. In 2022 came 8:16.21 at the Indian Grand Prix, 8:12.48 at the Rabat Diamond League and then, at the Commonwealth Games in Birmingham, 8:11.20 and a silver medal. He finished just 0.05 seconds behind Kenya's Abraham Kibiwot. It was India's first Commonwealth medal in the event and the first by a non-Kenyan since 1994.

The tenth record came on 7 July 2024 at the Paris Diamond League: 8:09.91.

Along the way he did things off the barriers too. In November 2020 he ran 1:00:30 at the Delhi Half Marathon, the first Indian under 61 minutes, and that remains the official national record. In 2022 he ran 13:25.65 for 5000 m, breaking a record Bahadur Prasad had set in Birmingham in 1992, and in 2023 he improved it to 13:19.30. At the 2023 Asian Games in Hangzhou he won steeplechase gold in a Games record of 8:19.50 and added silver in the 5000 m.

Coaches, and a change of air

The coaching behind those numbers was not a straight line. At the national camp, Sable worked with the Belarusian coach Nikolai Snesev, who had guided a generation of Indian distance runners. In 2018 Sable left him and returned to Amrith Kumar, and it was under Amrith that he ran 8:28.94 in 2019. Snesev left the Athletics Federation of India in February 2019 in what were reported as acrimonious circumstances, with Sable's return to his Army coach reported as one factor. Snesev was later reappointed and died in March 2021.

From 2022, Sable trained with the American coach Scott Simmons in Colorado Springs, at around 1,800 metres, as part of a camp that the federation and the government's sports schemes funded. Before the 2023 World Championships in Budapest he trained at altitude in Colorado Springs and in St Moritz in Switzerland, alongside Paul Chelimo. In July 2024, before the Paris Olympics, he was in St Moritz again.

His adaptation upon his return to altitude this year was almost instant.

— Scott Simmons, to Hindustan Times

That sentence, casual as it sounds, is a physiological claim. What exactly adapted? His breathing? His blood volume? His red-cell mass? How was "instant" measured, and against what baseline? Hold on to those questions. They are the questions of this whole Part.

In Paris in August 2024 Sable became the first Indian to reach the Olympic steeplechase final, finishing 11th in 8:14.18. In 2025 at Gumi, South Korea, he won the Asian Championships title in 8:20.92, India's first men's steeplechase gold at the championships in 36 years.

The knee

Then the engine met its limit, not in the heart or lungs but in a joint. At the Monaco Diamond League in July 2025, Sable injured the anterior cruciate ligament and meniscus of his right knee and needed surgery. He missed the World Championships in Tokyo that September, and in 2026 he missed the Commonwealth Games in Glasgow. He did much of his rehabilitation and rebuilding at the SAI Southern Centre in Bengaluru.

He has been candid about the comeback. "I think I might have rushed my return a bit, which caused the injury to recur," he told Olympics.com. His first steeplechase back, at the Indian Athletics Series final in September 2026, was 8:32.39, slower than the record he had first broken eight years earlier. Weeks later, at the Asian Games in Aichi-Nagoya, he ran 8:21.67, his best of the season, and finished fourth. He was entered in the 5000 m too, but did not start.

A window into the engine

Why begin a Part about physiology with one man's career? Because a career like this is an experiment run in public, even if most of the measurements were never published.

Look at what the record shows. An athlete who discovered serious running only in his twenties, after joining the Army, and who then lowered a national record by nearly twenty-one seconds in six years. An athlete whose results moved when his coaching, his training base and his exposure to altitude changed, and whose coach believed his body responded to that altitude almost immediately. An athlete who ran fast from 5000 m to the half marathon, which says something about both his aerobic capacity and how economically he could run. And an athlete whose progress was stopped not by oxygen but by tissue, a reminder that the endurance engine has a chassis as well as a fuel line.

Each of those observations points to a link in a chain. The heart that pumps the blood. The blood that carries oxygen, and the plasma it floats in. The red cells that altitude may or may not multiply. The ceiling of oxygen uptake, and the economy that decides how much of it each stride costs. The mitochondria that burn the fuel, the lactate that moves between fibres, the threshold that marks how hard a runner can go for how long, and the glycogen that decides how long the fast running can last.

Nobody can yet explain Avinash Sable's career with a single number, and nobody should try. But every coach can learn to ask better questions of the next boy who runs 6 kilometres to school, or the next soldier who turns up at a cross-country race. To ask those questions well, we need to open the engine and look at it one link at a time, starting with the chain itself.

Sources: <https://www.olympics.com/en/news/who-is-avinash-sable-india-athlete-steeplechase> ; <https://worldathletics.org/athletes/india/avinash-mukund-sable-14778911> ; <https://www.indiatoday.in/sports/other-sports/story/avinash-sable-shatters-national-record-for-10th-time-ahead-of-paris-olympics-2563569-2024-07-07> ; https://africa.espn.com/commonwealth-games/story/_/id/34355519/cwg-2022-india-athletics-avinash-sable-steeplechase-silver-just-005-seconds-short-gold-kenyan ; <https://www.hindustantimes.com/sports/others/steeplechaser-avinash-sable-gets-set-for-a-third-world-championships-101692209605101.html> ; <https://www.theweek.in/news/sports/2022/06/10/sub-8-minute-timings-not-impossible-record-breaking-steeplechaser-avinash-sable.html> ; <https://www.olympics.com/en/news/paris-2024-olympics-athletics-india-avinash-sable-3000m-steeplechase-result> ; <https://sportskeeda.com/summer-olympics/news-paris-olympics-2024-3000m-steeplechasers-avinash-sable-parul-chaudhary-train-hard-switzerland> ; <https://www.olympics.com/en/news/avinash-sable-injury-comeback-asian-games-2026-steeplechase> ; <https://m.rediff.com/sports/report/avinash-sable-triumphant-return-to-steeplechase-after-injury/20260910.htm> ; <https://www.thestatesman.com/sports/gulveer-singh-adds-1500m-silver-to-asian-games-haul-sable-finishes-fourth-1503644118.html> ; <https://www.olympics.com/en/news/nikolai-snesarev-the-late-belarusian-coach-s-contribution-to-indian-athletics> ; <https://www.deccanherald.com/sports/snesarevs-return>

leaves-two-indian-2176969 ; <https://thebridge.in/athletics/avinash-sable-parul-chaudhary-to-train-in-colorado-springs-45778>

CHAPTER 7

7. The Endurance System as a Chain

PART II — THE HUMAN ENDURANCE ENGINE

In the last week of September 2026, at the Asian Games in Aichi-Nagoya, Gulveer Singh did something that looks, on paper, like the work of three different athletes. He won silver in the 10,000 m in 28:29.38. He won silver in the 1500 m in 3:36.72. And on 29 September he took bronze in the 5000 m. Three medals, from a race that lasts well under four minutes to one that lasts nearly half an hour, won by a soldier from a farming family in Aligarh district who has said he "truly learned about competitive running only after joining the army." A 1500 m leans heavily on speed and anaerobic power. A 10,000 m depends on the steady, relentless delivery and use of oxygen. Yet it was one heart, one bloodstream and one set of muscles doing all of it. To understand how, we have to stop thinking of fitness as a single quality and start seeing it as a chain.

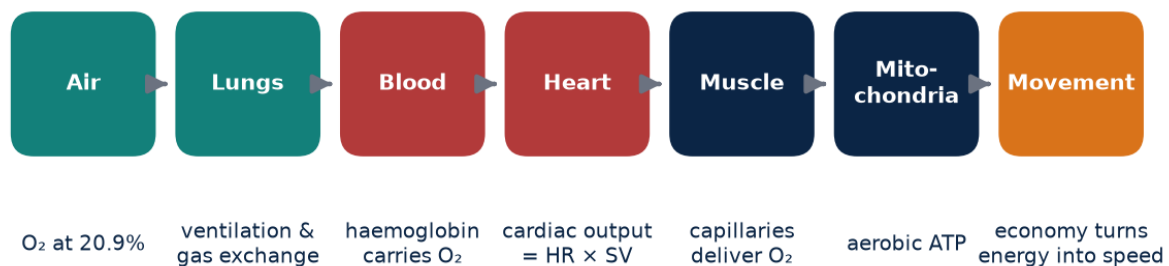


Figure 3. The endurance chain. A limitation at any link can affect performance, but improving one link does not guarantee a faster race.

Distance running is, at its core, an oxygen-and-energy delivery problem expressed through movement. A marathon runner leaving the Marina at dawn is solving that problem thousands of times a minute. Air must reach the lungs; oxygen must cross into the blood; haemoglobin must carry it; the heart must circulate that blood; the working muscle must receive the oxygen; mitochondria must use it, together with carbohydrate and fat, to generate ATP; and finally the athlete must convert that chemical energy into forward motion with as little waste as possible. Every one of those steps is a link.

Performance is what happens when all the links work together under the stress of pace, distance, heat and fatigue.

Following the oxygen

It helps to picture the chain in order, because each link has its own physiology, its own measurement and its own way of failing.

Link	What it does	Typical measure	Common confounder
Ventilation and lungs	Moves air and exchanges gas	Ventilation, SpO ₂ , spirometry	Altitude, airway conditions
Blood	Carries oxygen on haemoglobin	Hb mass, [Hb], ferritin	Plasma volume, hydration
Heart	Pumps blood to tissues	HR, stroke volume, cardiac output	Heat, dehydration, fatigue
Muscle and mitochondria	Extracts and uses oxygen	Lactate, muscle oxygenation, biopsy	Glycogen, training state
Mechanics	Turns energy into speed	Running economy, cadence, GCT	Footwear, surface, fatigue

The table is a simplification. In the living athlete these links interact continuously: a change in plasma volume alters the heart's filling, which alters stroke volume, which alters oxygen delivery, which alters how hard the muscle must work at a given pace. That interdependence is exactly why the chain metaphor is useful. It reminds us that the body does not have a single "fitness" dial.

Weak links and false promises

A limitation at any link can influence performance. An athlete with low iron stores may struggle to build haemoglobin; an athlete with poor mechanics may waste oxygen on vertical bounce; an athlete who has not practised race fuelling may run out of carbohydrate at 32 km regardless of how strong the heart is. In each case the weak link sets the ceiling on the day.

The opposite is also true, and it is where many coaching myths begin. Improving one link does not guarantee an improvement in race time. Plasma expansion after a block of heat training may increase stroke volume, yet the athlete still needs economical mechanics and sustainable metabolic control to turn that extra blood flow into faster kilometres. A higher VO₂max measured in a laboratory is encouraging, but if the athlete's economy has worsened because of a new shoe or an injury compensation, the race result may not move. The chain is only as useful as its weakest working link at the moment of competition.

KEY IDEA

Endurance performance is a chain of linked systems, from air to ATP to forward motion. Strengthening one link helps only if it is, or becomes, the link that limits the athlete.

MYTH VS EVIDENCE

Myth: One magic variable, such as VO_2max or haemoglobin, explains why some runners are faster.

Evidence: Performance emerges from several interacting systems; similar values in one link can coexist with very different race results because other links differ.

Why AEI is a profile, not a single dial

This is the reason the AKURA Endurance Index avoids a single-variable model. AEI is designed as a multidimensional profile with a performance anchor. The profile records the state of several links, such as cardiovascular response, blood markers where available, economy, threshold and fuelling behaviour. The anchor is real performance: a time trial, a race or a standardised field test. The profile helps explain the anchor; the anchor keeps the profile honest. If the physiology improves but performance does not, AEI should flag the mismatch rather than hide it.

For Indian coaches this matters practically. Many squads will have access to heart-rate monitors and GPS watches but not to carbon-monoxide rebreathing or laboratory gas analysis. A chain model lets the coach be explicit about which links are measured, which are estimated and which are unknown, instead of pretending that a watch's single "fitness score" captures the whole athlete.

The chain also has a time dimension. Some links adapt within days, such as plasma volume at the start of a training block or after the first week of heat exposure. Others, such as mitochondrial content and capillary density, build over weeks and months. Red-cell mass responds over weeks and depends on iron and the stimulus. Mechanics and economy may take years of consistent running and strength work. A coach who expects every link to respond on the same schedule will misread the data. AEI therefore records not only what was measured, but when, so that fast and slow adaptations can be separated and a short-term change is not mistaken for a long-term one.

COACH'S CORNER

When an athlete stalls, walk the chain before changing the plan. Ask which link is most likely limiting now: sleep and recovery, iron and blood, heat load, mechanics, threshold or fuelling. Change one major variable at a time so you can see what actually moved.

CHECKPOINT

For one athlete you coach, list the five links in the table above. Which ones do you currently measure, which do you estimate from feel, and which are completely unknown?

CHAPTER 8**8. The Heart: Pump, Not Just Heart Rate**

PART II — THE HUMAN ENDURANCE ENGINE

Two friends leave the Besant Nagar beach road together at six on a May morning, matched stride for stride at 5:30 per kilometre. The sea breeze has not arrived yet. The air is thick, the kind of Chennai morning when a cotton vest is wet before the first kilometre is done. Forty minutes later they compare watches. One shows an average of 148 beats per minute. The other shows 163. Same pace, same road, same sun, and a fifteen-beat difference. The second runner is worried: is his heart weaker? The first is quietly pleased: is he fitter? Neither conclusion follows. The number on the wrist counts how often the heart beats. On its own, it says nothing about how much blood leaves the heart with each beat, how much of that blood is being sent to the skin to shed heat, or how much oxygen the leg muscles are pulling out of it. For that, we need to see the heart as what it really is: a pump.

Same pace, similar cardiac output (~17.6 L/min), different heart rates

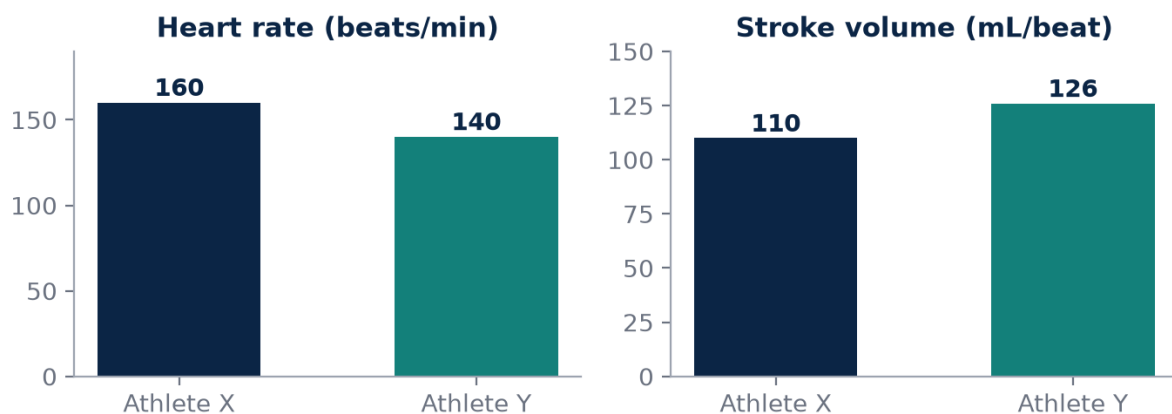


Figure 4. Illustrative example: cardiac output = heart rate × stroke volume. A larger stroke volume lets Athlete Y deliver similar blood flow at a lower heart rate.

Heart rate is the most familiar number in endurance sport. It sits on every wrist, it is cheap to record and it responds quickly to effort, heat and stress. It is a genuinely useful

field signal. But it is not the heart's entire story. The heart is a pump, and what matters for oxygen delivery is how much blood it moves each minute, not simply how often it beats.

Cardiac output and the Fick principle

Cardiac output is the volume of blood the heart pumps per minute. It equals heart rate multiplied by stroke volume, the volume ejected with each beat. The oxygen the body consumes is then described by the Fick principle: oxygen uptake equals cardiac output multiplied by the difference in oxygen content between arterial and mixed venous blood. In words, the body's oxygen use depends on how much blood is delivered and how much oxygen the tissues take out of each litre.

Quantity	Formula or meaning	Illustrative range (varies widely)
Cardiac output (CO)	$HR \times \text{stroke volume}$	About 5 L/min at rest; can exceed 30 L/min in elite endurance athletes at maximum
Stroke volume (SV)	Blood ejected per beat	Often 60–100 mL at rest; larger in trained hearts
a-vO ₂ difference	Arterial minus venous O ₂ content	Rises from about 50 mL/L at rest to well over 100 mL/L in hard exercise
VO ₂ (Fick)	$CO \times \text{a-vO}_2 \text{ difference}$	Whole-body oxygen uptake

These values are illustrative only; they differ with body size, sex, training status, posture and measurement method. The principle, however, is robust. Two athletes can share the same heart rate at the same pace while delivering very different amounts of oxygen, because one has a larger stroke volume or extracts more oxygen at the muscle.

KEY IDEA

Cardiac output = heart rate $\times$ stroke volume, and oxygen uptake = cardiac output $\times$ oxygen extraction. Heart rate is only one of the terms.

Why the same pace feels different

This explains a common coaching observation. During endurance training an athlete may improve performance while running at a similar heart rate, because more blood is pumped per beat or because the muscles extract oxygen more effectively. Conversely, the same pace can feel very different on a cool Bengaluru morning and a humid Chennai evening.

Heat can reduce effective stroke volume in two ways. First, blood is shifted towards the skin to release heat, so a larger share of cardiac output is diverted away from working muscle and less blood returns to fill the heart. Second, sweating reduces plasma volume, which again lowers filling. To maintain cardiac output with a smaller stroke volume, heart rate rises. Physiologists describe the gradual upward drift of heart rate

during prolonged exercise as cardiovascular drift, and heat and dehydration make it larger. Endurance training and heat acclimation can improve plasma volume and cardiac function, helping to maintain circulation under the same conditions.

MYTH VS EVIDENCE

Myth: A lower heart rate at the same pace always means the athlete is fitter.

Evidence: It often does, but it can also reflect cooler weather, better hydration, a different course or even a sensor error. Context is required before interpreting the change.

Recording heart rate properly in AEI

Because heart rate responds to so many inputs, AEI should store HR together with pace, temperature, humidity when available, body mass change and perceived effort, rather than treating HR as a standalone score. A 150 bpm run at 5:00 per km in 24 °C is not the same physiological event as a 150 bpm run at the same pace in 32 °C with high humidity. Stored with context, the two sessions become comparable; stored alone, they mislead.

Wrist-based optical sensors are convenient but can be inaccurate during intervals, in heavy sweat or with loose straps. Chest straps are usually more reliable for training analysis. Where exact stroke volume or cardiac output measurement is needed, that is laboratory or clinical work and should be done by qualified staff. Any unusual symptom such as chest pain, fainting or palpitations during exercise is a medical matter for a doctor, not a coaching variable.

FIELD PROTOCOL

1. Standardised submaximal HR check
2. choose a flat, familiar 2–3 km loop
3. warm up 10 minutes easy
4. run 3 × 1 km at a fixed pace (for example marathon pace) with 1 minute standing rest
5. record average HR of the last 30 seconds of each rep, air temperature, humidity, time of day and RPE
6. weigh the athlete before and after in minimal clothing
7. repeat every 2–4 weeks under similar conditions.

COACH'S CORNER

Do not change training zones on the basis of a single hot-day heart-rate reading. Look for trends in HR at fixed pace across several comparable sessions, and always note the weather and the athlete's sleep and hydration beside the number.

CHECKPOINT

On your athlete's last three easy runs, was the heart rate different because the athlete changed, or because the conditions changed? What data would let you tell the difference?

CHAPTER 9**9. The Lungs and the Oxygen Myth**

PART II — THE HUMAN ENDURANCE ENGINE

In Maroda, a village in Pauri Garhwal about 1,400 metres up in the Uttarakhand hills, a farmer's daughter used to run the hill roads with young men training for the Army's physical tests. "I decided to run with them ... They never slowed down for me," Ankita Dhyani later told ESPN. She went on to the national centre in Bhopal, to the Paris Olympics and, in August 2025, to a national record in the 2000 m steeplechase. Her coach put part of her strength down to geography: "because she was running in Uttarakhand, she had already been doing a lot of high-altitude training." It is an interesting sentence, mostly for what people tend to hear in it. Mountain villages, the popular story goes, have thinner air, and people who train in thin air grow bigger lungs. Almost every part of that belief needs checking. The air in Maroda, in Ooty and on Marina Beach contains the same fraction of oxygen. What differs is something else entirely.

Few ideas in endurance sport are repeated as confidently, or as inaccurately, as "there is less oxygen at altitude" and "hill running builds bigger lungs". Both contain a grain of truth wrapped in a misunderstanding. Getting the physics and physiology right matters, because a coach who misunderstands the stimulus will misjudge the dose.

Percentage versus pressure

The percentage of oxygen in normal atmospheric air is approximately 20.9% at sea level and remains approximately the same at altitude. The air in Ooty or Kodaikanal contains the same proportion of oxygen as the air in Chennai. What changes substantially with altitude is barometric pressure. Because the total pressure is lower, the partial pressure of oxygen falls, and it is partial pressure that drives oxygen from the lungs into the blood. A lower driving pressure reduces how readily oxygen loads onto haemoglobin, particularly during hard exercise when blood moves quickly through the lung capillaries.

At moderate altitudes of around 2,000–2,500 m, inspired oxygen pressure is roughly a quarter lower than at sea level. The body responds within minutes by breathing more, and over days to weeks with further adaptations. Performance in longer events is usually reduced at altitude, particularly on arrival.

MYTH VS EVIDENCE

Myth: There is less oxygen in the air at altitude.

Evidence: The fraction of oxygen stays about 20.9%. The barometric pressure falls, so the partial pressure of oxygen, and therefore the pressure pushing oxygen into the blood, is lower.

KEY IDEA

Altitude is a pressure stimulus, not a percentage stimulus. The same 20.9% oxygen at lower pressure means a lower partial pressure of oxygen and less driving force for gas exchange.

What hill running really trains

Hill running can make breathing harder because ventilation rises with workload, and it rises further at altitude. Repeated training can strengthen the respiratory muscles and improve tolerance to high ventilation. Hills also load the calves, quadriceps and hip extensors, and they can be a valuable tool for strength and economy. But ordinary hill running should not be described as automatically increasing lung size. In healthy adults, lung dimensions are largely set by body size and development, and training does not reliably enlarge them. What training changes more readily is how the athlete uses the lungs: breathing pattern, respiratory muscle endurance and the sensation of breathlessness at a given work rate.

In healthy, well-trained sea-level runners the lungs are often not the primary limiter of oxygen transport, although some highly trained athletes do show a fall in arterial oxygen saturation during very intense exercise. That phenomenon is individual and should be measured, not assumed.

Four things that are often confused

The useful distinction is between lung structure, ventilation, gas exchange and systemic oxygen transport. These are related, but they are not the same measurement.

Concept	Meaning	How it may be assessed
Lung structure	Size and mechanics of the lungs	Spirometry, lung volumes (clinical)
Ventilation	Volume of air moved per minute	Gas analysis during exercise
Gas exchange	Oxygen moving from alveoli into blood	Arterial saturation (SpO ₂ as a rough field proxy)
Systemic O ₂ transport	Oxygen delivered to tissues	Hb mass, cardiac output, VO ₂

A pulse oximeter reading at rest in Kodaikanal tells you something about gas exchange under those conditions. It does not tell you lung size, and it does not tell you how much oxygen is being delivered to the calf muscle at race pace. Keeping these categories separate prevents a common error: using one measurement to make claims about another.

For Indian athletes, air quality in some cities during certain seasons is also a relevant respiratory stressor. Coaches may reasonably move sessions to cleaner times or locations, but any persistent breathing symptom, wheeze or exercise-induced cough should be assessed by a clinician rather than trained through.

The body's early response to altitude is itself informative. Resting ventilation rises, resting heart rate is often higher for the first days, sleep can be disturbed and plasma volume may fall, which temporarily raises haemoglobin concentration without any new red cells being made. These early changes are responses to reduced oxygen pressure, not evidence that the lungs have grown or that the blood has already adapted. Distinguishing the first-week response from genuine longer-term adaptation is one of the reasons AKURA insists on repeated, standardised measurement at altitude rather than a single reading taken on the second morning of a camp.

COACH'S CORNER

Use hills for strength, economy and controlled intensity, and describe them honestly. If you take a squad to Ooty or Kodaikanal, expect paces to feel harder at first, reduce intensity for the first days, and record SpO₂, resting HR, sleep and RPE rather than assuming lungs are "growing".

CHECKPOINT

Can you explain to an athlete, in two sentences, why the air at altitude has the same percentage of oxygen but still makes running harder?

CHAPTER 10

10. Haemoglobin Mass

PART II — THE HUMAN ENDURANCE ENGINE

Sawan Barwal's Project 2:09 group normally spends about four months, from November to February, in the Nilgiris around Ooty and Wellington, then comes down to Bengaluru for final preparation. "For Sawan, most of the time he was staying in high altitude," his coach Ajith Markose told *The Bridge*. Four months of sleeping in thinner air is a large dose, far larger than the weekend hill trips many club runners take. A reader might expect, somewhere, a neat pair of numbers: how many grams of haemoglobin he carried in October, and how many in February. If such measurements were made, they are not in the public record. More broadly, the research behind this book could find no peer-reviewed study of Indian athletes' blood adaptations at Ooty at all. That is a remarkable gap, because the most important blood number for an endurance runner is not the one printed on an ordinary laboratory report.

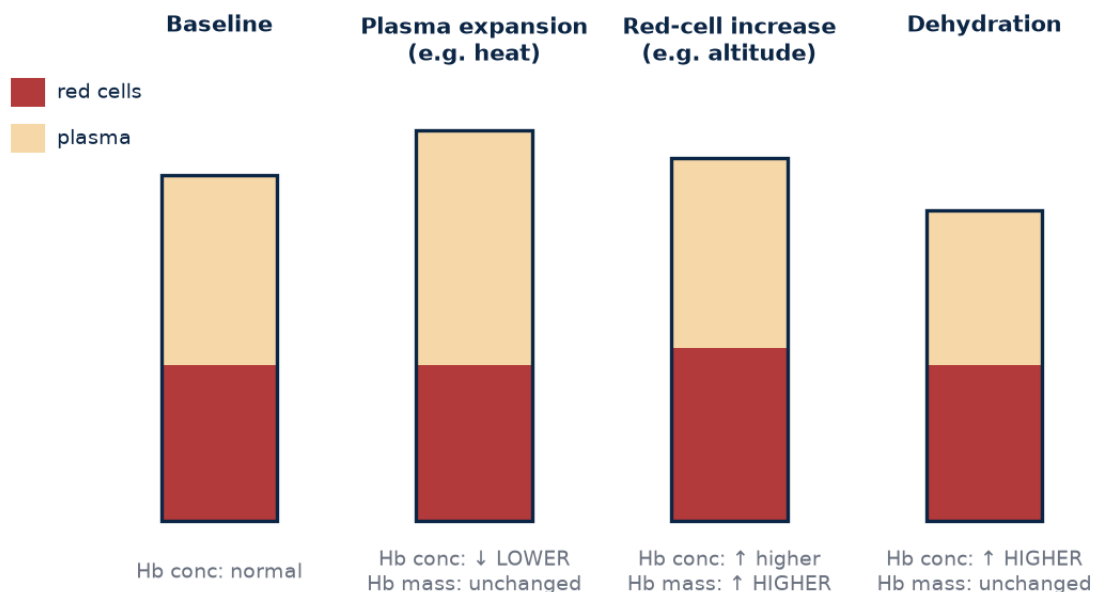


Figure 5. Schematic only: why haemoglobin concentration and haemoglobin mass can move in different directions. Concentration depends on plasma volume as well as red-cell mass.

Haemoglobin is the iron-containing protein inside red blood cells that binds oxygen. Each gram of fully saturated haemoglobin can carry roughly 1.34 mL of oxygen, so the total amount of haemoglobin in the body sets a large part of the blood's oxygen-carrying capacity. When the research question is oxygen-carrying capacity, haemoglobin mass,

the total grams of haemoglobin circulating in the body, is more informative than haemoglobin concentration alone.

Concentration is a ratio, mass is an amount

A routine blood test reports haemoglobin concentration, usually in grams per decilitre. That number is a ratio: haemoglobin divided by the volume of blood it is dissolved in. Concentration can rise because red-cell mass increases, because plasma volume falls, or both. It can fall because red-cell mass decreases, because plasma volume expands, or both. A single concentration value cannot tell which of these happened.

Scenario	Hb mass	Plasma volume	Hb concentration
Red-cell production after altitude	Up	Unchanged	Up
Dehydration after a hot long run	Unchanged	Down	Up (apparent)
Heat acclimation	Usually unchanged (short blocks)	Up	Down (apparent)
Altitude plus heat block	Up	Up	Could be unchanged
Iron deficiency, blood loss	Down	Variable	Usually down

The third and fourth rows are especially important for AKURA's heat-based research. An athlete returning from a Chennai summer training block may show a lower haemoglobin concentration, not because oxygen-carrying capacity has worsened but because plasma volume has expanded. Without a measure of mass or volume, a coach could misread a positive adaptation as a problem.

KEY IDEA

Haemoglobin concentration is a ratio that changes with plasma volume. Haemoglobin mass is the total amount, and is the better marker of oxygen-carrying capacity.

MYTH VS EVIDENCE

Myth: A higher haemoglobin value on a blood report always means better endurance capacity.

Evidence: A higher concentration can simply reflect dehydration or reduced plasma volume. Total haemoglobin mass, interpreted with performance, is the more meaningful variable.

How haemoglobin mass is measured

In research and elite sport, haemoglobin mass is commonly measured with the optimised carbon-monoxide rebreathing method, in which the athlete breathes a small, controlled dose of carbon monoxide and the change in carboxyhaemoglobin is used to calculate total haemoglobin. The method, developed and refined by Schmidt and Prommer and others, has good reliability in experienced hands. It must be performed by trained staff using established safety procedures; it is not a do-it-yourself field test.

Typical haemoglobin mass is higher in endurance-trained athletes than in untrained people when expressed per kilogram of body mass, and it differs between men and women and among individuals. Values should be interpreted relative to the athlete's own baseline rather than a single universal norm.

Altitude and the 1% rule

Altitude exposure can stimulate erythropoietin (EPO) and, over weeks, erythropoiesis, the production of new red blood cells. A meta-analysis by Gore and colleagues, published in the British Journal of Sports Medicine in 2013, pooled altitude studies that used carbon-monoxide rebreathing and estimated an average haemoglobin-mass increase of about 1.1% per 100 hours of exposure. The same analysis also showed substantial individual variability, and the authors estimated a wide prediction interval for any individual athlete's response.

Therefore "1% per 100 hours" should be treated as a research estimate from particular protocols, typically natural or simulated altitude of moderate elevation with sufficient daily hours and adequate iron, not as an athlete guarantee. Some athletes gain more, some gain little or nothing, and measurement error around a single test is not trivial. For an Indian squad considering a camp in Ooty or Kodaikanal, the practical translation is to calculate the expected hours of exposure, check iron status beforehand under medical supervision, and measure rather than assume the response.

FIELD PROTOCOL

1. Planning an altitude Hb-mass study
2. record baseline performance test and, if available, Hb mass by CO rebreathing
3. obtain clinician-supervised iron studies 4–6 weeks before
4. log daily hours at altitude, sleep altitude and training load
5. retest Hb mass within a few days of descent
6. repeat the performance test under standardised conditions
7. interpret change against the method's known measurement error.

It is worth adding a simple calculation to show why mass matters. If an athlete has 900 g of haemoglobin and each gram, fully saturated, binds about 1.34 mL of oxygen, the blood can carry roughly 1.2 litres of oxygen at any moment. A 3% increase in haemoglobin mass would add roughly 36 mL of carrying capacity, regardless of

whether the concentration on a lab report moved. These figures are illustrative, not targets, but they explain why researchers prefer to measure the total amount directly when they want to understand oxygen transport, and why concentration alone can hide or exaggerate genuine change.

COACH'S CORNER

Treat any Hb-mass gain as a potential that must still be converted into performance. Pair every blood measurement with a standard performance test, and resist announcing a camp as "successful" until both have been checked.

CHECKPOINT

If your athlete's haemoglobin concentration dropped after a heat block, what additional information would you need before deciding whether this was good, bad or neutral?

CHAPTER 11

11. Plasma Volume

PART II — THE HUMAN ENDURANCE ENGINE

It is May in Chennai, and the long run starts at 4:45 a.m. to beat the sun. It does not beat the humidity. By the tenth kilometre along the East Coast Road, the runners in a small club group have stopped talking. By the twentieth, their shirts are so heavy that one of them wrings his out beside a tea stall. Back at the car, the coach, who has started carrying a bathroom scale in the boot, weighs everyone. One runner is 1.8 kg lighter than when he started, despite drinking at every stop. Most of that loss is water that left his body as sweat, and some of it was drawn from the liquid part of his blood. Three weeks later, running the same route at the same pace, the same runner finishes with a lower heart rate and says the run felt easier. Nothing about his heart or lungs has visibly changed. Part of the explanation may be flowing through his veins.

Blood is not only red cells. Roughly half or more of it, depending on the individual, is plasma, the liquid portion of blood made mostly of water, together with proteins, electrolytes and other solutes. Plasma is easy to overlook because it does not carry much oxygen itself, yet it plays a central role in endurance performance and, for athletes training in India's heat, may be one of the most responsive parts of the whole system.

What plasma does for the endurance athlete

Expansion of plasma volume increases total blood volume. A larger blood volume can improve venous return, the flow of blood back to the heart, which helps fill the ventricles more completely before each beat. Better filling supports stroke volume, and a better-maintained stroke volume supports cardiovascular stability during long efforts. Plasma volume also contributes to thermoregulation: there is more fluid available both to send blood to the skin for cooling and to sustain sweating, without starving the working muscles or the heart.

Heat acclimation is particularly relevant because sweating and repeated thermal stress stimulate fluid-regulatory adaptations. Repeated exposure to exercise in the heat promotes retention of sodium and water and an increase in plasma proteins that help hold fluid within the circulation. Endurance training in cooler conditions also expands plasma volume, typically within days of starting a new training block, which is one reason heart rate at a fixed pace can fall quickly early in a training cycle.

KEY IDEA

Plasma volume supports venous return, stroke volume, cardiovascular stability and cooling. Heat acclimation and endurance training can both expand it, but by different amounts in different people.

Evidence, and its limits

Systematic reviews show that heat acclimation can change plasma volume and cardiovascular responses, but the size and speed of adaptation vary across protocols and individuals. Reviews such as that by Périard, Racinais and Sawka describe plasma expansion as one of the earlier heat adaptations, often apparent within the first week of a structured protocol, while also noting that some of it may be transient if heat exposure stops. Protocol details such as daily duration, core temperature achieved, hydration practice and training status strongly influence what is observed.

This matters for how AKURA frames its heat hypotheses. It is reasonable, on current evidence, to hypothesise that a controlled heat-acclimation block may expand plasma volume in many athletes. It is not reasonable to promise a specific percentage to any individual, or to assume that plasma expansion automatically produces a faster race in cool conditions. That is a separate research question that needs its own performance test.

MYTH VS EVIDENCE

Myth: A drop in haemoglobin concentration after heat training means the athlete has become anaemic.

Evidence: It may partly reflect plasma expansion diluting the red cells. Anaemia is a clinical diagnosis that requires proper medical assessment, including measures of red-cell mass or iron status where appropriate.

Measuring concentration and volume separately

This creates a central AKURA principle: measure concentration and volume separately whenever the research question requires it. A lower haemoglobin concentration after heat adaptation does not automatically mean worse oxygen-carrying capacity; it may partly reflect plasma expansion.

Approach	What it gives	Practical note
CO rebreathing with Hb and Hct	Hb mass, blood and plasma volume	Specialist equipment and trained staff
Change in Hb and Hct (Dill–Costill method)	Estimated percentage change in plasma volume	Needs standardised posture and timing
Body mass and urine colour or specific gravity	Hydration context only	Does not measure plasma volume
HR at fixed pace and conditions	Functional response	Indirect, many confounders

The Dill and Costill approach estimates the relative change in plasma volume from paired haemoglobin and haematocrit values. It is a practical research tool but is sensitive to posture, time of day and recent exercise, so blood samples must be taken under identical conditions. In a typical Indian club setting without laboratory access, the coach can still collect the functional evidence carefully: standardised heart rate at fixed pace, body mass, sweat observations and perceived effort, all logged with temperature and humidity.

There is also a practical interpretation problem in the days after a race or hard session. Prolonged exercise, especially in heat, can shift fluid out of the circulation and reduce plasma volume acutely, while recovery and training over subsequent days can expand it. Blood sampled on a Monday after a Sunday long run is not directly comparable with blood sampled after two easy days. For AKURA studies, the timing of the last hard session, fluid intake and even the posture in which the athlete waited before sampling should be recorded, because each of these can move plasma volume enough to change the apparent result.

COACH'S CORNER

If you run a heat block, schedule the post-test within a few days of finishing, because some plasma adaptations can fade without continued exposure. Keep hydration, caffeine and posture before blood tests the same at baseline and follow-up.

CHECKPOINT

For your next heat-training block, which marker will you use for concentration, which for volume, and which for performance?

CHAPTER 12

12. Red Blood Cells and Erythropoiesis

PART II — THE HUMAN ENDURANCE ENGINE

The blood report arrives on a phone in a sports hostel: a single page of numbers, a few of them flagged in red. The athlete is a seventeen-year-old middle-distance runner who has felt tired for weeks. Her haemoglobin is low. She is far from unusual. A venous-blood survey across eight Indian states found anaemia in 44% of adolescent girls and 41% of adult women. A 2025 study of 104 female athletes in Tamil Nadu, volleyball and ball badminton players rather than runners, found that about 53.8% were anaemic. For a distance runner, whose sport depends on carrying oxygen, figures like these should stop a coach cold. Yet the response in many squads is either to ignore the red flags or to hand out iron tablets without a doctor's advice. Neither approach respects how the body actually builds blood. Red cells are made slowly, from raw materials, in response to signals, and every one of those steps can fail.

Red blood cells are the delivery vehicles of the endurance system. They contain haemoglobin and transport oxygen from the lungs to the tissues, and help carry carbon dioxide back. They are produced in the bone marrow through erythropoiesis and, in healthy adults, circulate for roughly four months before being removed and recycled. That slow turnover is the first thing a coach must understand: changing the red-cell population takes weeks, not days.

How the body decides to make more

At altitude, reduced oxygen availability can increase erythropoietin signalling. Specialised cells in the kidney sense reduced oxygen delivery and increase production of the hormone erythropoietin, often within hours of arrival. EPO then stimulates the bone marrow to produce more red-cell precursors. Young red cells, called reticulocytes, appear in the blood over the following days, and a measurable increase in total red-cell and haemoglobin mass generally requires weeks of adequate exposure.

The response is not instantaneous and is affected by exposure dose, iron status, sleep, individual biology and the exact altitude protocol. Exposure dose includes both the altitude itself and the number of hours per day spent there, which is why the "live high, train low" model studied by Levine and Stray-Gundersen and others emphasises where the athlete sleeps as well as where the athlete trains. Illness, inflammation, inadequate energy intake and heavy training stress can all blunt the response.

KEY IDEA

Erythropoiesis is a slow, dose-dependent process. A rise in EPO is only the first step; building new red cells needs time, iron, energy, sleep and a sufficient hypoxic stimulus.

Iron: the raw material

Every new haemoglobin molecule needs iron. An athlete with insufficient iron stores may not respond as expected to an altitude intervention, and may simply become more fatigued. This is one reason ferritin and broader blood assessment may be relevant in a supervised sports-medicine programme. Endurance athletes, particularly female athletes, athletes with low energy availability and those following predominantly vegetarian diets, may be at greater risk of low iron status, although risk is individual and should be assessed rather than presumed.

Iron supplementation is not a harmless default. Excess iron can cause harm, and supplementation decisions, dose and duration belong with a qualified doctor or sports physician, based on proper blood tests. Ferritin in particular can rise with inflammation or recent hard exercise, so timing of blood draws and clinical interpretation matter.

MYTH VS EVIDENCE

Myth: Taking iron tablets before a camp will guarantee more red blood cells.

Evidence: Iron only helps if stores are actually insufficient, and the decision to supplement should be made by a clinician after testing. Iron is a necessary raw material, not a performance booster on its own.

Blood data is not a performance score

The coaching mistake is to treat a blood marker as a performance score. A higher red-cell count, haemoglobin or haematocrit is not the same as a faster 10 km. Blood markers fluctuate with hydration, posture, recent training and laboratory method. The correct approach is to connect blood data with performance and with longitudinal change in the same athlete.

Marker	What it may indicate	Interpret with caution because
Haemoglobin concentration	Oxygen-carrying ratio	Changes with plasma volume
Haematocrit	Proportion of blood that is cells	Changes with hydration and posture
Reticulocytes	Recent red-cell production	Short-lived, method-dependent
Ferritin	Iron stores (approximate)	Rises with inflammation
Hb mass	Total oxygen-carrying capacity	Needs specialist method

It is also worth stating plainly: artificially raising red-cell mass through EPO injection, blood transfusion or similar methods is prohibited in sport and carries serious health risks. AKURA's interest in erythropoiesis is entirely about understanding legitimate, natural adaptation to training and environment.

FIELD PROTOCOL

1. Supervised blood-monitoring routine
2. schedule tests at the same time of day after a rest or easy day
3. keep the athlete seated for a standardised period before sampling
4. record hydration, menstrual phase where relevant, recent illness and training
5. request ferritin, full blood count and any tests advised by the clinician
6. store results beside the nearest standardised performance test
7. review trends with the medical team, not in isolation.

Individual variation deserves emphasis. Two athletes on the same altitude camp, sleeping at the same elevation and following the same programme, may show very different red-cell responses. Some of this reflects measurement error, some reflects iron status, illness or training stress, and some appears to be genuine biological difference. This is why the phrase "responder" or "non-responder" should be used carefully. An athlete who does not respond on one camp may respond on another when iron, sleep and load are better managed. AEI records each exposure as an individual experiment rather than labelling the athlete permanently.

COACH'S CORNER

Build a relationship with a sports physician before planning altitude or heavy training blocks. Your job is to provide training context and performance data; the clinician's job is to interpret blood results and decide on any treatment.

CHECKPOINT

Does your athlete have a baseline ferritin and full blood count taken under standardised conditions, and do you know what the result was compared with their own previous values?

CHAPTER 13

13. VO₂max

PART II — THE HUMAN ENDURANCE ENGINE

Put two numbers side by side and they seem to tell a story on their own. In 1991, researchers testing Bengali female athletes on a cycle ergometer found that the long-distance runners among them had the highest relative VO₂max of any group: 43.0 mL/kg/min. In 1995, Bengt Saltin and colleagues reported that Kenyan senior runners averaged 79.9 mL/kg/min when tested at sea level. The gap looks enormous, and it is easy to draw a sweeping conclusion from it about who can and cannot run. That conclusion would be wrong in almost every way. The two studies tested different sexes, different generations and very different levels of training, and pedalling a bicycle is not the same task as carrying a body over the ground; many runners score lower on a cycle than they would running. Neither number says how anyone would perform on race day. VO₂max is a real and important measurement. It is also the most over-interpreted number in the sport.

VO₂max is probably the most famous number in endurance physiology. It describes the maximum rate at which the body can take up and use oxygen during severe exercise, usually expressed in millilitres of oxygen per kilogram of body mass per minute. It is important: elite endurance athletes almost always have high values, and a low VO₂max limits what an athlete can achieve at race pace. But it is not a complete description of running ability, and treating it as the master number leads to poor coaching decisions.

What VO₂max actually represents

Using the Fick principle from Chapter 8, VO₂max is the product of maximal cardiac output and maximal oxygen extraction. In trained athletes at sea level, much of the evidence suggests that oxygen delivery, especially maximal cardiac output and haemoglobin mass, is a major determinant of VO₂max, while muscle oxidative capacity also contributes. Reviews by Bassett and Howley, and by Joyner and Coyle in the *Journal of Physiology* in 2008, describe VO₂max as one of three key pillars of endurance performance, alongside the lactate or threshold characteristics and running economy.

VO₂max is measured in a laboratory with gas analysis during an incremental test to exhaustion. Values vary with protocol, equipment, motivation and whether the test is run on a treadmill or another ergometer. Values reported by watches are estimates derived from heart rate and pace; they can be useful for tracking trends in one person but should not be treated as laboratory measurements.

KEY IDEA

VO₂max sets the size of the aerobic ceiling. It does not tell you how close to that ceiling the athlete can race, or how much oxygen each kilometre costs.

Same engine, different results

Two athletes can have similar VO₂max values and different race performances because of running economy, threshold characteristics, body mass, fuel use, biomechanics and pacing. Consider a simplified illustration.

Variable	Athlete A	Athlete B
VO ₂ max (mL/kg/min)	70	70
Threshold as % of VO ₂ max	75%	85%
Oxygen cost at 16 km/h (mL/kg/min)	56	50
Likely marathon outcome	Slower	Faster

The numbers are hypothetical, chosen to show the logic. Athlete B can sustain a higher fraction of the same ceiling and needs less oxygen for the same speed, so Athlete B's sustainable race pace is higher despite an identical VO₂max. This is the same reasoning used in the Joyner and Coyle model, in which performance velocity reflects VO₂max, the fraction of it that can be sustained, and economy together.

Coaching software can add to the confusion. Jack Daniels' VDOT, for example, is a performance-based index derived from race times; its documentation explicitly distinguishes VDOT from laboratory VO₂max and notes that runners with similar VO₂max can produce different race results. VDOT is useful precisely because it folds economy and other factors into a single performance-derived value, but it should be labelled for what it is.

MYTH VS EVIDENCE

Myth: The athlete with the highest VO₂max will win the race.

Evidence: Among well-trained athletes of similar level, VO₂max alone predicts race results poorly. Economy, sustainable fraction of VO₂max, fuelling and pacing often separate them.

How AEI uses VO₂max

AEI therefore treats VO₂max as one physiological variable, not the master number. Where a valid laboratory test exists, AEI records the value together with the protocol, date, body mass and testing conditions. Where only a field or wearable estimate exists, AEI labels it as an estimate. In both cases, the value sits in the profile beside economy,

threshold and the performance anchor, so that a change in VO_2max can be checked against a change in real running.

VO_2max is trainable, especially in less-trained athletes, through a combination of consistent aerobic volume and intervals near maximal aerobic speed. It also has a substantial inherited component, and responses to the same training vary between individuals. For Indian athletes the practical message is not to fixate on a number that may be difficult to measure, but to build the aerobic system through progressive training and to measure it well when the opportunity exists.

There is also the question of how VO_2max is expressed. Relative values, per kilogram of body mass, are most relevant to running because runners carry their own weight. Absolute values, in litres per minute, better reflect the size of the cardiovascular engine. An athlete who loses body mass may see relative VO_2max rise without any change in the heart or blood, while an athlete who gains muscle may see it fall slightly despite becoming stronger and more robust. Neither change should be interpreted without body mass, performance and health in view, and deliberately chasing a higher relative number through weight loss is not a safe coaching strategy.

COACH'S CORNER

If your athlete's VO_2max estimate on a watch changes, check whether pace at a fixed heart rate or a time trial has also changed. If it has not, treat the watch number as noise until real performance confirms it.

CHECKPOINT

Do you know whether the VO_2max values you hear about for your athletes are laboratory measurements, field estimates or watch algorithms?

CHAPTER 14

14. Running Economy

PART II — THE HUMAN ENDURANCE ENGINE

On 29 March 2025, at The TEN meeting in San Juan Capistrano, California, Gulveer Singh ran 10,000 metres in 27:00.22. That is 25 laps of the track at an average of about 64.8 seconds each, or roughly 2 minutes 42 seconds for every kilometre, held for twenty-seven minutes. It made him, according to the Sports Authority of India, the third-fastest Asian of all time, and it left him just 0.22 seconds outside the 27-minute barrier he had been aiming for. At that speed, the difference between a good race and a great one is not only how much oxygen a runner can take in. It is also how little oxygen

each stride costs. A runner who wastes a few percent on every step, by bouncing too high, braking at each landing or tightening the shoulders, pays that tax thousands of times over. Across 10,000 metres, and even more across 42.2 kilometres, the tax adds up to minutes.

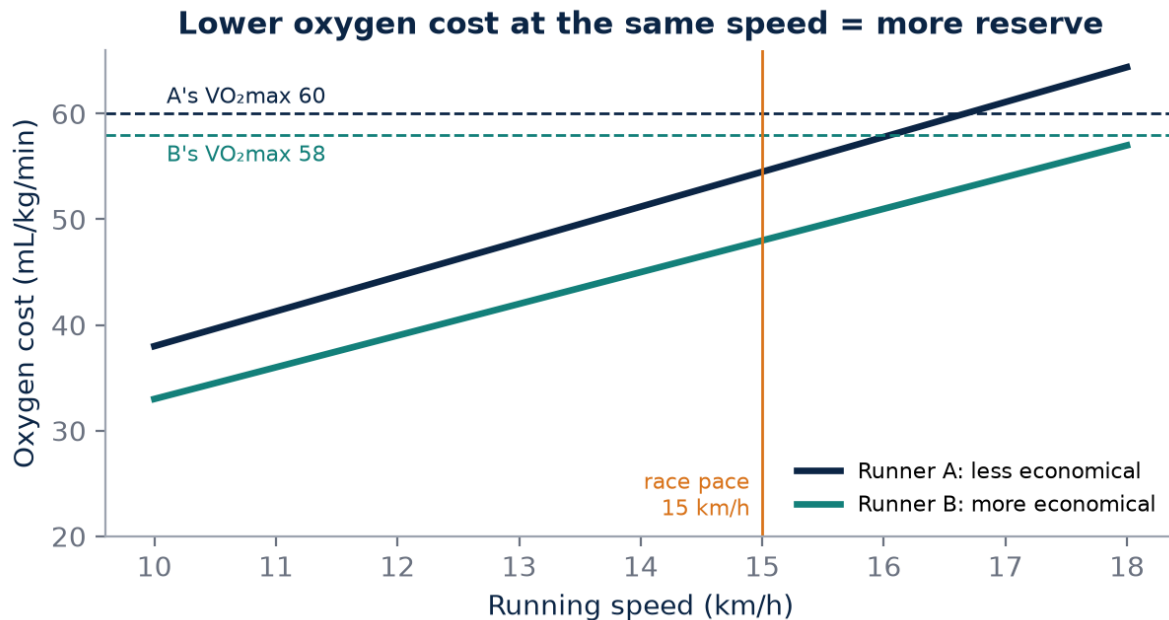


Figure 6. Illustrative values. Runner B has the lower VO_2max but uses less oxygen at race pace, so runs at a smaller fraction of maximum and may race faster.

Running economy is the oxygen cost of running at a given submaximal speed. If two athletes run at the same pace but one uses less oxygen, the more economical runner has more physiological reserve: more room below the ceiling, less heat produced for the same speed and potentially slower use of carbohydrate. The marathon is not only about how much oxygen you can take in. It is also about how little oxygen you need for the speed you want.

Measuring economy

In the laboratory, economy is measured by recording steady-state oxygen uptake while the athlete runs at one or more fixed submaximal speeds, usually below the lactate threshold so that aerobic metabolism dominates. It can be expressed as oxygen uptake per kilogram per minute at a given speed, or as oxygen cost per kilogram per kilometre, which allows comparison across speeds. Some researchers also express it as energy cost, taking into account the mix of fuels being used.

Expression	Units	Why used
VO_2 at fixed speed	mL/kg/min at, say, 14 or 16 km/h	Simple, protocol-specific
Oxygen cost of transport	mL/kg/km	Compares across speeds
Energy cost	kcal or kJ/kg/km	Accounts for fuel mix

Economy varies considerably even among athletes of similar ability, and small differences of a few percent can matter over 42.2 km. Measurement also has error: day-to-day variation, treadmill versus overground differences and footwear all influence results, so repeated tests should be standardised carefully.

KEY IDEA

Economy is how much oxygen the athlete spends to buy a given speed. A more economical runner has more reserve at race pace, even with the same VO_2max .

What shapes economy

Economy is influenced by biomechanics, body mass, stiffness and elastic return, neuromuscular coordination, footwear, fatigue and training history. Reviews such as that by Saunders and colleagues in Sports Medicine in 2004 describe many of these factors. Tendons and muscles that store and return elastic energy efficiently reduce the metabolic work the muscle must do. Coordinated recruitment reduces wasted co-contraction. Lighter shoes with compliant, resilient midsoles can lower oxygen cost for many runners, although responses differ individually. Body mass matters, but chasing weight loss carries serious health risks, including low energy availability, and is not a safe economy strategy.

Economy is one reason a runner can outperform another athlete despite having a similar or lower VO_2max . It is also a quality that can deteriorate late in a race: as fatigue accumulates, mechanics change, oxygen cost at the same speed may rise and the athlete moves closer to the ceiling without speeding up.

MYTH VS EVIDENCE

Myth: Economy is fixed by body type and cannot be trained.

Evidence: Economy reflects many trainable qualities. Years of running, strength and plyometric work, and technique refinement may improve it, although the size of change varies and some influences such as limb proportions are not trainable.

Building economy in an Indian context

Many Indian runners train on mixed surfaces: roads, dirt tracks, temple streets, synthetic tracks where available and hills. That variety can be an advantage, developing robust feet and adaptable mechanics. Economy development usually includes consistent volume over years, short relaxed strides and hill sprints, heavy and explosive strength training performed with good technique, and exposure to race-pace running so that the athlete becomes efficient at the speed that matters.

Heat is also relevant. Running in heat increases total physiological strain at the same speed, and some studies show a small rise in oxygen cost in the heat, although the effect is not consistent. A more economical runner produces less metabolic heat for the same pace, which may help in hot races, but that link needs individual testing.

FIELD PROTOCOL

1. Field economy proxy
2. on a flat, calm route or track, warm up 15 minutes
3. run 2 × 6 minutes at a fixed, well-below-threshold pace with 2 minutes easy between
4. record HR over the last 2 minutes of each, cadence and RPE
5. note temperature, humidity, shoes and body mass
6. repeat monthly with identical pace and shoes
7. a falling HR at the same pace and conditions may reflect better economy, plasma volume or fitness, so label it as a proxy.

Economy also interacts with pacing. Because oxygen cost rises with speed, an athlete who runs an uneven marathon, surging on inclines or chasing groups early, spends more oxygen and more carbohydrate than one who runs the same average pace evenly. In crowded Indian city marathons, with congested starts, sharp turns and flyovers, good pacing discipline can protect some of an athlete's economy advantage. AEI therefore treats pacing data from races, including split variation and response to gradients, as useful context when interpreting economy tests, not as a separate matter.

COACH'S CORNER

Protect economy late in races by building strength and by practising race pace when partially fatigued, for example the last kilometres of a long run. Film the athlete fresh and tired; changes in posture and ground contact often reveal where economy is lost.

CHECKPOINT

How many of your athlete's weekly sessions develop economy directly, through strength, strides, hills or race-pace running, rather than only aerobic volume?

CHAPTER 15**15. Mitochondria and Muscle Oxygen Use**

PART II — THE HUMAN ENDURANCE ENGINE

The training numbers Sawan Barwal's camp reported to The Tribune are not glamorous. About 200 kilometres a week. A long run of 40 to 42 kilometres every other week. Months of it, in camps in the Nilgiris, Kannur and Bengaluru. In most distance programmes, a large share of volume like that is run at an easy, conversational effort. Watch any group of distance runners on an ordinary easy morning and you might wonder what the point is. Nobody is racing. Nobody is gasping. Yet this is the kind of running that builds the deepest part of the endurance engine, the part no watch can display. Inside each working muscle fibre, week after week, the machinery that uses oxygen is expanded, and the network of tiny blood vessels that feeds it is extended. Whatever else went into Sawan's 2:11:37 for Asian Games silver in Nagoya, a marathon at that speed is impossible without that invisible machinery, built one unremarkable kilometre at a time.

The heart, lungs and blood deliver oxygen, but oxygen only creates energy when a muscle cell uses it. That happens inside mitochondria, the cellular machinery that supports aerobic ATP production. Mitochondria oxidise fatty acids and the products of carbohydrate breakdown, including pyruvate and lactate, to generate most of the ATP needed for sustained running. If delivery is the supply chain, mitochondria are the factory floor.

What endurance training does to muscle

Endurance training stimulates mitochondrial adaptations and improves the muscle's ability to use oxygen and oxidise substrates. The classic work of John Holloszy in the 1960s showed that endurance training increases mitochondrial enzyme activity and oxidative capacity in skeletal muscle. Since then, research has described increases in mitochondrial content and function, in capillary density around muscle fibres, and in the enzymes and transporters that handle fat and carbohydrate.

These adaptations increase the muscle's ability to extract oxygen from blood, widening the arteriovenous oxygen difference in the Fick equation. They also mean that, at a given submaximal pace, the trained muscle can meet its energy needs with less disturbance to its internal environment, relying relatively more on fat and producing less lactate accumulation in the blood.

Adaptation	Effect	Typical stimulus
Mitochondrial content and function	More aerobic ATP capacity	Aerobic volume, threshold and interval work
Capillary density	Shorter diffusion distance, longer transit time	Consistent aerobic training
Fat-oxidation enzymes	Greater fat use at submaximal pace	Aerobic volume, long runs
Lactate and glucose transporters	Better lactate exchange and use	Varied intensity training

KEY IDEA

A high-level endurance athlete is not a large oxygen tank but a coordinated system that delivers oxygen and uses it efficiently. Mitochondria and capillaries determine how well the muscle uses what the heart delivers.

Many roads to the same factory

This distinction helps explain why easy volume, threshold work, long runs and strength training can all contribute through different pathways. Easy volume accumulates time with aerobic signalling in many fibres and supports capillary growth. Threshold work challenges the muscle to sustain a high oxidative rate. Long runs deplete glycogen in working fibres and may amplify some metabolic signals, while also training durability. Strength training contributes mainly through economy, force production and injury resilience rather than through large mitochondrial gains, but it keeps the whole chain working.

The research literature continues to debate which combinations of volume and intensity best stimulate mitochondrial adaptation, and the answer likely depends on training history. For practical coaching, the consistent message is that repeated, well-recovered aerobic work over months and years builds the muscle's aerobic machinery, and that no single session type is sufficient.

MYTH VS EVIDENCE

Myth: Only high-intensity intervals build mitochondria, so easy running is wasted time.

Evidence: High-intensity work is a strong stimulus, but aerobic volume also drives mitochondrial and capillary adaptation, and it can be sustained over far longer periods with less risk. Most successful endurance programmes combine both.

Avoid chasing one number

The practical implication is to avoid chasing one physiological number at the expense of the system. A programme built entirely around raising VO_2max may neglect muscle endurance and economy; one built entirely around easy volume may leave the threshold underdeveloped. Mitochondrial adaptations are also not directly visible to most coaches. They are inferred from outcomes such as lower heart rate and lactate at a given pace, improved fat use and better late-race durability.

Near-infrared spectroscopy devices that estimate muscle oxygen saturation are increasingly available, but their readings depend on placement, skin and fat thickness and device algorithms. They may be useful for research within the same athlete under standardised conditions, and should not be treated as a direct measure of mitochondrial content.

Recovery is part of the mitochondrial story. The training session provides the signal, but the adaptation is built during the hours and days that follow, supported by adequate energy, carbohydrate and protein intake, and sleep. Athletes with low energy availability, chronic sleep restriction or frequent illness may accumulate training without the expected muscle adaptations. For Indian athletes balancing work or study, long commutes and early morning sessions to avoid heat, protecting sleep and nutrition may be as important for the muscle as adding another session. AEI should record these recovery inputs alongside training load, so that a lack of adaptation can be interpreted fairly.

COACH'S CORNER

Judge muscle adaptation by what the athlete can sustain, not by a single test. Track how pace, heart rate and perceived effort behave in the final third of long runs; better durability there is often the most practical sign that the muscle's aerobic machinery is improving.

CHECKPOINT

Over the last eight weeks, has your athlete's training included enough consistent aerobic volume and enough sustained threshold work to stimulate the muscle, or has it relied on one type of session?

CHAPTER 16**16. Lactate Is a Messenger**

PART II — THE HUMAN ENDURANCE ENGINE

On a school ground in Madurai, a PE teacher blows his whistle after the last of eight fast 400s. Twelve teenagers bend over, hands on knees, faces twisted. Their legs are burning. "Jog it out," he shouts. "Flush out the lactic acid, or you won't be able to walk tomorrow." The runners obediently shuffle a lap. It is advice that has been handed down on Indian grounds for decades, and at first glance it makes perfect sense. The burning is real, the soreness two days later is real, and the substance being blamed has a name that sounds suitably sinister. But almost every part of the explanation is wrong. The molecule in question is not the villain of the piece. It is not what makes the legs sore later in the week, and the body does not need to be rinsed of it like a dirty pan. In fact, trained muscle takes it up and uses it.

For much of the twentieth century, lactate was cast as the villain of endurance sport: a toxic waste product that made legs burn and caused fatigue. That picture is now considered outdated. Lactate is a normal metabolic intermediate, not simply a toxic waste product. It can be produced in active muscle, transported between tissues and used as a fuel. Understanding this changes how a coach thinks about threshold training, lactate testing and the language used with athletes.

From waste to shuttle

When glucose or glycogen is broken down through glycolysis, the end product pyruvate can either enter the mitochondria for oxidation or be converted to lactate. Lactate is produced continuously, even at rest, and production rises as exercise intensity and glycolytic rate increase. Crucially, lactate does not simply sit in the muscle. It moves out of cells through monocarboxylate transporters and can be taken up by neighbouring oxidative muscle fibres, the heart, the brain and the liver. Oxidative fibres and the heart can use it as a fuel; the liver can convert it back to glucose.

George Brooks and colleagues developed and extended this framework as the lactate shuttle concept, describing lactate as a link between glycolytic and oxidative metabolism and as a signalling molecule that may influence gene expression and adaptation. The details continue to be researched, but the broad idea that lactate is a useful fuel and messenger, not merely waste, is well supported.

KEY IDEA

Lactate is produced, moved and used throughout the body. Blood lactate concentration reflects the balance between production and clearance, not simply how much "waste" has been made.

MYTH VS EVIDENCE

Myth: Lactate causes muscle soreness the next day and must be flushed out after hard sessions.

Evidence: Blood lactate returns towards resting levels within roughly an hour after exercise, even without special cooling-down methods. Delayed soreness has other causes, mainly related to unfamiliar or eccentric loading.

What rising blood lactate means

During increasing exercise intensity, blood lactate concentration rises as production and clearance dynamics change. At low intensities, production and removal are closely matched, so blood lactate stays near resting values. As intensity rises, glycolytic flux increases, more fast-twitch fibres are recruited and clearance may fail to keep pace, so blood lactate begins to accumulate. At higher intensities it rises steeply.

Fatigue during hard exercise is associated with a complex mix of metabolic changes, including accumulation of hydrogen ions and inorganic phosphate, and changes in ion balance within the muscle. The older idea that lactate itself is the direct cause of the burning sensation and fatigue is now regarded as too simple. Blood lactate remains a useful marker of metabolic strain, precisely because it tracks the balance of these processes, but it should be seen as a signal rather than a poison.

Threshold training is therefore not about "removing all lactate". It is about improving the athlete's ability to sustain a high metabolic rate while controlling the accumulation and use of lactate and associated metabolites. Training may improve this through greater mitochondrial capacity, better lactate transport and oxidation, and changes in muscle fibre recruitment and fuel use.

Testing lactate responsibly

This is why lactate testing can be valuable when standardised, but a single lactate number should not be interpreted without workload, timing and protocol. A reading of 3 mmol/L means little unless you know the pace, the gradient, how long the athlete had been running, when the sample was taken, the analyser used and the athlete's recent diet and training.

Factor	Why it changes the reading
Stage duration	Short stages may not reach steady state
Sample timing	Lactate changes quickly after stopping
Glycogen status	Low glycogen can lower lactate at a given pace
Heat and hydration	Can raise physiological strain at fixed pace
Analyser and site	Devices and finger versus earlobe differ

Portable lactate meters are now affordable for some Indian academies and clubs. They are powerful tools when used carefully, and misleading tools when used casually. Hygiene is essential: sweat contamination, especially in humid conditions, can falsely raise readings, so the sampling site must be cleaned and dried, and sharps must be disposed of safely.

FIELD PROTOCOL

1. Standardised lactate step test
2. after a consistent diet and an easy day, warm up 10–15 minutes
3. run 5–7 stages of 4–5 minutes, increasing pace by a fixed step each stage
4. take a clean, dry earlobe or fingertip sample within about 30 seconds after each stage
5. record pace, HR, RPE, temperature and humidity
6. stop when RPE is very hard or lactate rises steeply
7. repeat on the same route or treadmill with the same analyser.

COACH'S CORNER

Stop describing lactate as poison to athletes. Explain it as a fuel and a messenger that the trained body learns to handle better. It changes how athletes relate to hard sessions and reduces fear of discomfort at threshold.

CHECKPOINT

If your athlete produced a lactate reading of 4 mmol/L today, what six pieces of information would you need to interpret it properly?

CHAPTER 17

17. Lactate Threshold

PART II — THE HUMAN ENDURANCE ENGINE

Tuesday is threshold day for a running club in Bengaluru, and three members arrive at Cubbon Park with three different ideas of what that means. The first has a laboratory report that puts her threshold at 4 mmol/L of lactate and a pace of 4:35 per kilometre. The second has a watch that announces his threshold pace every morning, a number that seems to move with his mood. The third has only her coach's instruction: run

comfortably hard, at the pace you could hold for about an hour in a race. All three are using the same word. They may be describing three different intensities. By the second repetition the first runner is fighting, the second is chatting, and the third is exactly where her coach wanted her. The problem is not that any of them is foolish. It is that one word has been asked to carry too many meanings, and each meaning comes from a different method.

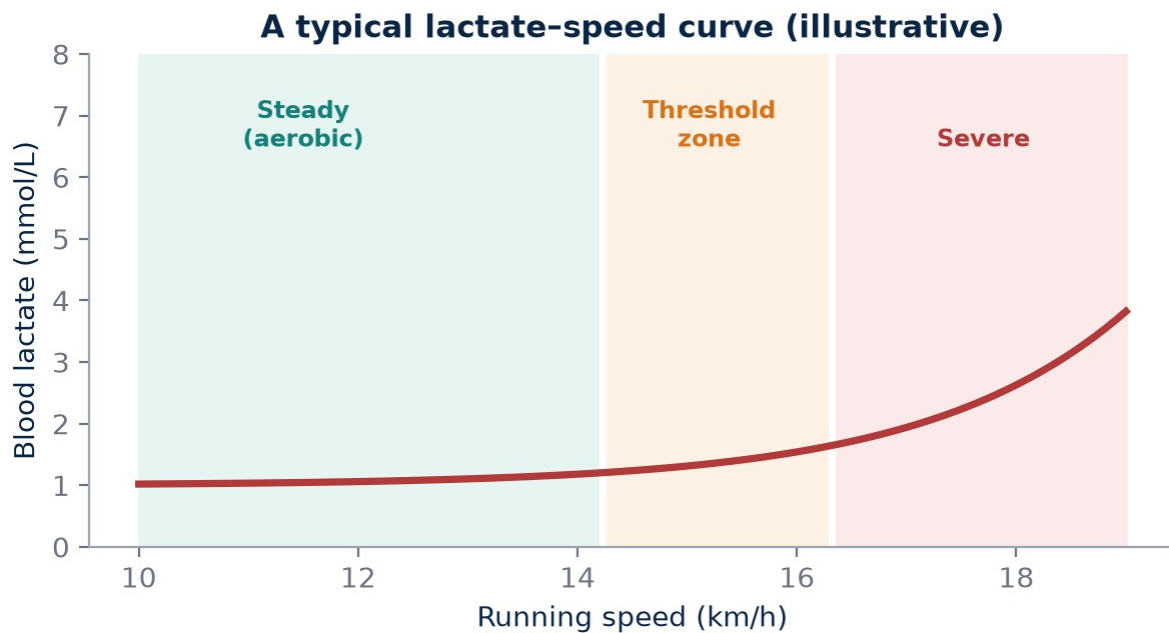


Figure 7. As speed rises, blood lactate stays low, then accumulates increasingly quickly. Threshold is best treated as a zone, and its definition depends on the test protocol.

Few words in coaching are used as loosely as "threshold". Athletes talk about "threshold runs", laboratories report "thresholds" and watches estimate "threshold pace", yet these often refer to different things. The term "threshold" can describe several related concepts, including fixed blood-lactate definitions and individual physiological breakpoints, and different laboratories may use different protocols and terminology.

Many thresholds, one idea

Broadly, two important transitions are commonly described as intensity rises. The first is the intensity at which blood lactate first rises meaningfully above resting or baseline values, sometimes called the first lactate threshold or aerobic threshold. The second is a higher intensity above which lactate can no longer be stabilised and keeps rising over time during constant-pace running, often associated with the maximal lactate steady state, the second threshold, or related concepts such as critical speed.

Term	Common definition	Caution
LT1 or aerobic threshold	First sustained rise above baseline lactate	Detection method varies
Fixed 2 mmol/L	Pace at a set concentration	Not individualised
Fixed 4 mmol/L (OBLA)	Pace at a set concentration	Individual MLSS may be higher or lower
MLSS	Highest pace with stable lactate over about 30 minutes	Time-consuming to determine
Critical speed	Boundary derived from time trials at several distances	Model and protocol dependent

Faude, Kindermann and Meyer, in a 2009 review in *Sports Medicine*, described the large number of lactate threshold concepts in use and cautioned that they are not interchangeable. Fixed concentrations such as 4 mmol/L are convenient but ignore the fact that individual maximal lactate steady state values differ considerably between athletes. A threshold number is therefore meaningful only alongside the method that produced it.

KEY IDEA

"Threshold" is a family of concepts, not one universal number. Every threshold value must be stored with the protocol and method used to find it.

Threshold as a training domain

For coaching, threshold is best treated as a training intensity domain rather than a magical pace. It should feel controlled-hard rather than maximal. Athletes often describe it as comfortably hard: breathing is deep and rhythmic, speaking is limited to short phrases, and the effort could be held for a sustained period in a race. It supports the ability to sustain a high fraction of aerobic capacity, which, as Chapter 13 showed, is one of the three pillars of endurance performance.

Because the domain has boundaries rather than a single point, sessions such as continuous tempo runs, cruise intervals and longer intervals with short recoveries can all target it. Heat, hills, fatigue and fuelling state all shift the pace at which the domain occurs, so on a humid Chennai evening the correct threshold effort may be noticeably slower than on a cool morning. Coaches who hold athletes to a fixed pace regardless of conditions risk turning a threshold session into a much harder one.

MYTH VS EVIDENCE

Myth: Threshold pace is a fixed number you can use all season.

Evidence: Threshold shifts with training, fatigue, fuel and environment. It is better treated as an effort domain checked by heart rate, RPE and, where available, lactate, and retested regularly.

Recording threshold in AEI

AEI should record the test protocol alongside the threshold result. A number without the protocol is incomplete data. For each threshold entry, AEI stores the method (lactate step test, fixed concentration, MLSS, critical-speed model or field time trial), the stage duration and step size, the surface or treadmill gradient, the environmental conditions and the analyser or device. Field estimates, such as the pace sustained in a 30-minute time trial, are useful and practical but should be labelled as such, and they are strongly influenced by motivation and pacing skill.

Changes over time are often more valuable than absolute values. A rightward shift of the lactate curve, where the same lactate appears at a faster pace, or a lower heart rate at the same threshold pace, is a meaningful sign of adaptation if the protocol has been kept the same.

FIELD PROTOCOL

1. Field threshold check
2. on a flat track or measured loop, warm up 15 minutes with strides
3. run a 30-minute evenly paced effort as hard as can be sustained
4. record average pace and HR for the final 20 minutes, RPE, temperature and humidity
5. cool down 10 minutes
6. repeat every 6–8 weeks in similar conditions
7. store as "field threshold estimate" with the protocol.

The relationship between threshold and heat deserves particular attention in India. Heat increases cardiovascular strain and, for many athletes, carbohydrate use at a given pace, so blood lactate and heart rate at a fixed speed may be higher on a hot day. A threshold test performed on a cool December morning in Chennai will not give the same result as one performed in May. Where possible, testing should be scheduled at a consistent time of day and season, or the conditions should be recorded carefully enough that the difference can be accounted for when results are compared.

COACH'S CORNER

Prescribe threshold by a combination of pace range, heart-rate range and perceived effort, and let conditions decide which one leads on the day. In heat, let effort and heart rate lead and allow the pace to slow.

CHECKPOINT

Which threshold definition have you been using for your athletes, and would another coach or laboratory reproduce the same number from your notes?

CHAPTER 18**18. Fuel, Glycogen and Fat Oxidation**

PART II — THE HUMAN ENDURANCE ENGINE

At the Rio Olympics in August 2016, O. P. Jaisha, India's national record holder in the marathon, collapsed after crossing the finish line of the women's race in 89th place. Afterwards she said Indian officials had not provided her with refreshments along the course. "All the [other] countries had their stalls at every two kilometres, but our country's stall was empty," she was quoted as saying by Scroll. The Athletics Federation of India disputed her account, and her teammate Kavita Raut said she herself had had enough water. What exactly happened at those drinks tables may never be settled. But the episode put a public spotlight on a question every marathoner eventually faces. Fitness gets a runner to the start line. What she has stored in her muscles, what she drinks and what she eats along the way decide how much of that fitness survives to the finish. A marathon is a fuel problem as much as a fitness problem.

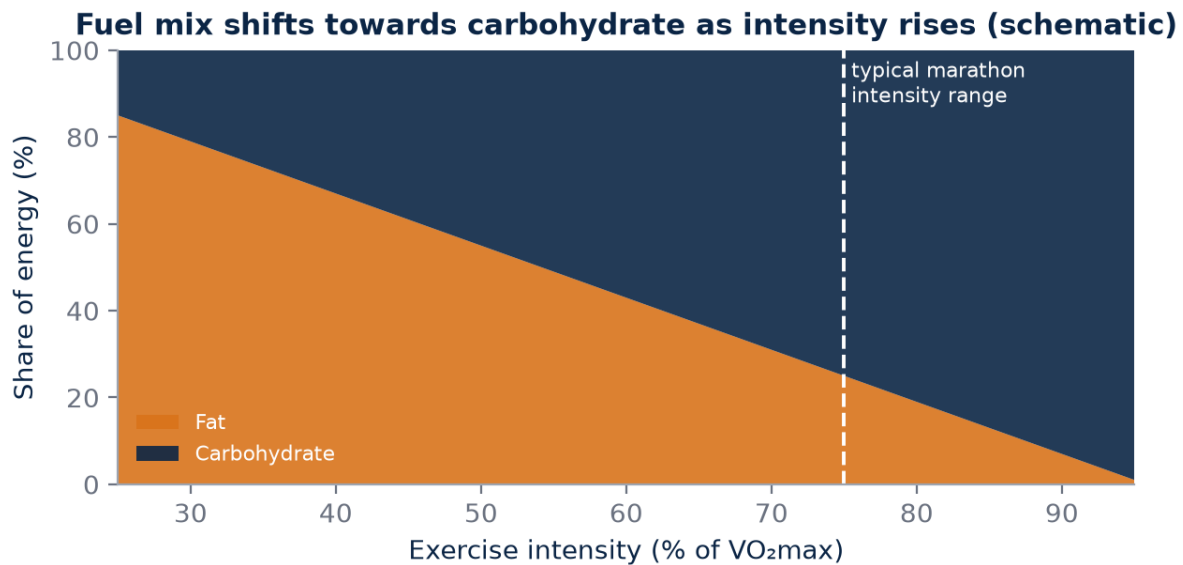


Figure 8. Schematic of the general pattern. Individual fuel use varies with training, diet, sex and duration, which is why marathon fuelling must be practised.

The marathon is where physiology meets the kitchen. Marathon performance depends heavily on substrate management: which fuels the athlete uses, how much is stored and how well it is replaced during the race. Carbohydrate supports high-intensity work efficiently, while fat oxidation contributes increasingly as intensity and duration change. Training can influence substrate use, but race-day fuelling remains essential.

Two fuel tanks

The body stores carbohydrate as glycogen in muscle and liver, and as a small amount of glucose in the blood. Glycogen stores are limited, on the order of a few hundred grams in muscle and around a hundred grams in the liver in a well-fed adult, although they vary with body size, training and diet. Fat stores, by contrast, are large even in lean athletes. Carbohydrate yields more ATP per litre of oxygen than fat, so it is the preferred fuel as intensity rises. Fat is abundant but oxidised more slowly and is the dominant fuel mainly at lower intensities.

Feature	Carbohydrate	Fat
Storage size	Limited (muscle and liver glycogen)	Large
Energy per litre of O ₂	Higher	Lower
Use at marathon pace	Major contributor for most runners	Contributes, varies by athlete
Main risk	Depletion late in race	Too slow to sustain high pace alone

As a marathon progresses, muscle glycogen in the working fibres falls. If carbohydrate availability becomes insufficient, the athlete may be unable to hold pace, a phenomenon runners call "hitting the wall". Training can increase the capacity to

oxidise fat at a given pace and increase glycogen storage, sparing carbohydrate, but for most runners racing a marathon at a competitive pace, carbohydrate remains central.

KEY IDEA

Carbohydrate is the high-performance fuel for racing, but its stores are limited. Fitness improves fat use and glycogen storage; race-day fuelling protects what remains.

Fitness and fuel need each other

The athlete who starts a marathon with excellent aerobic fitness but poor carbohydrate availability can still fail late. Conversely, high carbohydrate intake cannot rescue inadequate fitness. Current sports nutrition guidance, summarised in reviews by Burke, Hawley, Jeukendrup and colleagues, describes several practical tools: increasing carbohydrate intake for one to two days before a marathon to maximise glycogen stores, a carbohydrate-containing pre-race meal, and carbohydrate intake during the race, commonly in the range of about 30–60 g per hour and up to around 90 g per hour for longer events when using mixtures of glucose and fructose that the gut has been trained to tolerate. Exact amounts should be individualised, ideally with a qualified sports dietitian.

Gut tolerance is trainable. Athletes who try new gels or drinks on race day risk gastrointestinal distress, which is more likely in heat. In the Indian context, familiar carbohydrate foods such as rice, idli, bananas or other locally available options may form part of pre-race meals, provided they have been practised in training and suit the athlete's gut.

MYTH VS EVIDENCE

Myth: Fat-adapted runners do not need carbohydrate during a marathon.

Evidence: Low-carbohydrate adaptation can raise fat oxidation, but studies have shown it may impair the ability to use carbohydrate efficiently at high intensity. For most competitive marathoners, carbohydrate availability remains critical at race pace.

Fuel meets heat, pacing and hydration

The research model should therefore connect metabolic fitness with practical fuelling, hydration, pacing and environmental stress. Heat increases reliance on carbohydrate at a given pace in many individuals and makes the gut less tolerant, while dehydration reduces performance and increases strain. An aggressive early pace burns glycogen faster and may leave the athlete short in the final 10 km. These factors interact; a fuelling plan that works in a cool Bengaluru long run may fail in a humid Mumbai race.

Athletes should also avoid both extremes of drinking: significant dehydration impairs performance in heat, while drinking large amounts beyond sweat losses in long events can cause dangerous low blood sodium. Individual sweat-rate testing helps set sensible targets, and any athlete with symptoms such as confusion, severe headache or vomiting during or after a race needs urgent medical attention.

FIELD PROTOCOL

1. Race-fuel rehearsal long run
2. eat the planned pre-race meal 3–4 hours before
3. weigh the athlete nude or in minimal clothing before and after
4. run 25–32 km with the final 8–12 km at goal marathon pace
5. take the planned carbohydrate every 20–30 minutes at the planned hourly amount
6. record fluid intake, gut comfort, RPE, HR and pace drift
7. note temperature and humidity
8. adjust the plan in small steps.

COACH'S CORNER

Treat fuelling as a trained skill. Rehearse the race-day meal, gel or drink type, timing and quantity in at least three long runs, including one in conditions similar to the race, and record what happened in the athlete's log.

CHECKPOINT

Has your athlete practised the exact race-day fuelling plan, in similar heat, at goal marathon pace, and does AEI hold a record of how their pace and heart rate behaved late in that run?

PART II AT A GLANCE

The key idea from each chapter, for quick revision.

7

The Endurance System as a Chain

Endurance performance is a chain of linked systems, from air to ATP to forward motion. Strengthening one link helps only if it is, or becomes, the link that limits the athlete.

8

The Heart: Pump, Not Just Heart Rate

Cardiac output = heart rate × stroke volume, and oxygen uptake = cardiac output × oxygen extraction. Heart rate is only one of the terms.

9	The Lungs and the Oxygen Myth Altitude is a pressure stimulus, not a percentage stimulus. The same 20.9% oxygen at lower pressure means a lower partial pressure of oxygen and less driving force for gas exchange.
10	Haemoglobin Mass Haemoglobin concentration is a ratio that changes with plasma volume. Haemoglobin mass is the total amount, and is the better marker of oxygen-carrying capacity.
11	Plasma Volume Plasma volume supports venous return, stroke volume, cardiovascular stability and cooling. Heat acclimation and endurance training can both expand it, but by different amounts in different people.
12	Red Blood Cells and Erythropoiesis Erythropoiesis is a slow, dose-dependent process. A rise in EPO is only the first step; building new red cells needs time, iron, energy, sleep and a sufficient hypoxic stimulus.
13	VO₂max VO ₂ max sets the size of the aerobic ceiling. It does not tell you how close to that ceiling the athlete can race, or how much oxygen each kilometre costs.
14	Running Economy Economy is how much oxygen the athlete spends to buy a given speed. A more economical runner has more reserve at race pace, even with the same VO ₂ max.
15	Mitochondria and Muscle Oxygen Use A high-level endurance athlete is not a large oxygen tank but a coordinated system that delivers oxygen and uses it efficiently. Mitochondria and capillaries determine how well the muscle uses what the heart delivers.
16	Lactate Is a Messenger Lactate is produced, moved and used throughout the body. Blood lactate concentration reflects the balance between production and clearance, not simply how much "waste" has been made.
17	Lactate Threshold "Threshold" is a family of concepts, not one universal number. Every threshold value must be stored with the protocol and method used to find it.
18	Fuel, Glycogen and Fat Oxidation Carbohydrate is the high-performance fuel for racing, but its stores are limited. Fitness improves fat use and glycogen storage; race-day fuelling protects what remains.

PART III — HEAT

India is, for much of the year, a hot country, and for many coastal athletes heat is not an occasional problem but the daily condition of training. Part III treats heat as a measurable physiological stimulus rather than a test of character. It explains what heat does to the heart, blood and skin, how controlled heat acclimation works, why plasma expansion matters, how sweating and humidity interact, and how a city such as Chennai can become a genuine training laboratory. Throughout, safety comes first: heat can build adaptation, but uncontrolled heat can also cause serious illness. The aim is dose, data and judgement, not bravado.

IN THIS PART

- 19 What Heat Actually Does
- 20 Heat Acclimation
- 21 Plasma Expansion and Stroke Volume
- 22 Sweating and Thermoregulation
- 23 Chennai as a Training Laboratory

PART III • HEAT

Heat Is a Dose, Not Toughness

For coastal Indian athletes heat is the daily condition of training; controlled, measured exposure can build adaptation.

10–14 days

A controlled heat-acclimation block for healthy, medically cleared athletes



Two jobs, one circulation

The same pace costs more in heat because blood must serve both working muscles and the skin.



Dose

Dose-response block

Effects depend on how hot, humid, long, hard and often — and they fade without continued exposure.



Plasma ↑

Plasma, not red cells

Plasma expansion supports stroke volume and lowers HR at a given pace, but adds volume, not red cells.



Sweat must evaporate

Acclimation improves sweating; humidity limits evaporation. Hot, humid evenings demand the highest caution.



STOP

The non-negotiable boundary

Confusion, stumbling, collapse or chills in heat: stop at once, cool immediately, call medical help.



Chennai

A city laboratory

Chennai cannot provide hypoxia, but offers repeated, measurable heat and humidity — if logged as data.

THE PROCESS

Weigh, log
weather

10-min warm-up

Fixed easy pace

Record HR drift

Re-weigh, log
fluids

Heat can build adaptation, but uncontrolled heat causes serious illness: dose, data and judgement, not bravado.

THE STORY

The Road from Kavinadu

PART III — HEAT

It is 5:15 a.m. on the Marina, and the sea has not yet decided whether to be grey or silver. The air is already thick. A club runner in a cotton vest sets off along the beach road at the pace that felt easy in January, and by the second kilometre the vest is dark and heavy, clinging like a wet towel. Nobody here calls this heat training. They call it morning.

Chennai does not announce its heat with drama. There is no desert glare and no dust storm. There is a wet warmth that settles on the skin before sunrise and stays there, a humidity that turns sweat into something that drips rather than cools. Tea stalls near the lighthouse come alive to the hiss of kettles. Walkers in canvas shoes move in slow, patient loops. A few serious runners move faster, heads down, glancing at their watches as the heart-rate numbers creep upward while the pace stays the same.

Tamil Nadu has given India distance runners who grew up in air like this, though not on the Marina. Their story begins far to the south, beyond Tiruchirappalli, in the dry, hot country of Pudukkottai district, where the roads shimmer by mid-morning and the tar softens under the afternoon sun.

A club in a village

In Kavinadu village near Pudukkottai, a coach named S. Loganathan ran something called the Youth Sports Club. The name is modest. The results were not. According to *The Hindu*, one of the boys who came through it was Govindan Lakshmanan, from Sokkurani in Pudukkottai district, a runner the paper described as "born in a poor family". Loganathan was his first trainer, and the training ground was not a synthetic track. It was tar roads and village streets.

Think about what that means in this part of India. A village road in Pudukkottai is not a laboratory. There is no wet-bulb globe thermometer at the roadside, no sweat-rate chart, no ice bath waiting at the finish. There are bullock carts, two-wheelers, stray dogs, and a sun that has usually done its work on the tarmac before school starts. Whatever these runners learned about heat, they learned the way most Indian athletes do: by living in it.

Lakshmanan went into the Army, as so many Indian distance runners do. In 2015, at the Asian Championships in Wuhan, he won bronze in the 5000 m and silver in the 10,000 m. Two years later, at the Asian Championships in Bhubaneswar, he won both, gold in the 5000 m and gold in the 10,000 m, a double that carried him to the World Championships in London.

That path, from a village road to a regiment, is the most common one in Indian men's distance running. Nearly every leading Indian distance man of the past decade has been an Army soldier, and several, by their own accounts, found competitive running

only after they enlisted. The Army gave Lakshmanan a salary, a coach and a national camp. It did not give him cooler weather. The heat he grew up in came with him.

Then came Jakarta. At the 2018 Asian Games, Lakshmanan crossed the line third in the 10,000 m final. Soon afterwards the bronze was gone: he was disqualified for a lane infringement, and India's appeal was rejected. Distance running is often described as a pure test of the body. Jakarta was a reminder that it is also a test of rules, positioning and nerve, decided sometimes by a single step inside a line.

The record that will not move

The same Kavinadu club is linked, in reports, to another runner: Loganathan Suriya, born on 7 July 1990. Some accounts describe S. Loganathan as her father as well as her coach, though the primary source for that detail could not be confirmed, and it is recorded here only as reported.

What is not in doubt is what she did on 19 November 2017. At the Delhi Half Marathon, Suriya ran 1:10:31. It was a national record, and according to Olympics.com's national-records list, updated in September 2026, it is still the Indian women's half-marathon record nearly nine years later. It was two minutes and nineteen seconds faster than Kavita Raut's 1:12:50 from 2009, the last earlier record that could be verified. In a discipline where records usually move by seconds, that is a leap.

Suriya won the Indian elite women's race in Delhi in 2017 and again in 2019. Her World Athletics profile lists a 10,000 m best of 32:23.56 and a 5000 m best of 15:39.18, both run in New Delhi in 2018, and a short-track 3000 m of 9:09.81 in Hangzhou in 2016, itself a national record. The same profile records two top-eight finishes at the Asian Games, three at the Asian Championships and eight national titles.

For almost a decade nobody has caught her half-marathon time. The nearest attempt came from Seema, the Himachal runner, who ran 1:11:23 at the Delhi Half Marathon in October 2025, 52 seconds short. She said afterwards that she had missed the record "by a few seconds". It must have felt that way. On the clock, Suriya's mark still stands alone.

Where the fast times were run

Here the storyteller has to be careful, because the most interesting question about these runners is also the one the record cannot answer. The research dossiers behind this book found no documented, deliberate heat-acclimation protocol for India's elite distance runners. They found no peer-reviewed heat-stress study of an Indian marathon. What the public record holds is places, dates and times. It does not hold core temperatures, sweat rates or heart-rate drift.

But the places and dates are revealing in themselves. Suriya's record was set in Delhi in the second half of November, when the northern plains have usually shed the worst of their heat. Many of India's recent distance records were set even further from home, at fast, well-paced meets in California, Oregon, Japan and Europe. Runners raised in hot, humid country have often produced their very best times in cooler air.

That is not a paradox. It is physiology. Growing up in heat may help a runner tolerate it; the chapters that follow explain how the body can adapt. But the same pace still costs more on a humid morning than on a cool one, for everybody, however acclimatised. A runner can be at home in the heat and still be slower in it.

Thousands of runners, almost no data

Away from the elite, the heat question becomes a mass-participation question. The Chennai Marathon's 2026 edition, the fourteenth, drew 28,500 participants from 16 countries, and Ethiopian runners won the full-marathon titles. Every one of those runners, from the winners to the last walker, moved through the same coastal air. At the Mumbai Marathon in 2018, a press report surfaced by researchers stated that more than 2,320 participants needed medical assistance. The figure is only partly verified, but it points in the same direction as the international evidence: across large studies of marathons, a higher heat index is associated with more casualties, with one analysis reporting that each 1 °C rise in heat index was associated with a 22% higher rate of severe casualties.

So India holds, almost by accident, one of the world's great natural heat laboratories. It has runners who were raised in heat, races that are run in heat and cities where heat is simply the daily condition of training. What it lacks is the notebook. The club runner on the Marina can tell you how the run felt. Very few can tell you by how many beats their heart drifted, how much weight they lost in sweat, or whether that drift was smaller after two weeks of careful exposure than before.

The rest of this Part is about filling that notebook. To understand why the same pace costs more on a Chennai morning than on a Delhi one in November, we need to follow the blood, and watch the heart try to do two jobs at once.

Sources: <https://worldathletics.org/athletes/india/loganathan-suriya-14411795> ;
<https://www.olympics.com/en/news/athletics-list-of-indian-national-record-holders> ;
<https://vedantadelhihalfmarathon.procam.in/race-categories/half-marathon-elite/past-winners-and-record-holders> ;
<https://www.thehindu.com/news/cities/Tiruchirapalli/athletics/article19251278.ece> ; <https://scroll.in/fiield/891979/asian-games-2018-athletics-g-lakshmanan-disqualified-after-finishing-third-in-10000-metres-final> ;
<https://www.lokmattimes.com/other-sports/delhi-half-marathon-indian-elite-mens-winner-abhishek-not-satisfied-with-timing-seema-wins-to-break-national-record/> ; <https://theprint.in/india/ethiopian-runners-sweep-titles-at-freshworks-chennai-marathon-2026/2818631/> ;
<https://www.pressreader.com/india/the-free-press-journal/20180122/282291025653195> ;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12428376/>

CHAPTER 19

19. What Heat Actually Does

PART III — HEAT

It is 5:40 a.m. on Besant Nagar beach, and the sun is still below the Bay of Bengal. The sand is cool; the air is not. A club runner who ran this same loop in January at a gentle 5:30 per kilometre sets off at exactly the same pace in May. For the first ten minutes nothing seems different. Then her vest turns dark, her breathing climbs a notch, and the watch on her wrist begins to tell a strange story. The pace is unchanged, but her heart rate is drifting upward: five beats, eight, twelve. She has not become unfit in four months. She has not slept badly. Something invisible has been added to the run, a second job that her heart must do while her legs carry on with the first. By the time she turns at the far end of the promenade and heads home, the easy run no longer feels easy. What happened on that beach is the subject of this chapter.

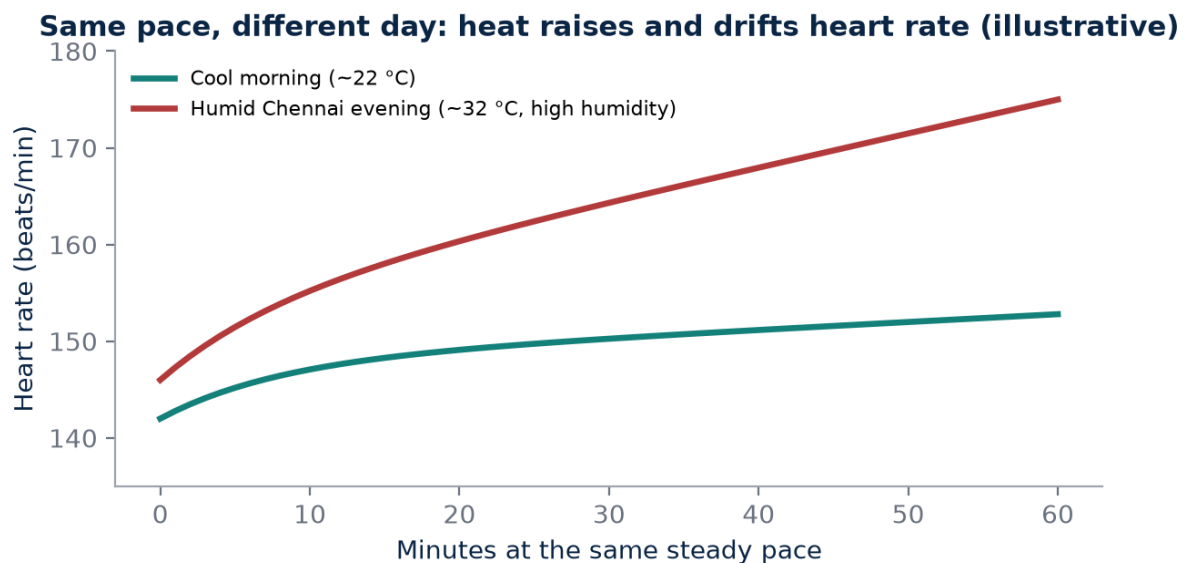


Figure 9. Illustrative example of cardiovascular drift. In the heat, blood is shared between muscle and skin and fluid is lost, so heart rate rises at the same pace.

Every runner in Chennai, Kochi, Mumbai or Visakhapatnam knows the feeling: a pace that feels comfortable on a cool January morning becomes laboured on a humid May evening. The legs have not changed overnight, and fitness has not vanished. What has changed is the job the body must do. Heat increases cardiovascular and thermoregulatory demand. Skin blood flow rises, sweating increases, core temperature is challenged and plasma volume can adapt with repeated exposure. At the same running pace, heart rate may be higher in the heat. Understanding why is the first step towards using heat intelligently rather than simply enduring it.

Two jobs for one circulation

During running, the heart must supply oxygen to working muscles. In the heat it must also carry heat from the body's core to the skin, where it can be lost to the environment. Both demands draw on the same finite blood volume and the same cardiac output. As skin blood flow increases, a larger share of blood sits in the expanded vessels of the skin, and sweating gradually reduces circulating fluid. Less blood returns to the heart with each beat, stroke volume tends to fall, and heart rate rises to compensate. This progressive rise in heart rate at a constant pace is often called cardiovascular drift, and it is usually more pronounced in hot and humid conditions.

The result is that the same external workload, for example 5:00 per kilometre, represents a greater internal load in the heat. Perceived effort climbs, the sustainable pace falls and, if the exposure is long enough, core temperature may rise to levels that impair performance and eventually threaten health.

Adaptation, not toughness

The body is not simply becoming "tougher" when it trains in heat. It is adapting to manage thermal strain. With repeated, controlled exposure, several changes can occur: plasma volume may expand, sweating may begin earlier and become more effective, resting and exercising core temperature may fall slightly, and heart rate at a given pace may decrease. These changes are specific responses to a specific stress. They are useful when the dose is controlled and recovery is sufficient, and they fade when exposure stops.

This matters for coaching language. Telling an athlete that suffering in the midday sun "builds character" confuses discomfort with adaptation. A well-designed heat session may feel uncomfortable, but its value comes from the physiological signal, the recovery that follows and the repetition over days, not from how much the athlete can tolerate on a single afternoon.

KEY IDEA

Heat stress is a training stimulus, not a badge of toughness. The same pace costs more in the heat because the circulation must serve both the muscles and the skin.

Heat illness: the non-negotiable boundary

Because heat is a real physiological stress, it carries real risk. Exertional heat illness ranges from heat cramps and heat exhaustion to exertional heat stroke, a medical emergency in which core temperature rises to dangerous levels and the brain is affected. Warning signs that should stop a session immediately include confusion, disorientation, irrational or unusual behaviour, stumbling or loss of coordination, collapse, severe headache, vomiting, dizziness, and chills or goosebumps while running in the heat. An athlete who is still sweating can still be developing heat stroke; sweating alone is not reassurance.

AKURA does not provide medical diagnosis. The coaching rule is simpler: when in doubt, stop, move the athlete to shade, begin cooling and seek medical help. International consensus guidance on exercise in the heat emphasises rapid whole-body cooling, ideally cold-water immersion, for suspected heat stroke, and any organised heat programme should have a medical plan agreed in advance with qualified professionals.

Warning sign	What the coach does
Confusion or odd behaviour	Stop at once, cool, call medical help
Stumbling or collapse	Emergency: cool immediately, call for help
Severe headache, nausea, vomiting	Stop the session; shade, cooling, medical review
Chills or goosebumps in heat	Stop; treat as a red flag
Unusual heart rate at easy pace	End the session; review next day

COACH'S CORNER

Before any hot-weather session, agree three things with the athlete: a stop rule (for example, any warning sign ends the session with no debate), a cooling plan (shade, cold water, ice towels or a tub available) and a contact plan (who calls for medical help and where the nearest facility is). Heat training without these is not a protocol; it is a gamble.

MYTH VS EVIDENCE

Myth: Drinking a lot of water guarantees protection from heat stroke.

Evidence: Hydration matters, but heat stroke is driven mainly by heat production and inadequate heat loss. A well-hydrated athlete running hard in humid heat can still overheat, and excessive drinking carries its own risks.

CHECKPOINT

When your athlete runs the same pace in cool and hot conditions, do you record the heart-rate difference? Do you have a written stop rule and cooling plan for every hot session?

CHAPTER 20

20. Heat Acclimation

PART III — HEAT

Before Avinash Sable became India's finest steeplechaser, he was a soldier. Olympics.com records that his early postings with the Mahar Regiment took him to the cold of Siachen and to the Rajasthan desert, with heat of up to 50 °C. It is a tempting origin story: the desert forged the runner. But the research dossiers behind this book found something more sobering about Indian distance running as a whole. There is no documented, deliberate heat-acclimation protocol in the public record of India's elite distance runners. Heat, for them, has mostly been incidental, something that happened to them on postings, on village roads and at races, rather than something a coach measured and dosed. That is not a criticism; it is an opening. The difference between being exposed to heat and being acclimated to it is the difference between weather and training. One is luck. The other is a plan with numbers attached.

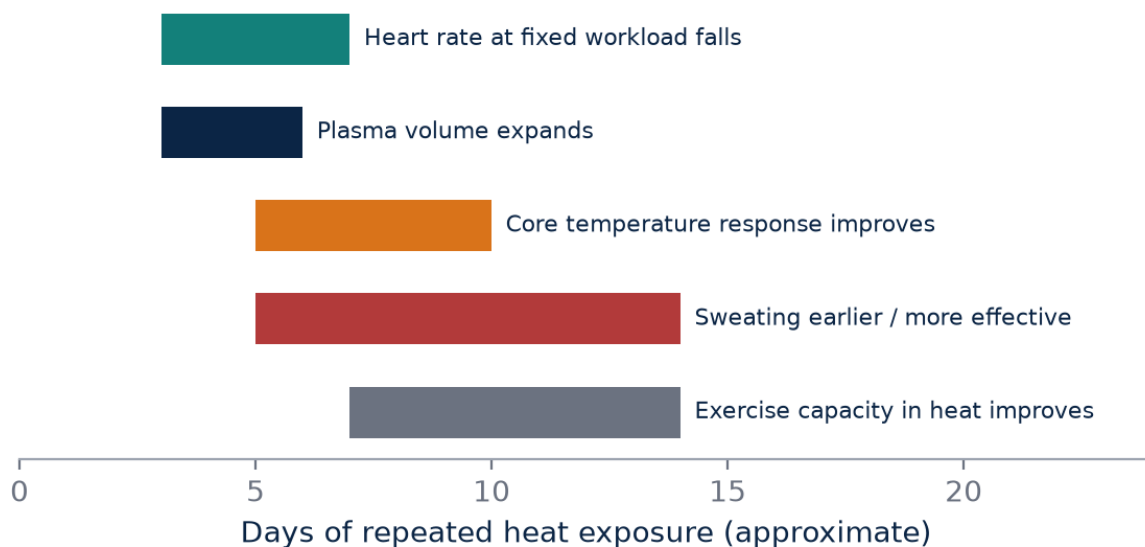
Heat acclimation: the usual order of adaptations (approximate)

Figure 10. Approximate time course summarised from heat-acclimation reviews. Timing varies with protocol, humidity, fitness and individual response.

If heat is a stimulus, then heat acclimation is the training block built around it. Heat acclimation generally involves repeated exposure to elevated environmental or body temperature during exercise, so that the body gradually becomes better at managing thermal strain. Strictly, "acclimation" refers to adaptation achieved in artificial or deliberately controlled conditions, while "acclimatisation" refers to adaptation in the natural environment. For an athlete living in Chennai, some acclimatisation happens

simply through daily life; a structured acclimation block adds deliberate dose, measurement and progression.

What the evidence suggests

Protocols differ in duration, intensity, humidity and whether exposure is active or passive. Some use running or cycling in a climate chamber; others use training in naturally hot conditions, sauna or hot-water immersion after exercise, or wearing extra clothing to raise body temperature. Evidence supports improvements in thermoregulatory and cardiovascular responses, with systematic reviews reporting changes in heart rate, core temperature, plasma volume and exercise performance depending on protocol.

Reviews such as that by Périard, Racinais and Sawka (2015) describe a broad pattern: many adaptations begin within the first several days of exposure, and a substantial portion is typically present after about one to two weeks of regular sessions. Heart rate at a given workload often falls relatively early, plasma volume can expand within the first week, and sweating responses and thermal comfort continue to improve with further exposure. Adaptations decay when exposure stops, although re-acclimation is generally quicker than initial acclimation. These are population tendencies, not promises for any individual.

KEY IDEA

Heat acclimation is a dose-response intervention. Its effects depend on how hot, how humid, how long, how hard and how often, and they fade without continued exposure.

Measuring the heat dose

A block cannot be evaluated if the dose is unknown. AKURA should record heat dose: temperature, relative humidity, session duration, workload, pre/post body mass, fluids, HR and perceived effort. Where available, wet-bulb globe temperature (WBGT) is preferable to air temperature alone because it incorporates humidity, radiant heat and wind. Core temperature measurement, for example with an ingestible sensor, is valuable in research settings but is not essential for a well-run field block; consistent recording of the variables above already gives a coach far more information than "it was hot".

Variable	How to record	Why it matters
Air temperature and humidity	Local sensor or nearby station, at start	Defines environmental load
WBGT (if available)	Handheld WBGT meter	Combines heat, humidity, sun, wind
Duration and workload	Minutes, pace or power	Separates dose from fitness
Body mass pre/post	Nude or minimal clothing, towel-dried	Estimates sweat loss
Fluid intake	Millilitres consumed	Corrects sweat-loss estimate
HR and RPE	Average and end-of-session	Internal response to the dose

FIELD PROTOCOL

1. Safe 10–14 day heat-acclimation block (healthy, medically cleared athletes only)
2. Days 1–3: 45–60 min easy running in warm conditions, heart rate capped at easy zone, full stop rule in place
3. Days 4–7: 60 min, same pace or slightly longer duration, still easy intensity, record HR drift and body mass change
4. Days 8–14: 60–90 min at easy to steady effort, with one or two short controlled strides only if the athlete is tolerating well
5. Schedule at least one lighter or rest day each week
6. Keep hard interval sessions in the cooler part of the day so training quality is protected
7. Weigh before and after each session and rehydrate to replace losses gradually
8. Stop immediately for any warning sign and have cooling and medical contact ready
9. Repeat a standard pace-HR test before and after the block

Active, passive and hybrid approaches

Not every heat dose must come from running in the midday sun. Some athletes combine normal training with post-exercise hot-water immersion or sauna, which raises body temperature without adding running load. Others use extra clothing layers during easy runs in mild conditions. Each method has advantages and risks, and none should be introduced without considering the athlete's health, recovery and overall stress. For a junior athlete, a recreational masters runner or anyone with a medical condition, professional medical clearance should come before any deliberate heat exposure.

COACH'S CORNER

Treat the first days of a heat block as a reduced-intensity phase. Lower the pace, cap the heart rate and let the environment provide the stress. If your athlete finishes a heat session unable to train normally the next day, the dose was too high.

MYTH VS EVIDENCE

Myth: The hotter and harder the session, the faster the adaptation.

Evidence: Adaptation depends on a sufficient, repeated rise in body temperature combined with recovery. Excessive single doses increase the risk of heat illness and can damage the rest of the training week.

Finally, remember that acclimation is not permanent. If an athlete completes a block in April and then trains mostly in air-conditioned gyms or cool early mornings, much of the adaptation may fade over the following weeks. For a target race in hot conditions, the timing of the block relative to race day matters, and brief top-up exposures in the final weeks may help maintain adaptation. The AEI should therefore store not only whether a block was completed, but when it finished and what exposure followed.

CHECKPOINT

For your last hot training week, could you state the temperature, humidity, duration, workload and body-mass change for each session? If not, what would you need to start recording?

CHAPTER 21**21. Plasma Expansion and Stroke Volume**

PART III — HEAT

A masters runner from Adyar walks into a diagnostic lab in late May, after three weeks of steady early-morning training through the hottest stretch of the Chennai summer. She feels fitter than she has all year. Her easy runs along the estuary have become smoother, and her heart rate is lower at the same pace. Two days later the report arrives, and one number is flagged: haemoglobin, slightly below the laboratory's reference range. A well-meaning friend suggests iron tablets. A cousin suggests she is overtraining. Neither has asked the obvious question. What if nothing has been lost at all? What if her blood has not grown weaker, but simply larger, the same red cells now carried in more fluid? The body's response to repeated heat

can quietly change the arithmetic of a blood test, and anyone who reads that test without knowing the athlete's recent heat exposure can reach the wrong conclusion in a hurry. Before anyone reaches for a supplement, the physiology deserves a look.

One of the most discussed effects of heat acclimation is an increase in plasma volume, the liquid portion of the blood. For a coach, the idea is intuitive: more fluid in the circulation should make it easier for the heart to fill and pump. The physiology broadly supports this, but it also carries important caveats. Plasma expansion can increase circulating volume. More effective venous return can support ventricular filling and stroke volume, although the complete response depends on vascular control, hydration, cardiac function and the exercise environment.

From blood volume to stroke volume

Stroke volume is the amount of blood the heart ejects with each beat. It depends partly on how much blood returns to the heart and fills the ventricle before contraction. In the heat, blood is redirected towards the skin and fluid is lost as sweat, so venous return can fall. A larger plasma volume offers a buffer: the circulation can supply the skin for cooling while still returning enough blood to the heart. The practical consequence, often seen after acclimation, is a lower heart rate at the same pace and a smaller rise in heart rate during prolonged exercise in the heat.

The mechanisms behind plasma expansion include changes in fluid- and sodium-regulating hormones and an increase in circulating protein that helps retain fluid within the vessels. Lorenzo and colleagues (2010), in a well-known study published in the *Journal of Applied Physiology*, reported that a 10-day heat-acclimation programme increased plasma volume and improved performance not only in hot conditions but also in cool conditions in trained cyclists. That finding generated wide interest in heat as a potential performance tool for athletes who do not compete in the heat, although later studies have not always reproduced the cool-condition benefit to the same degree.

KEY IDEA

Plasma expansion can support stroke volume and lower heart rate at a given pace, but it increases the volume of blood, not the number of red cells.

Why a sea-level athlete should care

This is one reason heat acclimation is interesting for sea-level athletes. It may improve cardiovascular stability without requiring a mountain camp. For an athlete in Chennai who cannot easily spend four weeks in Ooty, a carefully designed heat block is accessible, cheap and measurable. But plasma expansion is not equivalent to an increase in red-cell mass. More plasma without more red cells means each litre of blood carries slightly less haemoglobin, even though total circulating volume is larger.

This leads directly to a measurement trap. A blood test taken after a heat block may show a lower haemoglobin concentration. That result does not automatically mean the athlete is anaemic or has lost oxygen-carrying capacity; it may partly reflect dilution by expanded plasma. Concentration and total mass answer different questions. Haemoglobin concentration tells you how much haemoglobin is in each unit of blood; haemoglobin mass tells you how much haemoglobin the body contains in total. Only specialised methods, such as carbon-monoxide rebreathing, estimate total mass directly.

Measure	What it tells you	Effect of plasma expansion
Haemoglobin concentration	Hb per unit of blood	May fall (dilution)
Haematocrit	Share of blood that is red cells	May fall (dilution)
Plasma volume	Liquid part of the blood	Rises
Haemoglobin mass	Total Hb in the body	Not directly changed by plasma alone

COACH'S CORNER

Never interpret a single blood test in isolation after a heat block. Note the timing relative to training and heat exposure, and discuss unexpected results with a qualified sports-medicine professional rather than making dietary or supplement decisions on your own.

Separating the pathways

The research comparison should therefore separate plasma-related adaptation from erythropoietic adaptation. In AKURA terms, a heat block is expected to act mainly on fluid regulation, thermoregulation and cardiovascular stability. An altitude block is expected to act, at least in part, on red-cell production. Both might lower heart rate at a given pace or improve a time trial, but for different reasons. If the AEI records only the performance outcome, it cannot tell which pathway was responsible.

FIELD PROTOCOL

1. Plasma-sensitive pace-HR check
2. Choose a flat, familiar route or a treadmill
3. After an easy day, run 20 minutes at a fixed easy pace in similar conditions each time
4. Record average HR for minutes 10–20, the temperature and humidity, and body mass before and after
5. Repeat before a heat block, at its end and one to two weeks later
6. A lower HR at the same pace in similar conditions is consistent with improved cardiovascular stability, but interpret alongside sleep, illness and training load

MYTH VS EVIDENCE

Myth: If haemoglobin concentration drops after heat training, the athlete's blood has become weaker.

Evidence: A lower concentration can reflect a larger plasma volume diluting the same number of red cells. Total haemoglobin mass and clinical context are needed before drawing conclusions.

CHECKPOINT

If your athlete's heart rate at easy pace falls after a heat block, which explanations can you rule in or out with the data you currently collect?

CHAPTER 22**22. Sweating and Thermoregulation**

PART III — HEAT

In April 2026 the TCS World 10K wound through Bengaluru in conditions that reporters described as hot and humid. Bengaluru, at about 900 m, is supposed to be the pleasant city, the one Chennai runners escape to. Yet Rodrigue Kwizera of Burundi still won the men's race in 27:31, an event record, while the fastest Indian man, the Army runner Harmanjot Singh, clocked 29:13, reported to be one second outside the Indian event record. The gap between them is a story about many things: training, depth, racing experience. But on a humid morning, part of every runner's story is also written on the skin. Each athlete in that field was producing enormous amounts of heat, and each depended on the same cooling system: sweat, and the air's willingness to take it away. When the air is already full of water, that willingness shrinks. Runners on such mornings do not simply get hot. They get wet without getting cool.

Sweating is the body's most powerful cooling tool during running in warm conditions, but it only cools when sweat evaporates. That simple fact explains much of what Indian coastal runners experience. Repeated heat exposure can improve the timing and distribution of sweating and reduce the thermal strain associated with a given workload. The goal is not simply to sweat more; it is to regulate body temperature effectively while maintaining cardiovascular function.

How the body loses heat

A running body produces a large amount of heat as a by-product of muscular work. That heat can be lost through several routes: radiation, convection to moving air, conduction and, most importantly during hard exercise in warm weather, evaporation of sweat from the skin. When air temperature approaches or exceeds skin temperature, radiation and convection contribute little or may even add heat, and evaporation becomes the main route of heat loss.

With acclimation, sweating often begins at a lower core temperature, the sweat rate at a given workload may increase, and sweat may become more dilute as the body conserves sodium. Sweat can also be distributed more effectively across the skin. Together, these changes help the athlete lose heat earlier and more efficiently, which can lower core and skin temperature for the same pace and reduce the strain on the circulation.

KEY IDEA

Sweat cools only when it evaporates. Acclimation improves the timing and efficiency of sweating, but humidity limits how much of that sweat can do its job.

The humidity problem

Humidity complicates this because sweat evaporation becomes less efficient. When the air already holds a lot of water vapour, the gradient that drives evaporation from the skin is smaller. Sweat drips instead of evaporating, soaking clothing without providing proportional cooling. A runner in Chennai may experience a large thermal burden even when air temperature is not extreme. A 30 °C morning with very high humidity can be more stressful than a hotter but drier afternoon inland.

This is also why acclimation to humid heat and dry heat may not be identical in every respect, and why a coach should be cautious when transferring a protocol designed in a dry climate chamber to the streets of a coastal Indian city. Dripping sweat also means greater fluid loss for less cooling, so body-mass change can be substantial in long humid sessions.

Measuring the environment properly

This is why WBGT and local environmental measurement are more informative than temperature alone when designing heat research. Wet-bulb globe temperature combines air temperature, humidity (through the wet-bulb reading), radiant heat from sun and surfaces (through the globe reading) and, indirectly, air movement. It is the basis of many sports-medicine guidelines for modifying or cancelling activity in the heat. A weather application reporting "32 °C" says nothing about the humidity at the athlete's route, the sun on the tarmac or the lack of breeze between buildings.

Condition	Air temperature	Humidity	Likely evaporative cooling
Cool dry morning	Lower	Low	Good
Hot dry afternoon	High	Low	Good if well hydrated
Warm humid morning	Moderate	High	Limited
Hot humid evening	High	High	Poor; highest caution

COACH'S CORNER

Record where and when conditions were measured. A sensor on the balcony at 5:30 a.m. does not describe a sunlit beach road at 7:30 a.m. When you cannot measure WBGT, at least record temperature, relative humidity, cloud cover, wind and time of day, and adjust pace by heart rate and effort rather than by the watch.

Sweat, fluids and body mass

Sweat rates vary widely between athletes and with conditions. Weighing an athlete before and after a session, and correcting for fluid consumed and urine passed, gives a practical estimate of sweat loss. This helps individualise drinking plans without resorting to blanket rules. Large losses repeated day after day may compromise recovery, while drinking far beyond losses can also be harmful. Sodium needs, particularly for heavy salty sweaters in long events, are an individual matter best discussed with a qualified sports dietitian or physician.

FIELD PROTOCOL

1. Simple sweat-rate estimate
2. Athlete empties bladder and records body mass in minimal clothing
3. Run a normal 45–60 minute session and record all fluid consumed
4. Towel dry and weigh again in the same clothing
5. Sweat loss (litres) is approximately pre-mass minus post-mass plus fluid consumed (kg), minus any urine passed
6. Divide by hours to estimate hourly sweat rate
7. Record temperature, humidity and pace with the result and repeat in different conditions

MYTH VS EVIDENCE

Myth: A runner who sweats heavily is poorly conditioned.

Evidence: Well-trained and heat-acclimated athletes often sweat earlier and more. The key question is whether they regulate core temperature and maintain circulation, not how much they sweat.

There is one more practical lesson. Clothing, hair, sunscreen and even the colour and fabric of a vest affect how easily sweat can evaporate and how much radiant heat is absorbed. Light, loose, breathable clothing and running in shade where possible are simple, evidence-consistent choices. None of these replaces acclimation or sensible pacing, but together they reduce the thermal burden an athlete must manage on a humid Chennai morning.

CHECKPOINT

For your athletes, do you know their typical sweat rate in both humid and less humid conditions? Does your heat monitoring use WBGT, or temperature alone?

CHAPTER 23

23. Chennai as a Training Laboratory

PART III — HEAT

On 21 December 1995, Vally Satyabhama ran a marathon in Chennai in 2:39:10, an Indian women's national record at the time. It is a small line in the record books, easy to miss between the later marks O. P. Jaisha set in Mumbai and Beijing. But it is a reminder that serious distance running in this city is not new. Three decades on, the Chennai Marathon has become a mass event: its 2026 edition, the fourteenth, drew 28,500 participants from 16 countries, and Ethiopian runners won the full-marathon titles. Every one of those runners, from the winners to the last walker, ran through the same coastal air. Yet the research dossiers for this book found no published Indian elite results or course records for the race, and no peer-reviewed heat-stress study of any Indian marathon. Tens of thousands of bodies, running through one of the world's great natural heat chambers, and almost none of it measured. That is not a problem. It is an opportunity.

Every environment offers some stimuli and withholds others. Chennai provides a distinctive endurance environment: warm temperatures, high humidity for substantial portions of the year, dense urban running routes and relatively little natural elevation. That combination makes it unsuitable for copying a mountain camp, but potentially useful for studying heat and humidity adaptation. Rather than lamenting what the city lacks, the AKURA approach asks what the city can teach.

The Chennai environment

Chennai sits close to sea level on the Coromandel Coast. Its climate is hot for much of the year, with the peak heat typically in the pre-monsoon months, and it receives much of

its rainfall during the north-east monsoon towards the end of the year. Humidity is often high, especially near the coast and in the early morning, precisely when most runners train. Heat stored in roads and buildings, limited shade on some routes, traffic and air quality add further variables that a laboratory chamber would not reproduce.

For a coach, this means that the environmental dose varies by season, by time of day and by route. A beach-road run at dawn in January and the same route at dusk in May are, physiologically, different sessions. The city therefore offers a natural gradient of heat stress across the year, which can be used for research if it is recorded carefully.

KEY IDEA

Chennai cannot provide hypoxia, but it can provide repeated, measurable heat and humidity. Its value depends on turning daily exposure into controlled data.

Turning a challenge into controlled data

The scientific opportunity is to turn a challenge into controlled data. Compare the same athlete's pace-HR relationship in cool and hot conditions; record body mass change; standardise hydration; track sleep and recovery; and repeat after an acclimation block. None of these steps needs expensive equipment. A heart-rate monitor, a reliable scale, a basic temperature and humidity sensor, and a disciplined logbook are enough to generate useful individual data.

FIELD PROTOCOL

1. Chennai pace-HR heat test
2. Choose a flat, measured route of 3–5 km with some shade, or a track
3. Perform the test on a consistent day of the week after an easy or rest day
4. Record body mass, time, temperature, humidity and, if available, WBGT at the start
5. Warm up 10 minutes, then run 20–30 minutes at a fixed easy pace agreed in advance
6. Record average HR for the final 10 minutes, the HR drift (last 5 minutes minus minutes 5–10) and RPE
7. Weigh again after towel-drying and record fluid intake
8. Repeat in a cooler window (for example December–January dawn) and a hotter window, and again after a 10–14 day acclimation block
9. Stop immediately for any heat-illness warning sign

This test does not require the athlete to run hard, which keeps the risk low and the data comparable. Over months, a pattern emerges: how much the athlete's heart rate rises per degree of WBGT, how quickly drift appears, and whether an acclimation block reduces those responses. That pattern is exactly the kind of individual evidence the AEI is designed to store.

Protecting quality and safety

Training in Chennai also raises practical coaching questions. Hard interval sessions in severe heat may force the athlete to slow down so much that the intended stimulus is lost. Many coaches therefore schedule quality sessions in the coolest available window and use easy runs, when appropriate, as the controlled heat dose. Recovery may be affected by warm nights and poor sleep, so sleep duration and perceived sleep quality should be logged. Pollution, traffic safety and access to water and shade are part of the risk assessment.

Season (approximate)	Typical challenge	Research use
Cooler months	Milder mornings, lower heat load	Baseline pace-HR testing
Pre-monsoon hot months	Highest heat and heat illness risk	Cautious, closely monitored acclimation
Monsoon periods	Rain, humidity, disrupted routes	Humidity-focused observation

COACH'S CORNER

Build a local calendar. Mark the months when baseline testing is most reliable, the months when heat blocks are feasible with low risk, and the periods when heat is so severe that the priority is simply safe, reduced training. Revisit the calendar each year using your own data.

MYTH VS EVIDENCE

Myth: Athletes who grow up in hot cities are automatically heat-adapted and need no precautions.

Evidence: Living in a hot climate provides some acclimatisation, but many people spend much of the day indoors, and adaptation depends on exercise-related heat exposure. Every athlete still needs stop rules, hydration planning and medical support.

The goal is not to glorify heat. It is to understand the dose-response relationship. A Chennai athlete who knows exactly how their body responds to a WBGT of 22 °C versus 28 °C has a genuine advantage in pacing, preparation and safety, not because heat is magical, but because knowledge replaces guesswork.

CHECKPOINT

Which two months of the year would you choose for baseline testing of your Chennai athletes, and which for a monitored heat block? What data would convince you that the block worked?

PART III AT A GLANCE

The key idea from each chapter, for quick revision.

- | | |
|-----------|---|
| 19 | What Heat Actually Does
Heat stress is a training stimulus, not a badge of toughness. The same pace costs more in the heat because the circulation must serve both the muscles and the skin. |
| 20 | Heat Acclimation
Heat acclimation is a dose-response intervention. Its effects depend on how hot, how humid, how long, how hard and how often, and they fade without continued exposure. |
| 21 | Plasma Expansion and Stroke Volume
Plasma expansion can support stroke volume and lower heart rate at a given pace, but it increases the volume of blood, not the number of red cells. |
| 22 | Sweating and Thermoregulation
Sweat cools only when it evaporates. Acclimation improves the timing and efficiency of sweating, but humidity limits how much of that sweat can do its job. |
| 23 | Chennai as a Training Laboratory
Chennai cannot provide hypoxia, but it can provide repeated, measurable heat and humidity. Its value depends on turning daily exposure into controlled data. |

PART IV — ALTITUDE

Altitude is one of the most romanticised ideas in endurance sport. It is often described as a place with "less oxygen" that magically builds champions. Part IV replaces that story with physiology. It explains what actually changes as you climb, why the oxygen fraction of air stays close to 20.9%, how hypoxia can stimulate EPO and red-cell production, what live-high, train-low strategies attempt, how Indian hill stations such as Ooty and Kodaikanal can serve as research sites, and why athletes respond so differently. Altitude is treated here as a dose of hypoxia to be specified, monitored and measured.

IN THIS PART

- 24 What Changes at Altitude?
- 25 The 20.9% Oxygen Truth
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- 27 Living High, Training Low
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- 29 Altitude Responders and Non-Responders

PART IV • ALTITUDE

Altitude Is Pressure, Not Less Oxygen

Altitude is a dose of hypoxia to be specified, monitored and measured — not a magic place that builds champions.

20.9% Oxygen fraction of air — the same in Chennai and Ooty

112 mmHg

Lower inspired PO₂

Inspired PO₂ falls from about 150 mmHg at sea level to about 112 mmHg at Ooty (~2,240 m).

Days 1–4

Start easy

First days bring higher ventilation and HR and lower VO₂max: easy running and sleep first.

EPO

A slow red-cell response

Hypoxia can raise EPO and, over weeks, red-cell mass — modest, dose-dependent and highly variable.

LH TL

Live high, train low

Sleep high for the hypoxic stimulus, train low for quality. A rationale to test, not a guarantee.



“Ooty” is not a dose

The dose is sleeping elevation, hours spent there, training elevation and length of stay.



Responders vary

Iron, sleep, illness, exposure and measurement error shape response. A poor result is information, not a verdict.

THE PROCESS

Baseline at home

Easy days 1–4

Build quality gradually

Log dose daily

Retest after return

Altitude is a hypoxic dose: specify it, log it, and measure each athlete's response.

THE STORY

Above the Tea Line

PART IV — ALTITUDE

The bus leaves Mettupalayam in warm, heavy air, and for the next couple of hours the road does almost nothing but climb. Hairpin follows hairpin. Areca palms give way to eucalyptus, eucalyptus to the clipped green terraces of tea, and somewhere past Coonoor the windows begin to fog and passengers reach for sweaters. By the time the bus pulls into Udhagamandalam, the town everyone calls Ooty, the air smells of wet earth and wood smoke, and mist is drifting across the racecourse like something poured.

Ooty sits at 2,240 m above sea level. That is roughly the height of Iten in Kenya, the town that calls itself the "Home of Champions". Athletes from across India have gone there for altitude training for a long time, *The Hindu* reported in 2024. In June 2026 a ministry official told the *Hindustan Times* that the "Ooty centre is a preferred choice for middle and long distance runners".

A centre, and a question of ownership

In February 2024 the Tamil Nadu government opened a dedicated high-altitude training centre at Brecks Ground in Ooty, inaugurated by Chief Minister M.K. Stalin and Sports Minister Udhayanidhi Stalin. Its residents were 50 athletes selected from government schools, living in a hostel run by the Sports Development Authority of Tamil Nadu. *The Hindu* called it "south India's only high-altitude sports training centre".

Two years later, the *Hindustan Times* described things a little differently. It reported that SAI campuses at Shilaroo in Himachal Pradesh and at Ooty are India's only two high-altitude training centres, and that a third, at 1,496 m in Shillong, is planned with ₹150 crore of CSR funding. The same article quoted an official saying India is "short of roughly 750 coaches". The two newspapers frame the Ooty centre differently, one as a state facility and one as a SAI campus, and the full history of any older national facility in the town could not be confirmed. What is clear is that the hill has become a destination.

Not every Ooty story is an official one. Since 2018 the Indian Track Foundation, led by coach Karan Singh, has trained young runners in Ooty, including tribal youth from Jharkhand, after moving there from the Delhi region. In nearby Coonoor, the Army coach Yunus Khan has worked with the 10K runner Harmanjot Singh. The Nilgiris have become, quietly, a place where Indian distance running goes to think.

India's other mountain is further north. In 2022 the Sports Authority of India ran a 72-day high-altitude camp for junior middle- and long-distance runners at its Shilaroo centre near Shimla, at about 8,000 ft (roughly 2,450 m), with 27 athletes and 11 coaches.

The dossiers behind this book found no sourced record of national distance camps at Kodaikanal or Dharamshala, and no reliable report of an AFI or SAI distance camp in Iten or Kyrgyzstan. The main exception is the Bhopal-based Elite Distance Running Programme, launched by SAI and Procam, whose group under the Dutch coach Hugo van den Broek goes to Iten once a year. The documented Indian altitude map is smaller than the legend: the Nilgiris, Shilaroo, and a handful of foreign towns.

What a Nilgiri camp feels like, though, anyone can imagine. Mornings begin in mist so thick that the far side of the racecourse disappears. Runners move in single file along estate roads between walls of tea, breath showing in the cold, the slopes falling away into cloud. Afternoons are for sleep. Nights are cold enough for blankets, which in itself is a novelty for a runner from Chennai or Madurai. Every road goes either up or down, and the easy runs are rarely easy.

The marathon men

The clearest documented use of Ooty belongs to the marathoners. Before the 2016 Rio Olympics, Surendra Singh Bhandari of the Army Sports Institute ran an altitude block in Ooty for three Army runners who had all qualified at that year's Mumbai Marathon: Thonakal Gopi from Wayanad, Kheta Ram from Barmer and Nitendra Singh Rawat from Garur in Uttarakhand. ESPN described Gopi's preparation as a "ten-month training stint at a high-altitude Ooty facility".

Rawat had already credited the camp for his Mumbai run, which set the Indian course record of 2:15:48.

I was personally confident after the training we had been put through in the national camp at Ooty by our coach (Surinder Singh) Bhandari.

— Nitendra Singh Rawat (Sportskeeda/IANS, 2016)

In Rio, Gopi ran 2:15:25, a personal best, and finished 25th. Kheta Ram finished one second behind him, also in a personal best. Rawat, running with a hamstring injury, finished 84th.

Nearly a decade later, Gopi went back to the hills. Before the 2025 Valencia Marathon he spent four months at Ooty, and in Valencia he ran 2:12:23, only 23 seconds outside what was then the national record. "I think the high altitude at Ooty helped my performance," he said.

The runner who finally broke that record also came through the Nilgiris. Sawan Barwal, an Army runner from Mandi district in Himachal Pradesh, trains in a Reliance Foundation programme that, according to The Bridge, normally spends about four months, from November to February, in Ooty and nearby Wellington, then moves down to Bengaluru for pre-competition work. On 12 April 2026, on his marathon debut in Rotterdam, he ran 2:11:58 and erased Shivnath Singh's 2:12:00, a record that had

stood since 1978. In September he lowered it again, to 2:11:37, winning silver at the Asian Games in Aichi-Nagoya.

The American mountain

India's middle-distance and track runners went further. From 2019 onward, Colorado Springs in the United States, more than 1,800 m up in the Rocky Mountain foothills, became the national distance hub under the American coach Scott Simmons of the American Distance Project. The 1500 m record-holder Jinson Johnson went first. "I am hoping that training in Colorado Springs under coach Simmons would help in bettering my timing," he told the Times of India.

In 2022 the Athletics Federation of India ran a 16-week camp there for Parul Chaudhary, Shankar Lal Swami, Ajay Kumar Saroj and Avinash Sable. In January 2024, under the Target Olympic Podium Scheme, the sports ministry approved Sable and Parul to train there again, paying for airfare, boarding and lodging for athletes, coaches and physiotherapists. The same year Gulveer Singh moved his training base to Colorado Springs; in 2025 he ran 27:00.22 for 10,000 m, the national record.

Sable's coach saw something in the way his athlete handled the thin air. "His adaptation upon his return to altitude this year was almost instant," Simmons told the Hindustan Times in 2023.

St Moritz

Before the 2023 World Championships in Budapest, Sable combined Colorado Springs with St Moritz in the Swiss Alps, at 1,856 m, training alongside the American Paul Chelimo. In July 2024, before the Paris Olympics, he and Parul trained in St Moritz again. That month, at the Paris Diamond League on 7 July, Sable ran 8:09.91, his tenth national record in the steeplechase. At the Olympics he became the first Indian to reach the final of the event and finished 11th.

What nobody has written down

It is a remarkable map: the tea estates of the Nilgiris, the Rockies, the Engadine valley. Indian runners have spent months, sometimes most of a year, living high. And yet, when the researchers behind this book went looking for the numbers that would show what those months did inside the body, they found a gap. There is no peer-reviewed study of Indian athletes' haematology at Ooty. We know how long the athletes stayed. We know the times they ran afterwards. We do not know what happened to their blood.

That gap is where this Part begins. Before we can ask what months in Ooty do to a runner, we need to understand what actually changes when that bus climbs past Coonoor, and why the young runner who steps off it feels breathless on her very first jog, in air that looks exactly like the air at home.

Sources: https://www.sportsauthorityofindia.nic.in/sai/assets/news/1649157667_Middle%20and%20Long%20Distance%20runners%20to%20train%20for%20over%20two%20months%20at%20SAI's%20High%20Altitude%20Training

%20Centre%20in%20Shilatroo.pdf ; <https://worldathletics.org/news/feature/indian-endurance-project-kenya-netherlands-distance-running-broek-pal-murli> ;
<https://www.thehindu.com/news/national/tamil-nadu/udhagamandalam-boasts-south-indias-only-high-altitude-sports-training-centre/article68333440.ece> ; <https://www.hindustantimes.com/sports/others/high-altitude-training-centre-to-come-up-in-shillong-101780588041378.html> ; <https://theprint.in/sport/target-olympics-2032-coach-moves-from-gurugram-to-ooty-to-train-runners-from-jharkhand/715322/> ;
https://www.espn.com/athletics/story/_/id/17475379/india-marathon-runner-olympics-thonackal-gopi-talks-espn-exclusive-interview ; <https://sportskeeda.com/running/nitendra-rawat-credits-coach-success-mumbai-marathon> ;
<https://shyamgopan.com/2025/12/11/gopi-23-secs-short-of-national-record-sets-new-pb-at-valencia-marathon/> ;
<https://thebridge.in/athletics/ajith-markose-decode-sawan-barwal-48-year-old-record-breaking-run-56362> ;
<https://www.olympics.com/en/news/sawan-barwal-breaks-marathon-national-record-india> ;
<https://www.tribuneindia.com/news/sports/who-is-sawan-barwal-himachal-runner-who-challenged-marathon-myth-to-win-asian-games-silver/> ; <https://timesofindia.indiatimes.com/sports/more-sports/athletics/jinson-johnson-to-train-in-us-to-improve-timing-and-qualify-for-olympics/articleshow/70809839.cms> ; <https://sportskeeda.com/athletics/news-challenging-task-india-s-head-coach-scott-simmons-regroup-distance-running> ; <https://thebridge.in/athletics/avinash-sable-parul-chaudhary-to-train-in-colorado-springs-45778> ; <https://www.olympics.com/en/news/gulveer-singh-india-10000m-national-record-the-ten-2025-athletics-california> ; <https://sportskeeda.com/summer-olympics/news-paris-olympics-2024-3000m-steeplechasers-avinash-sable-parul-chaudhary-train-hard-switzerland> ;
<https://www.stmoritz.com/en/altitude-training/> ; <https://www.olympics.com/en/news/who-is-avinash-sable-india-athlete-steeplechase>

CHAPTER 24

24. What Changes at Altitude?

PART IV — ALTITUDE

The bus leaves Mettupalayam in warm, heavy air, and for the next couple of hours the road does nothing but climb. Hairpin follows hairpin. Areca palms give way to eucalyptus, eucalyptus to the clipped green terraces of tea, and somewhere past Coonoor the windows start to fog and passengers reach for sweaters. By the time the bus pulls into Ooty, at about 2,240 m, the air is cool and smells of pine and wet earth. A young runner from the plains, fresh off the bus, laces up for an easy evening jog around the lake. Within minutes, something is wrong. The pace is gentle, but she is breathing as if she were halfway through an interval session, and her heart is pounding. Nothing about the air looks different. If anything it seems cleaner and fresher than the air at home. Whatever has changed cannot be seen at all, and it begins with the weight of the sky above her.

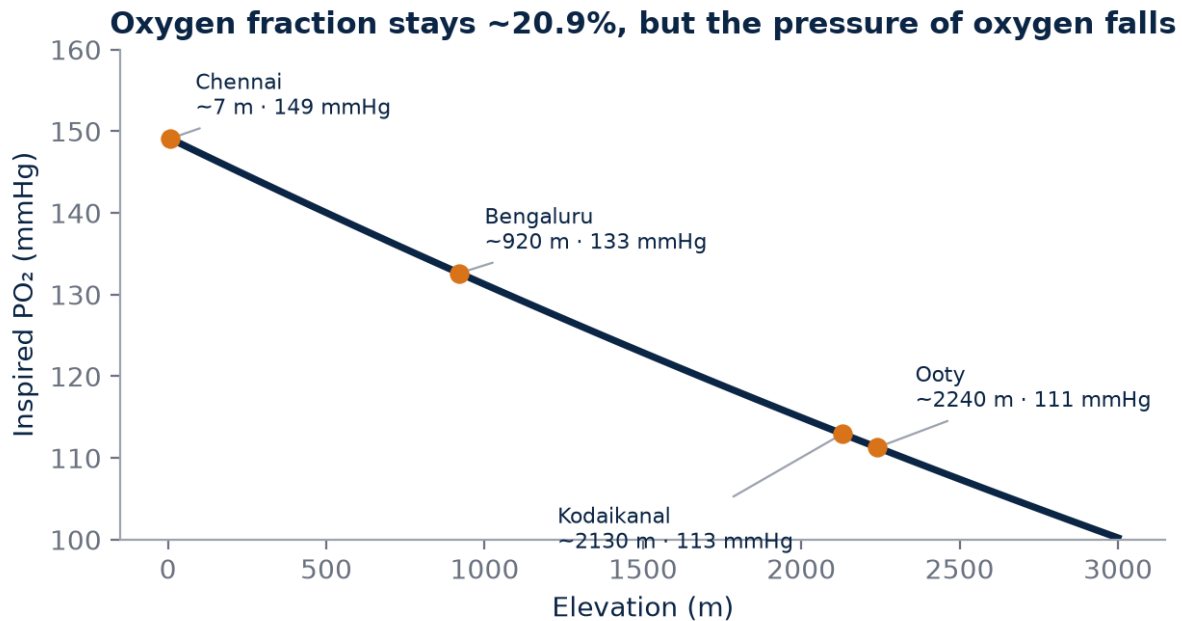


Figure 11. Approximate inspired oxygen pressure by elevation, calculated from the standard-atmosphere model (weather changes actual values). The Indian hill stations offer a moderate hypoxic stimulus.

Ask a group of runners what happens at altitude and most will say, "There is less oxygen." The intuition is understandable, but it hides the real mechanism. At altitude, barometric pressure falls. The fraction of oxygen in air remains approximately 20.9%, but the partial pressure of oxygen is lower. This reduces the oxygen pressure gradient and creates hypoxic stress. Everything an athlete experiences at altitude, from breathlessness on the first run to changes in blood over several weeks, flows from this reduced driving pressure.

The pressure gradient

Oxygen moves from the air in the lungs into the blood, and from the blood into the muscles, because of differences in partial pressure. Oxygen diffuses from higher to lower pressure. When the atmospheric pressure is lower, the partial pressure of oxygen in inspired air is lower, the pressure in the alveoli is lower, and the gradient driving oxygen into the blood is smaller. Arterial oxygen saturation may fall, particularly during hard exercise, and each litre of blood leaving the lungs may carry somewhat less oxygen than at sea level.

The magnitude of the change increases with altitude. The table below gives approximate values based on a standard atmosphere model. Real values vary with weather, season and latitude (tropical locations tend to have slightly higher pressure at a given altitude than the standard model), so these numbers illustrate the trend rather than exact local conditions.

Location (approx. elevation)	Barometric pressure (approx.)	Inspired PO ₂ (approx.)
Sea level, e.g. Chennai	760 mmHg	about 150 mmHg
~900 m, e.g. Bengaluru	about 680 mmHg	about 133 mmHg
~2,000 m	about 595 mmHg	about 115 mmHg
~2,200 m, e.g. Ooty	about 580 mmHg	about 112 mmHg
~2,400 m	about 565 mmHg	about 109 mmHg

Inspired PO₂ here refers to air warmed and saturated with water vapour in the airways, which is why it is lower than a simple 20.9% of barometric pressure. The key message is the direction and scale: at around 2,000–2,400 m, the oxygen pressure available to the lungs is roughly a quarter lower than at sea level.

KEY IDEA

Altitude is a hypoxic stimulus created by lower barometric pressure, not a mysterious source of "more oxygen" or a place where air contains a different percentage of oxygen.

Acute responses: the first days

Acute responses include increased ventilation and heart rate. Within minutes to hours of arrival, breathing increases to compensate for the lower oxygen pressure, and heart rate at rest and during submaximal exercise is often higher. Maximal aerobic capacity is typically reduced, and the reduction becomes more noticeable as altitude increases. As a result, a pace that was comfortable at sea level may feel harder, and interval sessions at sea-level target paces may be unattainable or excessively stressful.

Sleep can be disturbed in the first nights, and some people experience headache, poor appetite or unusual fatigue. At the moderate altitudes of Indian hill stations, serious altitude illness is uncommon in healthy athletes, but symptoms should not be ignored. Persistent or worsening headache, breathlessness at rest, confusion or unsteadiness warrant stopping training, descending if symptoms are severe, and seeking medical assessment.

Longer exposure: days to weeks

Over longer exposure, the body can alter fluid balance and stimulate erythropoietic pathways. In the first days, plasma volume often decreases, which raises haemoglobin concentration even before any new red cells are produced. Over subsequent weeks, erythropoietin signalling may increase red-cell production. Ventilatory acclimatisation continues, and many athletes find that training feels progressively more normal. The magnitude depends on altitude, duration, individual response and iron availability.

Time frame	Typical response	Coaching implication
First hours to days	Higher ventilation and HR; reduced VO ₂ max	Reduce intensity; prioritise easy running and sleep
First days	Plasma volume may fall; Hb concentration rises	Do not mistake concentration rise for new red cells
Weeks	Possible increase in red-cell production	Needs sufficient time, dose and iron availability
After return	Adaptations decline over weeks	Plan competition timing and retesting

COACH'S CORNER

Plan the first three to five days of any altitude camp as an adjustment phase. Reduce volume and intensity, monitor morning heart rate, sleep and wellbeing, and delay hard sessions until the athlete's responses stabilise.

MYTH VS EVIDENCE

Myth: Arriving at altitude gives an athlete an immediate boost.

Evidence: The immediate effect is usually a reduction in aerobic capacity and a higher heart rate at a given pace. Any benefit depends on sufficient exposure, recovery and adaptation over weeks.

For a coach in India, these principles translate directly into the design of a hill-station camp. The athlete's first runs in Ooty or Kodaikanal are not a fair test of fitness; they are a measure of acute hypoxic stress. Only after the adjustment phase does training at altitude begin to resemble normal training, and even then pace targets may need to be revised downwards for sustained efforts. Recording heart rate, pace and perceived effort from the first day allows the coach to see the adjustment curve rather than guessing it, and to compare it with future camps.

CHECKPOINT

If your athlete's heart rate is 10 beats higher at the same easy pace on day two of a camp, how will you adjust the plan, and what would you record?

CHAPTER 25

25. The 20.9% Oxygen Truth

PART IV — ALTITUDE

In June 2026 the Hindustan Times published three numbers that neatly map India's high-altitude ambitions. The Ooty centre sits at 2,240 m. Shilaroo, near Shimla, sits at about 2,450 m. A planned centre in Shillong, costed at ₹150 crore, would sit at 1,496 m. Ask a group of young athletes what separates those three places and most will answer with a single phrase: less oxygen as you go higher. It sounds right. It is the kind of thing coaches say on the bus up the ghat road and athletes repeat for the rest of their careers. And it is wrong in a way that matters. Take a sample of air from the Marina, from the Ooty racecourse and from the ridge above Shilaroo, and the share of oxygen in each is almost exactly the same. What differs is something the eye never sees and the athlete feels in every breath.

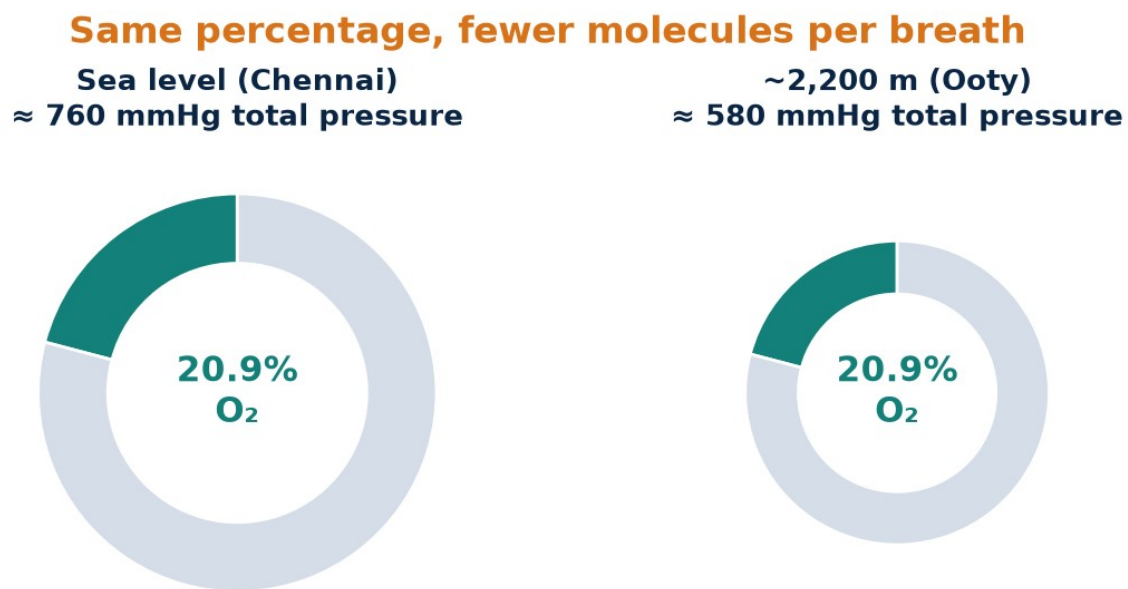


Figure 12. The ring size represents total air pressure. The oxygen fraction is unchanged at altitude; what falls is barometric pressure, and with it the partial pressure of oxygen.

Few phrases are repeated more often in endurance sport than "there is less oxygen at altitude". It sounds harmless, and in everyday conversation it communicates something real: breathing is harder up there. But the way it is usually understood, as a lower percentage of oxygen in the air, is wrong. The statement "there is less oxygen percentage at altitude" is a common shorthand, but scientifically it is inaccurate for normal atmospheric air. The oxygen fraction remains close to 20.9%.

Fraction versus pressure

The atmosphere is a well-mixed gas up to altitudes far higher than any runner will ever train. On a beach in Chennai, on the Ooty racecourse or on a Himalayan pass, roughly 20.9% of the molecules in dry air are oxygen. What falls is atmospheric pressure. Because partial pressure is related to gas fraction multiplied by total pressure, the partial pressure of oxygen declines. Each breath therefore contains fewer oxygen molecules per unit volume than at sea level.

A simple calculation makes the point. At sea level, with a barometric pressure of about 760 mmHg, the partial pressure of oxygen in dry air is roughly 0.209×760 , or about 159 mmHg. At around 2,200 m, with a barometric pressure of about 580 mmHg, it is roughly 0.209×580 , or about 121 mmHg. The percentage has not changed; the pressure, and therefore the number of oxygen molecules in each breath of a given volume, has.

KEY IDEA

The oxygen percentage of air stays near 20.9% at altitude. What falls is barometric pressure, and with it the partial pressure of oxygen that drives oxygen into the blood.

Why the distinction matters

This distinction is fundamental to understanding altitude physiology. It explains why the body responds by breathing more (to move a greater volume of thinner air), why blood oxygen saturation can fall during hard efforts, and why the stimulus to the kidneys that release erythropoietin is related to oxygen availability in the tissues rather than to a change in the composition of air.

It also clarifies how simulated altitude works. Hypoxic tents and rooms usually reduce the oxygen fraction of the air at normal pressure, creating "normobaric hypoxia", whereas real altitude is "hypobaric hypoxia", with a normal fraction at reduced pressure. Both lower the inspired partial pressure of oxygen, which is the main driver of the physiological response, although researchers continue to debate whether the two conditions are fully equivalent in every respect. For AKURA, this means that simulated and natural altitude should be recorded as different interventions, with the method, the equivalent altitude and the hours of exposure specified.

Situation	Oxygen fraction	Total pressure	Inspired PO ₂
Sea level, normal air	About 20.9%	Normal	Normal
Natural altitude (hypobaric)	About 20.9%	Reduced	Reduced
Hypoxic tent (normobaric)	Reduced	Normal	Reduced
Hill running at sea level	About 20.9%	Normal	Normal

Hills are not altitude

The same reasoning separates hill training from altitude training. A steep climb near sea level makes breathing hard because the workload is high, not because the air is thinner. Hill running can build strength, running economy on gradients and tolerance for high ventilation, but it does not create hypoxia. Ordinary hill running should not be described as automatically increasing lung size or producing altitude-type blood adaptations.

COACH'S CORNER

Teach athletes the correct language. Instead of "there's less oxygen up there", say "the air pressure is lower, so each breath delivers less oxygen". Correct language prevents incorrect training decisions, such as believing that a hilly loop in Chennai is a substitute for living in Kodaikanal.

MYTH VS EVIDENCE

Myth: Air at altitude contains a lower percentage of oxygen.

Evidence: The oxygen fraction of normal atmospheric air remains about 20.9% at altitude. The partial pressure of oxygen falls because total atmospheric pressure falls.

FIELD PROTOCOL

1. Explaining partial pressure to athletes
2. Show the formula: partial pressure $\approx$ oxygen fraction $\times$ barometric pressure
3. Calculate sea level: $0.209 \times 760 \approx 159$ mmHg
4. Calculate Ooty: $0.209 \times \text{about } 580 \approx 121$ mmHg
5. Ask athletes what changed, the fraction or the pressure
6. Link the answer to why easy pace feels harder on the first days of a camp

There is also a practical temperature point that often confuses athletes. Air at hill stations is usually cooler, and cool air is denser than warm air at the same pressure. This does not offset the effect of lower barometric pressure in any meaningful way for an athlete; the inspired partial pressure of oxygen is still lower in Ooty than in Chennai. Cooler air may make training feel more comfortable, but it does not cancel the hypoxic stimulus, which is precisely why altitude-plus-cool camps must be interpreted carefully.

CHECKPOINT

Can each of your athletes explain, in one sentence, why breathing is harder at altitude without saying "less oxygen percentage"? How will you record simulated versus natural altitude exposure?

CHAPTER 26

26. EPO and Red-Cell Adaptation

PART IV — ALTITUDE

Before the 2016 Rio Olympics, Thonakal Gopi, an Army runner from Wayanad, spent what ESPN described as a "ten-month training stint at a high-altitude Ooty facility". In Rio he ran 2:15:25, a personal best, and finished 25th; Kheta Ram, who had trained in the same coach's Ooty block, finished one second behind him. Nearly a decade later, before the 2025 Valencia Marathon, Gopi spent four more months at Ooty and ran 2:12:23, only 23 seconds outside the national record of the day. His own verdict was simple: "I think the high altitude at Ooty helped my performance." It may well have. But here is what the public record does not contain: a single peer-reviewed measurement of what months in Ooty do to the blood of Indian athletes. We know the months. We know the times. We do not know the haemoglobin mass. And haemoglobin mass is where the altitude story is supposed to live.

The most famous proposed benefit of altitude training is an increase in red cells. The logic is appealing: more red cells means more haemoglobin, more haemoglobin means more oxygen-carrying capacity, and more oxygen delivery should support faster running. Each link in that chain is plausible, but each is also conditional. Hypoxia can stimulate erythropoietin signalling. EPO acts on the bone marrow to increase red-cell production over time. Increased red-cell mass can increase the blood's oxygen-carrying capacity.

The EPO pathway

Erythropoietin is a hormone produced mainly by the kidneys in response to reduced oxygen availability. Within hours of arriving at altitude, EPO levels in the blood can rise. EPO then stimulates the bone marrow to produce more red cells. Because red cells take time to mature and the circulating red-cell population turns over slowly, measurable increases in total haemoglobin mass require weeks rather than days. Interestingly, EPO concentration often peaks early and then falls back towards baseline even while the athlete remains at altitude, so a single EPO measurement is a poor guide to eventual adaptation.

The most useful outcome measure for this pathway is total haemoglobin mass, which can be estimated with carbon-monoxide rebreathing methods in specialised laboratories. A meta-analysis by Gore and colleagues, published in the *British Journal of Sports Medicine* in 2013, pooled studies using this method and estimated an average

increase in haemoglobin mass of roughly 1% per 100 hours of altitude exposure, with considerable individual variability. That figure implies that a three-week camp with most of each day spent at altitude might, on average, produce a modest increase of a few per cent, and that short or partial exposures produce correspondingly less.

KEY IDEA

Hypoxia can increase EPO and, over weeks, red-cell mass. The average effect is modest, dose-dependent and highly variable between individuals.

Conditions for a response

But an athlete does not automatically become faster simply because EPO rises. The response needs sufficient exposure, adequate iron availability and enough time. The training programme must also maintain quality. Several factors influence whether an increase in red-cell production actually occurs and whether it translates into performance.

Factor	Why it matters
Hours of exposure	Hb-mass response is related to cumulative hypoxic time
Altitude level	Too low may give little stimulus; too high may impair sleep and training
Iron availability	Red-cell production requires iron
Illness and inflammation	Can blunt the response
Sleep and energy intake	Recovery and energy availability support adaptation
Training quality	Performance gains need maintained training stimulus

Iron deserves special mention. An athlete with insufficient iron stores may not respond as expected to an altitude intervention. Ferritin and broader blood assessment before a camp can be relevant in a supervised sports-medicine programme. Iron supplementation should be decided by a qualified clinician based on test results, not by a coach or athlete guessing, because excessive iron carries its own risks.

COACH'S CORNER

Four to six weeks before a planned altitude block, arrange for the athlete to discuss blood testing, including iron status, with a qualified doctor. Record the results in the AEI profile so that any later haemoglobin changes can be interpreted in context.

Asking the right question

It is tempting to ask, "Does altitude work?" The history of research on this question is full of apparently contradictory findings, partly because studies differed in altitude, duration, hours per day, training, athlete level and measurement methods. The correct

research question is not "Does altitude work?" but "For which athlete, at what dose, under what protocol, and with what measurable response?"

FIELD PROTOCOL

1. Red-cell adaptation monitoring plan
2. Baseline: blood assessment including iron status as advised by a doctor, and Hb mass if laboratory access exists
3. Record daily hours at altitude, sleeping elevation and training elevation
4. Retest Hb concentration only with the same laboratory, time of day and posture, noting that early rises may reflect plasma loss
5. Retest Hb mass, where available, near the end of the block and after return
6. Pair blood measures with a standard performance or pace-HR test

MYTH VS EVIDENCE

Myth: A higher haemoglobin concentration after one week at altitude proves new red cells have been made.

Evidence: Early rises in concentration often reflect a fall in plasma volume. New red-cell mass takes weeks and is best assessed by total haemoglobin mass.

Finally, red-cell adaptation is not permanent. After returning to sea level, the stimulus to produce extra red cells disappears, and haemoglobin mass is generally expected to return towards baseline over the following weeks as red cells reach the end of their natural lifespan. The timing of competition after a camp is therefore a practical question with no universal answer; coaches often balance the potential benefit of higher red-cell mass against the fatigue of travel and the need to re-establish sea-level race pace.

CHECKPOINT

For your next altitude plan, how many total hours of exposure will the athlete accumulate, and how will you know whether iron availability was adequate?

CHAPTER 27

27. Living High, Training Low

PART IV — ALTITUDE

Sawan Barwal, an Army runner from Mandi district in Himachal Pradesh, trains in a programme that, according to The Bridge, normally spends about four months, from November to February, in Ooty and nearby

Wellington, then comes down to Bengaluru for pre-competition work. "For Sawan, most of the time he was staying in high altitude," his coach Ajith Markose told The Bridge. On 12 April 2026, at the Rotterdam Marathon, he ran 2:11:58 and broke Shivnath Singh's national record of 2:12:00, a mark that had stood since 1978. Five months later, at the Asian Games in Aichi-Nagoya, he lowered it to 2:11:37 and won silver, India's first men's marathon medal at the Games since 1982. Look at the shape of that preparation: months high in the hills, then a descent to a lower base before racing. It was not a laboratory protocol. But it echoes one of the most famous ideas in altitude research, the attempt to live high and train low.

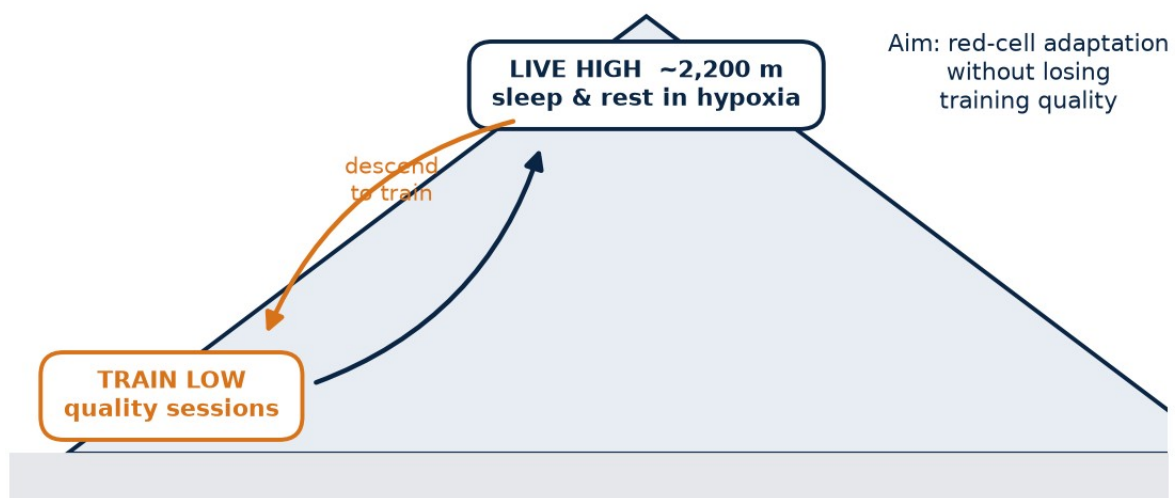


Figure 13. Live high, train low: the athlete sleeps at altitude for the hypoxic stimulus and descends for high-quality training. Responses vary between athletes.

One of the central problems of altitude training is that the environment which may stimulate red-cell production also makes hard training more difficult. At altitude, the athlete cannot usually run intervals at sea-level speeds. Over weeks, a lower training intensity might offset some of the benefit of any blood adaptation. Live-high, train-low strategies attempt to combine the physiological stimulus of hypoxia with the ability to perform high-quality running at a lower altitude. The rationale is to obtain erythropoietic adaptation without sacrificing training intensity.

The classic study

The best-known study of this approach was published by Benjamin Levine and James Stray-Gundersen in the *Journal of Applied Physiology* in 1997. Trained runners were divided into groups that lived and trained near sea level, lived and trained at moderate altitude, or lived at moderate altitude (about 2,500 m) while travelling down to train at a lower altitude (about 1,250 m). After four weeks, the live-high, train-low group improved 5000 m race performance compared with the sea-level control group. The

researchers attributed the benefit to a combination of increased red-cell mass and preserved training quality.

That study became the foundation of many national altitude programmes and later variants: living in hypoxic tents or rooms, training high for easy runs and low for key sessions, or using supplemental oxygen during hard sessions at altitude. Subsequent research from the same group and others also highlighted considerable individual variation in response.

KEY IDEA

Live-high, train-low aims to separate the hypoxic stimulus (sleeping and living high) from training quality (running hard sessions low). It is a rationale to test, not a guarantee.

Variable results

Results vary across protocols and athletes. The literature contains both positive and null findings, and response variability is important. Some well-controlled studies, including placebo-controlled designs in which athletes did not know whether their tents were hypoxic, have reported little or no performance benefit. Others report meaningful gains. Differences in hours per day, altitude, duration, athlete level, iron status and the precision of performance measurement all contribute. A change of one or two per cent in performance is important for elite athletes but is also close to the day-to-day variability of many tests, which makes detection difficult.

Variant	How it works	Main practical challenge
Classic LHTL	Live high, travel down to train	Requires two suitable sites close together
Live high, train high and low	Easy runs high, key sessions low	Travel logistics; tracking where each session occurs
Simulated LHTL	Hypoxic tent or room at home	Cost; sleep disruption; dose verification
Live and train high	All sessions at altitude	Intensity may need to fall

An Indian perspective

The Nilgiri and Palani hills offer interesting possibilities. An athlete could, in principle, live in Ooty or Kodaikanal and descend for key sessions, but the road descents are long and winding, so travel time and fatigue must be considered honestly. In practice, many Indian coaches may find it more realistic to reduce intensity at altitude and schedule the most important sessions before or after the camp, or to modify pace targets using heart rate and effort. Each of these is a different intervention and should be recorded as such.

AKURA should therefore avoid treating "live high, train low" as a guaranteed recipe. It is one intervention to be tested against the athlete's baseline and control condition. The athlete's own history, the logistics of the location and the goals of the season all determine whether it is sensible.

COACH'S CORNER

Before promising an athlete a live-high, train-low benefit, write down the sleeping elevation, training elevations, hours per day, number of days, planned key sessions and the tests you will use before and after. If you cannot specify these, you are not yet ready to run the intervention.

FIELD PROTOCOL

1. Altitude camp logging protocol
2. Daily: sleeping location and elevation, hours spent above target altitude, waking HR, sleep duration and quality, body mass, wellbeing score
3. Per session: training elevation, duration, distance, pace, average HR, RPE, temperature and humidity
4. Weekly: summary of total hypoxic hours and training load
5. Before and after: standard performance test and pace-HR test at the same location and conditions, blood tests as advised by a doctor
6. Note illnesses, travel days and missed sessions

MYTH VS EVIDENCE

Myth: Live high, train low always improves sea-level performance.

Evidence: Some studies show benefits and others show little or none. The outcome depends on dose, protocol, logistics and the individual athlete.

A further practical point concerns the return to lower altitude. Some athletes report feeling flat for several days after coming down from the hills, while others feel strong almost immediately. The reasons are not fully understood and likely include travel fatigue, changes in training load and individual physiology. Rather than relying on popular rules about the perfect number of days before a race, the AKURA approach is to log how each athlete performs in the days and weeks after returning, building an individual timing profile over several camps.

CHECKPOINT

If your athlete lived in Ooty for three weeks, where would the key sessions take place, and how would you know whether training quality was preserved?

CHAPTER 28

28. Ooty and Kodaikanal as Indian Research Sites

PART IV — ALTITUDE

In February 2024, Chief Minister M.K. Stalin and Sports Minister Udhayanidhi Stalin opened a dedicated high-altitude training centre at Breeks Ground in Ooty. Its first residents were not Olympians but 50 athletes selected from Tamil Nadu's government schools, housed in a hostel run by the Sports Development Authority of Tamil Nadu. The Hindu called it "south India's only high-altitude sports training centre". In 2026 a ministry official told the Hindustan Times that the "Ooty centre is a preferred choice for middle and long distance runners". So India has the altitude, the athletes and the centre. What it does not yet have, according to the research behind this book, is a published peer-reviewed study of Indian athletes' blood at Ooty. The two newspapers do not even describe the place the same way: The Hindu writes of a state centre, the Hindustan Times of a SAI campus. A mountain is not a laboratory until somebody writes things down.

India does not need to borrow mountains. The Western Ghats contain hill stations at elevations comparable with some well-known international altitude training sites, and many Indian coaches already take athletes there. Indian hill stations such as Ooty and Kodaikanal can provide useful altitude environments, but the exact physiological stimulus depends on elevation, sleeping altitude, time spent there and training load. A coach should never label a location "altitude training" without specifying the actual exposure.

The sites

Ooty (Udhagamandalam), in the Nilgiri Hills of Tamil Nadu, lies at roughly 2,200–2,240 m for the main town. Kodaikanal, in the Palani Hills, lies at roughly 2,100–2,130 m. Both fall within the moderate-altitude range often discussed for altitude training, typically described as roughly 1,800–2,500 m or a little higher, although the exact optimal altitude remains debated. For comparison, Bengaluru, at about 900–920 m, is too low to be considered a conventional altitude site but is not at sea level either; its milder climate may matter as much as its elevation.

Location	Approx. elevation	Environmental character
Chennai	Near sea level	Hot, often humid, coastal
Bengaluru	About 900–920 m	Milder, moderate elevation
Kodaikanal	About 2,100–2,130 m	Cool, hilly, moderate altitude
Ooty (Udhagamandalam)	About 2,200–2,240 m	Cool, hilly, moderate altitude

These figures describe the town centres. Accommodation, training routes and tracks within and around each town can lie hundreds of metres higher or lower, and routes often include steep gradients. A camp that sleeps at 2,000 m and trains on roads between 1,800 and 2,300 m provides a different stimulus from one that sleeps at 2,250 m and trains mainly on flat ground at the same height.

KEY IDEA

"Ooty" or "Kodaikanal" is not a dose. The dose is the sleeping elevation, the hours spent there, the training elevations and the duration of the stay.

Confounding factors

Travel itself can affect sleep, hydration and training. The journey up from the plains involves long road travel, and the first days may be disrupted by fatigue, new food and unfamiliar sleeping arrangements. Temperature can also change with elevation, creating a mixed altitude-plus-cooler-environment intervention. An athlete from Chennai who spends three weeks in Kodaikanal may improve partly because of hypoxia, partly because cooler conditions allow higher-quality training and better sleep, partly because of a dedicated camp environment with fewer distractions, and partly because of increased training volume on hilly terrain.

This is not a weakness of Indian hill stations; it is simply the reality of field research. It does, however, mean that performance improvements after a camp cannot automatically be attributed to altitude. A comparison group, or a comparison period at a cool, low-altitude location, would help separate these effects, though such designs are often difficult in practice.

Possible reason for improvement	How to check
Hypoxic adaptation	Hb mass change, where available; hours of exposure
Cooler training conditions	Compare session quality and HR with home training
Better sleep and fewer distractions	Sleep log; wellbeing scores
Increased hill training	Record elevation gain and terrain
Increased training load	Weekly load before and during camp

Recording the camp properly

A rigorous study records location, elevation, sleeping hours, training altitude and environmental conditions rather than simply writing "hill camp". A phone GPS or barometric watch can give a reasonable estimate of elevation, but readings should be checked against known reference points, because barometric altimeters drift with weather changes.

COACH'S CORNER

Keep a camp sheet for every athlete. At minimum it should show where they slept, for how long, at what elevation, where each session took place, and what the weather was. Two athletes at the same camp can receive very different doses if one spends afternoons in the town below or sleeps poorly.

FIELD PROTOCOL

1. Indian hill-station camp protocol
2. Two to four weeks before: baseline pace-HR test and performance test at home, and medical and blood assessment as advised
3. Travel day: easy or no training, focus on fluids and sleep
4. Days 1–4: easy running only, monitor waking HR, sleep and symptoms
5. Days 5 onwards: introduce quality gradually, adjusting pace targets by HR and effort
6. Log daily sleeping elevation, hours at altitude, training elevation and weather
7. Final days: repeat pace-HR test at altitude for within-camp comparison
8. After return: retest at home in comparable conditions at planned intervals

MYTH VS EVIDENCE

Myth: Any hill station is altitude training.

Evidence: A location becomes an altitude intervention only when elevation, sleeping hours, training altitude and duration are sufficient and recorded. A short weekend trip provides little hypoxic dose.

Hill stations also bring practical considerations that deserve planning. Monsoon rain and mist can make roads slippery and reduce visibility, some routes carry heavy tourist traffic in holiday seasons, and steep descents increase eccentric load on the legs. Accommodation, food quality and access to medical facilities vary. A camp calendar that avoids the heaviest rain and busiest tourist periods, and a clear plan for injury and illness, makes the altitude stimulus easier to deliver safely and consistently.

CHECKPOINT

For your last camp, could you state the sleeping elevation, total hours at altitude and training elevation for each athlete? Which other factors might explain the improvement you observed?

CHAPTER 29

29. Altitude Responders and Non-Responders

PART IV — ALTITUDE

In 2022 the Athletics Federation of India sent four of its best distance runners to Colorado Springs for a 16-week altitude camp under the American coach Scott Simmons: Parul Chaudhary, Shankar Lal Swami, Ajay Kumar Saroj and Avinash Sable. Same town, same coach, same thin air above 1,800 m. If altitude were a simple machine, four athletes would have gone in and four improved athletes would have come out. According to Sportskeeda's reporting that October, only Sable clearly benefited; that season he lowered his steeplechase record to 8:11.20 and won Commonwealth Games silver. Yet the same summer Parul ran 8:57.19 for 3000 m in Los Angeles, a national record. Which verdict is right depends on which marker, which event and which moment you choose. It would be easy, and unfair, to sort the four into winners and losers. But the camp poses the question at the heart of this chapter more sharply than any textbook: why does the same mountain give different athletes such different things?

Put twenty athletes through the same altitude camp and they will not return with twenty identical results. Some improve their haemoglobin mass and race times; others show little change; a few may perform worse. Athletes do not respond identically to altitude. Differences may arise from exposure dose, iron status, baseline Hb mass, genetics, sleep, ventilation and individual physiology. The variability is not noise to be ignored; it is one of the most important findings in altitude research.

Sources of variability

Chapman, Stray-Gundersen and Levine (1998), analysing runners in live-high, train-low research, described athletes whose performance improved as "responders" and those whose performance did not as "non-responders", and reported differences between the groups in EPO and red-cell responses. The Gore et al. (2013) meta-analysis also emphasised the wide spread of individual haemoglobin mass responses around the average.

Several factors plausibly contribute. An athlete with low iron stores may lack the raw material for red-cell production. An athlete who already has a high haemoglobin mass may have less room for improvement. Poor sleep at altitude may limit recovery, and differences in the ventilatory response may affect oxygen saturation. Illness during the camp, reduced energy intake or excessive training load can blunt adaptation. Inherited biology may also play a role, but it should be treated as background context rather than a convenient explanation for every disappointing result.

Possible contributor	What could be measured
Insufficient exposure	Hours at altitude, sleeping elevation
Low iron availability	Ferritin and blood assessment via a doctor
High baseline Hb mass	Hb mass, where available
Poor sleep	Sleep duration and quality log
Illness or overload	Symptom log; training load
Measurement error	Repeat tests; known test reliability

KEY IDEA

Response to altitude is individual and variable. A disappointing result is information about that athlete, dose and protocol, not a verdict on the athlete's potential.

Using the label carefully

The concept of a "non-responder" should be used carefully. Failure to improve a single marker may reflect insufficient exposure or measurement error rather than a permanent inability to adapt. Haemoglobin mass measurement has a typical error of a couple of per cent even in good laboratories, which is similar in size to the expected average adaptation from a short camp. Performance tests are also affected by weather, motivation, fatigue and timing. An athlete labelled a non-responder after one camp might respond to a longer exposure, better iron status, or a better-timed retest.

There is also a human risk. Telling an athlete they are "a non-responder" can become a self-fulfilling belief, discouraging effort or leading to the conclusion that their body is fundamentally limited. AKURA avoids fixed labels based on limited evidence.

Response as a trend

AEI should track response as a confidence interval and trend, not as a permanent label. In practice, this means recording each altitude exposure with its dose, the measured changes, and an honest estimate of uncertainty. Over several exposures, a pattern may emerge: perhaps an athlete consistently shows clear improvement after longer camps but not after short ones, or responds well only when iron status is adequate.

COACH'S CORNER

After each camp, describe the athlete's response in three parts: what changed, how confident you are that it was a real change beyond normal test variation, and what might explain it. Avoid the words "responder" or "non-responder" unless you have repeated evidence under well-controlled conditions.

FIELD PROTOCOL

1. Individual response record
2. List the dose: days, sleeping elevation, hours at altitude
3. List the pre and post values for each marker (pace-HR, performance test, Hb measures if available)
4. Note each test's typical error, estimated from repeated baseline tests if possible
5. Classify each change as likely real, unclear or likely noise
6. Note conditions that could explain the result: illness, iron, sleep, load
7. Compare with the athlete's previous camps before drawing conclusions

MYTH VS EVIDENCE

Myth: Some athletes simply cannot benefit from altitude.

Evidence: Single-camp results are affected by dose, iron status, illness and measurement error. Permanent non-response should not be assumed without repeated, well-controlled evidence.

CHECKPOINT

If an athlete showed no improvement after a camp, which three explanations would you investigate before concluding that altitude does not suit them?

PART IV AT A GLANCE

The key idea from each chapter, for quick revision.

- | | |
|-----------|---|
| 24 | <p>What Changes at Altitude?</p> <p>Altitude is a hypoxic stimulus created by lower barometric pressure, not a mysterious source of "more oxygen" or a place where air contains a different percentage of oxygen.</p> |
| 25 | <p>The 20.9% Oxygen Truth</p> <p>The oxygen percentage of air stays near 20.9% at altitude. What falls is barometric pressure, and with it the partial pressure of oxygen that drives oxygen into the blood.</p> |
| 26 | <p>EPO and Red-Cell Adaptation</p> <p>Hypoxia can increase EPO and, over weeks, red-cell mass. The average effect is modest, dose-dependent and highly variable between individuals.</p> |
| 27 | <p>Living High, Training Low</p> <p>Live-high, train-low aims to separate the hypoxic stimulus (sleeping and living high) from training quality (running hard sessions low). It is a rationale to test, not a guarantee.</p> |

28

Ooty and Kodaikanal as Indian Research Sites

"Ooty" or "Kodaikanal" is not a dose. The dose is the sleeping elevation, the hours spent there, the training elevations and the duration of the stay.

29

Altitude Responders and Non-Responders

Response to altitude is individual and variable. A disappointing result is information about that athlete, dose and protocol, not a verdict on the athlete's potential.

PART V — HEAT VS ALTITUDE

In Indian running circles, a question keeps returning: if heat can expand plasma volume and lower heart rate, can a Chennai summer do what a mountain camp does? Part V addresses that question directly and without slogans. It contrasts plasma-volume adaptation with red-cell adaptation, asks whether heat can replace altitude, and considers whether the two stresses should ever be combined. The answer is neither "yes" nor "no" but "define the adaptation you want, then measure it". Heat and altitude overlap in some outcomes and differ fundamentally in their mechanisms, and good coaching depends on knowing which is which.

IN THIS PART

- 30** Plasma Versus Red-Cell Adaptation
- 31** Can Heat Replace Altitude?
- 32** Combining Heat and Altitude

PART V • HEAT VS ALTITUDE

Define the Adaptation, Then Choose the Stress

Heat and altitude overlap in some outcomes but differ fundamentally in mechanism; good coaching knows which is which.

9:08.67

Parul Chaudhary's steeplechase NR, one of three finals in three days

Plasma

Heat targets plasma

Heat mainly raises plasma volume and directly improves thermoregulation, many changes within 1–2 weeks.

Hb mass

Altitude targets red cells

Altitude mainly raises red-cell (Hb) mass, which needs weeks of sufficient exposure.



Opposite Hb signals

Heat may lower Hb concentration by dilution; altitude often raises it early through plasma loss.



Match tool to goal

Hot race: heat acclimation. More red cells: altitude. Limited budget, no travel: heat plus well-planned training.



Stress is not stimulus

Combining heat and altitude raises total load: use only with clear purpose, monitoring and medical support.



10–14 d

Sequence, don't stack

Baseline, a 10–14 day heat block, washout, a 3–4 week altitude block, then retest.

THE PROCESS



Heat is a valuable sea-level tool, but not a copy of altitude: define the target first.

THE STORY

Three Finals in Three Days

PART V — HEAT VS ALTITUDE

Late September 2026, Aichi-Nagoya. On 26 September, Parul Chaudhary lines up for the 1500 m final at the Asian Games and finishes fourth in 4:09.46, a personal best, just outside the medals. The next day she is back on the track for the 3000 m steeplechase. Seven and a half laps, twenty-eight barrier jumps and seven water jumps. Yavi wins in 9:04.36; Tanui is second in 9:06.82. Parul is third in 9:08.67, a new Indian national record. On 28 September she runs a third final, the 5000 m, and takes another bronze in 15:14.61.

Three finals in three days, two medals and a national record. Afterwards she told ANI, as reported by Rediff, what the medals meant.

It feels really good. Last time in Hangzhou, I also took two medals, but this time the colour has changed...

— Parul Chaudhary (ANI, via Rediff, 2026)

At the previous Asian Games, held in Hangzhou in 2023, she had won gold in the 5000 m and silver in the steeplechase.

From the sugarcane paths

Parul was born on 15 April 1995 in Iklauta village near Meerut in Uttar Pradesh, into a farming family. Her first race was a school 800 m in 2011, run barefoot, at her father's suggestion, and she trained on the paths between the sugarcane fields. "Papa asked me if I wanted to run and I agreed. I had never practised before that school race," she told the Indian Express in 2023.

The job came before the fame, as it does for so many Indian runners. In 2015 she joined Western Railway as a travelling ticket examiner through the sports quota. In 2024 she became a Deputy Superintendent of Police in Uttar Pradesh.

A ladder of records

Between 2022 and 2026 Parul broke national records again and again, in three different events. In July 2022 in Los Angeles she ran 8:57.19 for 3000 m, the first Indian woman under nine minutes. In the 5000 m she ran 15:10.35 in California in 2023 and 15:04.26 at the Meeting Nikaïa in Nice in June 2026.

The steeplechase is the ladder that tells the story most clearly. At the 2023 World Championships in Budapest she ran 9:15.31 and finished 11th in the final. In May 2025 she ran 9:13.39 at the Doha Diamond League, then 9:12.46 to win silver at the Asian

Championships in Gumi, South Korea. In Nagoya she took more than three seconds off that.

Not every stop on the ladder went upward. At the Paris Olympics in 2024 she ran 9:23.39 in the steeplechase heats and 15:10.68 in the 5000 m heats, and did not reach either final. At the 2025 World Championships in Tokyo she was 20th overall in the steeplechase heats. Elite careers are not straight lines.

Put the steeplechase marks side by side and the shape is striking. From 9:15.31 in August 2023 to 9:08.67 in September 2026 is a gain of about 6.6 seconds over three years. Spread across seven and a half laps, that is less than a second a lap. At the elite end of the sport, that is how progress looks: not leaps, but small, stubborn increments, each one earned over a season or more.

Two environments

Look at where Parul trains and where she races, and two different environments appear.

The training is high. She has an altitude base in Colorado Springs, more than 1,800 m up, under the American coach Scott Simmons; she spoke to the Indian Express in 2023 "from her training base in Colorado Springs". She was part of the Athletics Federation of India's 16-week camp there in 2022. In January 2024 the sports ministry approved funding for her and Avinash Sable to train there under the Target Olympic Podium Scheme. In July 2024, before the Paris Olympics, the two of them trained in St Moritz in the Swiss Alps, at 1,856 m.

The racing is often in Asia, and often in warm places and warm seasons. Her Asian medals have come in Bangkok, in Hangzhou, in Gumi and now in Nagoya. The records do not tell us the temperature or humidity on any of those race days, and this book will not guess. But anyone who has stood in Bangkok in July, when the 2023 Asian Championships were held there and she was reported to have won steeplechase gold, knows that the air in much of Asia in the racing season is not the air of a Colorado morning.

So the questions write themselves. What does a runner carry down from the mountain, and how long does it last? What does a runner need on a hot evening in an Asian stadium that no mountain can give? If altitude builds oxygen-carrying capacity slowly and loses it slowly, and heat builds cooling and circulatory stability quickly and loses it quickly, where in the calendar should each one go?

What the record shows, and what it does not

It would be easy to turn Parul's season into a neat lesson: altitude base, hot-weather racing, medals. The honest version is more interesting. The published record shows where she trained and what she ran. It does not show her haemoglobin mass, her plasma volume, her sweat rate or whether she did any deliberate heat preparation before Nagoya. The research behind this book found no documented, deliberate heat-

acclimation protocol for Indian elite distance runners in general; for this generation, altitude has been the planned environment and heat, mostly, the incidental one.

She was not alone in Nagoya. Gulveer Singh, who also trains under Simmons in Colorado Springs, won three medals at the same Games: silver in the 10,000 m, silver in the 1500 m and bronze in the 5000 m. Sable, back from knee surgery, finished fourth in the steeplechase. Sawan Barwal, whose programme bases its winters in the Nilgiris and who also trained in Colorado Springs, took silver in the marathon. Different events, different bases, different outcomes. The environment is only one variable among many.

The steeplechase final itself carried a quieter contrast. Fourth, about ten seconds behind Parul, was Ankita Dhyani in 9:19.24, a personal best. Ankita was born in Maroda village in Pauri Garhwal, Uttarakhand, at about 1,400 m, and as a girl ran the hill roads with young men training for Army physical tests. Her coach Toppo told ESPN that "because she was running in Uttarakhand, she had already been doing a lot of high-altitude training." One Indian woman on that podium had flown to the Rockies and the Alps for altitude. Another, finishing fourth, had grown up with a version of it outside her front door. Neither fact, on its own, explains a result. Both are reminders that an athlete's environmental history starts long before the first funded camp.

Still, the question Parul's career frames is exactly the one this Part is built around. Heat and altitude are both stresses. Both can lower a runner's heart rate at a given pace once the body has adapted. Both have been sold as shortcuts. But they work through different pathways, on different timescales, for different targets. A coach who wants to use them well has to begin not with the mountain or the sun, but with a sentence: which adaptation does this athlete need, and when?

To answer that, we need to separate two things that are easily confused: more fluid in the blood, and more red cells.

Sources: <https://www.rediff.com/sports/report/asian-games-after-double-bronze-haul-parul-now-craves-family-time/20260928.htm> ; <https://www.deccanchronicle.com/sports/asian-games-2026-indias-parul-chaudhary-wins-bronze-in-womens-3000m-steeplechase-1991012> ; <https://worldathletics.org/athletes/india/parul-chaudhary-14497992> ; <https://indianexpress.com/article/sports/asian-games/running-sugarcane-field-parul-chaudhary-asian-games-2023-hangzhou-8943068/> ; <https://timesofindia.indiatimes.com/city/meerut/asian-games-gold-medalist-takes-charge-as-dsp-in-moradabad-dist/articleshow/121298908.cms> ; <https://www.tribuneindia.com/news/sports/parul-sets-new-3-000m-record-409357> ; <https://www.olympics.com/en/news/parul-chaudhary-national-record-5000m-meeting-nikaia-2026-athletics> ; <https://www.olympics.com/en/news/world-athletics-championships-2023-budapest-day-9-india-results-parul-chaudhary> ; https://asianathletics.com/wp-content/uploads/2025/05/417_3000SC-Women-Final-Result-Day-4.pdf ; <https://www.olympics.com/en/news/paris-2024-olympics-athletics-india-parul-chaudhary-jeswin-aldrin-qualification> ; <https://thebridge.in/athletics/avinash-sable-parul-chaudhary-to-train-in-colorado-springs-45778> ; <https://sportskeeda.com/summer-olympics/news-paris-olympics-2024-3000m-steeplechasers-avinash-sable-parul-chaudhary-train-hard-switzerland> ; <https://www.thestatesman.com/sports/gulveer-singh-adds-1500m-silver-to-asian-games-haul-sable-finishes-fourth-1503644118.html> ; <https://www.tribuneindia.com/news/sports/who-is-sawan-barwal-himachal-runner-who-challenged-marathon-myth-to-win-asian-games-silver/> ; https://www.espn.in/athletics/story/_/id/30859627/no-turning-back-ankita-dhyani-mastery-improbable-chases-makes-unbeatable

CHAPTER 30

30. Plasma Versus Red-Cell Adaptation

PART V — HEAT VS ALTITUDE

Few Indian runners have met both environmental extremes as directly as Avinash Sable. As a soldier he knew the Rajasthan desert, with heat of up to 50 °C, and the cold of Siachen. As an athlete he trained high in Colorado Springs and in St Moritz, 1,856 m up in the Swiss Alps. His coach Scott Simmons noticed something about how Sable's body handled the mountains: "His adaptation upon his return to altitude this year was almost instant," he told the Hindustan Times in 2023. Heat and altitude both leave marks on a runner's body, and both can make the heart work harder at first and then, with time, a little easier. From the outside, the two kinds of adaptation can look alike: a lower heart rate at the same pace, a faster time trial. Inside the circulation they are doing very different jobs. One widens the road. The other adds more lorries. Telling them apart is the point of this chapter.

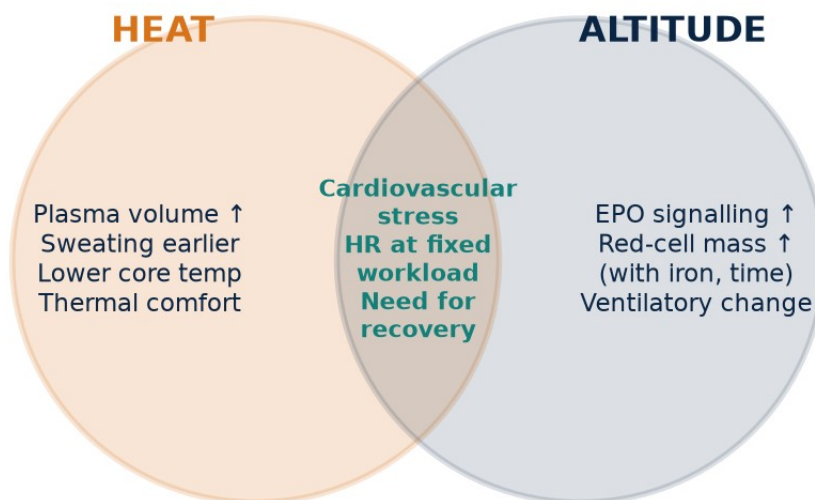


Figure 14. Heat and altitude overlap in cardiovascular stress but have different primary adaptations. Define the target adaptation before choosing the environment.

Heat and altitude are often discussed together because both are environmental stresses that can change an endurance athlete's blood and cardiovascular responses. Both can make running feel harder at first, both can lower performance acutely, and both may be followed by improved fitness. Heat and altitude can overlap in cardiovascular stress but differ in their primary stimuli. Heat strongly challenges thermoregulation and fluid balance; altitude reduces inspired oxygen pressure.

Two different signals

A useful conceptual contrast is: heat can produce plasma-volume and thermoregulatory adaptation, while sustained hypoxia can stimulate erythropoiesis. Both may affect cardiovascular function, but through different pathways. Heat acclimation asks the body to protect circulation while dumping heat, and the body's response includes expanding plasma, adjusting sweating and reducing cardiovascular strain. Altitude asks the body to deliver oxygen when the pressure driving it into the blood is lower, and the body's response, given sufficient time and dose, includes increased red-cell production.

Blood is made up mainly of plasma and cells, most of which are red cells. Total blood volume can therefore increase in two broad ways: by adding plasma, or by adding red cells (which brings along some additional plasma as well). The two routes have different consequences for oxygen transport.

Feature	Heat acclimation	Altitude exposure
Primary stimulus	Raised body and skin temperature	Lower inspired oxygen pressure
Main blood change	Plasma volume may rise	Red-cell (Hb) mass may rise over weeks
Typical time course	Many changes within 1–2 weeks	Hb mass needs weeks of sufficient exposure
Effect on Hb concentration	May fall (dilution)	Often rises early (plasma loss), then Hb mass may rise
Thermoregulation	Directly improved	Not a primary target
Accessibility in India	Widely available (e.g. Chennai)	Requires travel (e.g. Ooty, Kodaikanal) or simulation

KEY IDEA

Heat mainly targets plasma volume and thermoregulation; altitude mainly targets red-cell production. Both can influence cardiovascular function, but through different pathways.

What each adaptation can and cannot do

More plasma can support stroke volume, cooling and cardiovascular stability, especially in warm conditions. But plasma does not carry much oxygen; oxygen is carried by haemoglobin in red cells. More red cells can increase the blood's total oxygen-carrying capacity, but a thicker blood without adequate plasma might also increase viscosity, and red cells alone do not improve heat dissipation. The two adaptations are therefore complementary rather than interchangeable.

An analogy may help athletes. Plasma expansion is like widening the road and adding more lanes of traffic; red-cell expansion is like adding more delivery lorries carrying oxygen. Both can improve the supply chain, but they solve different problems. The analogy is simplified, and like all analogies it should not be pushed too far, but it captures why "heat equals altitude" misleads.

COACH'S CORNER

When an athlete asks, "Is heat training my altitude?", respond with a question: "What adaptation are we looking for?" If the answer is better tolerance of race-day heat, heat is the direct tool. If the answer is more red-cell mass, altitude or a well-designed simulated altitude programme is the more direct stimulus.

Measuring the difference

This is why "heat equals altitude" is too simple. The scientifically interesting question is where the adaptations overlap and where they do not. To answer that, the AEI must record separate markers: plasma-related measures (changes in pace-HR relationship, heart-rate drift, estimated plasma volume change where laboratory methods allow), thermoregulatory measures (sweat rate, core or skin temperature where available, thermal comfort) and red-cell measures (haemoglobin mass where available, with haemoglobin concentration interpreted cautiously).

FIELD PROTOCOL

1. Separating plasma and red-cell responses
2. Baseline: pace-HR test, sweat-rate estimate and blood assessment as advised by a doctor, with Hb mass if possible
3. After heat block: repeat all tests; expect possible HR and sweat changes and possible Hb concentration fall
4. After altitude block: repeat all tests; interpret early Hb concentration rise cautiously and look for Hb mass change
5. Record each result with the environmental conditions of testing
6. Summarise which pathway changed, not only whether performance changed

MYTH VS EVIDENCE

Myth: Heat training and altitude training produce the same blood adaptation.

Evidence: Heat mainly expands plasma volume and improves thermoregulation; altitude can stimulate red-cell production. Performance effects may overlap, but the mechanisms differ.

The time course also differs. Heat-related adaptations are relatively quick to appear and relatively quick to decay, often over a matter of weeks without exposure. Red-cell

adaptations to altitude take longer to build and, once exposure ends, decline as the extra red cells reach the end of their lifespan. These timelines matter for scheduling: a heat block might be placed close to a hot race, while an altitude block requires a longer lead time and careful planning of the return to sea level.

CHECKPOINT

Which markers in your current monitoring could distinguish a plasma-volume adaptation from a red-cell adaptation? Which are missing?

CHAPTER 31

31. Can Heat Replace Altitude?

PART V — HEAT VS ALTITUDE

Jyoti Gawate, a marathoner from Parbhani in Marathwada, ran 2:45:48 at the 2019 Mumbai Marathon, finishing as the second Indian woman. According to a detailed profile by the writer Shyam Gopan, she did it without a job and without access to the national camp, training in Parbhani with Ravi Raskatla, her coach since about 2005. She had taken a Mumbai Police job in 2014 and left after eight days to protect her training time. "I am still jobless; what more can I do?" she told Mid-Day. For runners like her, a month in Colorado Springs, or even in Ooty, is not a training decision. It is a financial one. Heat, on the other hand, arrives free every morning on the plains. So the question in this chapter is not academic. Can the stress that costs nothing do the work of the stress that costs a great deal? For thousands of Indian runners, it may be the only question that matters.

For a sea-level athlete in India, the question is practical and personal. Altitude camps cost money, time away from work or study, and family support. Heat is available on the doorstep for much of the year. If heat could deliver the same benefits, it would democratise a powerful training tool. Heat acclimation may improve endurance performance and can alter plasma volume and cardiovascular function. Altitude exposure can increase haemoglobin mass in appropriate protocols. These are not the same adaptation.

What the studies suggest

Interest in heat as an "altitude alternative" grew after studies such as Lorenzo et al. (2010) reported performance improvements in cool conditions following heat acclimation. Some studies compare heat and altitude interventions and show overlapping performance outcomes under certain conditions, while the underlying

physiology can differ. In some comparisons, both groups improved time-trial performance to a broadly similar extent, yet one group showed plasma expansion while the other showed changes in red-cell measures. Other studies have found little benefit from heat for performance in cool conditions, and results vary with protocol, athlete level and the test used.

The honest interpretation is that heat can improve some aspects of endurance performance for some athletes, particularly performance in the heat, and may help in cooler conditions in certain protocols. It does not reliably reproduce the red-cell response that altitude can provoke (a few studies of five or more weeks of heat training in elite cyclists have reported small haemoglobin-mass increases, but this is not yet established), and it should not be sold as a free equivalent of a mountain camp.

KEY IDEA

Heat can be a valuable sea-level tool, but it is not a copy of altitude. Define the target adaptation first, then choose the stimulus.

Start with the target adaptation

Therefore a coach should define the desired adaptation first. If the objective is thermoregulation, heat is directly relevant. If the objective is erythropoietic stimulation, altitude is the more direct environmental stimulus. If the objective is marathon performance, the outcome must be measured rather than assumed.

Coaching objective	More direct tool	Key measurement
Race in hot, humid conditions	Heat acclimation	Pace-HR in heat; sweat rate; thermal comfort
Increase red-cell mass	Altitude (natural or simulated)	Hb mass, where available
Improve cardiovascular stability at sea level	Heat, or consider both	Pace-HR drift; performance test
Improve marathon time	Depends on athlete and race	Race or time-trial performance, in comparable conditions
Limited budget, no travel	Heat, plus well-planned training	Pace-HR, performance trend

For many Indian marathoners, the race environment itself provides the answer. Several major Indian road races are run in warm and often humid conditions, even with early starts. An athlete preparing for such a race has a direct reason to be heat-acclimated. The same athlete preparing for a cool autumn marathon abroad may have more reason to consider altitude or simply high-quality training in the coolest available conditions.

COACH'S CORNER

Write the objective in one sentence before choosing heat or altitude, for example: "Reduce heart-rate drift during a 30 km run in Chennai conditions" or "Increase haemoglobin mass before the spring marathon". The sentence decides the intervention; the measurement decides whether it worked.

The measurement obligation

Whichever tool is chosen, the claim must be tested. A heat block followed by a faster race does not prove that heat caused the improvement; the athlete may also have been fitter, better tapered or racing in kinder weather. The AEI approach is to measure the outcome in comparable conditions, record the intervention dose and repeat interventions over time to build individual evidence.

FIELD PROTOCOL

1. Heat-versus-altitude decision test
2. Step 1: write the target adaptation in one sentence
3. Step 2: identify the most direct stimulus for that adaptation
4. Step 3: check feasibility, cost and safety, including medical clearance
5. Step 4: select one primary and two secondary measurements
6. Step 5: run a baseline test under recorded conditions
7. Step 6: deliver the intervention with full dose logging
8. Step 7: retest under comparable conditions and record the result with its uncertainty

MYTH VS EVIDENCE

Myth: A Chennai summer is a free altitude camp.

Evidence: Heat and humidity can produce valuable adaptations, particularly for racing in the heat, but they do not reliably reproduce the red-cell response to altitude. They are different tools for different targets.

Cost and access are legitimate coaching considerations, not signs of lower ambition. A Chennai athlete who cannot travel to the hills for a month can still build a high-quality aerobic base, improve running economy, develop threshold fitness and prepare for race-day heat with a careful acclimation block. These controllable elements often matter more than any environmental intervention. The mistake is not choosing heat over altitude; the mistake is making promises about either that the data cannot support.

CHECKPOINT

For your athlete's next target race, what is the likely race-day environment, and which adaptation matters most for that race?

CHAPTER 32**32. Combining Heat and Altitude**

PART V — HEAT VS ALTITUDE

Look at the 2026 calendars of India's leading distance runners and you can see two environments braided together. Sawan Barwal's programme winters in the Nilgiris, cool and high, before coming down to Bengaluru, and he flew to Rotterdam, where he broke the national marathon record in April. By late September he was racing the Asian Games marathon in Aichi-Nagoya. Parul Chaudhary went from a 5000 m national record in Nice in June to three finals in three days at the same Games. Gulveer Singh, based at altitude in Colorado Springs, won three Asian Games medals. Across the Indian team, altitude has been the deliberate part: camps, bases, months in the hills. Heat, as the research for this book found, has been mostly incidental: postings, village roads, the weather on race day. Nobody has yet published an Indian study that deliberately sequenced both, measured each and asked what the combination did. That leaves the coach with a real design problem, and a real opportunity.

If heat and altitude each offer something, it is tempting to use both, perhaps simultaneously. Some coaches have experimented with heat sessions at altitude or with alternating blocks. Combining environmental stresses can increase total physiological load. That can be useful, but it can also reduce training quality and recovery if the dose is poorly controlled.

Stress is not automatically stimulus

The first rule is not to stack stress simply because more stress sounds more scientific. The intervention should have a reason, a target adaptation and a monitoring plan. Every stressor draws on the same finite capacity to recover. An athlete doing hard training, adjusting to altitude, sleeping poorly and adding heat sessions may simply accumulate fatigue. At best, the extra stress adds little; at worst, it increases the risk of illness, heat injury or a poor training block.

There is also a physiological interaction to consider. Early in altitude exposure, plasma volume often falls, while heat acclimation tends to expand plasma volume. Some researchers have speculated that the two might complement each other; others worry

that combined stress could blunt one response or overwhelm recovery. The evidence on combined heat and hypoxia is limited and mixed, and it does not yet justify confident recommendations for routine practice.

KEY IDEA

Combining heat and altitude increases total load. Use it only with a clear purpose, a target adaptation, careful monitoring and medical support, not because more stress sounds more scientific.

Sequencing rather than stacking

For most athletes, sequencing is safer and easier to interpret than stacking. A practical research sequence is: baseline → heat block → washout/retest → altitude block or control → retest. Each block has a clear target, each is measured, and the washout period allows at least partial return towards baseline before the next intervention.

Phase	Purpose	Example duration
Baseline	Establish pace-HR, performance, blood markers	1–2 weeks of testing
Heat block	Target thermoregulation and plasma volume	10–14 days
Washout and retest	Allow partial decay; measure retention	Several weeks
Altitude block or control	Target red-cell response, or compare with no altitude	3–4 weeks
Retest	Measure combined and separate effects	Within 1–3 weeks after return

Durations are indicative and must be adjusted to the athlete, season and competition calendar. The washout interval in particular is uncertain; heat adaptations decline over weeks, but some benefits may persist longer than others.

Order effects and study design

The order should be randomised where feasible in a formal study, because adaptation from the first block can influence the second. An athlete who completes heat acclimation first may arrive at altitude with an expanded plasma volume and a different fluid balance. An athlete who does altitude first may begin the heat block with altered blood characteristics. In a group study, randomising the order (some athletes heat first, others altitude first) helps separate the effects of each intervention from the effect of sequence and from general improvements in fitness over time.

For an individual athlete, full randomisation may not be possible, but the coach can still learn by recording the order, the dose and the results carefully and comparing across seasons.

COACH'S CORNER

If you do combine or sequence heat and altitude, reduce training load during the first days of each new environmental block, keep your heat-illness stop rules active at every elevation, and check sleep, morning heart rate and wellbeing daily. The environment should add stimulus, not replace recovery.

FIELD PROTOCOL

1. Combined-intervention monitoring
2. Daily: waking HR, sleep, wellbeing, body mass, symptoms
3. Each session: environment (temperature, humidity, elevation), duration, pace, HR, RPE
4. End of each block: standard pace-HR test and performance test
5. Stop rules: any heat-illness warning sign, persistent unusual fatigue, illness or sharply elevated waking HR prompts reduced load and review
6. Medical: agree a plan with a qualified professional before the first block

MYTH VS EVIDENCE

Myth: Doing heat and altitude together doubles the benefit.

Evidence: Combined stresses add to total load and may interfere with recovery or with each other. The evidence is limited and mixed, and sequencing with measurement is usually safer and easier to interpret.

This is where the AKURA philosophy becomes concrete. The environment is a tool, the athlete's response is data, and the goal is understanding rather than spectacle. Whether an athlete trains in the humidity of Chennai, the cool air of Kodaikanal or both, the question remains the same: what did we give the body, how did it respond, and how certain are we?

CHECKPOINT

If you planned both heat and altitude for one athlete this year, what order would you choose, how long would the washout be, and what result would lead you to change the plan next year?

PART V AT A GLANCE

The key idea from each chapter, for quick revision.

30

Plasma Versus Red-Cell Adaptation

Heat mainly targets plasma volume and thermoregulation; altitude mainly targets red-cell production. Both can influence cardiovascular function, but through different pathways.

31

Can Heat Replace Altitude?

Heat can be a valuable sea-level tool, but it is not a copy of altitude. Define the target adaptation first, then choose the stimulus.

32

Combining Heat and Altitude

Combining heat and altitude increases total load. Use it only with a clear purpose, a target adaptation, careful monitoring and medical support, not because more stress sounds more scientific.

PART VI — MARATHON ENGINE

The marathon is where endurance physiology stops being theory and becomes a public examination. Aerobic power, sustainable intensity, fuel, fluid, heat and muscular resilience must all hold together for two to five hours, and the weakest link decides the outcome. This Part treats the marathon as a systems test rather than a longer 5K. It explains why marathon pace is a fraction of aerobic capacity, why that fraction differs between athletes, why pace so often collapses late in the race, and why the final 10 km reveals a quality short races cannot: durability. These ideas prepare the ground for AKURA's argument that an athlete needs distance-specific profiles, not one permanent number.

IN THIS PART

- 33** The Marathon Is a Systems Test
- 34** Fractional VO₂max and Marathon Pace
- 35** Why Marathon Pace Fails Late
- 36** The Last 10K

PART VI • MARATHON ENGINE

The Marathon Is a Systems Test

Aerobic power, sustainable intensity, fuel, fluid, heat and muscle must hold together for hours; the weakest link decides.

2:34:43

O. P. Jaisha's national marathon record, Beijing 2015 — still standing

3 factors

The performance model

Marathon speed = $VO_2\text{max} \times$ the fraction you can sustain, translated through economy. All three can be trained.

7

Seven systems

Aerobic power, sustainable intensity, fuel, fluid, thermoregulation, muscle-tendon and pacing: the weakest fails late.

A physiological budget

Fast starts, surges, low fuel, sweat loss and heat spend reserves you will miss in the final kilometres.

90 g/h

Fuel by duration

Up to about 30 g/h for 1–2 h, 60 g/h for 2–2.5 h, 90 g/h beyond 2.5 h.

Last 10K

The truth-teller

The final 10 km separates speed from durability: how well that speed survives fatigue.

5K ≠ M

Distance-specific profiles

A 5K tests aerobic power; a marathon tests fractional utilisation, fuel, heat and durability.

THE PROCESS

Export splits, HR

Pace 10–30 km

Pace 30–40 km

% pace change

Annotate conditions

A marathon is limited by the weakest system on the day, not the strongest test number.

THE STORY

Three Minutes Forty

PART VI — MARATHON ENGINE

It is January 2015 in Mumbai, and the city has not finished waking when the marathon field floods the road. The air off the Arabian Sea is soft and damp, the kind that settles on the skin before the sun has cleared the buildings. Somewhere in the pack is a runner from Kerala who has never raced this distance before.

Her name is Orchatteri Puthiyaveetil Jaisha, and until now her career has been written on the track. She won 5000 m bronze at the 2006 Asian Games. In August 2010, at the National Inter-State meet in Patiala, she set a national record in the 3000 m steeplechase, 10:03.05. A few months before this Mumbai morning, at the 2014 Asian Games, she took bronze in the 1500 m in 4:13.46, a race that is over in roughly the time it takes a kettle to boil.

This race will last more than two and a half hours.

When the results are posted, she is eighth overall and the first Indian woman home, in 2:37:29. It is her debut marathon, and it is a national record. The mark she has beaten, 2:39:10, was set by Vally Satyabhama in Chennai on 21 December 1995. It had stood for almost twenty years. Lalita Babar, who like Jaisha was coached by the Belarusian Nikolai Snesev, ran 2:38:21 at Mumbai that year as well.

How does that happen? How does an athlete built for four minutes of fury become, almost overnight, the fastest Indian woman ever over 42.195 km? And why does it not always happen twice?

Beijing, 30 August 2015

Seven months later Jaisha stands on the start line of the World Championships marathon in Beijing. This time there is no debut uncertainty. There is a target, a national team vest and the best women marathoners in the world spread across the road ahead of her.

She finishes 18th in 2:34:43. The national record falls again, this time by almost three minutes. Sudha Singh, better known as a steeplechaser, was reported 19th in the same race in 2:35:35. Two Indian women inside the top twenty of a world championship marathon: it had the feel of a door swinging open.

Pause on that number, because this Part of the book keeps returning to numbers like it. A time of 2:34:43 for 42.195 km works out at about 3 minutes 40 seconds per kilometre, held without a break for more than two and a half hours. Her 1500 m bronze the previous year had been run at roughly 2 minutes 49 seconds per kilometre. The marathon, in other words, was run at a little over three-quarters of her 1500 m speed, for about thirty-seven times as long.

That ratio is not a laboratory measurement. It does not tell us her VO₂max, her lactate threshold or her running economy, and nobody should pretend it does. But it points straight at the hidden subject of the next four chapters. No runner races a marathon at the limit of their aerobic engine. They race at a share of it, and the size of that share depends on threshold, economy, fuel, fluid, heat, the resilience of muscle and tendon, and the judgement with which the whole budget is spent. A 1500 m tells you a great deal about the size of the engine. A marathon tells you how well every other part of the machine protects it.

Eleven years on, the Beijing time is still the Indian women's marathon record. Sudha Singh came within 12 seconds of it at the Mumbai Marathon in January 2019, running 2:34:55, a course record. Nobody has gone faster. At the 2026 national championship, the New Delhi Marathon in February, Nirmaben Thakor won the elite women's race in 2:41:15, and Olympics.com noted plainly that the Indian women could not reach the Asian Games standard.

The Indian women's marathoners could not beat the Asian Games qualifying mark of 2:31:52.

— Olympics.com, New Delhi Marathon 2026 report

A record that survives more than a decade is a strange kind of monument. It tells you what was possible on one day in one city. It says nothing about how often it could be repeated, or what had to line up to make it possible.

Rio, August 2016

A year after Beijing, the marathon showed its other face.

Jaisha and Kavita Raut ran the women's marathon at the Rio Olympics. The results were a long way from Beijing. Kavita finished 120th. Jaisha finished 89th, and collapsed at the finish.

What followed became one of the most debated episodes in recent Indian athletics. Jaisha said that Indian officials had given her no refreshments during the race. Scroll.in reported her account in her own words.

All the [other] countries had their stalls at every two kilometres, but our country's stall was empty.

— O. P. Jaisha, as quoted by Scroll.in

The Athletics Federation of India disputed her version. Kavita Raut, who ran the same race, said she had had enough water. Scroll set out these positions side by side, and so does this book. It does not take a side, because the documented record does not allow one to be taken honestly.

Look at what that record contains: an athlete's account, a federation's rebuttal, and a teammate's different experience of the same course. Now look at what it does not contain. There is no fluid log. There is no body-mass reading from before and after. There is no core temperature, no heart-rate trace, no record of what was actually taken at each station or when. Without those, nobody reading the public record today can say with confidence which of the marathon's systems gave way that day, or why.

That absence is the lesson a coach should carry away from Rio, and it has nothing to do with blame. A marathon collapse is almost never a single event. It is the visible end of a chain: pace, fuel, fluid, heat, muscle and judgement, each link drawing on the others for hours. When the chain breaks, the only way to learn from it is to have measured the links while they were being tested. Most athletes, at every level of the sport, have not.

Two races, one runner

Put Beijing and Rio side by side and the puzzle of the marathon is laid bare. The same runner, in the same coaching system, produced a national record that still stands and, twelve months later, a race that ended on the ground past the finish line. The public record gives us the two results and the dispute that followed. It does not tell us which system held in Beijing and which failed in Rio.

This is not a story that belongs only to Olympians. Every winter Sunday in Mumbai, Chennai, Pune or Kolkata, a club runner who ran a lovely 10K in the cool of December discovers somewhere past 30 km that the marathon was asking a different question. Her 10K time was honest. It simply measured fewer things. The humidity rising off the sea, the gel she skipped, the first 5 km run twenty seconds per kilometre too fast because the crowd was loud: none of that shows up in a 10K, and all of it shows up in a marathon.

The coaching task, then, is not to find one magic number that predicts the marathon. It is to know which systems the marathon will test, to measure each of them where possible, and to design a race around the weakest. Jaisha's 2:34:43 tells us what an Indian woman has been able to run for 42.195 km. The rest of this Part asks what has to hold together for any runner to do something like it, and why, so often, it does not.

To understand why, we begin by treating the marathon not as a long race, but as a systems test.

Sources: <http://timesofindia.indiatimes.com/sports/more-sports/athletics/World-Athletics-Championships-Indias-Jaisha-Orchatteri-breaks-national-record-in-womens-marathon/articleshow/48728686.cms> ;
<https://www.sportskeeda.com/running/o-p-jaisha-karan-singh-winners-mumbai-marathon> ;
<https://scroll.in/field/814706/indian-marathon-runner-kavita-raut-seems-to-contradict-compatriot-s-version-about-callous-officials> ; <https://sportskeeda.com/athletics/rio-olympics-2016-athletics-op-jaisha-kavita-raut-put-up-dismal-show-womens-marathon-event> ; <https://web.archive.org/web/20121107223944/http://www.hindu.com/2010/08/09/stories/2010080952941500.htm> ; <https://shyamgopan.com/2019/02/04/jyoti-and-the-eight-minutes/> ;
<https://www.olympics.com/en/news/new-delhi-marathon-2026-results-winners> ;
<https://www.olympics.com/en/news/athletics-list-of-indian-national-record-holders>

CHAPTER 33**33. The Marathon Is a Systems Test**

PART VI — MARATHON ENGINE

It is 5:15 a.m. on a January Sunday in Mumbai, and the start area already smells of liniment, sea salt and wet tarmac. A club runner from Thane bounces on the spot, checking a watch programmed with one number: the pace she ran in a cool 10K last December. The number looks harmless. Over the next three and a half hours it will be tested against everything it never measured: the humidity rising off the harbour, the sun climbing over the buildings, the gel she has never tried before, the calf that tightened on the local train in. At 21 km she will feel wonderful. At 32 km she will find out whether she has been running a race or borrowing against one. The marathon is where training plans meet reality in public, with a clock running. Before the gun goes, it is worth asking what exactly is about to be examined.

The marathon is not a 42.195 km version of a 5K. A 5K asks how much oxygen an athlete can use and how much discomfort they can tolerate for a quarter of an hour. A marathon asks a different, broader question: can every system that supports running continue to cooperate for hours, while fuel stores fall, body water is lost, heat accumulates and muscle fibres fatigue? It is a long-duration systems test involving aerobic power, sustainable intensity, glycogen management, hydration, thermoregulation, musculoskeletal resilience and pacing. An athlete can be strong in five of these and still have a poor race because the sixth failed.

Seven systems, one finish line

It helps to name the systems explicitly, because each can be measured and each can be trained. Aerobic power sets the ceiling. Sustainable intensity, usually discussed through lactate threshold and fractional utilisation, determines how close to that ceiling the athlete can run for hours. Glycogen management determines whether carbohydrate lasts to the finish. Hydration influences plasma volume and cardiovascular strain. Thermoregulation decides how much blood must be diverted to the skin for cooling. Musculoskeletal resilience governs whether the legs can keep producing the same force with each step. Pacing is the decision-making system that spends all of these resources, wisely or badly.

System	What it contributes	How it fails late
Aerobic power	Upper limit of oxygen use	Rarely the late limiter itself
Sustainable intensity	Fraction of capacity held for hours	Pace set above true sustainable level
Fuel (glycogen)	Carbohydrate for high-intensity running	Depletion, rising effort, slowing
Fluid balance	Plasma volume, sweat capacity	Rising heart rate, heat strain
Thermoregulation	Heat dissipation	Core temperature rises, pace falls
Muscle and tendon	Force production each stride	Mechanics change, cost rises
Pacing	Allocation of all resources	Early overspending

KEY IDEA

A marathon is limited by the weakest system on the day, not by the strongest number on the athlete's test report.

Why the last 10 km tells the truth

The final 10 km exposes weaknesses that may not appear in a short race. Cardiac drift, dehydration, glycogen depletion, heat accumulation and muscular fatigue can all increase the cost of running. The athlete may be holding the same pace on the watch, yet the internal price of that pace is rising: heart rate creeps up, breathing deepens, perceived effort climbs, and stride mechanics become less efficient. When the internal cost exceeds what the athlete can pay, pace falls.

This is why two runners with similar 10K times can finish a marathon ten minutes apart. The shorter race measured mainly their aerobic capacity and threshold; the marathon measured their ability to keep those qualities stable as fatigue accumulated.

MYTH VS EVIDENCE

Myth: A fast 10K runner will automatically run a fast marathon.

Evidence: Short-race speed predicts marathon potential only when long-run conditioning, fuelling, heat tolerance and durability are also developed; the marathon adds systems that a 10K barely tests.

Central physiology and peripheral fatigue

A good marathon model therefore tracks both central physiology and peripheral fatigue. Central measures include heart rate, cardiac drift, breathing and core temperature, reflecting the heart, lungs and circulation. Peripheral measures describe what happens in the working muscles and tendons: local fatigue, glycogen depletion within fibres, and the gradual change in ground contact and stride that increases the oxygen cost of each kilometre.

In the Indian context, the marathon systems test is often run in warm, humid conditions, where fluid and heat systems are stressed earlier than in cool European races. A Mumbai, Chennai or Hyderabad marathon may therefore reward a different balance of qualities than a cool-weather race, and an athlete's profile should be interpreted with the environment recorded.

Measuring the systems test

If the marathon is a systems test, then preparation should include system checks. Not every athlete needs a laboratory. A coach can learn a great deal from simple, repeatable observations: the heart-rate drift during a steady 25 km run, the change in body mass across a long run, the pace of the last 5 km of a marathon-specific session, and the athlete's notes on gut comfort and leg feel. Recorded consistently, these observations show which system is lagging behind the others.

The aim is not to eliminate every weakness before race day; that is rarely possible. It is to know which systems are most likely to fail, so that pacing, fuelling and cooling strategies can be designed around them. An athlete with modest heat tolerance may choose a cooler race, start more conservatively or practise cooling strategies. An athlete with a sensitive gut may rehearse fuelling repeatedly. Knowing the weak link turns anxiety into a plan.

COACH'S CORNER

After each marathon, ask the athlete to review the race system by system rather than simply saying "I hit the wall". Was the start too fast, was fuel taken as planned, how much body mass was lost, how hot was it, and when did the legs start to feel different? This turns one disappointing race into a map of the next training block.

CHECKPOINT

For your last marathon, which of the seven systems do you believe failed first, and what evidence (splits, heart rate, fuel log, weather, body mass) supports that belief?

CHAPTER 34

34. Fractional VO_2max and Marathon Pace

PART VI — MARATHON ENGINE

Here is a sum worth doing on the back of a train ticket. When O. P. Jaisha set India's women's marathon record of 2:34:43 at the 2015 World Championships in Beijing, she averaged about 3 minutes 40 seconds per kilometre for 42.195 km. The year before, at the 2014 Asian Games, she had

won bronze in the 1500 m in 4:13.46, a speed of roughly 2 minutes 49 seconds per kilometre. Her marathon was therefore run at a little over three-quarters of her 1500 m speed, and held for about thirty-seven times as long. That ratio is not a laboratory measurement, and it says nothing precise about her VO_2max . But it captures the central puzzle of marathon running. Nobody races at their ceiling. The art lies in how large a fraction of it can be held for hours, and different athletes, or the same athlete on different days, hold very different fractions.

Every runner has a maximal aerobic capacity, but nobody races a marathon at it. Marathon pace is typically below maximal aerobic speed, run at a fraction of VO_2max that can be sustained for hours. That fraction is not fixed. It depends on fitness, durability, fuel, environmental conditions and individual physiology. Understanding fractional utilisation helps coaches move beyond the simple idea that a bigger VO_2max always means a faster marathon.

The three-part performance model

A widely cited framework, summarised by Joyner and Coyle in 2008, describes endurance performance as the product of three main factors: VO_2max , the fraction of VO_2max that can be sustained (closely related to lactate threshold), and running economy, which converts oxygen use into speed. The oxygen uptake an athlete can sustain, combined with economy, gives a performance velocity. A runner with a moderate VO_2max can outperform a runner with a higher VO_2max if they sustain a higher fraction or run more economically.

Factor	Question it answers	Typical assessment
VO_2max	How large is the aerobic ceiling?	Graded laboratory test
Fractional utilisation	How much of it can be held for hours?	Threshold testing, race analysis
Running economy	How much oxygen does each speed cost?	Submaximal steady-state test
Durability	Do these values hold as fatigue builds?	Pre- and post-long-run testing

Well-trained marathon runners are often described as racing at a high fraction of their VO_2max , commonly quoted around three-quarters or more for trained athletes and higher in elites, but the exact figure varies widely by individual, conditions and test method. The point for coaches is not to memorise a percentage but to understand that this fraction is trainable and individual.

KEY IDEA

Marathon speed is not VO_2max alone; it is VO_2max multiplied by the fraction you can sustain and translated through economy, and all three can change with training.

Same pace, different internal effort

The same nominal pace can represent different internal effort for different athletes. A 4:15 min/km pace might be a comfortable steady effort for one runner and close to threshold for another. Even for the same athlete, that pace represents a different internal load at 7 a.m. on a cool Ooty morning than at 7 a.m. on a humid Chennai beach road. Heat, humidity, hills, wind, sleep and accumulated training fatigue all shift the fraction of capacity required.

This is why pace prescriptions should be anchored to current fitness and adjusted for conditions. A pace table is a starting point; heart rate, perceived effort and environmental data tell the coach whether the prescription is being expressed as intended.

MYTH VS EVIDENCE

Myth: My marathon pace is a fixed number I should hold regardless of the weather.

Evidence: Marathon pace is an effort level expressed as a speed; in heat and humidity the same effort produces a slower pace, and holding the cool-weather pace raises the internal cost.

Performance-based anchors

VDOT, developed by Jack Daniels and presented in Daniels' Running Formula, is an established example of converting recent race performance into training intensities rather than treating VO_2max alone as the pace engine. Because it starts from what the athlete has actually run, it automatically includes the combined effect of VO_2max , fractional utilisation and economy as expressed in that race. This is a strength that AKURA respects and discusses in Part VII.

What a performance anchor cannot show on its own is which of those factors is responsible for the result. Two runners with the same race time may reach it with very different physiology, and they may need different training to improve. AKURA's interest is in adding that explanatory layer without discarding the clarity of performance-based prescription.

FIELD PROTOCOL

1. Marathon-effort calibration run
2. choose a flat, measured route
3. warm up 15 min easy
4. run 3×5 km at goal marathon pace with 2 min easy jog between
5. record pace, average heart rate, RPE, temperature and humidity for each block
6. compare heart rate and RPE drift between blocks
7. repeat in cooler and hotter conditions to see how the internal cost of the same pace changes.

Training the fraction

Fractional utilisation responds to training that develops sustained aerobic work: consistent easy volume, threshold sessions, and marathon-specific runs that hold goal effort for long periods. Improvements in mitochondrial density, capillary supply and the ability to use fat at moderate intensities help an athlete run faster before carbohydrate use and lactate accumulation rise steeply. These adaptations take months and years, which is one reason experienced marathon runners often improve their marathon times long after their 5K times have plateaued.

For coaches, this means marathon potential cannot be judged from a VO_2max test alone. A young athlete with a high laboratory value but little aerobic history may need several years of patient development before that ceiling becomes a fast marathon. An older athlete with a moderate value but deep aerobic conditioning may already be using a large share of their capacity.

COACH'S CORNER

When an athlete's marathon pace feels unexpectedly hard, check conditions before questioning fitness. Record temperature, humidity, sleep and the previous week's load; if these are unfavourable, the solution may be adjusting pace to effort rather than concluding that fitness has declined.

CHECKPOINT

What evidence do you have that your current marathon pace reflects your present fitness and today's conditions, rather than an old race or a hoped-for target?

CHAPTER 35

35. Why Marathon Pace Fails Late

PART VI — MARATHON ENGINE

Kilometre 32 of a coastal city marathon. The runner in the orange vest has been metronomic since the start: 5:10, 5:09, 5:11 per kilometre, splits so tidy her watch seems to be running itself. Then, without warning, 5:32. Then 5:48. Her legs feel as if someone has quietly swapped them for borrowed ones. Friends on the pavement shout that she is nearly there; she knows she is not. Afterwards she will say she hit the wall, as if a wall had been built overnight across the road at 32 km. It had not. It was laid brick by brick from the first kilometre: the eager start in the cool dark, the drink

station skipped because she felt fine, the gel that stayed in her pocket, the sun that rose over the sea at 19 km. The collapse was sudden. Its causes were not.

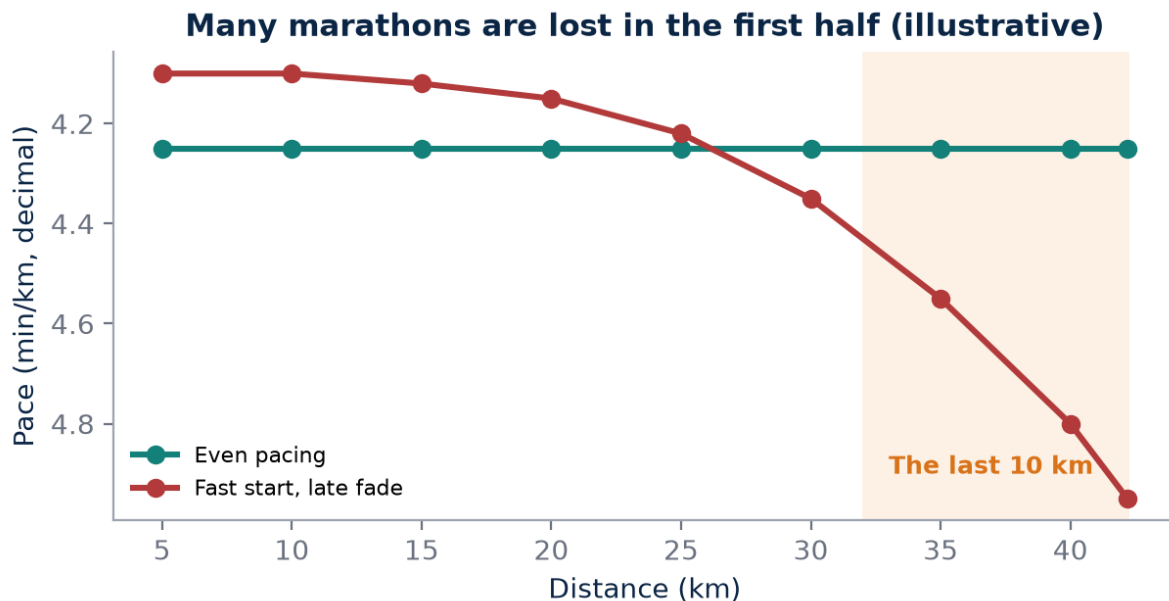


Figure 15. Illustrative split profiles. A fast start raises carbohydrate use and heat load, and the cost appears in the final 10 km.

Many runners describe the marathon as "going well until 32 km". The collapse feels sudden, but many late-race collapses begin long before the athlete feels the problem. By the time the legs refuse and pace falls by 20 or 30 seconds per kilometre, the causes have usually been accumulating for more than an hour. Understanding these hidden costs changes how athletes pace, fuel and prepare.

The hidden costs of the first half

Starting too fast raises carbohydrate use and thermal strain. At higher intensities the body relies more heavily on carbohydrate, so even a modestly aggressive first half draws down glycogen faster than planned. Higher intensity also produces more heat, which must be dissipated through sweating and skin blood flow. Inadequate fuelling reduces available substrate. Dehydration reduces plasma volume and can increase cardiovascular strain, so heart rate climbs for the same pace. Muscular fatigue changes mechanics: ground contact lengthens, the stride becomes less springy, and the oxygen cost of running rises.

None of these changes is dramatic on its own at 15 km. Together they create a debt that is repaid, with interest, in the final 10 km.

KEY IDEA

Marathon pacing is not merely a stopwatch problem. It is a physiological budgeting problem.

The physiological budget

It is useful to think of a marathon as a fixed budget that can be spent in different ways. Every second gained early by running above sustainable pace is purchased with glycogen, fluid, heat tolerance and muscular resilience that will be needed later.

Budget line	What spends it	How to protect it
Pacing	Fast start, surges, hills taken too hard	Even or slightly negative split by effort
Fuel	High intensity, low intake, poor gut tolerance	Planned carbohydrate intake, practised in training
Fluid	Sweat loss, heat, humidity	Individual drinking plan based on sweat testing
Heat	Air temperature, humidity, sun, intensity	Heat acclimation, cooling, pace adjustment
Mechanics	Accumulated muscle damage, downhill	Long runs, strength work, durable footwear choice

Fuel and fluid, practised not improvised

Carbohydrate intake during long events is well supported by sports-nutrition research, and guidance has become more individual. A commonly used summary, associated with work by Asker Jeukendrup, links recommended intake to event duration and the athlete's trained gut tolerance.

Event duration	Commonly recommended carbohydrate	Notes
Under about 1 hour	Little or none needed; mouth rinse may help	Performance rarely fuel-limited
About 1–2 hours	Up to about 30 g per hour	Simple sources are usually sufficient
About 2–2.5 hours	Up to about 60 g per hour	Gut training improves tolerance
Beyond about 2.5 hours	Up to about 90 g per hour	Usually multiple transportable carbohydrates

These are ranges, not targets to be reached on race day for the first time. The gut adapts to regular carbohydrate intake during training, and an athlete who has never taken 60 g per hour in a long run should not attempt it first in a race. Fluid should be individual too: the aim is to limit excessive dehydration without overdrinking, which carries its own risk of hyponatraemia. Sweat rates in Indian summer and humid coastal conditions can be substantial, so pre- and post-run body mass checks are valuable.

MYTH VS EVIDENCE

Myth: Hitting the wall is purely a matter of mental toughness.

Evidence: Late-race slowing usually reflects measurable causes such as early overpacing, carbohydrate depletion, dehydration, heat strain and muscular fatigue; mental skills matter, but they cannot replace an empty fuel tank.

FIELD PROTOCOL

1. Fuel and fluid rehearsal long run
2. weigh the athlete nude or in minimal clothing before the run
3. run 28–32 km including the final 8–10 km at marathon effort
4. take carbohydrate on the race schedule and log grams per hour
5. log all fluid consumed
6. weigh again after towelling dry
7. calculate body-mass change adjusted for fluid intake
8. note gut comfort, heart rate drift and pace in the final 10 km.

Durability as its own quality

The late-race story points to a quality that short-distance speed does not capture: the ability to maintain performance as exercise duration and fatigue increase. Recent sports-science work, including discussion by Maunder and colleagues in 2021 and by Andrew Jones and colleagues in studies of runners at sub-two-hour marathon pace, has highlighted durability or physiological resilience as an important addition to the traditional three-factor model. Research in this area is developing, and measurement protocols are still being standardised.

AEI should eventually estimate an athlete's durability separately from short-distance speed. A runner whose threshold and economy hold steady after two hours is a very different marathon prospect from one whose values degrade quickly, even if both share the same 10K time.

A practical sign of early overspending is a heart rate that rises faster than expected during the first half despite steady pace, or a perceived effort that already feels like threshold at 15 km. Athletes can be taught to recognise these signals and to adjust immediately, because a small correction at 15 km is far cheaper than a forced slowdown at 35 km.

COACH'S CORNER

Use the first 10 km of every marathon as a discipline exercise. Agree a target effort and a ceiling heart rate before the start, and treat any early pace faster than plan as a withdrawal from the final 10 km budget rather than "time in the bank".

CHECKPOINT

In your last long race, where did you spend your budget: pacing, fuel, fluid, heat or mechanics? Which line would you protect first next time?

CHAPTER 36**36. The Last 10K**

PART VI — MARATHON ENGINE

At the Rio 2016 Olympic marathon, two Indian runners crossed the line one second apart after 42.195 km. Thonakal Gopi, from Wayanad in Kerala, finished 25th in 2:15:25. Kheta Ram finished 26th in 2:15:26. Both times were personal bests. Both men had been part of the same high-altitude block at Ooty under coach Surendra Singh Bhandari, and both had qualified at the Mumbai Marathon that January. One second is less time than it takes to read this sentence. Spread across more than two and a quarter hours, it is almost nothing, yet it separated two places in an Olympic result. The public record keeps the finishing times; it does not tell us where that second was won or lost. For most marathon runners, though, the places that matter are decided in the same stretch of road: the final 10 kilometres.

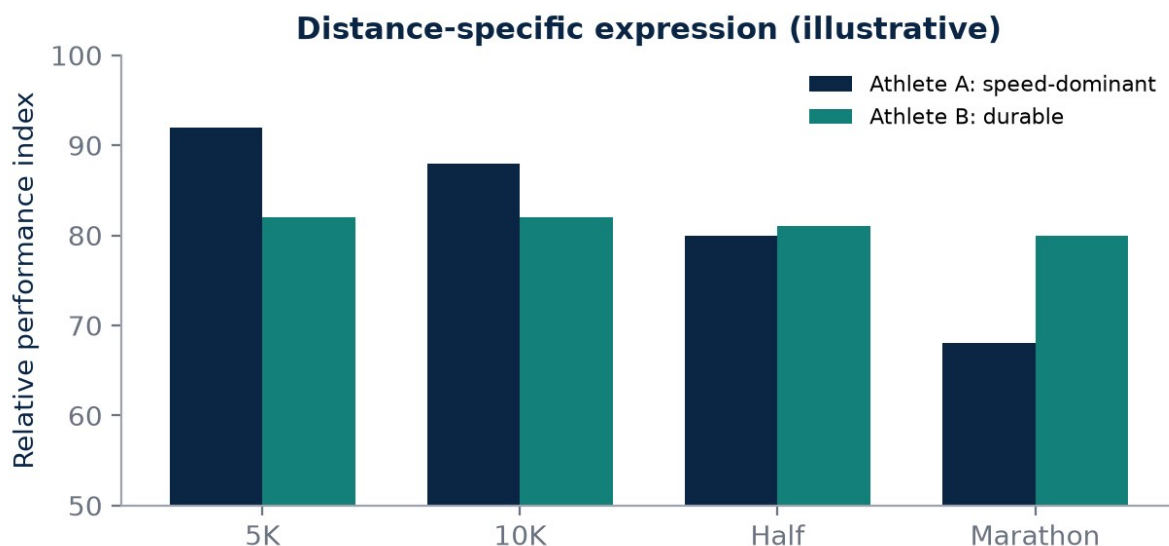


Figure 16. Illustrative only. The gap between short-distance and marathon expression shows where endurance breaks down, which one number would hide.

In a well-paced marathon, the last 10 km is where training becomes visible. Up to 30 km, many athletes look similar; after it, their paths separate. The final 10 km is where the interaction of fitness and durability becomes visible, and it provides coaches with some of the most honest information they will ever collect about an athlete.

Speed without durability

An athlete with excellent 5K speed but insufficient long-run conditioning may possess a high short-distance performance index but a weaker marathon profile. Their aerobic capacity and threshold are good, but these qualities are not yet protected against the fatigue of hours of running. By contrast, a runner with a modest 5K time but years of consistent volume may lose very little pace in the final 10 km and finish far higher up the results than their short-distance times would suggest.

Neither athlete is simply "better". They have different profiles, and they need different training. The first needs long runs, marathon-specific sessions and fuelling practice; the second may benefit from more speed and threshold work to lift the ceiling that their durability protects.

KEY IDEA

The final 10 km separates two qualities that short races blend together: how fast an athlete can run, and how well that speed survives fatigue.

Distance-specific expression

This suggests a useful AEI concept: distance-specific expression. The athlete should not receive one permanent number for all distances. A 5K profile, 10K profile, half-marathon profile and marathon profile can reveal where endurance breaks down. If an athlete's 5K and 10K performances imply a certain level, but the half-marathon and marathon imply progressively lower levels, the gap points towards durability, fuelling or heat tolerance rather than aerobic capacity.

Profile	Mainly tests	What a relative weakness may suggest
5K	Aerobic power, high-intensity tolerance	Limited top-end aerobic capacity
10K	Aerobic power, threshold	Threshold development needed
Half marathon	Threshold, early durability	Sustained-intensity conditioning
Marathon	Fractional utilisation, fuel, heat, durability	Long-run, fuelling or thermal limits

The difference between these profiles is itself useful information. A flat profile across distances suggests balanced development. A profile that falls steeply with distance suggests the athlete has speed to spare but lacks endurance-specific resilience. A profile

that is strong at the long end but weaker at 5K suggests a durable athlete who may need to raise the ceiling.

MYTH VS EVIDENCE

Myth: One performance number should describe an athlete across every distance.

Evidence: Race-equivalence models are useful guides, but individual athletes often deviate from them; the pattern of deviation across distances is diagnostic information in its own right.

Reading the final 10 km

Coaches can analyse the last 10 km with simple tools. Compare the average pace of 30–40 km with the average of 10–30 km. Compare heart rate at the same pace in the early and late race. Look at whether the athlete slowed gradually or suddenly; a gradual fade often reflects heat or cumulative fatigue, while a sudden drop may point to glycogen depletion or cramping. Record conditions, fuel taken and any gastrointestinal problems alongside the numbers.

FIELD PROTOCOL

1. Last-10K durability analysis
2. export race splits and heart rate
3. calculate mean pace for 10–30 km and for 30–40 km
4. calculate percentage pace change
5. compare mean heart rate for the same segments
6. annotate temperature, humidity, fuel intake and any stops
7. repeat after every marathon and key long run to build a personal durability trend.

The last 10 km also carries emotional and tactical weight. Athletes who have practised finishing long runs at marathon effort know how that fatigue feels and are less likely to panic when it arrives. Those who have only ever run long runs at easy pace may interpret normal late-race discomfort as disaster and slow more than their physiology requires. Specific preparation therefore trains both body and judgement.

Over a career, the pattern of performance over the final 10 km becomes one of the most useful records an athlete can keep. It shows whether durability is improving from one marathon cycle to the next, independent of course, weather and finishing time. In AKURA's terms, it is an early, practical estimate of the AEI-D Durability domain.

COACH'S CORNER

Build at least two long runs per block that finish with the final 8–10 km at marathon effort. They rehearse the exact situation that decides races, and the pace and heart-rate change in those final kilometres is an early indicator of marathon readiness long before race day.

CHECKPOINT

If you plotted your 5K, 10K, half-marathon and marathon performances as equivalent levels, where would the line drop, and what would that drop tell you about your next training priority?

PART VI AT A GLANCE

The key idea from each chapter, for quick revision.

- | | |
|-----------|--|
| 33 | The Marathon Is a Systems Test
A marathon is limited by the weakest system on the day, not by the strongest number on the athlete's test report. |
| 34 | Fractional VO₂max and Marathon Pace
Marathon speed is not VO ₂ max alone; it is VO ₂ max multiplied by the fraction you can sustain and translated through economy, and all three can change with training. |
| 35 | Why Marathon Pace Fails Late
Marathon pacing is not merely a stopwatch problem. It is a physiological budgeting problem. |
| 36 | The Last 10K
The final 10 km separates two qualities that short races blend together: how fast an athlete can run, and how well that speed survives fatigue. |

PART VII — VDOT AND AEI

Good coaching frameworks turn complexity into decisions. Jack Daniels' VDOT is one of the clearest examples in running: it converts a recent performance into a measure of current ability and a set of training intensities. This Part begins by describing what VDOT does and why it has earned its place. It then explains why AKURA proposes a different, broader layer, the AKURA Endurance Index, which asks what physiological systems support performance, how the athlete responds to environment and training, and how confident we are in each measurement. AEI is presented as a research prototype that sits beside, not above, established tools, with separate Fitness, Readiness and Trend views.

IN THIS PART

- 37 What VDOT Does
- 38 Why AKURA Needs a Different Layer
- 39 AEI Architecture
- 40 Proposed AEI Domains
- 41 AEI-Fitness
- 42 AEI-Readiness
- 43 AEI Trend

PART VII • VDOT AND AEI

Beside VDOT: A Broader Endurance Layer

VDOT turns a race into ability and training paces; AEI asks why, how the athlete is responding, and how sure we are.

27:00.22

Gulveer Singh's 10,000 m national record, The TEN, California 2025

VDOT

Performance first

VDOT measures current running ability from a race — not a laboratory VO₂max — and sets E, M, T, I, R paces.

AEI

A research prototype

AEI adds physiology, environment, recovery and confidence. Not validated; it sits beside, not above, established tools.

Fitness

What can they do?

AEI-Fitness changes over weeks to months, anchored in races, threshold, economy and durability.

Readiness

How much today?

AEI-Readiness changes over days: sleep, HRV trend, resting HR, fatigue, soreness, load, illness, heat.

10

Ten domains

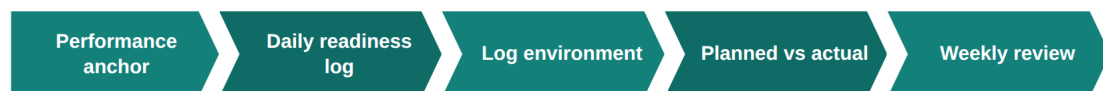
Performance, Aerobic, Threshold, Economy, O₂ Transport, Cardiovascular, Recovery, Heat, Altitude, Durability: evidence, not scores.

7·28·90

Rolling trend windows

7-day for readiness, 28-day for block response, 90-day for fitness and adaptation.

THE PROCESS



Fitness is the engine; readiness is whether it can be used fully today. Keep them separate.

THE STORY

Twenty-Two Hundredths

PART VII — VDOT AND AEI

The race is on a track in San Juan Capistrano, southern California, on 29 March 2025, and it has been built for one thing: fast times. The meet is called The TEN, and its whole reason to exist is the 10,000 m. Twenty-five laps. A field packed tight on the inside lane, chasing the clock more than each other.

One of them is an Indian Army soldier named Gulveer Singh.

He finishes sixth. The clock stops at 27:00.22. It is a national record, and according to the Sports Authority of India it makes him the third-fastest Asian of all time over the distance. AFI's own website, reporting the run, notes that he narrowly missed his goal of going under 27 minutes. Twenty-two hundredths of a second. Less than a stride.

It is worth sitting with that for a moment. A man runs 10,000 metres at about 2 minutes 42 seconds per kilometre, faster than any Indian has ever run it, and the headline includes the word "misses". Numbers do that. They can make a triumph and a near-miss out of the same fraction of a second.

Where the numbers came from

Gulveer was born on 1 June 1998 in Sirsa village, in the Aligarh district of Uttar Pradesh, into a farming family. He joined the Army through the sports quota in 2018, with The Grenadiers, and rose to Junior Commissioned Officer. He was posted in Arunachal Pradesh, on patrolling duties at altitude. He did not arrive in the Army as a finished runner, and he has said as much.

I truly learned about competitive running only after joining the army.

— Gulveer Singh, quoted by [Olympics.com](https://olympics.com)

His breakthrough came with a 10,000 m bronze at the 2022 Federation Cup. From the Army Sports Institute he moved into the SAI national camp, and from 2024 he was based at altitude in Colorado Springs under the American coach Scott Simmons.

Then the numbers began to fall.

Forty-nine years and twelve months

To understand what 27:00.22 means, you have to look at the line of numbers behind it.

In 1976, at the Montreal Olympics, Hari Chand ran 28:48.72 in his 10,000 m heat. He ran barefoot. That time stood as the Indian record for 32 years. In 2008, in Vigo, Spain, Surendra Singh lowered it to 28:02.89. That stood for sixteen more.

Then, in March 2024, at the same California meet, Gulveer ran 27:41.81. In November 2024, at Hachioji in Japan, he ran 27:14.88. In March 2025 he ran 27:00.22.

The first 32 years of that history took 46 seconds off the record. The last twelve months took more than a minute. If Hari Chand's Montreal run and Gulveer's California run could be staged on the same track, at those exact paces, Gulveer would cross the line with Hari Chand still more than 600 metres from home, over a lap and a half behind.

Around those 10,000 m marks, the rest of the page filled up. He ran 12:59.77 for 5000 m on a short indoor track in Boston in February 2025, the first Asian to break 13 minutes indoors and an Asian record. He ran 7:34.49 for 3000 m in Budapest in August 2025, another national record. He ran 1:12:06 for 25 km in Kolkata that December. In May 2026 he ran 13:03.93 outdoors in Los Angeles, the outdoor national best, finishing second.

And on 15 March 2026, in the New York City Half Marathon, he ran 59:42 and finished third, the first Indian under an hour for the distance. Even that number came with a footnote. Because the course runs point to point, it is not eligible as a record, and Avinash Sable's 1:00:30 from Delhi in 2020 remains the official mark. The fastest half marathon ever run by an Indian is not the Indian record. Numbers, again, come with conditions attached.

The other column

Now look at a different column of the same athlete's results.

At the 2025 World Championships in Tokyo, the same season as the California record, Gulveer finished 16th in the 10,000 m final in 29:13.33. In his 5000 m heat he was ninth, in 13:42.34, and missed the final by 0.19 seconds. At the 2026 Commonwealth Games in Glasgow he won 10,000 m silver in 27:49.78, behind Ky Robinson, and 5000 m bronze in 13:24.95, becoming the first Indian to win two track medals at a single Games. At the Asian Games in Aichi-Nagoya in September 2026 he won silver in the 10,000 m in 28:29.38, silver in the 1500 m in 3:36.72, and bronze in the 5000 m in 13:32.75.

These are medals, several of them historic. Yet almost none of those championship times is close to his best. The Tokyo 10,000 m was more than two minutes slower than California. The Nagoya 10,000 m silver was about a minute and a half slower.

Which number is the real Gulveer Singh?

A coach who uses a performance-based system such as Jack Daniels' VDOT has a sensible answer. Take the best recent, fair performance, the one run in conditions designed to let ability show, and prescribe training from that. The California record is precisely that kind of race. It is a good anchor.

But the results sheet will not tell that coach why Tokyo was slower. It will not say whether the difference lay in tactics, in the weather, in the stage of the season, in the athlete's state that week or in the simple truth that championship finals are races for places, not for times. The public record does not say, and this book will not guess. What a coach can see is the shape of the problem: one athlete, one event, one year, and two numbers more than two minutes apart.

What numbers can and cannot tell

Here is the lesson of Gulveer's numbers, and it is one every coach should carry into the next seven chapters.

A race time is a superb summary. It includes, in one figure, the athlete's aerobic capacity, the fraction of it they could hold, their economy, their durability and their tactics on the day. That is exactly why systems built on performance work so well for setting training intensity.

But a summary is not an explanation. It cannot tell you which of those ingredients changed when the time changed. It cannot tell you whether the athlete who runs 29:13 in one race and 27:00 in another has a different engine or the same engine used differently. And it cannot tell you how confident to be in the number itself: whether the course was measured, the pace aided, the air cool or the track fast.

Consider, too, how small the margins in this story are, and how differently they read. Twenty-two hundredths of a second kept Gulveer from a sub-27 in California. Nineteen hundredths kept him out of a world championship final in Tokyo. On a results sheet those two fractions look almost identical. To the athlete, one was a national record with a near-miss attached; the other was the end of his 5000 m campaign. The same kind of number carried completely different meanings, depending on what the race was for.

The twenty-two hundredths of a second in California were measured to the finest precision a stadium clock can offer. The reasons behind them were not measured by any clock at all.

That is the gap this Part explores. First we look at what VDOT does, and why it deserves its reputation. Then we ask what a second layer, the AKURA Endurance Index, would need to add to explain the numbers rather than merely record them.

Sources: <https://www.olympics.com/en/news/gulveer-singh-india-10000m-national-record-the-ten-2025-athletics-california> ; <https://indianathletics.in/gulveer-singh-improves-national-10000m-record-narrowly-misses-his-goal-of-sub-27-minutes/> ; <https://www.olympics.com/en/news/who-is-gulveer-singh-india-athlete-long-distance-runner> ; <https://worldathletics.org/athletes/india/gulveer-singh-15023839> ; <https://www.olympics.com/en/news/hari-chand-indian-athlete-runner-olympian-dies> ; <https://www.hindustantimes.com/sports/others/after-asian-games-10-000m-silver-kartik-targets-national-record-101697292334456.html> ; <https://www.olympics.com/en/news/world-athletics-championships-2025-india-annu-rani-gulveer-singh-report> ; <https://khelnow.com/commonwealth-games/gulveer-singh-track-and-field-history-commonwealth-games-202608> ; <https://www.thestatesman.com/sports/gulveer-singh-adds-1500m-silver-to-asian-games-haul-sable-finishes-fourth-1503644118.html> ; <https://www.tribuneindia.com/news/sports/gulveer-singh-wins-third-medal-at-2026-asian-games-clinches-bronze-in-mens-5000m/> ; <https://www.olympics.com/en/news/athletics-list-of-indian-national-record-holders> ;

<https://www.tribuneindia.com/news/sports/with-a-silver-at-us-event-indian-army-athlete-breaks-5000-metre-national-record/>

CHAPTER 37

37. What VDOT Does

PART VII — VDOT AND AEI

Late afternoon at a municipal track in Pune. The concrete stands are empty except for a coach with a biro, a notebook swollen by humidity and a phone propped against a water bottle. His group of eleven runners has just finished a 10K time trial, and he is doing what coaches all over the world do next: turning a finishing time into a training plan. One runner's 41:30 goes into a calculator, and out come an easy pace, a threshold pace and an interval pace. It takes seconds. No mask, no treadmill, no blood sample. By tomorrow morning the athlete will know exactly how fast to run on Tuesday. There is something quietly radical about that. It treats the race itself as the test, and it trusts that a well-run 10K already contains most of what a laboratory would want to know about a runner's current ability.

Any honest discussion of a new endurance index must start by acknowledging what already works. VDOT, developed by the American coach and exercise physiologist Jack Daniels and set out in Daniels' Running Formula, is an established running-performance framework that represents current running ability and provides training paces from recent or estimated race performance. It has been used by coaches and runners around the world for decades, and its enduring popularity reflects a real achievement: it makes training intensity simple, consistent and actionable.

Performance first

The central idea of VDOT is that a recent race result already contains a great deal of physiological information. A runner's 10K time reflects their aerobic capacity, the fraction of it they can sustain, and their running economy, all expressed together under race conditions. Rather than requiring laboratory testing, VDOT uses that performance to estimate a single value of current running ability. Resources associated with Daniels' methods describe VDOT as a measure of current running ability and distinguish it from laboratory-measured VO_2max . Two runners can have similar VO_2max values yet different VDOT values because economy, threshold and other factors differ.

KEY IDEA

VDOT is a measure of current running ability derived from performance, not a laboratory measurement of VO_2max .

From ability to training intensity

VDOT is valuable because it translates performance into actionable training intensity. From one value, a runner receives guidance for different kinds of training and estimates of equivalent performances at other distances. Daniels organises training into clearly defined categories, each with a distinct purpose.

Category	Broad purpose in Daniels' system
Easy (E)	Aerobic development, recovery, volume
Marathon (M)	Marathon-specific pace and endurance
Threshold (T)	Raise sustainable high intensity
Interval (I)	Stress aerobic power
Repetition (R)	Speed, economy and mechanics

This book deliberately does not reproduce VDOT pace tables. Coaches and athletes who wish to use VDOT should consult Daniels' Running Formula and its official resources directly, where the method and its assumptions are explained by its author.

MYTH VS EVIDENCE

Myth: VDOT is just another name for VO_2max .

Evidence: VDOT is estimated from race performance and reflects current running ability, including the effects of economy and sustainable intensity; laboratory VO_2max measures maximal oxygen uptake only.

What AKURA learns from VDOT

The clarity of VDOT offers several lessons. First, a framework should begin with something athletes and coaches can observe: real performance. Second, it should produce decisions, not just descriptions; a number that does not change the next session has limited coaching value. Third, it should be updated as the athlete changes, rather than fixed for a season. Fourth, it should be simple enough to use daily on the track or road.

AKURA should learn from this clarity rather than copy the same purpose. VDOT already does its job well. There is little value in building a second system that merely produces training paces from race times. AKURA's research question is different, and Chapter 38 explains why.

Using VDOT in the Indian context

Performance-based systems assume that the race used as an anchor was a fair expression of ability. In India, many races are run in heat and humidity, on courses with variable measurement, and sometimes with crowded starts. A coach using VDOT

or any similar framework should prefer recent, well-measured, reasonably controlled performances, and should interpret results from very hot races cautiously. This is not a weakness of VDOT; it is a reminder that every input needs context.

The value of simplicity

Part of VDOT's success is that it is easy to explain. A runner with no laboratory access can enter a recent race result and immediately understand how hard their easy runs, threshold sessions and intervals should be. Coaches working with large groups can assign intensities consistently. As the athlete improves and races faster, the value rises and the training intensities progress with it. This self-correcting link between performance and prescription is one of the framework's most practical features.

Like any model, VDOT relies on assumptions. Its race equivalences represent typical relationships between distances, and individual athletes may be relatively stronger at short or long distances. Daniels himself emphasises that training should be adjusted to the individual and to conditions. Recognising these assumptions is not a criticism; it is how every model should be used, with judgement.

COACH'S CORNER

If your athlete already trains with VDOT, keep it. Use AKURA's measurements to explain why their VDOT is moving or not moving, for example whether an improvement came with better economy, better heat tolerance or simply a cooler race day.

CHECKPOINT

Which recent race would you use as your performance anchor today, and was it run in conditions fair enough to represent your current ability?

CHAPTER 38

38. Why AKURA Needs a Different Layer

PART VII — VDOT AND AEI

In 2025 Gulveer Singh ran two very different 10,000 m races. On 29 March, at The TEN in California, he finished sixth in 27:00.22, a national record that, according to SAI, made him the third-fastest Asian of all time. Later that year, in the World Championships final in Tokyo, he finished 16th in 29:13.33. Same athlete, same event, same season, and a gap of more than two minutes. A performance-based system does something sensible with that pair: it trusts the best recent, fair expression of ability and prescribes

training from it. But a coach looking at the two results wants to ask other questions too. Was the difference in the engine, the tactics, the conditions, the stage of the season or the athlete's state that week? The results sheet does not say. It records what happened, not why, and that gap between what and why is where this chapter begins.

If VDOT works, why propose anything else? The answer lies in the question each framework is designed to answer. VDOT begins primarily with performance: what can the athlete run now, and how should they train as a result? AKURA's proposed Endurance Index begins with a broader research question: what is the athlete's current performance expression, what physiological systems support it, how is the athlete responding to environment and training, and what is the confidence in each measurement?

Different questions, different tools

A performance-based system is intentionally efficient. It compresses many physiological factors into one ability value and does not need to know which factor produced the result. For daily training prescription, that compression is a strength. For research and long-term athlete development, particularly in an environment as varied as India's, it can hide the very information a coach needs. When an athlete's race times stagnate, the coach wants to know whether the limitation lies in aerobic capacity, economy, threshold, heat tolerance, recovery or durability. A single ability value cannot distinguish between these.

Feature	VDOT (Daniels)	AEI (AKURA prototype)
Primary input	Recent race performance	Performance plus physiology, environment, recovery
Main output	Current ability and training paces	Domain profile, fitness, readiness, trend
Status	Established, widely used	Research prototype, not validated
Environment	Handled through coach judgement	Logged as explicit data
Confidence	Implicit in quality of race used	Attached to each measurement

KEY IDEA

VDOT answers "what can this athlete run and how should they train?"; AEI asks "why can they run it, how are they responding, and how sure are we?"

Beside, not instead of

AEI is therefore not proposed as a replacement for VDOT. It is a research framework that could sit above or beside pace-based systems. An athlete might continue to receive training paces from an established method while AEI tracks whether their economy is

improving, whether a heat block changed their cardiovascular response, and whether their durability supports a marathon attempt. In that arrangement the two layers complement each other: one prescribes, the other explains and monitors.

MYTH VS EVIDENCE

Myth: A new index must replace older methods to be useful.

Evidence: Frameworks answering different questions can coexist; AEI is designed to add explanatory and monitoring information, while established pace systems continue to guide daily intensity.

A conservative first version

The first version should be deliberately conservative: fewer validated claims, more transparent data. It is tempting to produce a striking single score, a 0–100 value on a colourful dashboard. But any number that combines measurements with unvalidated weights may look precise while being scientifically fragile. AEI v0.1 therefore prioritises three principles: keep raw measurements visible; report domains separately until a combined model is validated; and attach a confidence level to every value.

For Indian athletes, the additional layer is particularly relevant. Training may alternate between hot plains, humid coasts and hill stations such as Ooty or Kodaikanal. Travel, exams, work shifts and irregular access to laboratory testing all influence the data. A framework that records environment and confidence explicitly is better suited to this reality than one that assumes stable conditions.

What the extra layer must earn

An additional layer adds complexity, and complexity has a cost. Athletes must record more data, coaches must interpret more information, and there are more opportunities for error. AEI must therefore earn its place by improving decisions: identifying the limiting system earlier, showing whether an intervention such as a heat block produced a response, flagging when readiness is too low for a key session, or predicting marathon durability more accurately than short-race times alone.

If, after careful study, a domain does not improve decisions, it should be simplified or removed. This is the discipline that separates a research framework from a marketing score. AKURA's aim is not to build the most elaborate index possible, but the most useful one that the evidence supports.

A practical example

Consider two Indian club runners, each with the same recent 10K time. A performance-based system would give them similar training intensities, and that is a sensible starting point. AEI might then reveal that the first runner's easy pace at a fixed heart rate deteriorates sharply in humid conditions, while the second runner's pace holds steady but their long-run durability is poor. The same pace prescription remains useful

for both; what changes is the emphasis of their next training block, heat preparation for one and marathon-specific endurance for the other.

COACH'S CORNER

Before adding any new metric to an athlete's profile, write down the decision it could change. If you cannot name a decision, record the metric for research but do not let it influence training yet.

CHECKPOINT

When your performance stalls, which question would help you more today: "what pace should I train at?" or "which system is limiting me?" Which tool do you currently use to answer each?

CHAPTER 39

39. AEI Architecture

PART VII — VDOT AND AEI

An overnight express from the south pulls into the platform at 4:40 a.m., and a young athlete climbs down with a kit bag, a stiff neck and perhaps three hours of broken sleep. She is fit. Her time trial three weeks ago was the best of her life. Today her plan says 6 × 1 km at threshold pace, and the plan knows nothing about the train, the heat of the upper berth, the canteen dinner or the scratchy throat she noticed somewhere after midnight. Nothing about her engine has changed overnight. Everything about her ability to use it has. Her coach, meeting her on the track at 6:30, faces one of the oldest questions in the sport: does he trust her fitness, or her face? Good coaches have always answered that by instinct. A research system has to answer it on purpose, with two separate records.

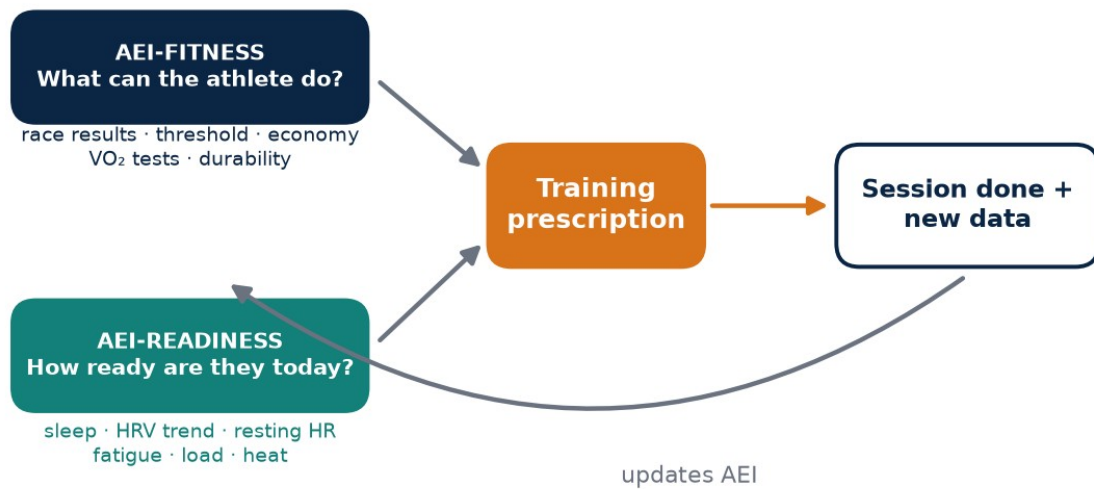


Figure 17. AEI architecture. Stable fitness and daily readiness are separate layers that together inform the prescription; every session feeds new data back.

A framework needs structure before it needs numbers. The proposed AEI architecture has two levels, reflecting a distinction that experienced coaches already make intuitively: what an athlete is capable of, and what they can express today. Separating these levels prevents two common errors, over-reacting to a single bad day and ignoring warning signs because the athlete is fit.

Level one: AEI-Fitness

AEI-Fitness describes relatively stable performance capacity: recent race performance, aerobic markers, threshold characteristics, economy and endurance durability. These qualities change over weeks and months in response to training, not overnight. Fitness is the athlete's current engine.

Level two: AEI-Readiness

AEI-Readiness describes today's ability to express that fitness: sleep, HRV trend, resting heart rate, fatigue, recent load, soreness, illness flags and environmental stress. These signals can change daily. A fit athlete who has slept badly after an overnight train journey, trained in 38 °C heat for three days and developed a sore throat may be unable to express their fitness in a hard session, even though their engine is unchanged.

Level	Time scale	Main inputs	Coaching question
AEI-Fitness	Weeks to months	Races, threshold, economy, aerobic markers, durability	What can this athlete do?
AEI-Readiness	Days	Sleep, HRV trend, RHR, fatigue, soreness, load, illness, heat	How much can they express today?

KEY IDEA

Fitness is the engine; readiness is whether the engine can be used fully today. AEI keeps them separate so that one does not distort the other.

The closed loop

The conceptual loop is: performance + physiology + environment + recovery → AEI → training prescription → actual session → new data → updated AEI. Each training session is therefore both an intervention and a measurement. The planned session reflects the current AEI; the executed session, with its pace, heart rate, perceived effort, conditions and completion status, becomes new evidence that updates the profile.

1. Collect performance, physiological, environmental and recovery data.
2. Update domain values and confidence levels in AEI.
3. Generate a training prescription, reviewed by the coach.
4. Record what the athlete actually did and how it felt.
5. Compare planned and actual responses, then update AEI.

This loop is what separates a research system from a static test report. A test tells the coach where the athlete was; a loop shows how the athlete is changing and whether training is working.

No weights before validation

No component should be given a final weight until validated on real athlete data. It would be easy to declare that performance counts for 40%, economy for 15% and recovery for 10%, and to produce a neat total. But such weights would be guesses, and guesses presented as numbers can mislead coaches and athletes. Until longitudinal data show which components genuinely predict performance and injury risk in the populations AKURA serves, each component should be reported separately with its confidence.

MYTH VS EVIDENCE

Myth: A single combined score is always more useful than several separate numbers.

Evidence: A combined score is only as good as its weights; with unvalidated weights, separate domain values with clear confidence levels give coaches more honest and more usable information.

FIELD PROTOCOL

1. Setting up the AEI loop for one athlete
2. record a recent standardised race or time trial as the performance anchor
3. record easy pace at a fixed heart rate on a flat route
4. log daily sleep, morning resting heart rate, HRV if available, fatigue and soreness
5. log temperature and humidity for every session
6. after each key session, record planned versus actual pace, heart rate and RPE
7. review the loop weekly with the coach.

Why two levels matter in practice

In the proposed AKURA ecosystem, the readiness layer also connects to AISRi, the daily training-decision layer, while AEI-Fitness informs longer-term planning. Keeping the interfaces separate makes it clear which information is driving which decision. Many training errors come from mixing the two levels. An athlete who feels flat for three days may conclude they have lost fitness and train harder to compensate, deepening their fatigue. Another athlete may feel excellent one morning after a rest day and attempt a session far beyond their current fitness. By keeping fitness and readiness in separate views, AEI encourages the right response to each: patience with readiness fluctuations, and evidence before revising fitness.

COACH'S CORNER

Keep a simple rule when the two levels disagree. If fitness is high and readiness is low, protect the athlete and modify the day. If readiness is high but fitness is low, do not mistake a good day for a breakthrough; progress the plan as designed.

CHECKPOINT

Can you describe your current fitness and your readiness today as two separate statements? Which one are you more likely to confuse with the other?

CHAPTER 40**40. Proposed AEI Domains**

PART VII — VDOT AND AEI

Spread a national-camp athlete's file across a table and it looks like an archaeological dig. Here is a race result from the Federation Cup. Here, a blood report from a laboratory in another city, its reference ranges printed

too small to read. Here, three months of watch data, a physio's note about a tight Achilles, a lactate curve from a winter camp, and a handwritten sleep diary that stops halfway through exam season. Every piece is real evidence. None of it, alone, describes the athlete. Pile it all together and add it up, and you get a number that looks precise and means very little. Sort it instead into labelled trays, one for the heart, one for the blood, one for heat, one for how the legs hold up after two hours, and the file starts to speak. That sorting is what the AEI domains are for.

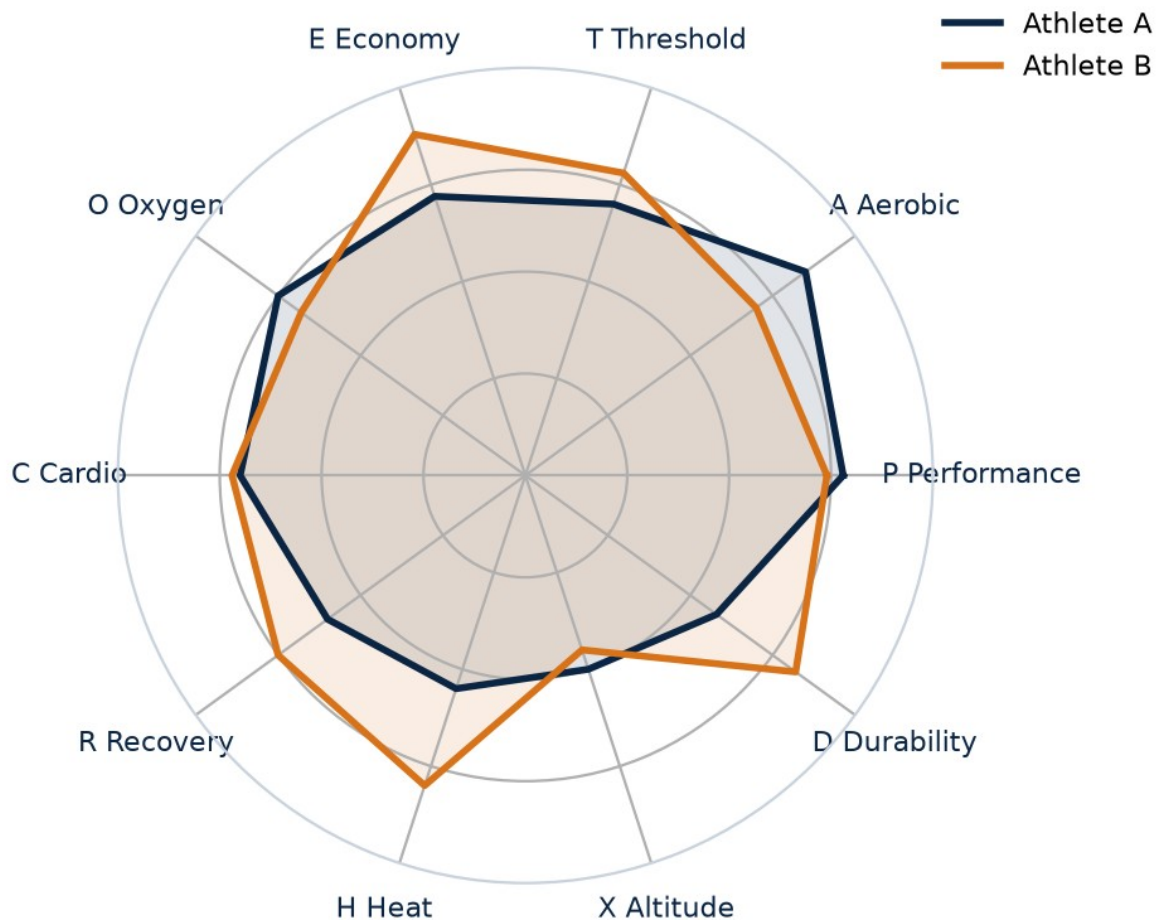


Figure 18. Two illustrative AEI domain profiles. The shape, not a single total, tells the coach where to look. Values are invented and not validated scores.

The two-level architecture needs content. A research prototype can store ten domains, each describing one part of the endurance system that earlier chapters of this book explained: performance, aerobic capacity, threshold, economy, oxygen transport, cardiovascular response, recovery, heat adaptation, altitude response and durability. Together they form a map of the athlete rather than a single verdict.

The ten domains

Domain	Primary examples	Interpretation	Confidence note
AEI-P Performance	5K, 10K, half-marathon, marathon results	Current race expression	High when recent and standardised
AEI-A Aerobic	VO ₂ -related tests, submaximal HR-pace	Aerobic capacity	Protocol dependent
AEI-T Threshold	Lactate or threshold pace	Sustainable high intensity	Protocol dependent
AEI-E Economy	O ₂ cost at a fixed speed	Movement efficiency	Needs repeatability
AEI-O Oxygen Transport	Hb mass, Hb concentration	O ₂ transport context	Hb mass needs specialist testing
AEI-C Cardiovascular	HR, stroke-volume-related measures	Circulatory response	Context dependent
AEI-R Recovery	Sleep, HRV trend, RHR, fatigue	Today's expression capacity	Baseline dependent
AEI-H Heat Adaptation	Heat exposure, thermal response	Heat adaptation	Environment must be logged
AEI-X Altitude Response	Sleeping altitude, Hb mass response	Hypoxic adaptation	Dose dependent
AEI-D Durability	Long-run degradation	Marathon-specific resilience	Needs standardised protocol

KEY IDEA

The ten AEI domains are a structured way to store evidence about an athlete; they are not yet validated scores.

Domains, not scores

These are domains, not yet validated scores. Each should have raw measurements, normalisation rules, confidence and missing-data handling. Raw measurements must remain visible so that a coach can always trace a conclusion back to the test that produced it. Normalisation rules describe how a value is compared, whether against the athlete's own history or, where appropriate, a reference population. Confidence reflects how reliable the measurement is: a laboratory economy test repeated twice under controlled conditions deserves more confidence than a single estimate from a wrist device. Missing-data handling defines what happens when a domain has no recent measurement, so that absence of data is never mistaken for a low value.

Some domains overlap. Aerobic capacity, threshold and economy all contribute to performance, and recovery influences how every other domain is expressed on the

day. Oxygen transport and altitude response are related but distinct: one describes the current state of the blood, the other how it changed after hypoxic exposure. Heat adaptation connects to the cardiovascular domain through plasma volume and heart-rate responses. Recording these domains separately allows the research programme to study their relationships rather than assuming them.

Not every athlete needs every domain

A club runner in Coimbatore may have reliable performance, recovery and heat data but no laboratory tests. A national-camp athlete may have regular haemoglobin mass and lactate testing. AEI should work in both cases by reporting what is known, with confidence, and clearly marking what is not. The minimum useful profile is often performance, recovery, a simple submaximal HR–pace marker and durability from long runs; laboratory domains add depth when available.

MYTH VS EVIDENCE

Myth: A polished 0–100 score is more scientific than a table of separate values.

Evidence: A beautiful 0–100 score can be scientifically misleading if its weights are arbitrary; transparent domain values with stated confidence are more honest until a combined model has been validated.

FIELD PROTOCOL

1. Building a first domain profile
2. list the ten domains
3. for each, write the most recent measurement, date and method
4. mark confidence as high, moderate or low
5. mark missing domains clearly as not measured
6. choose the two missing domains that would most change coaching decisions
7. plan how and when to measure them.

Confidence deserves its own display. A domain value measured last week in a laboratory and one estimated six months ago from a single field session should not look identical on a dashboard. Simple labels, such as high, moderate and low confidence, together with the date of the last measurement, allow coaches to weigh each domain appropriately.

How domains link back to the book

Each domain corresponds to physiology explained earlier. AEI-A and AEI-T draw on the chapters on VO_2max , lactate and threshold. AEI-E reflects running economy. AEI-O and AEI-X connect to haemoglobin mass, red-cell adaptation and altitude. AEI-C and AEI-H link to the heart, plasma volume and heat acclimation. AEI-R reflects recovery

monitoring, and AEI-D draws on the marathon chapters in Part VI. This mapping keeps AEI grounded in established physiology rather than invented categories, and it makes clear which scientific literature each domain should be validated against.

When a domain changes, the coach should be able to ask which physiological mechanism might explain the change and which intervention preceded it. A domain that cannot be linked to a mechanism, a measurement and a decision is not yet ready to be part of the index.

COACH'S CORNER

Resist the pressure to fill every domain. An honest profile with four well-measured domains is more useful than a complete profile with six guessed values. Ask the athlete to help keep the raw data clean, because they are the source of most daily inputs.

CHECKPOINT

Which AEI domains can you measure reliably today, which require help from a laboratory or clinician, and which missing domain matters most for your next goal race?

CHAPTER 41

41. AEI-Fitness

PART VII — VDOT AND AEI

On 15 March 2026 Gulveer Singh ran the New York City Half Marathon in 59:42, finishing third and becoming the first Indian to run under an hour for the distance. By any ordinary measure it was the fastest half marathon ever run by an Indian. Yet it is not the national record. The course runs point to point, with its start and finish about 11.6 km apart, and Olympics.com notes that it is not recognised for records. The official mark remains Avinash Sable's 1:00:30 from the Delhi Half Marathon in 2020. This is no slight on either man. Point-to-point courses can be helped by a following wind or a net drop, which is why record rules treat them differently, however fast the runner. Even the most striking performance must be read with its conditions. Record-keepers make that judgement with rules; AEI-Fitness has to make it with metadata.

AEI-Fitness should answer one question: "What can this athlete currently do under standardised conditions?" It is the slow-moving, evidence-heavy layer of the index, the

description of the engine that training has built. Getting it right matters because fitness is the foundation for goal-setting, race selection and long-term planning.

The performance anchor

The strongest anchor is recent performance under reasonably controlled conditions. Just as VDOT begins with performance, AEI-Fitness gives recent, well-measured results the greatest influence. A certified-course half marathon on a cool morning, or a carefully run time trial on a measured track, provides a clear expression of current capacity. A hot, crowded race or an estimated course distance provides weaker evidence and should carry lower confidence.

Supporting measures

Supporting measures can include threshold pace, VO_2 -related testing, running economy, long-run durability and physiological response. These do not replace performance; they help explain it and fill the gaps between races. An athlete who races only three or four times a year can still show fitness progression through a stable submaximal marker, such as easy pace at a fixed heart rate on the same route in similar conditions, or through a repeated threshold session.

Input	Role in AEI-Fitness	Update frequency
Recent race or time trial	Primary anchor	Every valid race or trial
Threshold session or test	Sustainable intensity marker	Every 4–8 weeks
Submaximal HR–pace	Aerobic trend marker	Weekly, same route and conditions
Running economy test	Efficiency marker	When laboratory access permits
Long-run durability	Marathon-specific resilience	Key long runs

KEY IDEA

AEI-Fitness is anchored in real performance and supported by physiological markers; it describes capacity, not how the athlete feels today.

Slow to change, by design

Fitness should update more slowly than daily readiness. A single bad night's sleep should not erase months of training adaptation. Conversely, a single breakthrough race should not permanently redefine the athlete without confirmation. A new personal best in ideal weather with a strong pacing group may overstate typical capacity; a disappointing race in 34 °C heat may understate it. AEI-Fitness should respond to consistent evidence across several data points, and it should record the conditions of each so that environmental effects are not mistaken for changes in fitness.

Feature	Fitness	Readiness
Question	What can the athlete do?	How much can they express today?
Time scale	Weeks to months	Hours to days
Primary evidence	Races, tests, long-run durability	Sleep, HRV trend, RHR, soreness, load
Typical change	Gradual, training-driven	Rapid, recovery- and life-driven
Main use	Goals, race choice, training phase	Daily session decision

MYTH VS EVIDENCE

Myth: One brilliant race proves a new fitness level.

Evidence: A single result may reflect ideal conditions, pacing or course; fitness should be confirmed by consistent evidence from races, tests and training markers.

FIELD PROTOCOL

1. Standardised fitness check
2. select a flat measured loop or 400 m track
3. record temperature, humidity and time of day
4. warm up 15 min easy
5. run 20 min at a fixed heart rate near the top of the easy zone and record average pace
6. after 5 min recovery, complete a controlled 3 km or 5 km time trial
7. repeat every 4–6 weeks under similar conditions and compare.

Fitness also has a direction. An athlete returning from injury may have a modest current anchor but a steeply rising trend, while a veteran may hold a high anchor that is slowly declining. Both facts matter for planning, which is why AEI-Fitness should always be read alongside the trend view described in Chapter 43.

Fitness is multidimensional

Although AEI-Fitness may one day be summarised in a single figure, it is built from several components. Two athletes with the same performance anchor may have different supporting profiles: one with excellent economy but modest threshold, another with a high aerobic ceiling but poor durability. Keeping these components visible allows training to target the actual limiter. In the prototype stage, AEI-Fitness is best presented as the performance anchor plus a short list of supporting markers, each with its date and confidence.

For Indian athletes who train through long hot seasons, conditions deserve special attention. Fitness markers collected in April and May may look weaker than those

collected in December, even when underlying fitness has improved. Recording temperature and humidity with every standardised test allows AKURA to compare like with like and, over time, to model environmental effects directly.

COACH'S CORNER

Update your athlete's fitness picture after patterns, not events. When a surprising result appears, ask whether conditions, pacing or measurement explain it, and look for a second confirming data point before changing training zones or goals.

CHECKPOINT

What is your current performance anchor, how old is it, and what supporting marker would confirm that your fitness has really changed since then?

CHAPTER 42

42. AEI-Readiness

PART VII — VDOT AND AEI

In July 2025, at the Monaco Diamond League, Avinash Sable, India's steeplechase record holder, injured the ACL and meniscus of his right knee and needed surgery. He missed that year's World Championships in Tokyo and could not compete at the 2026 Commonwealth Games in Glasgow. His first steeplechase back was 8:32.39 at the Indian Athletics Series final in September 2026; he then finished fourth at the Asian Games in 8:21.67, a season's best. Looking back on his injury, he told Olympics.com: "I think I might have rushed my return a bit, which caused the injury to recur." It is an unusually honest sentence from an elite athlete, and it describes a problem every coach knows. Fitness can say yes before the body is ready to. Readiness is the layer that listens for that difference, day by day.

Readiness asks a different question: "How well can the athlete express current fitness today?" The engine may be strong, but on a given morning it may be cold, under-fuelled, overheated or strained. AEI-Readiness is the daily layer that helps the coach decide whether the planned session should go ahead, be modified or be replaced.

Inputs, not verdicts

A readiness layer can use HRV trend, resting heart rate, sleep quantity and quality, subjective fatigue, muscle soreness, recent training load and environmental strain. No single metric should dominate. Heart-rate variability is a useful, context-dependent signal of autonomic state, and research such as that by Plews and colleagues has shown

the value of interpreting it as a rolling trend rather than a single morning value. Resting heart rate, sleep and subjective wellbeing add further perspectives. Simple self-reported measures of fatigue and soreness are often as informative as device data and should never be dismissed.

Environmental strain belongs in readiness because it changes the cost of the day's session. A threshold workout planned for 6 a.m. may be appropriate in cool conditions and inappropriate in a humid heatwave, even for a well-recovered athlete.

Signal	What it may reflect	Common confounder
HRV trend	Autonomic balance and recovery	Alcohol, illness, stress, measurement timing
Resting heart rate	Fatigue, illness, heat, dehydration	Caffeine, poor sleep, measurement posture
Sleep	Recovery opportunity	Travel, shift work, exams, family demands
Fatigue and soreness	Perceived recovery	Mood, motivation, life stress
Recent load	Accumulated training stress	Unrecorded activity, heat
Environmental strain	Cost of today's session	Microclimate, time of day

KEY IDEA

Readiness is a confidence-adjusted judgement built from several signals; no single wearable metric is a diagnosis.

From signals to decisions

The practical output is not "train hard" or "rest" from one number. It is a confidence-adjusted recommendation: planned session, modified session, easy session or recovery, with coach review when risk signals conflict. When most signals agree that the athlete is well recovered, the planned session proceeds. When some signals are unfavourable, the session can be modified, for example by reducing volume, moving it to a cooler time or converting intervals to steady running. When several signals are poor, an easy session or recovery day is preferable. When signals conflict, or when an illness flag appears, a human review is required.

1. Planned session: signals broadly favourable, no risk flags.
2. Modified session: one or two unfavourable signals or high heat strain.
3. Easy session: several unfavourable signals, no illness flags.
4. Recovery or coach review: illness symptoms, conflicting risk signals or sharp unexplained change.

MYTH VS EVIDENCE

Myth: A high readiness score means the athlete is safe to train hard.

Evidence: Readiness scores cannot detect every problem; symptoms such as chest pain, fever, dizziness or signs of heat illness always override any score and require appropriate medical attention.

FIELD PROTOCOL

1. Morning readiness routine
2. on waking, before caffeine, record resting heart rate and HRV in the same posture
3. record sleep duration and a 1–5 sleep quality rating
4. rate fatigue, soreness and mood 1–5
5. note any illness symptoms
6. check forecast temperature and humidity for the session time
7. compare with the athlete's 7-day and 28-day norms rather than with other athletes.

Recording the decision itself is equally important. When the coach modifies a session, the reason and the outcome should be logged. Over months, these records show whether readiness-based modifications were justified, and they become part of the data that will eventually be used to validate or revise the readiness model.

Readiness in real Indian life

Readiness is shaped by more than training. Long commutes, night shifts, examination seasons, family responsibilities, festival travel and poor sleep in hot, humid bedrooms all influence how an athlete arrives at a session. A readiness system designed for Indian athletes should make it easy to record these factors, because they often explain changes that physiological data alone cannot. The coach's knowledge of the athlete's life remains one of the most important readiness inputs.

Readiness data should also be interpreted against each athlete's own baseline. One athlete's normal resting heart rate may be another's warning sign. For this reason, the first few weeks of monitoring are best treated as a baseline-building period, during which recommendations rely more heavily on coach judgement.

COACH'S CORNER

Teach athletes that modifying a session is a skill, not a failure. A hard session moved by one day or shortened by a third on a poor readiness morning usually costs nothing in fitness, while forcing it can cost a week.

CHECKPOINT

Which readiness signal do you trust most for yourself, and have you ever checked whether it actually predicts your good and bad training days?

CHAPTER 43**43. AEI Trend**

PART VII — VDOT AND AEI

Line up India's men's 10,000 m records and they read like a slow film that suddenly speeds up. In 1976, at the Montreal Olympics, Hari Chand ran 28:48.72 barefoot in his heat. That mark stood for 32 years. In 2008, in Vigo, Surendra Singh lowered it to 28:02.89, and it stood for another sixteen. Then Gulveer Singh ran 27:41.81 in California in March 2024, 27:14.88 at Hachioji in Japan that November, and 27:00.22 back in California in March 2025. The first 32 years of that history took 46 seconds off the record. The last twelve months took more than a minute. Any single one of those times is a photograph. Only laid side by side do they show direction and speed of change, and raise the question every trend raises: what changed? The records list cannot answer. A well-kept training record sometimes can.

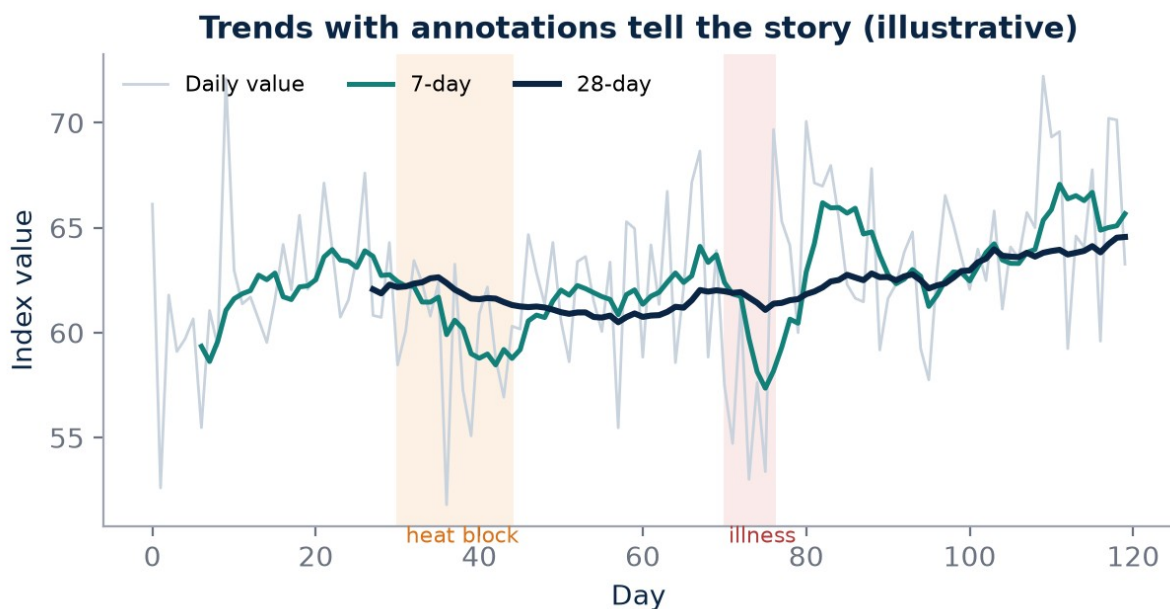


Figure 19. Rolling 7- and 28-day trends with annotations for interventions and illness make cause and effect visible. A single day's number hides this.

A single measurement is a photograph; a trend is a film. A trend is often more informative than a point, because it shows direction, speed of change and response to interventions. Many of the most important coaching questions, whether a heat block worked, whether fitness is still improving, whether fatigue is accumulating, can only be answered by looking across time.

When the trend and the race disagree

A runner's pace at a fixed heart rate can improve gradually while race performance remains unchanged because durability or pacing is limiting. Without a trend view, the coach might conclude that training has failed. With it, the coach can see that aerobic fitness is rising and look for the real limiter elsewhere, perhaps in the final 10 km of races, in fuelling, or in pacing discipline. The trend therefore protects good training from being abandoned because of one unrepresentative result.

KEY IDEA

Trends reveal whether an athlete is changing, and in which system, even when race results have not yet caught up.

Rolling windows

AEI Trend should therefore display rolling changes rather than only today's score. Useful views include 7-day, 28-day and 90-day trends, with annotations for races, illness, travel, heat blocks and altitude exposure.

Window	Best for	Typical signals	Caution
7-day	Short-term readiness and fatigue	HRV trend, RHR, sleep, soreness, load	Noisy; sensitive to single events
28-day	Training-block response	Submaximal HR–pace, threshold, load pattern	Affected by heat or travel within block
90-day	Fitness and adaptation	Race anchors, economy, durability, Hb markers	Slow to reveal sudden problems

Annotations are essential. A falling 28-day HR–pace trend might mean improved fitness, or it might mean the weather cooled. A rising resting heart rate might signal fatigue, or it might coincide with a week of travel to a hot city. By marking races, illness, travel, heat blocks, altitude camps, exams, work shifts and equipment changes, the coach can separate cause from coincidence.

MYTH VS EVIDENCE

Myth: A single green number on a dashboard tells you the athlete is on track.

Evidence: A single number hides direction and context; rolling trends with annotations show what is changing, how fast, and what might be causing it.

A dashboard that asks questions

A research dashboard should make cause-and-effect questions visible rather than hiding them behind a single green number. Did easy pace at a fixed heart rate improve after the Chennai heat block? Did resting heart rate settle after the athlete returned from Kodaikanal? Did long-run durability improve over the last 90 days while 5K speed stayed flat? A well-designed trend view invites these questions and allows the coach and athlete to test their hypotheses against the data.

For AKURA, trend data are also the raw material for validation. Only by collecting consistent, annotated measurements over months can the research programme learn which domains truly predict performance, which interventions produce meaningful responses, and how Indian athletes adapt to the specific environments in which they train.

FIELD PROTOCOL

1. Monthly trend review
2. export 90 days of training, readiness and race data
3. plot submaximal HR–pace, resting heart rate, HRV trend and weekly load
4. mark races, illness, travel, heat and altitude exposure on the timeline
5. identify one improving, one stable and one declining signal
6. write a hypothesis explaining each
7. agree one training adjustment and one measurement to test it next month.

Trend windows should also match the question. A coach deciding whether to hold tomorrow's intervals needs the 7-day view. A coach evaluating a four-week heat block needs the 28-day view with the block clearly annotated. A coach planning the next marathon cycle needs the 90-day view and the history of previous cycles. Showing all three windows side by side prevents short-term noise from being mistaken for long-term change, and prevents slow long-term decline from being hidden by a good week.

COACH'S CORNER

Review trends with the athlete, not just for them. When athletes see their own HR–pace line improving through a hard humid summer, they understand their training, and understanding improves adherence far more than any score.

CHECKPOINT

Looking at your last 90 days, which signal has changed most, what annotation might explain it, and what measurement would confirm or reject that explanation?

PART VII AT A GLANCE

The key idea from each chapter, for quick revision.

- | | |
|-----------|--|
| 37 | What VDOT Does
VDOT is a measure of current running ability derived from performance, not a laboratory measurement of VO ₂ max. |
| 38 | Why AKURA Needs a Different Layer
VDOT answers "what can this athlete run and how should they train?"; AEI asks "why can they run it, how are they responding, and how sure are we?" |
| 39 | AEI Architecture
Fitness is the engine; readiness is whether the engine can be used fully today. AEI keeps them separate so that one does not distort the other. |
| 40 | Proposed AEI Domains
The ten AEI domains are a structured way to store evidence about an athlete; they are not yet validated scores. |
| 41 | AEI-Fitness
AEI-Fitness is anchored in real performance and supported by physiological markers; it describes capacity, not how the athlete feels today. |
| 42 | AEI-Readiness
Readiness is a confidence-adjusted judgement built from several signals; no single wearable metric is a diagnosis. |
| 43 | AEI Trend
Trends reveal whether an athlete is changing, and in which system, even when race results have not yet caught up. |

PART VIII — TESTING LAB

Measurement is where the AKURA Endurance Index stops being an idea and becomes a record. This Part moves from the field to the laboratory: the minimum baseline every athlete should have, the blood markers that matter and why they must be read by a clinician, the difference between haemoglobin concentration and total haemoglobin mass, and the practical protocols for VO_2 , lactate and running-economy testing. It closes with cardiac screening, where the coach's role is to refer, not to diagnose. Throughout, one principle holds: a test is only as useful as its protocol, its metadata and our understanding of its error.

IN THIS PART

- 44 The AKURA Athlete Baseline
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- 50 ECG and Echocardiography

PART VIII • TESTING LAB

Every Test Needs Its Protocol

From field baseline to blood, Hb mass, VO₂, lactate and economy testing: each value needs protocol, metadata and known error.

2% vs 3%

An economy gain smaller than the test's typical error means nothing

✓ 7–10 d

Field baseline battery

Resting HR, HRV, sleep, a submax run, a threshold session and a long run, spread over 7–10 days.

☀ 7 fields

Metadata is half the test

Protocol ID, date and time, device, operator, environment, units and athlete state make values comparable.



Question first, test second

Blood markers such as ferritin and haemoglobin must be interpreted by a qualified clinician.

💧 Hb mass

Mass, not concentration

To know if red-cell mass rose, measure total Hb mass by optimised CO-rebreathing under supervision.

🏃 Submax

Beneath the ceiling

VO₂max is one output; submaximal oxygen cost at fixed speeds often reveals adaptations VO₂max hides.

❤ Refer

Coaches refer, clinicians clear

Exertional chest pain, fainting or unexplained palpitations: stop and see a doctor.

THE PROCESS



A value without its protocol, device, conditions and units cannot safely be compared over time.

THE STORY

The Coach Who Counted

PART VIII — TESTING LAB

Guangzhou, 21 November 2010. The women's 10,000 m at the Asian Games. Twenty-five laps, and at the end of them two Indian women cross the line less than a second apart.

Preeja Sreedharan wins gold in 31:50.47. Kavita Raut takes silver in 31:51.44. Ninety-seven hundredths of a second separate them after ten kilometres.

Sixteen years later, Preeja's time is still the Indian national record. Nobody, in all the years of improved shoes, foreign camps and new coaching systems since, has run faster.

Behind both women stood the same man.

The arrival

Nikolai Snesev came to India from Belarus in 2005 to coach distance runners. According to The Bridge, he arrived to work with Preeja and Kavita, and five years later he guided them to that one-two finish in Guangzhou.

They were not privileged athletes. Preeja was born in 1982 at Mullakkanam in Idukki, Kerala. It has been reported that her father, a manual labourer, died when she was eight, and that her brother left school to support the family. She went on to work as a superintendent with Southern Railway. Kavita came from Nashik district in Maharashtra, where her coach was Vijender Singh. He later described the distances she had to cover as a child.

Remember, Kavita had to walk so many kilometres to reach school or fetch water.

— Vijender Singh, quoted by The Indian Express

Consider what that finish demanded. A 10,000 m in 31:50 means holding a little over 3 minutes 11 seconds for every kilometre, for more than half an hour, and still having enough left at the end to race a teammate to the line. The stopwatch caught the result to the hundredth. It did not catch anything that made it possible.

Guangzhou was not an isolated peak. A month earlier, at the Commonwealth Games in Delhi, Kavita had won 10,000 m bronze in 33:05.28, the first individual track medal by an Indian woman at the Games. Five days after the 10,000 m in Guangzhou, Preeja took 5000 m silver in 15:15.89, a national record that lasted until 2023, and Kavita took bronze. At the same Games, Sudha Singh won the women's 3000 m steeplechase in 9:55.67.

The steeplechase years

Sudha Singh, born in Raebareli, Uttar Pradesh, came to the steeplechase through her employer rather than a stadium. She worked for Central Railway, and a Railways officer pointed her towards the event.

Madhu Shivdutt, my railway officer introduced me to Steeplechase.

— Sudha Singh, quoted by Mumbai Live

Lalita Babar came from Mohi, a village in the drought-prone Satara district of Maharashtra, from a farming family. As a child she ran four kilometres each way to school and fetched water for a household of seventeen. At fifteen, in 2004, she won her first prize money, 10,000 rupees, in a 10 km road race. She made her name on the roads first, and won the Indian women's title at the Mumbai Marathon three years in a row, from 2012 to 2014. Snegarev introduced her to the steeplechase around 2010. She committed to it fully in 2014, and eight months later won bronze at the Incheon Asian Games in a national record of 9:35.37.

Then came the kind of sequence a statistician loves. In June 2015 Lalita won the Asian Championships in Wuhan in 9:34.13. At the World Championships in Beijing that August she ran 9:27.86 in her heat and went on to finish eighth in the final, the first Indian woman to reach a world steeplechase final. In late April 2016 she ran 9:27.09 at the Federation Cup in New Delhi. In May, at the Shanghai Diamond League, Sudha ran 9:26.55 and took the record back. That year both women trained together in a Bengaluru camp under Snegarev. Two athletes in the same group, trading a national record back and forth within weeks.

In August 2016, in her Olympic heat in Rio, Lalita ran 9:19.76, another national record. In the final she was tenth in 9:22.74, the first Indian to reach an Olympic track final since P. T. Usha in 1984.

The steeplechase is an unusually hard event to read from a time alone. Over 3000 m the athlete clears twenty-eight barriers and seven water jumps, so a result blends aerobic power, economy, hurdling technique, rhythm and the ability to keep landing cleanly as the legs tire. When Lalita's time fell by more than fifteen seconds between Incheon and Rio, it is natural to ask which of those qualities improved. The results sheet cannot say.

Every one of those numbers was measured to the hundredth of a second. They are precise, public and permanent.

What was counted

Snegarev's reputation rested on those numbers, and on something harder to count. Olympics.com described him as an advocate of "natural methods" over performance-enhancing drugs. That is a characterisation by a writer, not a quotation from the coach, but it is how he was remembered.

His time with India did not end smoothly. He left the Athletics Federation of India in February 2019, in circumstances that were reported as acrimonious. One reported factor was the departure of steeplechaser Avinash Sable, who left Snesev's group in 2018 and went back to his Army coach, Amrish Kumar. At the Federation Cup in Patiala in March 2019, Sable broke the national record with 8:28.94.

Snesev was re-appointed in early 2021. He returned to India on 2 March and was based in Bengaluru. On 5 March 2021, during Indian Grand Prix 3, he was found dead in his hostel room at the National Institute of Sports in Patiala. He was 72. The cause was not known at the time, and a post-mortem was pending. This book records the fact and goes no further.

The limits of the stopwatch

Here is what the public record of the Snesev era preserves in abundance: finishing times, placings, medals, national records, the dates on which they fell. Here is what it preserves almost nothing of: the physiology beneath them.

We do not know, from anything in the public domain, how these athletes' haemoglobin mass changed through a training cycle, what their lactate curves looked like in winter and in summer, how their running economy moved from one season to the next, or how their bodies responded to a camp in Bengaluru compared with one elsewhere. If such records were kept, they have not been published. The legacy that survives in public is almost entirely a legacy of times.

Imagine, instead, a different archive. Next to each national record, a dated haemoglobin-mass value. Next to each camp, the altitude, the days spent there and the before-and-after blood work. Next to each season, a lactate curve run to the same protocol, on the same analyser, at the same time of year. With that archive, a young coach in Nashik or Idukki today could ask not only what Preeja and Lalita ran, but what their bodies were doing when they ran it.

That is not a criticism of one coach or one era. It is true of most of Indian distance running, and much of the world's. But it has a cost. When Preeja's 31:50.47 has stood for sixteen years, the obvious question is why. Was it something about the athletes, the training, the altitude, the racing, the era? A stopwatch cannot answer. A stopwatch can only tell you that nobody has run faster.

Measurement is how a coaching tradition outlives its coach. Times tell the next generation what was done. Only systematic, well-documented measurement of the body can tell them how, and whether it can be done again.

The Testing Lab begins there: with what should be measured, how it should be measured, and how much any single number can really be trusted.

Sources: <https://worldathletics.org/athletes/india/preeja-sreedharan-14283083> ; <https://worldathletics.org/athletes/india/kavita-raut-14283028> ; <https://thebridge.in/athletics/india-athletics-coach-nikolai-snesev-dead-19834> ; <https://www.olympics.com/en/news/nikolai-snesev-the-late-belarusian-coach-s-contribution-to-indian-athletics> ; <https://www.olympics.com/en/news/india-s-athletics-coach-nikolai-snesev-found>

dead-in-hostel-room ; https://www.espn.com/athletics/story/_/id/31010198/athletics-coach-nikolai-snesarev-found-dead-nis-patala-hostel-room ; <https://www.deccanherald.com/sports/snesarevs-return-leaves-two-indian-2176969> ; <https://indianexpress.com/article/sports/sport-others/another-middle-distance-star-emerges-from-nashik-4739302/> ; <https://indianexpress.com/article/sports/sport-others/indian-long-distance-runner-to-retire-from-international-events/> ; <https://web.archive.org/web/20111002054457/http://hindu.com/2010/10/09/stories/2010100956762200.htm> ; <https://www.mumbailive.com/en/sports/exclusive-interview-jakarta-asian-games-2018-steeplechase-silver-medalist-sudha-singh-29667> ; <https://www.thequint.com/sports/sports-buzz/from-running-in-fields-to-the-rio-tracks-lalita-babars-story> ; <https://www.hindustantimes.com/other-sports/national-record-a-gift-for-india-s-support-steeplechaser-sudha-singh/story-zUIqWcXotQy21aqUfvaiXK.html> ; <http://indianexpress.com/sports/rio-2016-olympics/lalita-babar-finishes-10th-in-3000m-steeplechase-india-2977275/> ; <https://www.olympics.com/en/news/who-is-avinash-sable-india-athlete-steeplechase>

CHAPTER 44

44. The AKURA Athlete Baseline

PART VIII — TESTING LAB

In 2011 a schoolgirl from Iklauta village near Meerut lined up for an 800 m race barefoot, because her father had suggested she try it. "Papa asked me if I wanted to run and I agreed. I had never practised before that school race," Parul Chaudhary later told The Indian Express. She went on to train on the paths between sugarcane fields. Fifteen years later, on 27 September 2026 at the Asian Games in Nagoya, she won steeplechase bronze in 9:08.67, a national record. Between that barefoot 800 m and the Nagoya track lies an entire physiological biography, and the public record holds only a few of its pages. That is normal. Most Indian athletes enter structured programmes with no baseline at all: no resting heart rate on file, no recorded training history, no first blood test. By the time anyone thinks to measure, the most interesting starting point has usually gone.

A serious athlete research programme starts with a baseline. Without a clear picture of where an athlete begins, every later number floats free of context: a threshold pace that looks impressive may simply reflect a cool morning, and a heart rate that looks alarming may be that athlete's normal. The AKURA baseline is not a single test but a structured snapshot of the athlete's physiology, training history and daily life, recorded in a way that can be repeated months or years later. It has two layers: a minimum field baseline that any coach in India can collect with a watch, a scale and a notebook, and a laboratory baseline added when it is clinically and logistically appropriate.

The minimum field baseline

The minimum field baseline can include body mass, a recent race result, resting heart rate, the heart-rate variability (HRV) trend, sleep, training history, the easy pace-HR relationship, a threshold session and long-run durability. None of these requires a laboratory, yet together they describe much of what the AEI needs. The race result

anchors performance; resting HR and HRV describe the autonomic background; sleep and training history explain what the athlete has been able to absorb; the easy pace-HR relationship describes aerobic efficiency; the threshold session estimates sustainable intensity; and long-run durability shows how well that efficiency holds as fatigue builds.

The laboratory baseline may include blood testing, body composition, ECG, echocardiography, VO₂ testing and lactate testing. These add depth and precision, but they are not a prerequisite for a meaningful AEI. A well-run field battery, repeated consistently, often tells the coach more than a single expensive laboratory visit that is never repeated.

Domain	Field measure	Laboratory measure
Performance	Recent race or time trial	Incremental test to exhaustion
Aerobic efficiency	Easy pace-HR relationship	Submax VO ₂ and running economy
Threshold	Threshold session pace and HR	Lactate step test
Durability	Long-run pace-HR decoupling	Prolonged steady-state testing
Health background	Resting HR, HRV, sleep, body mass	Blood markers, ECG, echocardiography
Body composition	Body mass trend	DXA or other validated method

FIELD PROTOCOL

1. AKURA field baseline battery (spread over 7–10 days)
2. Days 1–7: record waking resting HR, HRV (same device, same posture), sleep duration and morning body mass after toilet, before food
3. Day 2: standardised submax run – 10 min easy warm-up, then 20 min at a fixed easy pace on a flat, known route, recording mean HR for minutes 10–20
4. Day 4: threshold session – for example 3 × 10 min at estimated threshold effort with 2 min jog, recording pace, HR and RPE for each rep
5. Day 7: long run of the athlete's usual duration at easy effort on a flat route, recording pace and HR for pace-HR decoupling
6. Log temperature, humidity, time of day, surface and shoes for every session

Metadata: the invisible half of every test

Every test needs a protocol, date, device, operator, environmental conditions and units. Without those fields, longitudinal comparison becomes fragile. A resting heart rate taken lying down on waking is not comparable with one taken seated after tea; a lactate value from a fingertip is not directly comparable with one from an earlobe; a submax run at 6 a.m. in Pune in January is not the same test as the same run at 4 p.m. in Chennai in May. Metadata is what allows AEI to compare like with like, and to flag comparisons that should not be made.

Metadata field	Example entry	Why it matters
Protocol ID	SUBMAX-20-v1	Ensures the same test is repeated
Date and time	2026-03-14, 06:10	Circadian and seasonal effects
Device and firmware	Chest strap, watch model	Device changes alter values
Operator	Coach or lab technician name	Operator technique varies
Environment	Temp, humidity, WBGT, wind	Heat alters HR and lactate
Units	mmol/L, ml/kg/min, min/km	Prevents conversion errors
Athlete state	Sleep, illness, last meal	Explains outliers

KEY IDEA

A baseline is not a single number but a reproducible snapshot. The test and its metadata together form the measurement; a value without its protocol, device, conditions and units cannot safely be compared over time.

Typical error and the smallest worthwhile change

Every measurement contains noise. Hopkins's influential 2000 review in Sports Medicine described how the typical error of a measurement, the within-athlete variation when a test is repeated under stable conditions, determines how confidently we can detect real change. Alongside typical error sits the smallest worthwhile change: the smallest difference that would matter in practice, often estimated from the variation between athletes or from what would change a race result. A change is interesting only when it is clearly larger than the noise and at least as large as the smallest worthwhile change. The practical way to estimate typical error for an individual athlete is simple: repeat the same test two or three times, close together, under controlled conditions, and see how much the results vary when nothing real has changed.

COACH'S CORNER

In the first month, run the standardised submax test twice, a week apart, before any major training change. The difference between those two runs gives you a rough sense of that athlete's day-to-day noise. Later, treat only changes clearly beyond that range as signals worth acting on.

CHECKPOINT

Which items in your athletes' current baseline lack a recorded protocol, device or environment? If you repeated your main field test tomorrow, how much would you expect the result to differ with no real change in fitness?

CHAPTER 45**45. Blood Testing**

PART VIII — TESTING LAB

Seven in the morning at a sports hostel in Tamil Nadu. A queue of teenage athletes in tracksuits waits outside the sick-room door for a blood draw, some chattering, some turning pale at the sight of the needle. Few of them will have seen their own haemoglobin value before. When the reports come back, the coaches may be surprised. In a 2025 study of 104 female athletes in Tamil Nadu, volleyball and ball-badminton players rather than runners, about 53.8 per cent were anaemic. The national background is similar: a venous-blood survey across eight Indian states found anaemia in 44 per cent of adolescent girls and 41 per cent of adult women. For a distance runner, whose sport is built on carrying oxygen, those figures are not abstract. Nor do they justify handing out iron tablets on a hunch. A blood report is a clinical document, and it needs a clinical reader.

Blood tests are powerful because they reach beneath performance into physiology: oxygen-carrying capacity, iron stores, vitamin status, thyroid and kidney function. They are also easy to misuse. A panel ordered without a question produces numbers without meaning, and a coach who interprets them without clinical training can cause real harm, either by missing something serious or by creating anxiety and inappropriate supplementation. In the AKURA system, blood testing should be clinically supervised and interpreted by qualified professionals. The coach's role is to recognise when a question exists, to help the athlete arrive at the test in a standardised state, and to record the clinician's interpretation alongside the training data.

Markers and why they are measured

Depending on the athlete and research question, relevant markers can include haemoglobin, haematocrit, ferritin, iron indices, B12, folate, vitamin D, thyroid markers, renal function and electrolytes. Each has a purpose, and each has confounders that a clinician must weigh.

Marker	Why it may be measured	Note
Haemoglobin	Oxygen-carrying capacity; anaemia screening	Interpret with clinician
Haematocrit	Proportion of red cells; plasma-volume context	Interpret with clinician
Ferritin	Iron stores; fatigue or altitude planning	Rises with inflammation; clinician
Iron indices (serum iron, transferrin saturation)	Iron availability for red-cell production	Interpret with clinician
Vitamin B12 and folate	Red-cell formation; diet-related deficiency	Interpret with clinician
Vitamin D	Bone health and general status	Interpret with clinician
Thyroid markers	Unexplained fatigue or weight change	Interpret with clinician
Renal function and electrolytes	Kidney health; hydration and heat context	Interpret with clinician

Question first, test second

The list should not become a routine shopping list for every runner. Tests should be linked to a clinical or research question. "Why is this athlete's easy-run heart rate rising while pace falls?" is a question. "Let us check everything" is not. Question-led testing reduces cost, avoids false alarms from the many markers that will fall slightly outside a laboratory's reference range by chance, and makes interpretation sharper because the clinician knows what is being asked.

Standardisation also matters. Many markers vary with hydration, time of day, recent hard exercise, illness and menstrual cycle. Plasma volume can shift after heavy training or heat exposure, altering concentrations without any change in total red-cell mass. A clinician can only interpret a result well when the sampling conditions are known and consistent.

FIELD PROTOCOL

1. Preparing an athlete for a clinically ordered blood test
2. Confirm the clinical or research question with the supervising clinician
3. Schedule a morning draw after an easy day or rest day, not within 24 hours of a hard or very long session
4. Follow the clinician's instructions on fasting and supplements
5. Record sleep, recent illness, menstrual-cycle day where relevant, and recent heat or altitude exposure
6. Use the same laboratory for repeat tests where possible
7. Store the report and clinician interpretation in separate AEI fields from the coach's notes

Iron: the endurance athlete's recurring question

In endurance athletes, iron status is particularly relevant when investigating erythropoietic response, fatigue or unexplained performance change. Iron is needed to build new red cells, so an athlete heading to altitude with depleted stores may not gain the expected haematological benefit. Endurance training itself can increase iron losses and alter absorption, and dietary patterns common in India, including predominantly vegetarian diets, mean iron intake and bioavailability deserve attention. Female athletes face additional menstrual losses. Yet iron supplementation is not harmless and should never be started on guesswork; the decision belongs with a qualified clinician after appropriate testing.

Longitudinal records add a further layer of value. A single ferritin value tells a clinician relatively little; a series of values taken under similar conditions, alongside training load, diet notes and menstrual history, can reveal a slow decline long before it becomes a clinical problem. This is where AKURA's data discipline helps the medical team. The coach does not interpret the numbers, but by keeping clean, dated, well-annotated records, the coach gives the clinician the context needed to interpret them well. In Indian programmes where athletes may use different laboratories in different cities, recording the laboratory name, the method where reported and the reference range printed on each report is especially important, because values and ranges are not always directly comparable between laboratories.

Privacy also matters. Blood results are sensitive health information. They should be stored with the athlete's informed consent, shared only with those who need them, and never posted in group chats or displayed on team noticeboards. For minors, parental or guardian consent and the involvement of a paediatric clinician are essential.

MYTH VS EVIDENCE

Myth: Every tired runner needs an iron tablet.

Evidence: Fatigue has many causes, including under-recovery, poor sleep, low energy availability and illness. Iron should be assessed with appropriate tests and supplementation decided by a qualified clinician, not assumed by a coach.

KEY IDEA

A blood test is only as useful as the question behind it. Markers must be interpreted by a qualified clinician who knows the sampling conditions and the athlete's history.

COACH'S CORNER

When an athlete reports persistent fatigue or an unexplained drop in performance, first check training load, sleep and illness records, then refer to a doctor with a written summary. Do not suggest a diagnosis or a supplement; describe what you have observed and let the clinician decide what to test.

CHECKPOINT

For each blood marker your athletes have had tested in the past year, can you name the question it was meant to answer? Who interpreted it, and is that interpretation stored with the result?

CHAPTER 46**46. Measuring Haemoglobin Mass**

PART VIII — TESTING LAB

Ooty sits 2,240 m above sea level in the Nilgiris, and on winter mornings the mist hangs so thickly over the eucalyptus that runners on the hill roads appear and vanish like rumours. Athletes from across India have long gone there for altitude. Before the 2016 Olympics, marathoner Thonakal Gopi spent what ESPN described as a ten-month stint at a high-altitude Ooty facility; before his 2:12:23 at the Valencia Marathon in December 2025, he spent four more months there. "I think the high altitude at Ooty helped my performance," he said afterwards, as reported by the running writer Shyam Gopan. He may well be right. But here is the uncomfortable part: research for this book found no peer-reviewed study of Indian athletes' haematology at Ooty. The belief that the Nilgiris build red cells is widely

held and physiologically plausible. For Indian athletes, it has rarely been tested in public with the one measurement that could settle it.

Altitude training is often justified by its effect on red cells, so it is natural to reach for a blood test to see whether it "worked". The difficulty is that the most common blood value, haemoglobin concentration, answers a different question from the one being asked. This chapter explains why, and why the AKURA research rule is simple: use the measurement that actually answers the question.

Concentration is not mass

Haemoglobin concentration from a standard blood sample is not identical to total haemoglobin mass. Concentration is the amount of haemoglobin per unit volume of blood; total haemoglobin mass is the absolute amount in the whole circulation. Plasma-volume changes can alter concentration without proportional changes in total Hb mass. When an athlete arrives at altitude, plasma volume often falls in the first days, so concentration rises even though no new red cells have been made. Conversely, heat acclimation and heavy endurance training typically expand plasma volume, diluting haemoglobin so that concentration may fall even while total mass is stable or increasing. An endurance athlete can look mildly "anaemic" on concentration while carrying a large, healthy total haemoglobin mass.

Measure	What it describes	Main limitation
Haemoglobin concentration	Hb per unit blood volume	Changes with plasma volume
Haematocrit	Red-cell share of blood volume	Changes with plasma volume
Total haemoglobin mass	Absolute Hb in circulation	Needs specialist method
Ferritin	Iron stores for red-cell production	Inflammation raises it

Direct measurement: optimised CO-rebreathing

When the research objective is altitude response, direct Hb-mass measurement methods such as optimised carbon-monoxide rebreathing can provide a more specific measure. The method most widely used in sport was described by Schmidt and Prommer in 2005. In simple terms, the athlete rebreathes a small, precisely controlled dose of carbon monoxide for a short period; because carbon monoxide binds to haemoglobin, the change in blood carboxyhaemoglobin after a known dose allows total haemoglobin mass to be calculated.

The method requires specialist equipment and safety procedures. Carbon monoxide is toxic, so dosing must be calculated carefully, exhaled and ambient levels monitored, and the procedure run by trained staff under qualified medical supervision. It is not a field test, and it must never be improvised. Even in expert hands it has a measurement error that must be known, which is why duplicate measurements or a well-documented laboratory typical error are important before claiming an altitude effect.

KEY IDEA

Haemoglobin concentration can change simply because plasma volume changes. If the question is whether red-cell mass increased, measure total haemoglobin mass with a validated method under medical supervision.

Designing an altitude Hb-mass study

A useful altitude study measures total haemoglobin mass before the camp, towards the end of it, and after return, with iron status checked beforehand so that low stores do not limit the response. Timing, hydration and the laboratory's own typical error must be documented. The AEI record should store the raw values, the method, the laboratory, and whether the observed change exceeds measurement error, rather than only "Hb up" or "Hb down".

It also helps to understand why the distinction matters for performance. Total haemoglobin mass is closely related to the body's capacity to transport oxygen, and it tends to be higher in endurance-trained athletes. A training or altitude intervention that genuinely increases total mass may support a higher VO_2max , whereas a rise in concentration caused only by plasma-volume loss does not add oxygen-carrying capacity to the circulation as a whole. Confusing the two can lead to wrong conclusions: a coach may believe a camp was successful when it simply dehydrated the athlete, or may abandon a useful camp because concentration fell after plasma volume expanded on return.

For most Indian programmes, direct Hb-mass measurement will be available only through research collaborations or specialist centres. That is acceptable. The AKURA approach is not to demand every measure for every athlete, but to be honest about what each available measure can and cannot show. When total mass cannot be measured, the record should say so, and conclusions about red-cell adaptation should be framed cautiously.

FIELD PROTOCOL

1. Outline for an altitude Hb-mass research block (clinically supervised)
2. 3–4 weeks before the camp: clinician reviews iron status and advises on any action
3. Baseline Hb-mass by optimised CO-rebreathing in an accredited or experienced laboratory, ideally duplicate
4. Log altitude, days at altitude, training load, illness and sleep during the camp
5. Repeat Hb-mass near the end of the camp using the same laboratory and protocol
6. Repeat after return to assess retention
7. Compare changes against the laboratory's typical error before drawing conclusions

COACH'S CORNER

If an athlete shows you a post-camp blood report with a higher haemoglobin concentration, do not celebrate or worry yet. Ask whether plasma volume, hydration or timing could explain it, and whether a total Hb-mass measure is available. Frame the result as a question for the medical team, not a verdict.

CHECKPOINT

In your last altitude or heat block, which measurement did you use to judge the haematological response? Did it answer the question you were actually asking?

CHAPTER 47**47. VO₂ Testing**

PART VIII — TESTING LAB

In 1991 a team of Indian physiologists put 67 young Bengali female athletes on a cycle ergometer and measured how much oxygen they could use. The long-distance runners came out on top, with an average relative VO₂max of 43.0 ml/kg/min. Set that beside the 79.9 ml/kg/min that Saltin and colleagues reported for senior Kenyan runners at sea level, and it looks like a chasm. Look closer. One group pedalled; the other ran. The tests were done in different laboratories, with different equipment and protocols, on athletes at very different stages of development. A cycle test typically gives a runner a lower value than a treadmill test would. The comparison is tempting and close to meaningless. That is the first lesson of VO₂ testing: the number cannot be separated from the way it was obtained.

VO₂max is the most famous number in endurance physiology, and for that reason it is also one of the most over-interpreted. A VO₂ test can provide maximal or submaximal oxygen-uptake information, but the value depends heavily on protocol, equipment calibration, athlete effort and whether the athlete reaches a valid maximal criterion. In AKURA, the VO₂ test is treated as a source of raw physiological data, of which the headline VO₂max is just one output.

What makes a maximal value valid

A maximal test typically uses an incremental protocol on a treadmill until volitional exhaustion. The number reported depends on the gas analyser's calibration, the sampling and averaging interval, the treadmill grade and speed increments, the athlete's motivation and familiarity, and the environment. Traditional criteria for

confirming a true maximum include a plateau in oxygen uptake despite increasing workload, a high respiratory exchange ratio, near-maximal heart rate and high post-test blood lactate. Midgley and colleagues reviewed these criteria in 2007 and showed that none is perfect, which is why many laboratories add a verification bout at a higher workload. A value obtained without clear criteria should be labelled "peak" rather than "max".

Factor	Effect on the result	What to record
Analyser calibration	Systematic error in all values	Calibration time and gases
Averaging interval	Short intervals give higher peaks	Interval used (e.g. 30 s)
Protocol	Stage length and grade alter peak	Protocol ID and increments
Athlete effort	Early stop underestimates max	Criteria met, RPE, reason for stopping
Environment	Heat raises HR and lowers peak	Temperature and humidity

Submaximal testing: seeing adaptation beneath the ceiling

Submaximal testing can be especially useful for economy and fixed-workload comparisons. If the athlete uses less oxygen at the same speed after training, that can reveal an adaptation even when VO_2max is unchanged. Many well-trained athletes improve for years with little change in VO_2max ; the improvement lies in running economy, in the fraction of VO_2max that can be sustained, and in durability. A submaximal test is also less demanding, so it can be repeated more often and with less disruption to training.

FIELD PROTOCOL

1. Combined submax and max treadmill test outline (laboratory, qualified staff)
2. Confirm medical clearance and analyser calibration
3. 10 min warm-up at easy pace, 1% treadmill grade
4. Submax phase: 3–4 stages of 4 min at fixed speeds below threshold, recording steady-state VO_2 , HR and RPE in the last minute of each stage
5. Short rest
6. Max phase: incremental stages (e.g. grade or speed increase each minute) to volitional exhaustion
7. Optional verification bout after recovery at a workload above the final stage
8. Record criteria met, averaging interval, environment and the athlete's recent training and sleep

KEY IDEA

VO_2max is one output of a VO_2 test, not the whole test. Submaximal oxygen cost at fixed speeds often reveals adaptations that VO_2max hides.

Field estimates and their limits

Many watches now report an estimated VO_2max from pace and heart-rate data. These estimates can be useful for tracking broad trends in a single athlete using the same device, but they are not measurements of oxygen uptake. They depend on the device's algorithm, the accuracy of heart-rate recording, terrain, heat and the athlete's recorded maximum heart rate. A hot, humid week in Chennai can lower a watch estimate without any change in true aerobic capacity, and a firmware update can shift every value overnight. AKURA therefore stores watch estimates in a separate field from laboratory measurements, labels them clearly as estimates, and never mixes the two in a single trend line. Where a laboratory test and a watch estimate disagree, the laboratory value, collected under a documented protocol, takes priority, while the watch trend can still be used to monitor week-to-week direction.

Safety is part of protocol too. A maximal test is a demanding effort, and athletes should be medically cleared, free from illness and supervised by staff trained in emergency procedures.

Storing raw data

AEI should store raw workload and oxygen data, not only the headline number. The breath-by-breath or averaged export, the speed and grade at each stage, heart rate, respiratory exchange ratio and lactate samples allow later reanalysis, comparison across devices and detection of errors. A single headline VO_2max in a spreadsheet cannot be corrected, reprocessed or compared fairly with a test done on a different system; the raw data can.

Finally, remember that VO_2max is only one of several determinants of endurance performance. The fraction of VO_2max an athlete can sustain, their running economy and their durability over long distances often explain more of the difference between athletes of similar aerobic capacity. AEI treats VO_2 data as one input among these, not as a ranking of talent.

COACH'S CORNER

When an athlete brings a VO_2max number from a gym or clinic, ask for the protocol, the averaging method and the raw export. Use the submaximal stages to set a baseline for future comparisons, and be cautious about comparing numbers from different laboratories or watches.

CHECKPOINT

If your athlete's VO_2max stayed the same over a season but race times improved, what submaximal measures would help explain why?

CHAPTER 48

48. Lactate Testing

PART VIII — TESTING LAB

A winter morning at a track in Patiala. The coach wears a woollen cap; the athlete wears shorts and a halo of steam. Every five minutes the runner stops at a folding table beside lane one and holds out a fingertip. A gloved technician wipes it, pricks it, discards the first drop and touches a test strip to the second. A small analyser beeps. 1.4. Then 1.6. Then 2.1. Then 3.8. Then a jump to 6.2, and the runner's shoulders start to rise. On a clipboard those numbers become a curve, and the curve is the story: how fast this athlete can run before the balance between producing and clearing lactate tips over. Run the same test in May heat, after a hard day, with a different analyser or a slower sample, and it might tell a different story. The beeping box is precise. Whether it is accurate depends on everything around it.

Blood lactate has a reputation it does not deserve. For decades it was blamed for fatigue and soreness; in reality it is a metabolic intermediate that is produced and cleared continuously, and a useful window into how the body is coping with a given workload. Lactate testing can map the relationship between workload and blood lactate. Used well, it turns lactate from a mysterious number into a performance relationship: a curve showing how much work the athlete can do before the balance between production and clearance tips.

Why protocol consistency is everything

Protocol consistency is essential: stage duration, sample timing, sampling site, analyser, warm-up and nutrition can all influence results. Short stages may not allow lactate to reach a steady state, shifting the curve to the right. A sample taken 30 seconds late after a stage may read differently from one taken immediately. Fingertip and earlobe samples can differ, portable analysers differ from one another, and sweat contamination on the skin can raise a reading. An athlete who has not eaten carbohydrate, or who trained hard the day before, may produce a lower lactate curve that looks like improved fitness but actually reflects depleted glycogen. Faude, Kindermann and Meyer reviewed the many lactate threshold concepts in 2009 and highlighted how different definitions and protocols produce different "thresholds" from the same data, which is exactly why the method must be fixed and documented.

Variable	How it can distort results	Standardise by
Stage duration	Too short: no steady state	Fixed length, e.g. 4–5 min
Sampling site	Finger vs earlobe differ	Same site every test
Analyser	Devices read differently	Same analyser, check calibration
Pre-test nutrition	Low glycogen lowers lactate	Same meal timing and content
Prior training	Fatigue shifts the curve	Easy day before testing
Environment	Heat raises HR and lactate	Record and match conditions

An incremental step test outline

FIELD PROTOCOL

1. Incremental lactate step test (track or treadmill, trained operator, hygienic sampling)
2. Standardise the previous 24 hours: easy training, usual carbohydrate intake, same meal 2–3 h before
3. 10–15 min easy warm-up, then take a resting or post-warm-up baseline sample
4. Run 5–7 stages of 4–5 min each, starting well below marathon pace and increasing by a fixed step (e.g. 0.5–1.0 km/h or 10–15 s/km)
5. At the end of each stage stop briefly (30 s or less), clean and dry the site, discard the first drop, sample, and record HR and RPE
6. Stop when lactate rises steeply or the athlete reaches a pre-agreed limit
7. Record temperature, humidity, surface, shoes, analyser and site
8. Plot lactate and HR against speed

Sampling blood, even from a fingertip, must follow hygiene and sharps-disposal procedures, and should be done by a trained person. On a track, holding a steady pace is harder than on a treadmill, so use pacing lights, a pacer or clear lap splits.

Asking the right question

A useful research design repeats the same test after an intervention. The question is not whether lactate is "high" or "low", but whether the athlete can sustain a higher workload at a comparable physiological response. If, after an eight-week block, the curve shifts to the right, so that the speed at a given lactate concentration or at a fixed rise above baseline is faster, the athlete has improved in a way that matters for racing. If only the peak lactate changed, that may reflect motivation, glycogen or the analyser more than fitness.

Interpretation also depends on context outside the curve. The same athlete may show a different curve in the hot season than in the cool season, because heat increases cardiovascular strain and can raise lactate at a given speed. Carbohydrate availability,

recent racing, travel and illness all matter. That is why every lactate test in the AKURA system is stored with its full metadata, and why test-retest comparisons should ideally be made at a similar time of year, time of day and phase of training. When conditions cannot be matched, the difference should be noted and the conclusion softened.

It is also worth resisting the temptation to prescribe training from a single lactate number. Thresholds derived from different methods, such as a fixed concentration, a rise above baseline or a curve-fitting approach, can give quite different speeds from the same data. The value of the test lies less in a precise threshold and more in repeated, comparable curves that show whether the athlete's relationship between workload and lactate is moving in the right direction. For many coaches, the speed and heart rate at a fixed lactate level, tracked over months, are more useful than any single named threshold.

KEY IDEA

Lactate is not good or bad. The useful information lies in the relationship between workload and lactate, compared under identical conditions before and after training.

COACH'S CORNER

Choose one threshold definition and one protocol for your group and never change it mid-season. When comparing tests, look first at the speed and HR at a fixed lactate level, then check whether the change is larger than your test's typical error before adjusting training zones.

CHECKPOINT

If your athlete's lactate curve moved to the right, what other explanations besides improved fitness would you rule out first?

CHAPTER 49

49. Running Economy Testing

PART VIII — TESTING LAB

Imagine two club runners in Bengaluru who share a training group and, according to a laboratory test they took together last year, the same VO_2max to the decimal point. On the road they are not equals. Over 10 km one routinely beats the other by more than a minute, and on the long Sunday runs around Cubbon Park the slower one works visibly harder at the same pace, breathing louder and landing a little heavier as the

kilometres pile up. Their engines are the same size. What differs is how much fuel each one spends to cover a kilometre. It is an unglamorous quality, easy to overlook because it rarely produces a headline number, yet among well-trained runners with similar aerobic capacity it is often the one that decides who finishes ahead.

Two runners with identical VO_2max can race very differently, and running economy is often the reason. Economy describes how much oxygen, or energy, an athlete needs to run at a given speed. Saunders and colleagues' 2004 review in *Sports Medicine* described how training, physiology, biomechanics, footwear and environment all influence it, and why it is one of the strongest distinguishing features among well-trained distance runners with similar aerobic capacity. For AKURA, economy is a key efficiency component of the AEI, but only if it is measured well.

Standardising the test

Running economy testing requires a standardised speed, grade, footwear condition and steady-state collection period. Economy can be expressed as oxygen cost per unit distance at a given speed, for example millilitres of oxygen per kilogram per kilometre, which makes values at different speeds easier to compare. Some laboratories also express it as energy cost, which accounts for changes in fuel use via the respiratory exchange ratio.

The test must be submaximal: speeds should be comfortably below the lactate threshold so that oxygen uptake reaches a true steady state and anaerobic contribution is small. Shoes matter; a change from a worn trainer to a new carbon-plated racer can alter economy, so footwear must be recorded or held constant. Treadmill familiarisation matters too, because unfamiliar athletes often run less economically on their first visits.

FIELD PROTOCOL

1. Running economy steady-state protocol (laboratory, calibrated gas analysis)
2. Standardise the previous 24 hours, meal timing and caffeine
3. Record body mass in running kit and shoe model and condition
4. Familiarise the athlete with the treadmill if new
5. 10 min warm-up at easy pace
6. Run 3 stages of 5–6 min at fixed submaximal speeds below threshold, with treadmill grade set at 1%
7. Average VO_2 over the final 2 min of each stage, confirming a steady state and RER below 1.0
8. Express economy as ml O_2 per kg per km at each speed
9. Record HR, RPE, room temperature and humidity
10. Repeat the identical protocol for follow-up tests

Change versus noise

Small changes matter, but measurement error must be understood. If the observed change is smaller than the test's typical error, it should not be treated as a real adaptation. Running economy has a typical error of a few per cent in well-controlled laboratories, which is similar in size to the improvements that training or footwear might produce. That makes the logic of Hopkins's work essential: estimate the typical error from repeated tests, identify the smallest worthwhile change, and interpret an observed change in relation to both.

Observed change vs noise	Interpretation	AEI confidence
Smaller than typical error	Cannot distinguish from noise	Low
Similar to typical error	Possible change, needs repeat	Moderate
Clearly larger than typical error and above smallest worthwhile change	Likely real and meaningful	High

KEY IDEA

A 2% improvement in running economy cannot be distinguished from noise if the test's typical error is 3%. Every economy result needs a confidence label based on known measurement error.

Beyond the laboratory

Laboratory economy testing is not available to most Indian athletes. Field proxies, such as the heart rate at a fixed easy pace on a standard flat route, can offer a rough window into efficiency, but they mix economy with cardiovascular and thermal factors. A lower heart rate at the same pace may reflect better economy, a larger stroke volume, cooler weather or simply better sleep. AKURA therefore labels such measures as efficiency indicators rather than economy measurements, and interprets them cautiously and only when conditions are matched.

Body mass, footwear and surface also deserve attention in field comparisons. Because economy is often expressed relative to body mass, a change in weight can alter the number without any change in running mechanics. Shoes with different foam or plates, and surfaces from synthetic tracks to uneven roads, can all change the oxygen cost of a given speed. Recording these details is what allows a real improvement to be separated from a change in equipment or ground.

Building measurement confidence into AEI

AEI should therefore include measurement confidence rather than pretending every change is meaningful. Each economy result can carry a confidence flag derived from the laboratory's typical error, the number of repeat trials and whether conditions were

matched. A trend supported by several tests carries more weight than a single striking result.

Economy also changes with fatigue. An athlete may be economical in the first minutes of a fresh laboratory test yet lose efficiency late in a marathon. Where resources allow, repeating economy stages after a prolonged run can show how well economy is preserved, linking this chapter to the durability domain of the AEI.

COACH'S CORNER

If a new shoe or strength block seems to improve economy, ask for a repeat test before changing race plans. Report the result to the athlete as "probably improved", "unclear" or "no change", not as a precise percentage that suggests false certainty.

CHECKPOINT

Do you know the typical error of the economy or submax test you use? If not, how could you estimate it with your athletes over the next month?

CHAPTER 50

50. ECG and Echocardiography

PART VIII — TESTING LAB

Anyone who has run a big Indian city marathon has passed them: the medical tents near the finish, the stretchers folded against a railing, the volunteers in bright bibs scanning faces in the crowd. At the 2018 Tata Mumbai Marathon, more than 2,320 participants reportedly needed some form of medical assistance. At any marathon most such cases are minor: cramps, blisters, exhaustion, heat. A tiny number are not. In 2018 a group of Indian clinicians writing in the Indian Heart Journal proposed pre-participation screening and better medical infrastructure for athletes and marathoners in India. The heart of a trained endurance runner differs from an untrained heart, and most of those differences are healthy adaptations. Telling them apart from a rare dangerous finding is expert work. For a coach, the most important skill is not interpretation. It is knowing when to stop, and whom to call.

The endurance athlete's heart adapts to training, and those adaptations can look unusual to anyone unfamiliar with them. A slow resting heart rate, certain ECG patterns and a larger heart can all be normal consequences of training. Rarely, similar-looking findings signal disease that can be dangerous. This chapter sets out the AKURA

position clearly: cardiac screening and diagnostic tests belong under qualified medical supervision, and the coach's job is to observe, record and refer.

What the tests show

ECG can identify rhythm and electrical patterns; echocardiography can evaluate structure and function. A resting 12-lead ECG records the heart's electrical activity and can reveal rhythm disturbances and patterns associated with some inherited conditions. Echocardiography uses ultrasound to image the chambers, walls, valves and pumping function. Other investigations, such as exercise testing, ambulatory monitoring or cardiac MRI, are decided by cardiologists when needed.

Test	What it evaluates	Who performs and interprets
12-lead resting ECG	Rhythm and electrical patterns	Trained staff; physician or cardiologist interprets
Echocardiography	Heart structure and function	Sonographer or cardiologist; cardiologist interprets
Exercise ECG	Response to exertion	Under medical supervision
Ambulatory monitoring	Rhythm over 24 h or longer	Ordered and read by physician

Athlete's heart or pathology

Endurance athletes may develop physiological cardiac adaptations, but interpretation must distinguish normal training-related findings from pathology. The international recommendations for electrocardiographic interpretation in athletes, published by Sharma and colleagues in 2017, grouped ECG findings into those considered normal in athletes, borderline findings and abnormal findings that warrant further evaluation. These criteria are tools for physicians trained to use them, and they must be applied with the athlete's age, sex, ethnicity, symptoms and family history in view.

A coach should not diagnose from a watch, ECG screenshot or isolated heart-rate value. Consumer devices can mis-detect rhythm, and a low resting heart rate on its own is common in trained runners. What a coach must never ignore are warning symptoms: chest pain or pressure, fainting or near-fainting especially during or just after exercise, unexplained breathlessness out of proportion to effort, palpitations or a racing irregular heartbeat, and a family history of sudden cardiac death at a young age. These require the athlete to stop training and be assessed by a doctor promptly.

Context is essential in the Indian setting. Many athletes come from regions where formal pre-participation screening is uncommon, and some may never have had an ECG before joining a structured programme. Family history is often under-reported because relatives' deaths may not have been investigated. A careful, respectful conversation at registration, supported by a standard questionnaire, is therefore an important first step. Where screening is offered, it should be organised through qualified clinicians with access to cardiology referral, and the athlete should understand what the tests can and cannot rule out.

It is also important to avoid the opposite error. Over-reading normal athletic adaptations can lead to unnecessary anxiety, repeated tests and lost training. The purpose of expert interpretation is to reassure where reassurance is appropriate and to investigate where it is not. Neither job belongs to a coach, but a coach who understands the difference will refer calmly and support the athlete through the process.

MYTH VS EVIDENCE

Myth: A coach can clear an athlete by looking at their smartwatch ECG.

Evidence: Consumer ECG features are limited and are not a cardiac screening. Only a qualified clinician can interpret cardiac tests and decide on fitness to train or compete.

KEY IDEA

Coaches observe and refer; clinicians interpret and clear. Any exertional chest pain, fainting or unexplained palpitations means stop and see a doctor.

Recording clearance in the AKURA system

The research system should record medical clearance and clinician interpretation as separate fields. The clearance field states whether a qualified clinician has cleared the athlete for training or competition, when and with what restrictions; the interpretation field holds the clinician's written summary. Keeping these apart from coach observations prevents a coach's note such as "HR felt high today" from being mistaken for a medical finding, and makes it clear who is responsible for each judgement. Access to these records should be restricted and handled with the athlete's consent.

FIELD PROTOCOL

1. AKURA cardiac referral pathway for coaches
2. Ask about symptoms and family history at registration using a standard questionnaire
3. If any warning symptom occurs, stop the session and do not resume training that day
4. Refer the athlete to a doctor with a written, factual description of what happened
5. Store the clinician's clearance and interpretation in separate, restricted fields
6. Resume training only after documented clearance, following any restrictions

COACH'S CORNER

Keep a simple emergency plan for every training venue, including the location of the nearest automated external defibrillator and how to call for help. Encourage pre-participation screening organised by qualified clinicians, and never let an athlete talk their way out of a medical review after a fainting episode.

CHECKPOINT

Does your athlete record keep medical clearance separate from coaching notes?
Could every coach in your group name the cardiac warning symptoms and the next step?

PART VIII AT A GLANCE

The key idea from each chapter, for quick revision.

- | | |
|-----------|--|
| 44 | The AKURA Athlete Baseline
A baseline is not a single number but a reproducible snapshot. The test and its metadata together form the measurement; a value without its protocol, device, conditions and units cannot safely be compared over time. |
| 45 | Blood Testing
A blood test is only as useful as the question behind it. Markers must be interpreted by a qualified clinician who knows the sampling conditions and the athlete's history. |
| 46 | Measuring Haemoglobin Mass
Haemoglobin concentration can change simply because plasma volume changes. If the question is whether red-cell mass increased, measure total haemoglobin mass with a validated method under medical supervision. |
| 47 | VO₂ Testing
VO ₂ max is one output of a VO ₂ test, not the whole test. Submaximal oxygen cost at fixed speeds often reveals adaptations that VO ₂ max hides. |
| 48 | Lactate Testing
Lactate is not good or bad. The useful information lies in the relationship between workload and lactate, compared under identical conditions before and after training. |
| 49 | Running Economy Testing
A 2% improvement in running economy cannot be distinguished from noise if the test's typical error is 3%. Every economy result needs a confidence label based on known measurement error. |
| 50 | ECG and Echocardiography
Coaches observe and refer; clinicians interpret and clear. Any exertional chest pain, fainting or unexplained palpitations means stop and see a doctor. |

PART IX — ENVIRONMENT

India offers almost every endurance environment on one subcontinent: humid coastal cities, dry northern plains in summer, cool hill stations and high Himalayan terrain. The same session can therefore produce very different physiological strain depending on where and when it is run. This Part explains why temperature alone is not enough, how wet-bulb globe temperature captures the combined thermal load, how pace-HR decoupling can signal cardiac drift and heat strain, and why humidity deserves to be treated as a separate variable. The aim is simple: every major AEI session should carry its environmental context, so that fitness is never confused with weather.

IN THIS PART

- 51** WBGT and Heat Load
- 52** Pace-HR Decoupling
- 53** Humidity Is Not Just “More Heat”

PART IX • ENVIRONMENT

Never Confuse Fitness with Weather

Temperature alone is not enough: WBGT, pace-HR decoupling and humidity reveal the true cost of a session.

28°C WBGT

Above this, consider postponing or moving hard sessions

3 parts

What WBGT combines

Natural wet-bulb (evaporation), black-globe (radiant sun) and dry-bulb air temperature in one heat-load index.

18·23·28

Four risk categories

Below 18°C low, 18–23 moderate, 23–28 high, above 28 very high. Adapt to local medical rules.

~7%

Pace-HR decoupling

Same 200 m/min, HR 140 then 150 bpm: efficiency falls from 1.43 to 1.33, about 7%.

Humidity

Not just “more heat”

Humidity limits sweat evaporation; a warm, very humid day can strain more than a hotter dry one.



India's full spectrum

Humid coasts, dry plains, cool hill stations, Himalayan valleys: the same session, very different strain.

15 min

Log at the site

Measure within 15 minutes of the start, at athlete height, in the same sun or shade as the route.

THE PROCESS

Run 45–60 min

Split into halves

Speed ÷ HR

Compare halves

Log conditions

Record temperature, humidity and heat load separately, so AEI can tell which one limits the athlete.

THE STORY

The Weather Is Not in the Results

PART IX — ENVIRONMENT

Mumbai, 17 January 2016, before sunrise. The start of the Mumbai Marathon has the particular smell of a coastal city in winter: salt, diesel, damp tarmac, and the sharp tang of pain-relief spray. By Mumbai standards, January is the kindest month. By the standards of a cool European marathon morning, the air is still warm and heavy with moisture, and it will get warmer the moment the sun clears the rooftops.

Among the elite men that morning are three Indian runners who will all come away with tickets to the Rio Olympics: Nitendra Singh Rawat, Thonakal Gopi and Kheta Ram.

Rawat, an Army runner, runs 2:15:48. It is a new Indian course record, beating Ram Singh Yadav's 2:16:59 from 2012, and it qualifies him for Rio. Afterwards he credits the preparation his group had done far from the sea, up in the Nilgiris.

I was personally confident after the training we had been put through in the national camp at Ooty by our coach (Surinder Singh) Bhandari.

— Nitendra Singh Rawat (Sportskeeda/IANS, 2016)

Up the hill

Ooty sits at 2,240 m above sea level. The Hindu describes athletes from across India going there for altitude training over many years, and a Hindustan Times report in 2026 quoted a ministry official saying the Ooty centre is "a preferred choice for middle and long distance runners". Mornings there can be cold enough to see your breath; the eucalyptus drips with mist, and the hill roads bend and climb in a way that no marathon course on the plains ever will.

Before Rio, according to ESPN, Gopi spent a ten-month stint at a high-altitude Ooty facility. Kheta Ram was part of the same block under Bhandari. The logic is familiar to any student of physiology: thinner air, a stronger stimulus to make red cells, more oxygen-carrying capacity when the athlete comes back down.

Then came Rio, in August. Gopi finished 25th in 2:15:25. Kheta Ram was 26th in 2:15:26, one second behind. Both were personal bests. Rawat, running with a hamstring injury, finished 84th in 2:22:52.

Three men. One training camp in the mountains. One qualifying race on a humid coast. One Olympic race in a different hemisphere and a different season. The results sheet records the times and places with perfect precision. It records nothing about the air.

The desert and the glacier

Indian distance runners come to the sport through a remarkable range of climates. Avinash Sable, the steeplechase record holder, did not begin as a runner at all. His early Army postings, Olympics.com reports, exposed him to the cold of Siachen and to Rajasthan desert heat of up to 50 °C. Gopi grew up in Sulthan Bathery, in Wayanad, which sits about 1,000 m above sea level.

It would be easy to assume that so much exposure adds up to a national expertise in heat. The research compiled for this book did not find that. Among India's elite men, it found altitude training to be standard, first in the Nilgiris and later in Colorado Springs, but it found no documented, deliberate heat-acclimation protocol. Heat and cold were things that happened to these athletes, on postings and in hometowns, rather than things that were measured and planned.

The same gap shows up in the science. A search for this book found no peer-reviewed study of heat stress in Indian marathons. There is general evidence from elsewhere that a higher heat index raises marathon casualty rates. There is a 2018 paper in the Indian Heart Journal proposing pre-participation screening and medical infrastructure at Indian marathons. And there is a press report that more than 2,320 participants needed medical assistance at the 2018 Mumbai Marathon. But the systematic Indian data that would connect a given morning's conditions with a given runner's physiology largely does not exist in public.

Imagine what a proper record of that Mumbai morning in January 2016 might have held: the temperature and humidity at the start line, the readings two hours later, the wet-bulb globe temperature on the exposed stretches of road, the sun and wind. Now imagine the same record for the Rio course in August. Side by side, they might explain a great deal about why some runners ran their best in one and struggled in the other, or they might show the conditions were similar and send the search elsewhere. Either answer would be useful. Neither is available.

Two adjectives

Fast forward to April 2026, and the TCS World 10K in Bengaluru, a World Athletics Gold Label race. Rodrigue Kwizera of Burundi wins the men's race in 27:31, an event record. Army runner Harmanjot Singh is the first Indian man home in 29:13, reported to be one second outside the Indian event record. Sanjivani Jadhav wins the Indian women's race in 35:01, her third straight title.

And the conditions? The reports describe the race as run in hot, humid conditions. That is the whole of it. Two adjectives. No temperature. No humidity figure. No measure of the radiant heat coming off the road, no wind, no time-stamped reading at the start and the finish.

Consider what that means for anyone who tries to use those results later. A coach comparing an athlete's 29-minute 10K in Bengaluru in April with a 29-minute 10K on a cool track in Europe is comparing two very different physiological tests. A selector

deciding between two runners who raced in different cities, months apart, has nothing to correct for the weather. A research programme trying to understand how Indian athletes adapt to heat has almost nothing to work with. The clock has been recorded to the second. The environment has been recorded as a mood.

Contrast the half marathon records set in Delhi: L. Suriya's 1:10:31 in November 2017 and Avinash Sable's 1:00:30 in November 2020, both in the cooler weeks of the northern Indian autumn, and both still standing as official national records. Runners and race organisers talk constantly about fast seasons and slow seasons. Everyone knows the weather matters. Almost nobody writes it down properly.

Club runners live the same problem every week without a stopwatch culture to hide it. A runner in Kochi finishes a tempo run in the monsoon and decides she has lost fitness. A runner in Jaipur runs the same session on a dry winter morning and decides she has found some. Neither has changed as much as she thinks. The air has.

What a number is worth

Every time in this book, every threshold pace, every heart-rate trend, every AEI domain, is only as meaningful as the conditions it was measured in. A 2:15 marathon in Mumbai in January, a 2:15 in Rio in August and a 2:15 in a cool European spring are three different performances wearing the same number. A heart rate of 150 at a given pace means one thing at dawn in Ooty and another on a humid Chennai road at nine in the morning.

India, more than almost any other running nation, offers the full spectrum on one subcontinent: humid coasts, dry plains, cool hill stations and high Himalayan valleys. That is a gift for research, if the environment is measured. It is a trap if it is not, because fitness and weather will be confused again and again.

The fix begins with a better thermometer, one that knows the difference between heat you feel and heat the body must get rid of. Fittingly, it was invented for soldiers. It is called the wet-bulb globe temperature.

Sources: <https://sportskeeda.com/running/nitendra-rawat-credits-coach-success-mumbai-marathon> ;
https://www.espn.com/espn/story/_/id/25815512/i-ensure-the-best-marathon-runner-india-says-resurgent-nitendra-rawat-mumbai-win ; https://www.espn.com/athletics/story/_/id/17475379/india-marathon-runner-olympics-thonackal-gopi-talks-espn-exclusive-interview ; <https://www.firstpost.com/sports/rio-olympics-2016-thanackal-gopi-and-kheta-ram-clock-personal-best-to-finish-25th-26th-in-marathon-2969066.html> ;
<https://timesofindia.indiatimes.com/sports/rio-2016-olympics/india-in-olympics-2016/athletics/rio-olympics-indias-thanackal-gopi-kheta-ram-clock-personal-best-in-marathon/articleshow/53800667.cms> ;
<https://www.thehindu.com/news/national/tamil-nadu/udhagamandalam-boasts-south-indias-only-high-altitude-sports-training-centre/article68333440.ece> ; <https://www.hindustantimes.com/sports/others/high-altitude-training-centre-to-come-up-in-shillong-101780588041378.html> ; <https://www.olympics.com/en/news/who-is-avinash-sable-india-athlete-steeplechase> ; <https://www.deccanherald.com/sports/tcs-world-10k-triple-delight-for-sanjivani-harman-surprises-domestic-field-3981750> ; <https://www.topafricanews.com/2026/04/27/rodrigues-event-record-and-florences-maiden-victory-highlight-tcs-world-10k/> ; <https://pubmed.ncbi.nlm.nih.gov/articles/PMC5903013/> ;
<https://pubmed.ncbi.nlm.nih.gov/articles/PMC12428376/> ;

<https://www.pressreader.com/india/the-free-press-journal/20180122/282291025653195> ;

<https://vedantadelhihalfmarathon.procam.in/race-categories/half-marathon-elite/past-winners-and-record-holders>

CHAPTER 51

51. WBGT and Heat Load

PART IX — ENVIRONMENT

April in Bengaluru rarely matches the city's pleasant-weather reputation. At the TCS World 10K in 2026, reports described the race as run in hot, humid conditions. Burundi's Rodrigue Kwizera still won the men's race in 27:31, an event record. Among the Indian men, Army runner Harmanjot Singh ran 29:13 and was reported to be one second outside the Indian event record. Sanjivani Jadhav won the Indian women's race in 35:01 for a third straight title. Those numbers will live on in the results. The conditions survive only as two adjectives, hot and humid, in a newspaper report. There is no temperature, no humidity figure, no measure of the sun's radiant load on the tarmac. Anyone who later compares these times with a cool-morning race will be comparing two very different physiological tests, with almost nothing to correct for the difference.

Example WBGT risk bands: follow local sports-medicine guidance

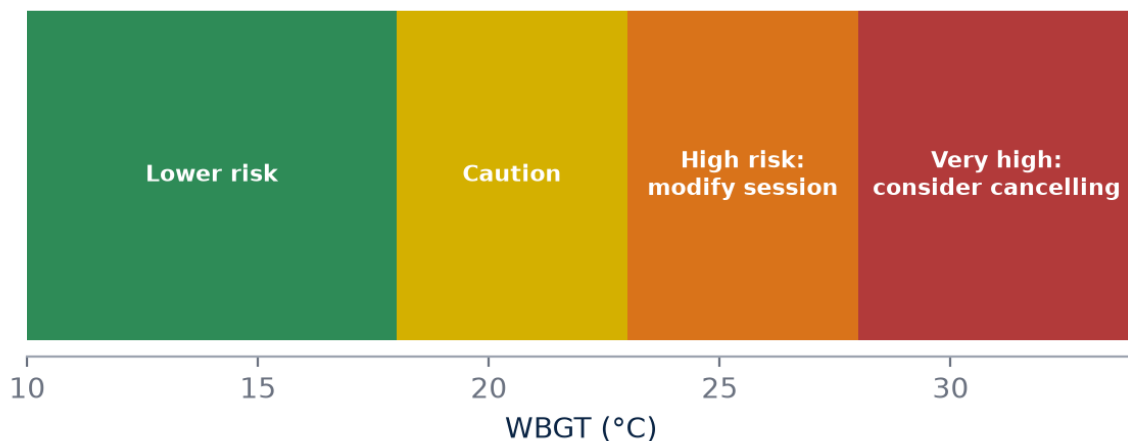


Figure 20. Example categories in the style of widely used athletic heat-safety flag systems. Local federation and medical guidelines take precedence.

Temperature alone does not describe heat stress. A thermometer reading tells us how warm the air is, but the body gains and loses heat through several routes at once: through the air, through solar and ground radiation, through airflow over the skin and, most importantly for runners, through the evaporation of sweat. Humidity influences evaporative cooling, while radiant heat and wind can alter the total thermal

environment. Wet-bulb globe temperature (WBGT) is one commonly used environmental index that tries to capture this combined load in a single number.

What WBGT combines

WBGT was developed for military training in the 1950s and is now widely used in occupational health and sport. Outdoors, it combines three readings: a natural wet-bulb temperature, which reflects the cooling power of evaporation and so responds strongly to humidity and airflow; a black-globe temperature, which reflects radiant heat from sun and surfaces; and ordinary dry-bulb air temperature. The wet-bulb component carries the greatest weight, which is why a humid day can have a high WBGT even when the air temperature does not look extreme.

Component	What it reflects	Why it matters for runners
Natural wet-bulb temperature	Evaporative cooling potential	Humidity limits sweat evaporation
Black-globe temperature	Radiant heat from sun and ground	Direct sun adds heat load
Dry-bulb air temperature	Air temperature	Convective heat exchange

Proper WBGT measurement uses a dedicated instrument placed in the conditions the athletes experience, not a phone app's city estimate. Values can differ markedly between a shaded park at dawn and an exposed road at mid-morning.

Example categories and guidance

Many race-medicine and sports-medicine bodies use flag or category systems based on WBGT. The table below shows example categories in the style of the widely used American College of Sports Medicine flag system for endurance events. They are illustrative only: thresholds differ between organisations, sports, and levels of acclimatisation, and local sports-medicine and federation guidelines always apply.

Example WBGT category	Illustrative risk	General guidance
Below 18 °C	Low	Normal training; still hydrate and monitor
18–23 °C	Moderate	Monitor closely; adjust pace and hydration
23–28 °C	High	Reduce intensity and duration; extra cooling; watch vulnerable athletes
Above 28 °C	Extreme	Postpone, cancel or move sessions indoors; follow medical guidance

Individual factors shift risk within any category. Athletes who are not yet heat-acclimatised, who are unwell, sleep-deprived or dehydrated, who are returning from injury, or who are running at high intensity for long durations can be at greater risk even in a moderate category. Young athletes and masters athletes may respond

differently, and body size, clothing and fitness also matter. The category is a starting point for decisions, not a permission slip.

In practice, WBGT guidance should shape the whole session plan: when to start, how long to run, how hard to go, where to place fluid and shade, and who will watch for signs of serious heat illness such as confusion, unsteady running, collapse or unusual behaviour. Exertional heat stroke is a medical emergency, and every hot-weather programme should have a clear plan for rapid recognition, cooling and medical help, developed with qualified sports-medicine professionals.

KEY IDEA

WBGT combines humidity, radiant heat and air temperature into one heat-load index. Category thresholds are guides that must be adapted to local medical and federation rules and to each athlete's acclimatisation.

Logging environment in Indian training

For Indian endurance training, the environment should be logged whenever possible. A 60-minute run at the same pace can create very different physiological stress at 20 °C and at 34 °C with high humidity. Heart rate may be many beats higher, sweat loss much greater and perceived effort far harder, even though the watch shows the same pace. If AEI ignored this, it would read a hot-season plateau as lost fitness and a cool-season improvement as a breakthrough.

This is why AEI should carry an environmental context tag for major sessions and races: at minimum air temperature, relative humidity and time of day; ideally WBGT, sun or shade, and wind. With that context, the system can compare sessions run in similar conditions and adjust expectations for sessions that were not.

FIELD PROTOCOL

1. Environmental logging for key sessions
2. Measure at the training site within 15 minutes of the start, at athlete height, in the same sun or shade exposure as the route
3. Record air temperature, relative humidity, WBGT if a meter is available, wind estimate, cloud cover and time
4. Re-measure at the end of sessions longer than 60 minutes
5. Tag the session with its environmental category
6. Compare only sessions within similar categories when judging fitness trends

COACH'S CORNER

Schedule quality sessions in the coolest part of the day during hot months, and treat pace targets as flexible when WBGT is high: run by effort and heart rate instead. Make heat-illness recognition and a cooling plan part of every hot-season training camp.

CHECKPOINT

For your last five key sessions, could you state the temperature, humidity and time? How many of your "bad" sessions coincided with high heat load?

CHAPTER 52**52. Pace-HR Decoupling**

PART IX — ENVIRONMENT

It is 5:10 a.m. on Marina Beach, and the sand is still grey. A runner sets off along the promenade at an easy, conversational pace, a chest strap snug under his vest. For the first half hour the numbers on his watch are dull in the best way: pace steady, heart rate settled in the low 140s. Then the sun lifts clear of the Bay of Bengal. The breeze drops. The sweat that was evaporating from his arms starts to run off them instead. His pace has not changed, but his heart rate is quietly drifting, 146, 149, 152, as if the same run were getting harder to afford. He does not feel dramatically worse. The watch has noticed before he has. That slow slide, in which the heart works harder for the same speed, is one of the most useful and most misread signals in endurance training.

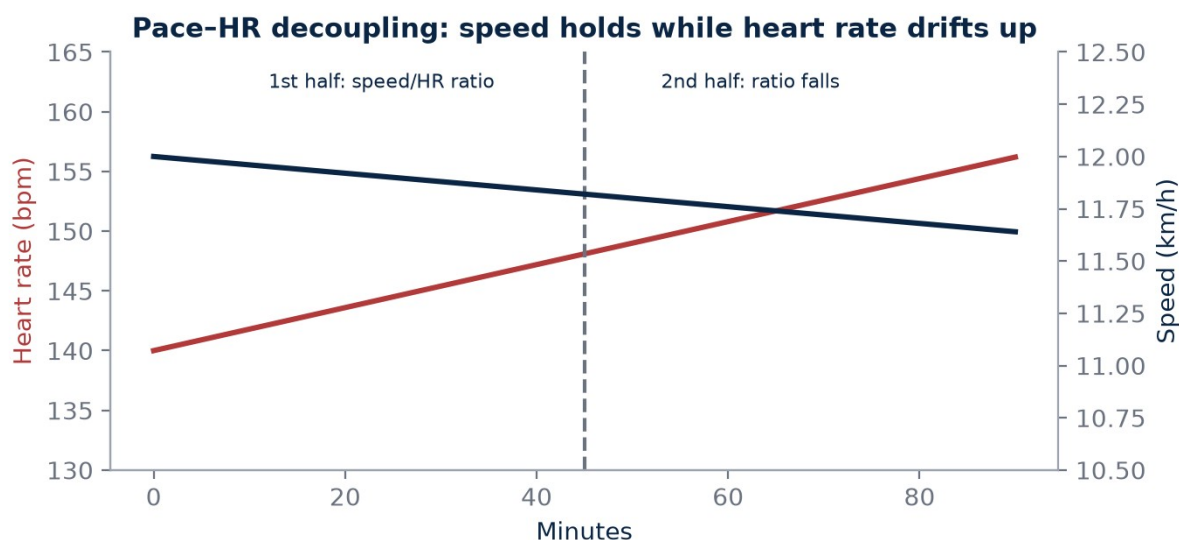


Figure 21. Illustrative long run. $\text{Decoupling (\%)} = (\text{first-half speed:HR} - \text{second-half speed:HR}) \div \text{first-half ratio} \times 100$. It is a signal to interpret, not a diagnosis.

On a steady run, pace and heart rate usually travel together. As the run goes on, that partnership can loosen. If heart rate rises while pace remains constant, or pace falls while heart rate remains similar, the relationship may indicate cardiac drift, heat strain, dehydration, fatigue or pacing changes. Pace-HR decoupling measures this loosening, and it is one of the most useful field signals available to AKURA because it needs nothing more than a heart-rate strap and a GPS watch.

Why the relationship drifts

During prolonged exercise, especially in the heat, heart rate tends to rise gradually even at a constant workload. Coyle and González-Alonso described this cardiovascular drift and linked it to rising body temperature, fluid loss and the increasing demand for skin blood flow, which together reduce stroke volume so that heart rate must climb to maintain output. Fatigue of the running muscles, glycogen depletion and subtle changes in running mechanics can add to the effect. A well-conditioned, well-hydrated athlete in cool weather typically shows little drift; the same athlete in humid heat may show a lot.

Calculating decoupling

The most practical method compares the first half and the second half of a steady run. For each half, calculate an efficiency ratio: average speed divided by average heart rate. Decoupling is the percentage change in this ratio from the first half to the second half. A falling ratio means the athlete is getting less speed for each heartbeat as the run progresses.

FIELD PROTOCOL

1. Pace-HR decoupling calculation
2. Choose a steady run of at least 45–60 min on flat or consistent terrain, after a 10–15 min warm-up that is excluded from analysis
3. Split the analysed portion into two equal halves by time
4. For each half, compute average speed (e.g. metres per minute) and average HR
5. Efficiency ratio = average speed ÷ average HR for each half
6. Decoupling (%) = (first-half ratio – second-half ratio) ÷ first-half ratio × 100
7. Record temperature, humidity, fluid intake, terrain and any pace changes
8. Store the result against the athlete's own history, not a universal cut-off

For example, suppose the first half averages 200 m/min at 140 bpm, giving a ratio of about 1.43, and the second half averages 200 m/min at 150 bpm, giving about 1.33. The decoupling is roughly 7%. Whether that is large depends on the conditions and on what is normal for this athlete.

Half of run	Average speed	Average HR	Efficiency ratio
First half	200 m/min	140 bpm	1.43
Second half	200 m/min	150 bpm	1.33
Decoupling	—	—	About 7%

A signal, not a diagnosis

Decoupling is not a diagnosis. It is a signal. The athlete's hydration, temperature, terrain and effort must be considered. A hilly second half, a headwind, a stop at a traffic signal, a poorly fitting strap or a sunrise that turned a cool start into a hot finish can all create apparent decoupling without any change in fitness. Conversely, a runner who slowed down deliberately may show little decoupling despite significant strain.

Repeated standardised runs can make this signal more useful. Over time, AEI can learn an athlete's normal response instead of using a universal threshold. If an athlete usually shows 2–4% decoupling on a standard 60-minute run in the cool season, a value of 10% under similar conditions deserves attention, while the same 10% on a humid May morning may simply be expected.

A few technical points improve the quality of the signal. Chest straps usually give more reliable heart-rate data than wrist sensors during running, particularly in sweat and heat. Very short runs produce noisy ratios, so the analysed period should be long enough to reveal drift. Runs that include intervals, big hills or long stops are unsuitable. On undulating routes, some coaches use grade-adjusted pace, but even then the most reliable comparisons come from the same route under similar conditions. Finally,

decoupling on a treadmill and outdoors are not directly comparable, because airflow, surface and cooling differ.

Decoupling can also help with race preparation. A marathon runner who shows modest decoupling on a long run at goal pace in conditions similar to race day has useful evidence of durability, while one who shows large decoupling may need to reconsider pace targets, fuelling or hydration.

KEY IDEA

Decoupling compares speed per heartbeat in the first and second halves of a steady run. Read it against the athlete's own history and the day's conditions, never as a stand-alone verdict.

COACH'S CORNER

Use one standard route and duration for decoupling checks every few weeks. Falling decoupling on matched days suggests improving aerobic durability; a sudden rise should prompt questions about sleep, illness, hydration and heat before any change to training.

CHECKPOINT

What decoupling does your athlete usually show on a standard steady run in cool conditions? How would you tell whether a higher value reflected fatigue or simply the weather?

CHAPTER 53

53. Humidity Is Not Just “More Heat”

PART IX — ENVIRONMENT

Avinash Sable's early Army postings, Olympics.com reports, took him from the cold of Siachen to Rajasthan desert heat of up to 50 °C. Now imagine a runner hardened in that kind of dry, furnace-like heat stepping out on a Mumbai monsoon morning at 29 °C. On paper it is more than 20 degrees cooler. On the road it can feel far worse. In the desert, sweat vanishes almost as fast as it appears, and each drop carries heat away with it. On a saturated coastal morning the sweat pours down the face, soaks the vest, drips from the elbows and does very little cooling at all. The thermometer says the coast is the gentler place to run. The body may disagree.

Understanding why means looking past temperature to the variable most weather apps treat as an afterthought.

It is tempting to think of humidity as simply making a hot day feel hotter. Physiologically, it does something more specific. Humidity changes how effectively sweat evaporates. Two environments with the same air temperature can create very different thermal strain, because the runner's main cooling mechanism works well in one and poorly in the other.

Evaporation: the runner's radiator

A running body produces a great deal of heat. In warm conditions, most of that heat must be removed by evaporating sweat from the skin. Evaporation depends on the difference between the water vapour pressure at the skin and in the surrounding air. In dry air, sweat evaporates readily and cools effectively. In humid air, the gradient is small, sweat drips rather than evaporates, and much of the fluid lost provides little cooling. The athlete sweats heavily, dehydrates faster and still heats up. Heart rate climbs as blood is diverted to the skin, and perceived effort rises.

Condition	Sweat evaporation	Typical consequence
Hot and dry	Efficient	Cooling works; fluid loss high
Hot and humid	Limited	Core temperature rises faster; HR drift larger
Warm and very humid	Limited	Strain can exceed a hotter dry day
Cool and humid	Adequate at low heat production	Usually manageable

The coastal Indian runner

This matters particularly for coastal India. An athlete who can run comfortably at 32 °C in dry conditions may struggle at a lower temperature when humidity is high. Runners in Chennai, Mumbai, Kochi, Visakhapatnam or Kolkata face long stretches of the year when the air is warm and saturated, and early mornings may be the most humid part of the day even though they are the coolest. An athlete who moves from a dry inland city to the coast, or travels to a humid race, can find familiar paces unexpectedly hard.

Heat-index or apparent-temperature numbers reported by weather apps attempt to combine temperature and humidity into a single feels-like value. They can be helpful as a quick warning, but they were not designed for athletes producing large amounts of heat, and they do not include radiant load or wind at the training site. For AKURA, the raw temperature and humidity values, plus WBGT where available, are more useful than a single feels-like figure, because they allow the analysis to separate the contributions of each variable.

Clothing and practical choices also interact with humidity. Light, breathable kit, shaded routes, earlier start times and regular cooling with water or ice can reduce strain, but

none of them fully overcomes the physics of limited evaporation. Humid conditions therefore call for honest adjustment of expectations rather than simply trying harder.

MYTH VS EVIDENCE

Myth: If the temperature is lower, the run will be easier.

Evidence: A warm, very humid day can impose more thermal strain than a hotter, drier one, because humidity limits sweat evaporation, the body's main cooling route during running.

Designing heat-acclimation research

Heat-acclimation research should therefore distinguish temperature, humidity and total heat stress rather than treating them as one variable. An acclimation block in dry heat may not fully prepare an athlete for humid heat, and vice versa. Consensus recommendations on training and competing in the heat, such as those led by Racinais and colleagues in 2015, emphasise matching acclimatisation to the expected competition environment where possible. For AKURA, that means recording air temperature, relative humidity and, where available, WBGT separately, so that the analysis can ask whether an athlete's response is driven by heat, by humidity, or by their combination.

FIELD PROTOCOL

1. Humid-heat readiness check
2. Choose a standard 45–60 min easy run performed in both drier and more humid conditions over several weeks
3. Record air temperature, relative humidity, WBGT if available and time of day
4. Weigh the athlete before and after (minimal clothing, towel-dried) to estimate sweat loss, adjusting for fluid taken
5. Calculate pace-HR decoupling for each run
6. Compare decoupling, sweat loss and RPE between drier and more humid days at similar temperatures
7. Use the results to individualise pacing and hydration plans for humid races

Acclimatisation itself is specific and reversible. Adaptations such as an expanded plasma volume, earlier and greater sweating, and a lower heart rate at a given workload develop over roughly one to two weeks of repeated heat exposure and fade if exposure stops. Recording the type of heat an athlete has trained in, and for how long, allows AEI to judge how prepared they are for a particular race environment.

KEY IDEA

Temperature, humidity and total heat stress are different variables. Record them separately so AEI can tell which one is limiting the athlete.

COACH'S CORNER

Before a humid race, rehearse race-pace segments in similar humidity and adjust targets by effort and heart rate. Plan fluid and cooling strategies with a qualified professional where possible, and remind athletes that heavy sweating in humidity is not a sign of efficient cooling.

CHECKPOINT

Does your training log record humidity separately from temperature? Which of your athletes struggle most on humid days, and how would you test whether humidity rather than temperature is the main cause?

PART IX AT A GLANCE

The key idea from each chapter, for quick revision.

51**WBGT and Heat Load**

WBGT combines humidity, radiant heat and air temperature into one heat-load index. Category thresholds are guides that must be adapted to local medical and federation rules and to each athlete's acclimatisation.

52**Pace-HR Decoupling**

Decoupling compares speed per heartbeat in the first and second halves of a steady run. Read it against the athlete's own history and the day's conditions, never as a stand-alone verdict.

53**Humidity Is Not Just “More Heat”**

Temperature, humidity and total heat stress are different variables. Record them separately so AEI can tell which one is limiting the athlete.

PART X — TRAINING DESIGN

Measurement only matters if it changes what happens on Monday morning. Part X turns the AKURA Endurance Index from a profile into a planning tool. It explains how performance, readiness, environmental and recovery information can shape intensity, how six broad intensity domains give coach and athlete a shared language, how the 3–3–1 weekly rhythm can organise load without becoming a rigid formula, how progression can be planned yet reversible, and how strength work can support running economy and resilience without stealing from key sessions. Throughout, the principle is the same: the algorithm informs, the coach decides, and the athlete's response is the final judge.

IN THIS PART

- 54 From AEI to Training
- 55 Training Intensity Domains
- 56 The 3–3–1 Concept
- 57 Progressive Overload Without Blind Progression
- 58 Strength and Running Economy

PART X • TRAINING DESIGN

The Algorithm Informs, the Coach Decides

AEI becomes a planning tool: six intensity domains, a 3–3–1 weekly rhythm, earned progression and supportive strength work.

3–3–1 Weekly rhythm: three development, three complementary, one reset day

5 layers

From AEI to training

Performance sets targets; readiness and environment adjust them; recovery protects them; execution closes the loop.

6

Six intensity domains

Recovery, Aerobic, Endurance, Threshold, Power, Speed: each defined by purpose, not by pace alone.



Absorb, then progress

Progress when key sessions are hit and HR/RPE are stable; hold if costly; reduce if not absorbed.



Pain means reduce

New or worsening localised pain, limping or altered gait: reduce load and seek professional assessment.

2–3 reps

Strength for runners

Squats, deadlifts, split squats, calf raises, pogo hops: stop 2–3 reps short of failure.

1,800 m

The Colorado lesson

Some of India's best runners trained above 1,800 m: altitude as a base, one coach planning the year.

THE PROCESS

Set targets

Check readiness

Adjust for
environment

Execute session

Update plan

Progression is earned by adaptation, not scheduled by the calendar alone.

THE STORY

The Colorado Experiment

PART X — TRAINING DESIGN

It is just after sunrise in Colorado Springs, and the first hill tells a runner where they are. The city sits at more than 1,800 m on the eastern edge of the Rocky Mountains, with Pikes Peak standing over it, and the air is dry and thin. A pace that feels like jogging at sea level takes a surprising amount of breathing here. For most of the past decade, some of India's best distance runners have trained in this town, half a world away from the sugarcane fields, drought-hit villages and Army barracks they came from.

To see why it mattered, start with how long India's records had stood still. Gopal Saini ran the 3000 m steeplechase in 8:30.88 at the 1981 Asian Championships in Tokyo, and nobody in India ran faster for 37 years. Bahadur Prasad's 13:29.70 for 5000 m, set in Birmingham in 1992, lasted about 30 years. Hari Chand ran 28:48.72 for 10,000 m, barefoot, in a heat at the 1976 Montreal Olympics, and that mark lasted 32 years. Generations of Indian runners trained, raced, won Asian medals and retired without touching those numbers.

India had not lacked coaching. The Belarusian Nikolai Snegarev arrived in 2005 and built a national distance group that produced a gold and silver in the 10,000 m at the 2010 Asian Games and took Lalita Babar to the Rio 2016 steeplechase final. But he left the Athletics Federation of India in February 2019 "in acrimonious circumstances", as Olympics.com put it. He was re-appointed in 2021 and found dead in his hostel room at NIS Patiala on 5 March 2021, aged 72, three days after returning to India. By then the centre of gravity of Indian distance running was already shifting west.

The first arrivals

In 2019 the middle-distance runner Jinson Johnson, an Army athlete, arranged to train in Colorado Springs. "I am hoping that training in Colorado Springs under coach Simmons would help in bettering my timing," he told The Times of India. That September, at ISTAF Berlin, he ran 1500 m in 3:35.24. Seven years later it is still the national record. A severe Achilles injury followed from 2019, and he announced his retirement in January 2026.

The coach Jinson named was Scott Simmons, an American who runs the American Distance Project in Colorado Springs and, according to his own X bio, also coaches for the US Army. The AFI appointed him to oversee India's distance programme; by October 2022 Sportskeeda was calling him India's distance head coach. That year the federation sent four athletes to him for a 16-week camp: Avinash Sable, Parul Chaudhary, Shankar Lal Swami and Ajay Kumar Saroj. Sportskeeda's verdict was blunt. Only Sable, it reported, had clearly benefited.

The soldier from Beed

Sable is a useful place to begin because his story shows how late an Indian distance runner can start. He was born in 1994 in Mandwa village, Beed district, Maharashtra, to farming parents, and ran 6 km to school. After Class 12 he joined 5 Mahar Regiment. His postings took him to Siachen, Rajasthan and Sikkim. He came to running only in 2015, through Army cross country, and in 2017 the Army coach Amrish Kumar moved him to the steeplechase.

In September 2018 he finally broke Gopal Saini's record. He would break it ten times in all. The 2022 season, his first with Simmons, is where the steps got bigger: 8:16.21 in March, 8:12.48 at the Rabat Diamond League in June, and 8:11.20 at the Commonwealth Games in Birmingham in August. That run won silver, just 0.05 seconds behind Kenya's Abraham Kibiwot. It was India's first steeplechase medal at the Games and the first by a non-Kenyan in the event since 1994. The same year he ran 13:25.65 for 5000 m in California, finally erasing Bahadur Prasad's mark from 1992.

In 2023 he won Asian Games gold in Hangzhou in a Games record of 8:19.50. Before the Budapest World Championships that summer he trained at altitude in Colorado Springs and St Moritz alongside Paul Chelimo, and Simmons noticed how quickly he settled.

His adaptation upon his return to altitude this year was almost instant.

— Scott Simmons, *Hindustan Times* (2023)

In January 2024 the sports ministry's TOPS scheme agreed to pay for Sable and Parul to train in Colorado Springs, covering airfare, boarding and lodging for athletes, coaches and physios. In July he ran 8:09.91 at the Paris Diamond League, the current national record. Weeks later he became the first Indian to reach an Olympic steeplechase final, and finished 11th.

The girl from the sugarcane fields

Parul Chaudhary was born in 1995 in Iklauta village near Meerut, Uttar Pradesh, into a farming family. Her running began almost by accident, with a barefoot 800 m school race in 2011. "Papa asked me if I wanted to run and I agreed. I had never practised before that school race," she told *The Indian Express*. She trained on the paths between the sugarcane fields, took a Western Railway ticket-examiner's job in 2015 and, in 2024, became a Deputy Superintendent of Police in Uttar Pradesh.

With the Colorado group, and training alongside Sable, she began rewriting the women's record book. In July 2022 she ran 8:57.19 for 3000 m in Los Angeles, the first Indian woman under nine minutes. In August 2023 she reached the World Championships steeplechase final in Budapest and ran 9:15.31, breaking a national record Lalita Babar had held since Rio. She won 5000 m gold and steeplechase silver at the Hangzhou Asian Games. In 2025 she lowered the steeplechase record twice more. In

June 2026 she ran 15:04.26 for 5000 m in Nice, and on 27 September 2026, at the Asian Games in Nagoya, she won bronze in 9:08.67. Between 2022 and 2026 she set seven national records.

The man who learned to race

Gulveer Singh was born in 1998 in Sirsa village, Aligarh district, Uttar Pradesh, into a farming family. He joined the Army through the sports quota in 2018, with The Grenadiers, and did patrol duties at altitude in Arunachal Pradesh. His own summary of that path is striking: "I truly learned about competitive running only after joining the army," he said in an interview quoted by Olympics.com.

In 2024 he moved his base to Colorado Springs under Simmons. The 10,000 m record, 28:02.89 by Surendra Singh since 2008, fell to 27:41.81 in March 2024, then 27:14.88 in Japan that November, then 27:00.22 at The TEN in California in March 2025, which SAI said made him the third-fastest Asian of all time. In a Boston indoor race in February 2025 he ran 12:59.77 for 5000 m, the first Asian to break 13 minutes indoors. At the Glasgow Commonwealth Games in 2026 he won 10,000 m silver and 5000 m bronze, the first Indian to win two track medals at a single Games. Weeks later, in Nagoya, he added three Asian Games medals.

What changed, and what did not

What did the Colorado years actually change? Not talent: these athletes came from the same villages and the same Army units as their predecessors. What changed was the system around them. Altitude became a base rather than a trip. A single coach planned the year. Government money paid for the flights, the physios and the rent. And the racing moved abroad: almost every national record since 2022 was set at a fast, well-paced overseas meet, in California, Oregon, Japan, Boston, Budapest or Paris. Sable himself had made the point, as Olympics.com reported, that nobody in India could set a record pace for him.

A system is not a guarantee, though. Of the four athletes in the 2022 camp, only one clearly improved that year. Sable injured the ACL and meniscus in his right knee at the Monaco Diamond League in July 2025, and by August 2026 Deccan Herald reported that Simmons was no longer coaching him. And the model was not immune to the sport's oldest problem. Kartik Kumar, who spent three months in Colorado Springs in 2023 and won Asian Games 10,000 m silver that year, accepted a three-year USADA ban in November 2025 after testing positive for anabolic steroids, and his results from February 2025 were disqualified. Moving athletes abroad does not move them beyond anti-doping rules.

What the Colorado group shows is that Indian distance running did not transform because of one workout or one secret. It changed because many ordinary decisions were made more deliberately and more consistently: where to live, how hard to run each day, when to race, when to back off. Those decisions are the subject of Part X. To

make them well, the numbers from the laboratory and the watch have to become an actual plan for Monday morning.

Sources: <https://www.olympics.com/en/news/who-is-avinash-sable-india-athlete-steeplechase> ; <https://www.olympics.com/en/news/who-is-gulveer-singh-india-athlete-long-distance-runner> ; <https://sportskeeda.com/athletics/news-challenging-task-india-s-head-coach-scott-simmons-regroup-distance-running> ; <https://thebridge.in/athletics/avinash-sable-parul-chaudhary-to-train-in-colorado-springs-45778> ; <https://www.hindustantimes.com/sports/others/steeplechaser-avinash-sable-gets-set-for-a-third-world-championships-101692209605101.html> ; <https://timesofindia.indiatimes.com/sports/more-sports/athletics/jinson-johnson-to-train-in-us-to-improve-timing-and-qualify-for-olympics/articleshow/70809839.cms> ; <https://indianexpress.com/article/sports/asian-games/running-sugarcane-field-parul-chaudhary-asian-games-2023-hangzhou-8943068/> ; <https://www.rediff.com/sports/report/asian-games-after-double-bronze-haul-parul-now-craves-family-time/20260928.htm> ; <https://www.deccanherald.com/sports/how-scott-simmons-helped-turn-gulveer-singhs-records-into-cwg-medals-4112643> ; <https://www.olympics.com/en/news/gulveer-singh-india-10000m-national-record-the-ten-2025-athletics-california> ; <https://www.olympics.com/en/news/nikolai-snesarev-the-late-belarusian-coach-s-contribution-to-indian-athletics> ; <https://www.usada.org/sanction/kartik-kumar-accepts-doping-sanction/> ; <https://www.olympics.com/en/news/athletics-list-of-indian-national-record-holders>

CHAPTER 54

54. From AEI to Training

PART X — TRAINING DESIGN

It is 6:40 p.m. at a district stadium outside Pudukkottai, and the floodlights have not come on yet. A school coach sits on the concrete steps with a notebook on one knee and a phone on the other. The phone holds everything: heart-rate graphs from a donated watch, a sleep score, a colourful readiness dial that turned amber this morning. The notebook holds one line for tomorrow: 6 × 1 km. Between the two is the real question of coaching, the one no app answers by itself. Should the amber dial change the line in the notebook? If so, by how much? If not, why collect it at all? Twenty teenagers are jogging their cool-down lap in the dusk, and every one of them will turn up tomorrow expecting a plan. This chapter is about the bridge between the screen and that plan.

The purpose of an athlete index is not to create a pretty dashboard. A number that looks impressive on a screen but never alters a session, a pace target or a recovery day is decoration, not science. The AKURA Endurance Index (AEI) earns its place only if it improves decisions: which session to run, how hard to run it, when to hold back and when to push. This chapter describes how the different layers of AEI connect to the training plan, and why the coach remains the final point of responsibility.

Five layers, five kinds of decision

Each layer of AEI answers a different practical question. A performance score describes current capacity and can guide intensity: it tells us what paces or power outputs are

realistic for threshold, long-run or interval work. A readiness layer describes today's ability to express that capacity and can modify the session: shorten it, lower the target, or swap it for an easier day. Environmental data—temperature, humidity, altitude, air quality—can alter pace expectations, so that a threshold run in Chennai humidity is judged against the conditions rather than against a cool-morning benchmark. Recovery data such as sleep, resting heart rate and subjective fatigue can reduce unnecessary intensity before it becomes accumulated fatigue. Finally, actual workout execution—what was really run, at what heart rate and with what perceived effort—feeds back into the model and updates the next decision.

AEI layer	Question it answers	Typical training decision
Performance	What can the athlete do?	Set pace or HR targets for each domain
Readiness	Can they express it today?	Keep, modify or replace the session
Environment	How hard are the conditions?	Adjust pace expectations, hydration, timing
Recovery	Is fatigue accumulating?	Reduce intensity or add a reset day
Execution	What actually happened?	Update the model and next week's plan

KEY IDEA

AEI is a decision-support loop, not a scoreboard. Performance sets the targets, readiness and environment adjust them, recovery protects them, and execution data closes the loop.

The feedback loop

A training plan is a hypothesis: "If this athlete does this work, they will adapt in this way." Each session is a small experiment, and execution data is its result. When an athlete completes a planned threshold session at the expected pace with a stable heart rate and moderate perceived effort, the model gains confidence in its estimate of threshold. When the same session produces an unusually high heart rate, a sharp rise in perceived exertion or an early stop, something in the model—fitness, readiness, environment or execution—needs to be re-examined. Over weeks, these small signals accumulate into a clearer picture than any single test can offer.

This loop also protects against a common error: treating the plan as more important than the athlete. A plan written three weeks ago did not know about last night's poor sleep, a sudden rise in humidity or an exam week. The loop allows the plan to bend without breaking.

Judgement remains with the coach

The coach remains responsible for context. An algorithm can notice that resting heart rate is elevated; it cannot know that the athlete travelled overnight by train, is fasting

for a festival, is caring for a sick family member or has just received good news that will lift motivation. Algorithms work with the variables they are given. Coaches work with the whole person.

For this reason, AEI outputs should be framed as recommendations with confidence levels, not commands. A sensible default is that the system proposes and the coach disposes: the coach accepts, modifies or overrides the suggestion, and the override itself is recorded so that the model can learn which contexts it handles poorly. Over time, a well-designed system should make the coach more consistent and better informed—not more passive.

COACH'S CORNER

Before each training block, write down which AEI outputs will influence which decisions—for example, "readiness below the athlete's usual range on a key day means we shorten intervals, not cancel them." Agreeing the rules in advance reduces emotional decisions on the day and makes your overrides easier to review later.

Avoiding dashboard drift

Many monitoring systems begin with good intentions and slowly become cluttered with metrics that nobody acts upon. A useful discipline is to ask of every metric: "What decision would change if this number moved?" If the answer is "none", the metric may still have research value, but it should not compete for attention on the daily training screen. In Indian club and academy settings, where coaches may manage large groups with limited staff, fewer and clearer signals are usually more valuable than a comprehensive but unread report.

CHECKPOINT

For each metric you currently collect, can you name the specific training decision it influences? Which metric would you remove if you could only keep five?

CHAPTER 55

55. Training Intensity Domains

PART X — TRAINING DESIGN

Say "easy" to a group of runners on Marina Beach at 5:30 a.m. and watch what happens. The veteran coming back from a calf strain drops to a shuffle. The seventeen-year-old who wants to impress the senior group settles into something that is not easy at all. The college runner with a new watch checks a number and runs to it, whatever the rising humidity is

doing to her heart rate. One word, three different sessions. Repeat that across a season and it becomes a serious problem: easy days that are secretly moderate, hard days that are never quite hard, and a coach who cannot tell from the log what the athlete actually did. Endurance coaching has always needed a shared language for effort, one that still works when the city, the season or the watch changes. Without it, even a perfect plan gets lost in translation.

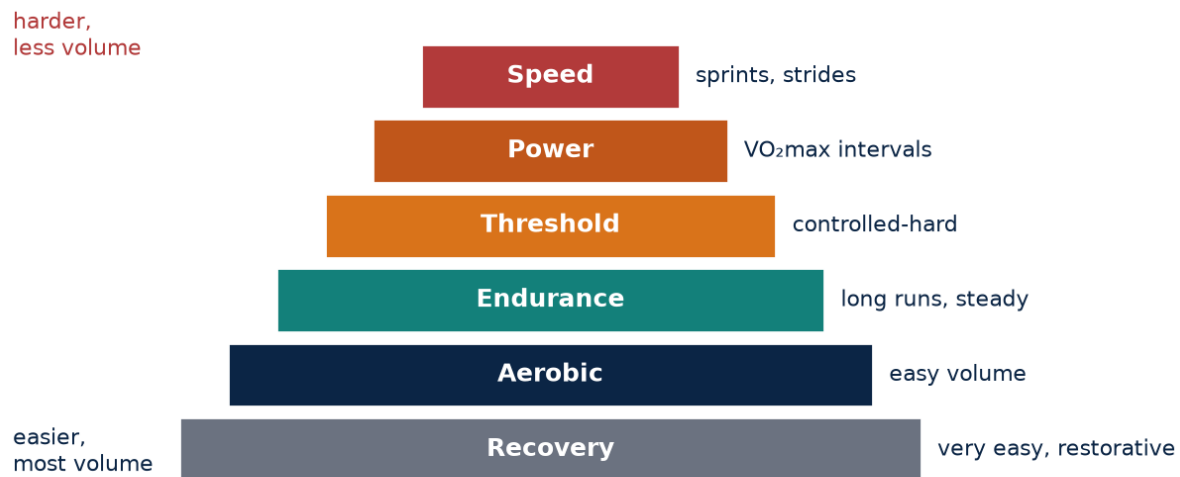


Figure 22. The six AKURA intensity domains. Most weekly volume sits at the base; harder domains are used in smaller, planned doses.

Coaches and athletes need a shared language for effort. "Go easy" means different things to a teenager chasing a club record and a veteran marathoner returning from injury. AKURA uses broad intensity domains—Recovery, Aerobic, Endurance, Threshold, Power and Speed—that can be mapped to pace, heart rate, rating of perceived exertion (RPE) or workout structure depending on the athlete, the equipment available and the purpose of the session.

The six AKURA domains

The domains are deliberately broad. They are categories of training purpose, not precise physiological boundaries, and the transitions between them are gradual rather than sharp. The table below gives illustrative descriptions; each athlete's anchors must be established from their own testing and training history.

Domain	Main purpose	Feel (RPE 0–10)	Typical session examples
Recovery	Promote circulation and restoration	1–2, very easy, full conversation	20–40 min jog or brisk walk; easy cycling
Aerobic	Build aerobic base and capillarity	3–4, easy, conversational	45–75 min steady easy run
Endurance	Durability and fuel use over time	4–5, comfortable but sustained	Long run 90–150 min; marathon-pace segments
Threshold	Raise sustainable pace near lactate turn-point	6–7, comfortably hard	3 × 10 min or 20–30 min tempo
Power	Develop VO ₂ max and high aerobic power	8–9, hard, few words	5 × 3 min or 6 × 1 km with recoveries
Speed	Neuromuscular speed and running mechanics	Fast but relaxed; not maximal strain	8 × 100 m strides; 6 × 200 m with full recovery

KEY IDEA

Intensity domains describe the purpose of a session, not only its pace. The same pace can belong to different domains on a cool morning and a humid afternoon.

Anchoring domains to the individual

Established systems already show how current fitness can define training categories. Jack Daniels' VDOT approach, for example, derives Easy, Marathon, Threshold, Interval and Repetition paces from a recent race performance, and it remains a practical and widely used reference. Heart-rate zones based on laboratory lactate or ventilatory thresholds offer another route. RPE, meanwhile, is available to every athlete at no cost and captures internal strain that pace alone may miss.

AKURA does not need to replace these tools. The six domains can sit on top of any of them: a well-resourced athlete may have laboratory-derived heart-rate anchors, while a school athlete in a district programme may use recent race times and RPE. What matters is consistency—using the same anchoring method over time so that changes can be interpreted.

Research on intensity distribution adds useful perspective. Seiler and Kjerland's 2006 analysis of elite junior cross-country skiers found that much of the training volume was performed at low intensity, with a smaller proportion at high intensity. Such findings do not prescribe an exact ratio for every runner, but they caution against the common habit of doing most runs at a "moderately hard" pace that is too fast for recovery and too slow for high-quality stimulus.

Adding environmental and readiness context

AEI should add environmental and readiness context rather than simply creating another pace table. A threshold pace established in December in Bengaluru may be unrealistic in May in Chennai. If the athlete insists on hitting the same pace in higher heat stress, the session silently moves from Threshold into Power, with a very different cost. AKURA's approach is to hold the domain constant and let pace flex: the session's purpose is protected, and heart rate and RPE help confirm the athlete is in the intended domain.

Readiness works the same way. On a low-readiness day, the athlete may still train in the Aerobic domain but at a slower pace than usual, or a planned Threshold session may be shortened. The domain language helps the coach explain the change: "Today's purpose is still threshold—we are just protecting it."

COACH'S CORNER

When conditions are hot or humid, prescribe threshold and endurance sessions by heart rate and RPE with a pace range as a guide, not a target. Tell athletes in advance that slower pace in heat is expected and does not mean lost fitness.

Common misuses

Three misuses appear repeatedly. First, the grey zone: easy days drift into moderate effort, leaving the athlete too tired for quality days. Second, the chase: an athlete forces a pace from a watch rather than respecting the session's purpose. Third, the false precision: treating a zone boundary as a hard line, so that a heart rate two beats above the limit is judged a failure. Domains are tools for communication and planning, not tests of obedience.

CHECKPOINT

Over the last two weeks, what proportion of your athlete's runs were genuinely in the Recovery or Aerobic domains? Were any "easy" days actually moderate?

CHAPTER 56

56. The 3–3–1 Concept

PART X — TRAINING DESIGN

When Sawan Barwal broke India's longest-standing athletics record in 2026, the headlines were about the time. The training behind it was about rhythm. The Tribune reported that his weeks reached about 200 km, with long runs of 40–42 km on alternate weeks, not every week. The Bridge

reported that his Reliance Foundation "Project 2:09" group normally spends about four months, from November to February, at altitude in Ooty and Wellington before moving to Bengaluru for pre-competition work. Notice what is missing: a huge long run every single Sunday. Even at the top of the sport, the biggest sessions are spaced out, alternated and surrounded by easier days so the body can absorb them. Most runners reading this will never run 200 km in a week. But every runner lives inside a week, and the shape of that week decides whether hard training becomes fitness or just fatigue.

An illustrative 3-3-1 week (individualise every element)

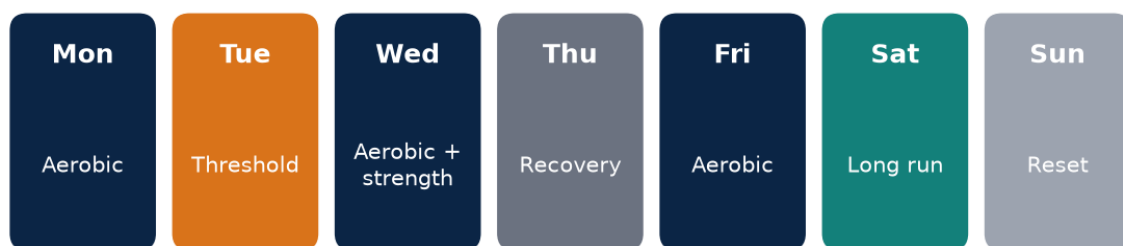


Figure 23. One possible reading of the 3-3-1 rhythm. The value lies in how load is distributed, not in the numbers themselves.

Weekly structure gives training its rhythm. Athletes like to know what Tuesday and Saturday will bring, and coaches need a framework that balances stress and recovery. AKURA uses a 3-3-1 concept as one possible weekly architecture. It is a starting template for thinking, not a rule to be applied identically to every athlete.

What 3-3-1 means

A 3-3-1 weekly rhythm can be interpreted as three days emphasising aerobic development, three days supporting complementary qualities or recovery, and one key recovery or reset day. The three aerobic-development days usually carry the week's main endurance stimulus: a long run, a threshold session and a steady aerobic run, for example. The three complementary days might include strength work, speed or strides, easy running, mobility or cross-training. The single reset day allows physical and psychological recovery—complete rest for some athletes, a short easy jog or walk for others.

The exact structure must be individualised. A school athlete balancing examinations, a working professional training before office hours and a full-time academy runner will not share the same week. Some athletes need two reset days; some advanced athletes may combine elements on the same day.

Day	Illustrative emphasis	AKURA domain	Example content
Mon	Reset	Recovery	Rest or 20–30 min very easy jog; mobility
Tue	Aerobic development	Threshold	Warm-up; 3 × 10 min threshold, 2 min jog; cool-down
Wed	Complementary	Aerobic + strength	45 min easy; runner strength session (Chapter 58)
Thu	Aerobic development	Aerobic	60–70 min steady aerobic run
Fri	Complementary	Recovery + speed	30–40 min easy; 6–8 × 100 m relaxed strides
Sat	Complementary	Aerobic + strength	40 min easy; short strength maintenance
Sun	Aerobic development	Endurance	Long run 90–120 min, early morning in heat season

This example week is illustrative only. It must be individualised to the athlete's training age, injury history, race calendar, climate and life commitments before use.

KEY IDEA

The value of 3–3–1 lies in how it distributes stress and recovery, not in the numbers themselves. A week is well designed when the athlete can absorb it and arrive at key sessions ready to perform.

Load distribution over labels

Two athletes with the same weekly kilometres can experience very different stress depending on intensity, hills, heat and strength work. Sixty kilometres of flat, easy running on cool mornings is not the same as sixty kilometres containing two threshold sessions, a hilly long run and afternoon runs in 34 °C humidity. Kilometre totals are easy to count and easy to compare on social media, but they hide much of what the body actually experiences.

For this reason, AEI should track internal load as well as external kilometres. A practical and validated approach is session RPE: the athlete rates the whole session on a 0–10 scale about 30 minutes after finishing, and this rating is multiplied by the duration in minutes to produce a session load in arbitrary units. Session RPE, described by Foster and colleagues, requires no equipment and captures the combined effect of pace, terrain, heat and fatigue. Heart-rate-based measures can complement it where reliable data are available.

Spacing the key days

The order of days matters. Hard sessions placed back to back leave little time for recovery unless the athlete is highly trained and the second session is deliberately modest. In the illustrative week, the threshold day and long run are separated by lower-stress days, and the reset day follows the long run. Strength work sits on days when the legs can tolerate it without compromising the next key running session.

Environmental timing also belongs in the design. In much of India, the long run is best placed early on a weekend morning, before heat and humidity rise and before traffic makes roads unsafe. During monsoon months, flexible scheduling helps—moving a threshold session by a day to avoid waterlogged roads is sensible planning, not indiscipline.

COACH'S CORNER

Record session RPE for every session for four weeks, including strength and cross-training. Compare weekly load with weekly kilometres. You will often find that the "hardest" week by kilometres was not the hardest week for the athlete.

When 3–3–1 does not fit

Rigid adherence to any template can cause problems. An athlete returning from injury may need a 2–4–1 week with fewer aerobic-development days. An athlete in a taper will reduce volume while preserving some intensity. Travel weeks, exam weeks and festival periods may call for a simpler structure. The concept is useful precisely because it can flex: it reminds the coach to balance development, support and recovery, whatever the exact numbers.

CHECKPOINT

If you compared two of your athletes with similar weekly kilometres, would their internal load be similar? What in their weeks—heat, hills, intensity, strength—might explain the difference?

CHAPTER 57

57. Progressive Overload Without Blind Progression

PART X — TRAINING DESIGN

Avinash Sable did not break the national steeplechase record once. He broke it ten times. Between September 2018 and July 2024 the mark came down in steps: 8:29.80 in Bhubaneswar, 8:21.37 in a World Championships final in Doha, 8:18.12 at the Tokyo Olympics, 8:11.20 for Commonwealth silver in Birmingham, and finally 8:09.91 at the Paris Diamond League. No single leap, just a staircase. Then, in July 2025, came a knee injury at the Monaco Diamond League, damage to the ACL and meniscus, and surgery. When he returned to the steeplechase in September 2026, he ran 8:32.39. Looking back on his injuries, he told Olympics.com: "I think I might have

rushed my return a bit, which caused the injury to recur." Progression, it turns out, runs in two directions, and the way down needs as much planning as the way up.

Adaptation requires progressive challenge. If the training stimulus never changes, the body has little reason to change either. But progression that ignores the athlete's response is not a strategy; it is a hope. This chapter argues for progression that is planned, monitored and reversible.

The problem with fixed percentages

A fixed percentage increase every week—such as the familiar "add 10% per week"—is easy to remember but not always appropriate. For a beginner running 15 km per week, 10% is a tiny change; for an athlete already at 120 km, it may be a large one. Percentages also ignore what the kilometres contain: adding 10% of easy running is very different from adding 10% of threshold work or hill repetitions.

The athlete's current load, recovery, injury history, race calendar and environmental stress all matter. An athlete who has recently come through a stress injury needs more conservative progression than a healthy athlete with years of consistent training. An athlete entering the hottest months may need to hold volume while the body adapts to thermal strain, even if the calendar says it is a "build" week.

MYTH VS EVIDENCE

Myth: Increasing weekly mileage by exactly 10% is the safe way to progress.

Evidence: There is no universal safe percentage. Safe progression depends on current load, training history, session content, recovery and environment, and should be adjusted to the athlete's observed response.

Load, capacity and the training–injury paradox

Gabbett's 2016 review on the training–injury prevention paradox argued that well-developed physical qualities, built through appropriately high training loads, can be protective, while rapid spikes in load relative to what the athlete is prepared for are associated with higher injury risk. The practical message is not "train less" or "train more", but "build load steadily and avoid sudden jumps". Methods used to express this relationship, such as acute:chronic workload ratios, have since been debated statistically, so they are best treated as prompts for attention rather than precise risk predictions.

KEY IDEA

The key question is not "Did mileage increase?" but "Did the athlete absorb the load and produce a better response?" Progression is earned by adaptation, not scheduled by the calendar alone.

A decision matrix for progression

Progression should be planned, monitored and reversible. At the end of each week or block, the coach can review a small set of signals: whether key sessions were completed as intended, how heart rate and RPE compared with previous weeks at similar paces, trends in sleep and resting heart rate, soreness or pain, and the athlete's own sense of how the week went. The matrix below offers an illustrative way to turn those signals into a decision.

Absorbed-load signals	Interpretation	Decision
Key sessions hit; HR/RPE stable or lower; sleep and mood normal	Load absorbed, positive response	Progress: modest increase in volume or one quality variable
Sessions completed but RPE higher than usual; mild fatigue	Load absorbed at a cost	Hold: repeat similar week before progressing
Mixed: one key session missed or cut; readiness below usual for 2–3 days	Load at the edge of tolerance	Hold or small reduction; review sleep, heat, life stress
Multiple sessions compromised; persistent fatigue; rising resting HR	Load not absorbed	Reduce: lower volume and intensity for several days
New or worsening localised pain; limping; altered gait	Possible injury warning	Reduce and seek assessment from a qualified professional

These signals and thresholds are illustrative and should be calibrated to each athlete's own baseline.

Reduction is part of adaptation

A reduction week is not failure; it is part of adaptation. Many athletes—and some coaches—feel that a lighter week wastes fitness. In reality, recovery is when much of the adaptation to the preceding training is consolidated. Planned reduction weeks, for example every third or fourth week, allow fatigue to dissipate and provide a cleaner window in which to test or observe the athlete's response. Unplanned reductions, triggered by the decision matrix, are equally legitimate: they are the plan responding intelligently to the athlete.

Reversibility also means communicating early. Athletes who know that the plan will adapt to their response are more likely to report fatigue honestly, rather than hiding it to protect their "streak".

COACH'S CORNER

At the end of each week, ask three questions: Were the key sessions completed as intended? Did the effort feel appropriate for the pace? Is anything hurting that did not hurt last week? Let the answers—not the calendar alone—decide whether next week progresses, holds or reduces.

CHECKPOINT

Look back at your athlete's last build phase. Was each increase in load preceded by clear evidence that the previous load had been absorbed? Where would a hold week have helped?

CHAPTER 58**58. Strength and Running Economy**

PART X — TRAINING DESIGN

The gym at a state sports hostel is usually the least glamorous room on campus. A few rusted plates, a squat rack with one wobbly safety bar, a smell of monsoon damp that never quite leaves. Distance runners often walk past it on the way to the track, and for years many were told to keep walking. Lifting, the old advice went, adds bulk, and bulk is the enemy of a 10,000 m runner. So the rack went to the throwers and sprinters, and the runners did another set of strides. Then the research started pointing the other way. Runners who added short, heavy and explosive work often became more economical, not slower. The question stopped being whether distance runners should lift. It became how much, when and why, so that the gym helps the track instead of competing with it.

For many years, distance runners were told that strength training would make them heavy and slow. The evidence now points in a different direction. Strength training can support force production, neuromuscular coordination and resilience. It should be integrated with running rather than treated as a separate sport—and it should be judged by what it does for running, not by how much weight the athlete can lift.

What the evidence suggests

Several reviews have examined strength training in endurance athletes. Beattie and colleagues' 2014 review in *Sports Medicine* concluded that strength training, including heavy and explosive methods, can improve performance-related outcomes in endurance athletes, with running economy among the most frequently reported benefits. Balsalobre-Fernández and colleagues' 2016 meta-analysis of controlled trials in highly trained runners found that strength training programmes of several weeks were associated with improvements in running economy. Earlier, Paavolainen and colleagues showed that replacing part of endurance training with explosive-strength work improved 5 km time in well-trained runners, linked to improved economy and muscle power.

These findings are encouraging but should be read carefully. Study samples are often small, programmes vary, and effects on actual race performance are less consistently measured than effects on economy. The likely mechanisms—improved neuromuscular recruitment, tendon and muscle stiffness characteristics, and better force application—are plausible but not fully settled.

KEY IDEA

For distance runners, the objective is not bodybuilding volume. It is sufficient strength, control, stiffness management and movement quality without compromising key running sessions.

Designing strength for runners

Effective strength work for runners tends to share a few features: relatively low volume, emphasis on quality of movement, progressive loading of major lower-limb muscles, some plyometric or elastic work once basic strength is established, and attention to the calf–Achilles complex and hip stability that running places under repeated demand. Sessions are usually short—30 to 45 minutes—and performed two times per week in general preparation, reducing to one maintenance session during competition phases.

Exercise	Illustrative prescription	Purpose	Coaching note
Goblet or back squat	3 × 5–6, controlled	Hip and knee extensor strength	Stop 2–3 reps short of failure
Romanian deadlift	3 × 6	Posterior chain, hamstrings	Neutral spine; hinge at hips
Rear-foot elevated split squat	3 × 6 each leg	Single-leg strength and control	Knee tracks over mid-foot
Single-leg calf raise (straight and bent knee)	3 × 10–12 each	Calf–Achilles capacity	Slow lowering; full range
Pogo jumps or low hurdle hops	3 × 10 contacts	Elastic stiffness, ground contact	Quiet, quick contacts; add later
Side plank with hip abduction	2 × 20–30 s each side	Lateral hip and trunk control	Stable pelvis

This session is illustrative and should be adapted to the athlete's training age, technique, injury history and available equipment. Beginners should start with bodyweight or light loads and learn movement patterns before adding plyometrics; athletes with current pain should be assessed by a qualified professional first.

Integration with the running week

Strength should be placed so that it supports, rather than competes with, key running sessions. A common approach is to schedule strength on the same day as an easy run or after a quality session, leaving the following day easy, so that the hard days stay hard and the easy days stay easy. Heavy lower-body work immediately before a threshold session or long run is usually unwise. During the hottest months, strength sessions performed indoors can also provide useful training stimulus when outdoor running volume is limited by heat.

Rønnestad and Mujika's 2014 review on optimising strength training for running and cycling highlighted the importance of maintaining some strength work during the competition season, since gains can fade when strength training stops entirely.

COACH'S CORNER

Start runners with two short sessions per week built around four or five movements. Progress load only when technique is stable and the running sessions that follow are unaffected. If key runs are suffering, reduce strength volume before reducing running quality.

What AEI should—and should not—track

AEI can track strength adherence and selected performance markers—for example, sessions completed per week, a single-leg calf raise count, a countermovement jump height or a simple loaded squat estimate—alongside changes in running economy or submaximal heart rate. These records help the coach see whether strength work is being done consistently and whether it coincides with improvements in running.

However, AEI should not assume that more strength always means faster running. Beyond a sufficient level, additional strength may offer diminishing returns for a distance runner, and extra training time spent in the gym is time not spent recovering or running. The relationship is likely non-linear and individual. The useful question is not "How strong is the athlete?" but "Is the athlete strong enough to run well, stay healthy and express their aerobic fitness efficiently?"

MYTH VS EVIDENCE

Myth: Strength training will make distance runners bulky and slow.

Evidence: Low-volume, well-programmed strength and plyometric work is associated with improved running economy in several controlled studies, with little change in body mass when combined with endurance training.

CHECKPOINT

Is strength work in your athlete's plan scheduled to protect key running sessions?
Which strength marker would you track to know whether it is helping?

PART X AT A GLANCE

The key idea from each chapter, for quick revision.

- | | |
|-----------|---|
| 54 | From AEI to Training
AEI is a decision-support loop, not a scoreboard. Performance sets the targets, readiness and environment adjust them, recovery protects them, and execution data closes the loop. |
| 55 | Training Intensity Domains
Intensity domains describe the purpose of a session, not only its pace. The same pace can belong to different domains on a cool morning and a humid afternoon. |
| 56 | The 3–3–1 Concept
The value of 3–3–1 lies in how it distributes stress and recovery, not in the numbers themselves. A week is well designed when the athlete can absorb it and arrive at key sessions ready to perform. |
| 57 | Progressive Overload Without Blind Progression
The key question is not "Did mileage increase?" but "Did the athlete absorb the load and produce a better response?" Progression is earned by adaptation, not scheduled by the calendar alone. |
| 58 | Strength and Running Economy
For distance runners, the objective is not bodybuilding volume. It is sufficient strength, control, stiffness management and movement quality without compromising key running sessions. |

PART XI — CASE STUDIES

Principles become clearer when applied to an athlete. Part XI presents five hypothetical case studies, each designed to show how the AKURA Endurance Index can move a coach beyond a single headline number. We meet a runner with a high $VO_2\text{max}$ who fades in the marathon, a runner with modest $VO_2\text{max}$ but excellent economy, an athlete who responds strongly to heat acclimation, an athlete whose altitude exposure is measured rather than assumed, and an athlete whose fitness is high but whose readiness is low. All profiles and numbers are illustrative, created for teaching; they do not describe real individuals and must not be used as norms or diagnostic thresholds.

IN THIS PART

- 59** Case Study A: High $VO_2\text{max}$, Weak Marathon
- 60** Case Study B: Moderate $VO_2\text{max}$, Excellent Economy
- 61** Case Study C: Strong Heat Response
- 62** Case Study D: Altitude Response
- 63** Case Study E: Strong Fitness, Poor Readiness

PART XI • CASE STUDIES

Five Athletes, Beyond One Number

Five illustrative profiles show how AEI moves a coach past a headline number. Teaching examples, not norms.

9:19.76

Lalita Babar's national record in her Rio 2016 heat

 **66**

High VO₂max, fading marathon

VO₂max 66, but threshold ~76% and second half 8 min slower: durability, fuel and heat limit.

 **55**

Modest VO₂max, great economy

VO₂max 55 with economy ~185 ml O₂/kg/km and threshold ~87%: a 1:21 half marathon.

 **-13 bpm**

Strong heat response

HR at a fixed pace in 32°C fell from 164 to 151 bpm; plasma volume about +6%.

 **+4%**

Measured altitude response

~24 days at 2,100–2,200 m: Hb mass 820 to 855 g; 10,000 m 31:05 to 30:40.

 **5.4 h**

Fit but not ready

Fitness at a personal high, but sleep fell from 7.2 to 5.4 h and resting HR rose 48 to 55.



Illustrative only

All five profiles were created for teaching; never use them as norms or diagnostic thresholds.

THE PROCESS



No single metric defines an athlete; read fitness and readiness separately.

THE STORY

Lalita Babar and the Long Road to Rio

PART XI — CASE STUDIES

August 2016, Rio de Janeiro. The women's 3000 m steeplechase final is about to start in the Olympic Stadium: seven and a half laps, 28 barriers, seven water jumps. On the start line is a 27-year-old from a village in Satara district, Maharashtra, wearing India's colours. Nobody from India has reached an Olympic track final since P. T. Usha in 1984. Lalita Babar has got there by running 9:19.76 in her heat, a national record. She will finish tenth in the final, in 9:22.74. To understand how she got to that start line, you have to go back to the dry fields of Satara, and to a lot of water carried by hand.

Mohi

Lalita Shivaji Babar was born on 2 June 1989 in Mohi village, in a drought-prone part of Satara district, into a farming family. As a child she ran 4 km each way to school, and she fetched water for a household of 17 people. None of that was training. It was ordinary daily life, the kind that sets up a body long before any coach sees it.

At 15, in 2004, she won a 10 km road race and earned her first prize money: Rs 10,000. For a farming family in a drought-prone district that was a meaningful sum, and it made a point that many Indian distance runners learn early: running could pay. She later credited her family: "My parents inspired me, continuously pushing me to devote my life to athletics after having watched my passion," she told *The Bridge*.

The next step was the one that supports so many Indian athletes: a job. Lalita joined Indian Railways, whose sports quota has long paid Indian distance runners to keep training. For years she was a road runner, and a successful one. She won the Indian elite women's race at the Delhi Half Marathon in 2010, 2011 and 2015, and she won the Indian women's title at the Mumbai Marathon three years in a row, from 2012 to 2014. Her marathon best, 2:38:21, came in Mumbai in 2015.

The switch

Around 2010 the national distance coach Nikolai Snegarev introduced her to the steeplechase. The event was new for Indian women. The first women's steeplechase at the Asian Games was held only in 2010, and in May that year the national record stood at 10:09.56. Lalita kept racing on the roads for several more years. In 2014 she switched to the steeplechase full time.

She was not the only Indian woman being moved into the event. O. P. Jaisha, also coached by Snegarev, had taken up the steeplechase only in 2008 and set a national record of 10:03.05 at Patiala in August 2010. Sudha Singh had been steered into it by a

Railways officer. In such a young event, nobody could say in advance who would take to the barriers and the water. The only way to find out was to try, race and watch.

Eight months after her full-time switch, Lalita won bronze at the 2014 Asian Games in Incheon in 9:35.37, a national record. In June 2015 she won the Asian Championships in Wuhan in 9:34.13. That August, at the World Championships in Beijing, she ran 9:27.86 in her heat, again a national record, and finished eighth in the final. She was the first Indian woman to reach a World Championships steeplechase final. At the same championships Jaisha ran 2:34:43 in the marathon, a women's national record that still stands. For the Snesev group, Beijing 2015 was a remarkable week.

In 2016 the record passed back and forth. Lalita ran 9:27.09 at the Federation Cup in New Delhi in late April. In May, Sudha Singh, a Central Railway athlete from Raebareilly and the 2010 Asian Games champion, ran 9:26.55 at the Shanghai Diamond League. That year the two trained together in a Bengaluru camp under Snesev, so the national record was being pushed by two women in the same group. Then came Rio: 9:19.76 in the heat and a place in the final.

Reading her career as a case study

What would a coach see if Lalita Babar's career were laid out as an athlete profile? The honest answer starts with what we cannot see. We have no public data on her VO_2max , running economy, lactate threshold, haemoglobin mass or weekly training load. Anyone who claims to know why she ran 9:19.76 in physiological terms is guessing. If she were starting today in an AKURA-style system, a coach would want exactly those missing numbers, plus the context around them: the heat of a Satara summer, the travel between camps and races, the sleep and recovery around a Railways job. None of that was recorded in a form we can read now. What we do have is the performance record, and a performance record, read carefully, still says a good deal.

First, a long aerobic foundation came before specialisation. Years of daily running as a child, then years of road racing up to the marathon, built the base she later took into a technical track event. Second, the event switch was matched to her strengths. The steeplechase rewards aerobic strength, rhythm and resilience over barriers and water, and a road runner already has the aerobic strength. Third, the progression was fast but not sudden: 9:35.37 in 2014, 9:27.86 in 2015, 9:19.76 in 2016. That is about 16 seconds in two seasons, roughly eight seconds a year, from an athlete who was already mature.

Fourth, and most unusual, she ran her best times when they mattered most. Most of her national records came at championships: the Asian Games, the Asian Championships, a World Championships heat and an Olympic heat. Many athletes peak at the wrong time; her record shows repeated peaks at the championships. Fifth, context: a stable job, a single long-term coach and a training partner of similar level pushing the same record. No laboratory test measures those factors, yet they run through almost every successful Indian distance career in this book.

Her 9:19.76 stayed the national record for seven years, until Parul Chaudhary ran 9:15.31 in the Budapest World Championships final in 2023. Snesev left the AFI in 2019 and died in March 2021. A report from about 2019 said Lalita was set to leave the Railways for a post as a deputy collector with the Maharashtra government.

The lesson for the chapters ahead

Lalita Babar's story is real, and that is exactly why it cannot be dissected the way a coach needs to dissect a case. There is no heart-rate file from Incheon and no blood test from before Rio. The chapters that follow do the opposite. They build hypothetical athletes, with invented but plausible numbers, so that we can see every layer at once: the high-VO₂max runner who fades, the moderate runner with superb economy, the heat responder, the altitude responder, and the fit athlete who is not ready. They are not portraits of anyone. Each is presented the same way: the profile, what the index shows, the likely contributors, an intervention plan, what to retest, and, most important, what evidence would change our mind. That last heading is the discipline Lalita's career cannot give us. With a real athlete and no data, every explanation fits. With a measured athlete, some explanations can be ruled out. They are practice for the moment when a real athlete's data are on the table and the coach has to decide what to do next.

Sources: <https://www.thequint.com/sports/sports-buzz/from-running-in-fields-to-the-rio-tracks-lalita-babars-story> ; <https://thebridge.in/athletics/lalita-babar-feels-proud-by-making-her-village-girls-established-athletes> ; <http://indianexpress.com/sports/rio-2016-olympics/lalita-babar-india-3000m-steeplechase-result-final-medal-2973964/> ; <http://indianexpress.com/sports/rio-2016-olympics/lalita-babar-finishes-10th-in-3000m-steeplechase-india-2977275/> ; <https://thebridge.in/athletics/india-athletics-coach-nikolai-snesev-dead-19834> ; <https://www.hindustantimes.com/other-sports/national-record-a-gift-for-india-s-support-steeplechaser-sudha-singh/story-zUIqWcXotQy21aqUfvaiXK.html> ; <http://media.aws.iaaf.org/competitiondocuments/pdf/4875/AT-3KSC-W-h----RS4.pdf> ; <https://www.sportskeeda.com/running/o-p-jaisha-karan-singh-winners-mumbai-marathon> ; <https://vedantadelhihalfmarathon.procam.in/race-categories/half-marathon-elite/past-winners-and-record-holders> ; <https://www.olympics.com/en/news/nikolai-snesev-the-late-belarusian-coach-s-contribution-to-indian-athletics> ; <https://www.olympics.com/en/news/world-athletics-championships-2023-budapest-day-9-india-results-parul-chaudhary>

CHAPTER 59

59. Case Study A: High VO₂max, Weak Marathon

PART XI — CASE STUDIES

At 5:30 a.m. in Chennai the air already feels like a wet towel. Along a city marathon route, volunteers are cutting open crates of water sachets, and the lead pack passes 30 km looking comfortable. Then the race changes. Somewhere between 30 and 35 km, runners who looked unstoppable at halfway start to shorten their stride, grab two sachets instead of one, and check their watches as if the numbers might be wrong. Every big Indian

city marathon has a version of this scene, and nearly every club coach has an athlete who lives it: brilliant over 5 km, fast on the track, yet unable to carry that speed through the second half of a marathon. The usual instinct is to train the engine harder. The better question is why the engine stops delivering. Consider one such athlete, invented for teaching but familiar to anyone who has coached marathoners.

This is a hypothetical case study. Athlete A (Chennai, 28, club marathoner) has excellent 5K performance and a high VO_2max from a laboratory test, but shows a large fall in pace over the final 10–12 km of the marathon. On paper, the athlete looks capable of a much faster marathon than they have run. The coach's first instinct—and the athlete's—is often to chase even more speed. AEI suggests a different line of enquiry.

Metric	Illustrative value	Interpretation
VO_2max (lab treadmill)	66 ml/kg/min	High aerobic ceiling for a club athlete
5K personal best	16:20	Strong short-distance performance
Marathon best	2:58, second half 8 min slower	Large positive split; poor durability
Threshold as % of VO_2max	~76%	Relatively low fractional utilisation
Longest regular long run	24 km, done at mixed paces	Limited race-specific durability
Race-day carbohydrate intake	Irregular, gels only after 25 km	Possible fuelling shortfall
Race conditions	Start 5:30 a.m., 27 °C, high humidity	Meaningful thermal strain

All values are illustrative and invented for teaching purposes.

What AEI shows

AEI would not reduce Athlete A to "needs more VO_2max ." Their ceiling is already high; the gap lies in how much of that ceiling can be sustained, for how long, in race conditions. The profile directs attention towards durability, pacing, substrate management and race-specific conditioning. In AEI terms, the performance layer shows high capacity, but indicators of fractional utilisation and late-race resilience lag behind—a mismatch rather than a deficiency. This reflects a point made by Joyner and Coyle in their 2008 review: endurance performance depends on VO_2max , the fraction of it that can be sustained, and running economy acting together.

KEY IDEA

A high VO_2max describes the size of the engine, not how long it can run near full power. Marathon performance depends on sustaining a large fraction of capacity for hours, often in heat.

Likely contributors

Possible contributors include insufficient long-run durability, fuelling errors, excessive starting intensity, low threshold relative to maximal capacity and environmental strain. These are hypotheses, not a diagnosis, and more than one may be operating.

- **Durability:** long runs have been relatively short and inconsistent, so late-race fatigue resistance may be under-developed. Maunder and colleagues have argued that "durability"—how physiological markers drift with prolonged exercise—deserves a place in athlete profiling.
- **Fuelling:** taking carbohydrate only after 25 km may leave the athlete short of fuel at the point where glycogen becomes limiting.
- **Pacing:** a fast early pace, encouraged by a strong 5K background, may produce an unsustainable first half.
- **Threshold:** threshold at a modest percentage of VO_2max suggests room to raise sustainable pace.
- **Heat and humidity:** Chennai race mornings impose thermal strain that increases cardiovascular drift late in the race.

Intervention plan (8–12 weeks)

- **Weeks 1–4:** extend the long run progressively from 24 km towards 30 km in the Endurance domain, early morning, with steady pacing; introduce one threshold session per week (for example 3×12 min).
- **Weeks 1–12:** practise race-day fuelling in every long run, starting carbohydrate early and aiming for a consistent intake the athlete tolerates, developed with a qualified sports nutritionist where possible.
- **Weeks 5–8:** add marathon-pace segments late in long runs (for example the final 8–10 km) to train pace control under fatigue.
- **Weeks 5–10:** progress threshold volume; reduce high-intensity Power sessions to one short session per week to preserve speed without excessive fatigue.
- **Throughout:** two short strength sessions per week; heat-aware pacing, with heart-rate and RPE ceilings on hot days.
- **Weeks 11–12:** taper; rehearse race-morning routine, including pacing plan with an even or slightly negative split target.

What to retest

- Threshold pace and heart rate (field test or laboratory) at weeks 0, 6 and 12.
- Pace–heart-rate decoupling in the last third of long runs of similar length and conditions.
- Gut tolerance and energy in the final third of long runs (athlete log).
- Pacing profile and second-half slowdown in a tune-up half marathon and the target marathon.

COACH'S CORNER

Keep a "late-race" column in the athlete's log: pace, heart rate and RPE for the final 30 minutes of every long run. Improvement in this column is often more meaningful for a marathoner than a faster 5K.

What would change our mind

If the athlete completes longer, well-fuelled, evenly paced long runs with little decoupling but still slows sharply in the marathon, the explanation may lie elsewhere—perhaps race-day heat management, a gastrointestinal problem or a musculoskeletal issue emerging late in the race. If threshold does not improve despite consistent work, the coach may need to reconsider training distribution or recovery. And if the athlete runs an evenly paced marathon only in cool conditions, heat may be a larger contributor than assumed.

CHECKPOINT

For an athlete like Athlete A, which single change—long-run length, fuelling, pacing or threshold work—would you prioritise first, and what evidence would tell you it worked?

CHAPTER 60**60. Case Study B: Moderate VO₂max, Excellent Economy**

PART XI — CASE STUDIES

Every running group in Pune seems to have one. The runner who never wins the Tuesday interval session, gets dropped on the fast 400s and tested lower than half the group on the laboratory treadmill. Yet on race morning, somewhere past 15 km of the half marathon, this runner quietly moves up through the field while the "fitter" runners fade. Training partners joke about it. Coaches shrug and call it "strength" or "experience". But there is physiology hiding in that shrug. How fast a runner can race over a long distance depends not only on the size of the aerobic engine but on how much oxygen it uses at a given pace, and how close to its limit it can hold. A modest engine that sips oxygen can outrun a large one that gulps it. This athlete teaches coaches to look beyond the headline number.

This is a hypothetical case study. Athlete B (Pune, 31, amateur half-marathoner and marathoner) has only a moderate VO₂max, yet runs longer races better than several

training partners with far higher laboratory values. The athlete shows a very good oxygen cost at submaximal running speeds and strong threshold performance. This case demonstrates why a single physiological metric cannot define the whole athlete.

Metric	Illustrative value	Interpretation
VO ₂ max (lab treadmill)	55 ml/kg/min	Moderate for a competitive amateur
Running economy at 14 km/h	~185 ml O ₂ /kg/km	Low oxygen cost; efficient
Threshold as % of VO ₂ max	~87%	High fractional utilisation
Half-marathon best	1:21	Strong relative to VO ₂ max
1500 m / 5K times	Modest; limited finishing speed	Low speed reserve
Training history	8 years consistent, mostly aerobic	Deep aerobic base

All values are illustrative and invented for teaching purposes.

What AEI shows

AEI shows a profile in which economy and fractional utilisation compensate for a moderate ceiling. This athlete may outperform a higher-VO₂max runner at longer distances, because racing speed at the marathon depends on the oxygen that can be sustained and how economically it is used—not just on the maximum. Jones's analysis of the physiology of the then women's marathon world-record holder (Paula Radcliffe) illustrated how improvements in running economy and threshold, rather than VO₂max alone, can accompany performance progression.

KEY IDEA

A single physiological metric cannot define the whole athlete. Moderate VO₂max combined with excellent economy and a high threshold can produce outstanding endurance performance.

Likely contributors

- Long training history building aerobic and musculoskeletal adaptations.
- Favourable running mechanics and possibly anthropometric factors that reduce oxygen cost.
- Consistent threshold-focused training that raises sustainable pace.
- Limited exposure to Speed and Power domains, restricting finishing kick and speed reserve.

These contributors are plausible, but economy is influenced by many factors—footwear, fatigue, measurement conditions and body mass—so a single laboratory value should be interpreted with caution.

Intervention plan (8–12 weeks)

The training focus might be to preserve economy while improving speed reserve, rather than endlessly chasing maximal oxygen uptake.

- Maintain the established aerobic volume and long run; avoid dramatic changes that could disrupt what already works.
- Add regular Speed-domain work: strides after easy runs two times per week and short relaxed repetitions (for example 8×200 m) with full recovery.
- Include one Power session every 7–10 days (for example 5×3 min) to nudge aerobic power without large volume.
- Maintain two strength sessions per week, adding progressive plyometrics once basic strength is stable.
- Monitor economy-related markers so that added intensity does not degrade efficiency.

What to retest

- Running economy at the same standard speeds and conditions, ideally the same laboratory and footwear.
- VO_2max and threshold, noting whether threshold remains a high percentage of maximum.
- Short-distance performance (1500 m or 5K time trial) as a marker of speed reserve.
- Half-marathon race performance and pacing profile.

COACH'S CORNER

When an athlete's strength lies in economy, protect it. Introduce speed in small, relaxed doses and check that easy-run heart rate at familiar paces does not drift upwards as intensity is added.

What would change our mind

If added speed work increases fatigue and worsens economy or submaximal heart rate, the intervention may be too aggressive, or the athlete may benefit more from continued aerobic development. If the "excellent economy" does not reproduce on repeat testing, the initial result may reflect measurement error or favourable test conditions rather than a stable characteristic. And if longer-race results stall while speed improves, the balance may need to shift back towards threshold and endurance.

CHECKPOINT

Among your athletes, who performs better than their VO_2max would predict? What might their economy and threshold profiles reveal?

CHAPTER 61

61. Case Study C: Strong Heat Response

PART XI — CASE STUDIES

Pre-monsoon Kochi does not so much warm up as close in. By 7 a.m. the backwaters are steaming, the air over the roads is heavy with salt and diesel, and a steady run soaks a runner's vest within ten minutes. Anyone who trains on this coast knows the weather well. But knowing the weather is not the same as adapting to it, and adapting is not the same as racing faster. A runner who feels better in the heat after two weeks of deliberate exposure has changed something real, but what exactly? The heart, the blood volume, the sweat glands, the mind, or simply the calendar moving towards cooler mornings? Heat acclimation is one of the most reliable adaptations in sports science, and one of the easiest to over-interpret. Here is a hypothetical runner who shows how to read it carefully.

This is a hypothetical case study. Athlete C (Kochi, 24, 10K and half-marathon runner) completes a controlled heat-acclimation block before a humid coastal race season. Afterwards, the athlete shows a lower heart rate at a standard submaximal workload, improved thermal comfort and a measurable increase in plasma-volume-related markers. The case is useful not because the response is dramatic, but because it shows how to interpret a response carefully.

Metric	Before (illustrative)	After (illustrative)	Interpretation
HR at 10 km/h, 32 °C, 70% RH, 30 min	164 bpm	151 bpm	Lower cardiovascular strain
RPE at same workload	7	5	Session feels easier
Thermal comfort (1–7 scale)	6 (hot)	4	Improved perceived tolerance
Estimated plasma volume change	—	about +6%	Consistent with acclimation
Sweat rate (L/h)	1.1	1.3	Possible earlier, greater sweating
Body-mass loss in session	0.8%	1.0%	Hydration needs rising

All values are illustrative and invented for teaching purposes. Plasma-volume changes estimated from haemoglobin and haematocrit carry measurement error and depend on hydration status at testing.

What AEI shows

AEI's environmental layer records a clear, measurable response to a specific intervention. Heart rate and RPE fall at an identical workload in identical conditions, and markers linked to plasma-volume expansion move in the expected direction. These changes match the adaptations commonly described in heat-acclimation research, summarised by Périard, Racinais and Sawka in 2015: lower heart rate, earlier and greater sweating, plasma-volume expansion and improved thermal comfort.

The correct conclusion is not "heat made the athlete elite." It is that the athlete demonstrated a measurable response to a specific intervention. Whether that response improves race performance in the heat is a separate question requiring a separate measurement.

KEY IDEA

A good heat response is evidence of adaptation to heat, not proof of general performance improvement. Record the intervention, the test conditions and the response, then test again.

Likely contributors

- The heat-acclimation protocol itself: repeated sessions producing sustained elevation of body temperature over roughly 10–14 days.
- Standardised testing conditions that make the before–after comparison meaningful.
- Possible contributions from improved hydration habits and familiarity with the test (a learning effect).
- Seasonal change: if outdoor temperatures rose during the block, natural acclimatisation may have contributed.

Intervention plan (8–12 weeks)

- Weeks 1–2: structured acclimation block—for example 60–90 minutes of mostly Aerobic-domain running on most days in hot, humid conditions, with heart-rate and RPE ceilings and a safety plan.
- Weeks 3–8: maintenance exposures two to three times per week to help retain adaptations, since they decay without continued exposure.
- Throughout: individual hydration plan based on measured sweat rate and body-mass change; avoid large body-mass losses and also avoid over-drinking.
- Weeks 6–10: race-specific sessions in heat with heat-adjusted pacing; practise cooling strategies such as pre-cooling and ice towels.
- Weeks 11–12: taper, maintain light heat exposure, rehearse race-day routine.

COACH'S CORNER

Run heat sessions in a buddy system, with water, shade and a clear stop rule for symptoms such as dizziness, confusion, nausea or stopping sweating. Heat acclimation is a training stimulus with real risk; any athlete showing warning signs should stop and receive appropriate care immediately.

What to retest

- The same standard heat test (speed, duration, temperature, humidity, time of day, hydration state) at 2 weeks and 6–8 weeks.
- Plasma-volume-related markers under consistent conditions.
- Sweat rate and body-mass change to update the hydration plan.
- A heat-affected performance marker, such as a time trial in controlled warm conditions.

Repeat testing is required before treating the response as stable.

What would change our mind

If the heart-rate reduction disappears on a second test with stricter standardisation, the first result may have reflected day-to-day variability, hydration differences or cooler conditions. If laboratory-style markers improve but performance in hot races does not, the adaptation may be real but insufficient to change the race outcome—or other limiters such as pacing or fuelling may dominate. And if the athlete's response is much smaller in a second block, individual variability, not a stable trait, may be the better explanation.

CHECKPOINT

If Athlete C races well in Kochi next month, how would you separate the effect of heat acclimation from general fitness gains made during the same period?

CHAPTER 62**62. Case Study D: Altitude Response**

PART XI — CASE STUDIES

At 2,240 m, Ooty is only a little lower than Iten, the Kenyan town that calls itself the Home of Champions. For decades Indian distance runners have climbed the Nilgiri ghat roads to train in its thin, misty air. Thonakal Gopi spent a ten-month stint at altitude in Ooty before the 2016 Rio Olympic marathon, ESPN reported, and four months there before running 2:12:23 in

Valencia in December 2025. "I think the high altitude at Ooty helped my performance," he said afterwards, as reported by the running writer Shyam Gopan. He may well be right. But look at the shape of the claim: the athlete went up, came down and ran fast. What happened in between, in the blood, is the part that is almost never measured. The research for this book found no peer-reviewed study of Indian athletes' blood changes at Ooty. This hypothetical case shows what measuring it would look like.

This is a hypothetical case study. Athlete D (Bengaluru, 26, 5000 m and 10,000 m track athlete) completes a documented altitude camp. After returning, haemoglobin mass (Hb mass) has increased and a standardised submaximal test improves after sufficient recovery. The case illustrates how measurement strengthens a causal story that is often told with no data at all.

Metric	Before (illustrative)	After (illustrative)	Interpretation
Altitude exposure	—	~24 days at ~2,100–2,200 m, ~20 h/day	Documented dose
Hb mass (CO rebreathing)	820 g	855 g (+4%)	Plausible haematological response
Serum ferritin (pre-camp screen)	Within adequate range	Monitored	Iron status supports adaptation
HR at standard submax speed (sea level)	158 bpm	152 bpm	Lower cardiovascular strain
Blood lactate at same speed	2.8 mmol/L	2.3 mmol/L	Improved submaximal response
10,000 m time (2–3 weeks post-camp)	31:05	30:40	Performance improvement

All values are illustrative and invented for teaching purposes. Retests assume similar conditions, time of day and recovery status.

What AEI shows

This is a stronger causal story than simply saying "the athlete trained in the hills and ran faster." The measurement separates environmental exposure from the performance outcome. We have a documented dose (time and altitude), a physiological mediator (Hb mass), a standardised submaximal response and a performance result. Each link can be examined on its own. The observed Hb-mass increase is in line with what Gore and colleagues' 2013 meta-analysis reported for similar exposures using the optimised carbon-monoxide rebreathing method, though individual responses vary widely.

KEY IDEA

Measuring the exposure, the physiological response and the performance separately turns a story into evidence. Even then, the result belongs to that athlete and that protocol.

Likely contributors

- Hypoxic stimulus sufficient in duration and altitude to increase red-cell production.
- Adequate iron availability, energy intake and recovery supporting that response.
- Training performed during the camp, which may itself have improved fitness regardless of altitude.
- Focused camp environment: fewer distractions, more sleep, structured routine.
- Race selection and conditions after the camp.

The last three contributors matter: without a comparison condition, some of the improvement may have occurred with equivalent training at home.

Intervention plan (8–12 weeks)

- Weeks 1–2 (pre-camp): baseline Hb mass, submaximal test and iron screen, with any supplementation decided by a qualified medical professional.
- Weeks 3–6 (camp): reduce intensity for the first days at altitude, rebuild volume gradually, monitor resting heart rate, sleep and oxygen saturation trends, and keep high-intensity work controlled.
- Weeks 7–8 (return): allow a period of readjustment before key tests or races; avoid judging the camp on the first few days back.
- Weeks 8–12: repeat submaximal test and Hb mass; race or time-trial in suitable conditions.

COACH'S CORNER

Log the altitude "dose" as carefully as you log training: altitude of sleeping and training locations, hours per day and number of days. Without the dose, you cannot compare one camp with another.

What to retest

- Hb mass using the same method and laboratory, allowing for its typical measurement error.
- The standardised submaximal test at sea level, controlling time of day, temperature and preceding training.

- Performance in a comparable race or time trial.
- Whether gains persist 4–6 weeks after return.

What would change our mind

If Hb mass rises but submaximal and race performance do not improve, the haematological adaptation may not have been the limiting factor. If performance improves without any change in Hb mass, other mechanisms—training quality, rest or motivation—may explain the result. If a second camp produces no Hb-mass change, the athlete's response may be less reliable than the first result suggested. Even with a clear response, this result should not be generalised to everyone: other athletes, altitudes and protocols may produce different outcomes.

CHECKPOINT

Before your athlete's next altitude or hill camp, which three measurements will you take before and after, and how will you account for the training done during the camp?

CHAPTER 63

63. Case Study E: Strong Fitness, Poor Readiness

PART XI — CASE STUDIES

It is 11:40 p.m. in Hyderabad's HITEC City, and the last video call of the day has just ended. On the laptop, a training app is celebrating its owner's best fitness score of the year. At the desk, a 35-year-old marathoner rubs tired eyes, sets an alarm for 4:45 a.m. and thinks about tomorrow's threshold session, the one the plan calls the most important of the week. In India's technology hubs, from Hyderabad to Bengaluru to Pune, many masters runners live this double life, with working hours set by clients in other time zones and training hours set by the sun and the heat. Fitness built over months is real. So is a week of five-hour nights. The hard part is deciding which one should decide tomorrow's session.

This is a hypothetical case study. Athlete E (Hyderabad, 35, masters marathoner working in the IT sector) has an AEI-Fitness score that is high by their own standards: recent threshold tests and long runs show the best form in years. Yet over the past week sleep has been poor, resting heart rate is elevated relative to baseline, and subjective fatigue is high. A key threshold session is scheduled for tomorrow. The case shows why fitness and readiness must be kept separate.

Metric	Athlete baseline (illustrative)	Current week (illustrative)	Interpretation
AEI-Fitness	Personal high	Unchanged	Capacity is strong
Sleep duration (7-day mean)	7.2 h	5.4 h	Substantial sleep deficit
Morning resting HR	48 bpm	55 bpm	Elevated relative to baseline
Subjective fatigue (1–10)	3	7	High perceived fatigue
Muscle soreness (1–10)	2	4	Moderately raised
Context	—	Project deadline; late calls	Life stress increasing

All values are illustrative and invented for teaching purposes. Thresholds for concern should be set from each athlete's own baseline and variability.

What AEI shows

A readiness layer can prevent the coach from confusing capacity with today's ability to express capacity. The performance layer says the athlete can run a strong threshold session; the readiness layer says the athlete may not be able to run it well, or absorb it well, today. Several independent signals—sleep, resting heart rate and subjective fatigue—point in the same direction, which strengthens the case for caution. Reviews of athlete monitoring, including Saw, Main and Gastein's 2016 systematic review, have found that subjective self-reported measures often respond to training load and wellbeing changes at least as sensitively as common objective markers.

KEY IDEA

Fitness is what the athlete has built; readiness is what the athlete can express today. Treating them as the same number leads to hard sessions on the wrong days.

Likely contributors

- Sleep restriction linked to work deadlines and late calls.
- Accumulated training load from a strong build phase.
- Psychological stress, which adds to physical load.
- Possible early illness—which the coach should not diagnose, but should consider when advising the athlete to seek medical advice if symptoms appear or persist.
- Environmental strain if recent training has been in afternoon heat.

Intervention plan (8–12 weeks)

The correct intervention may be a modified workout rather than a complete programme change. The next 24–72 hours of data become part of the decision.

- Tomorrow: replace 3 × 10 min threshold with 40–45 min Aerobic-domain running plus 6 relaxed strides, or shorten the threshold work to 2 × 8 min if the athlete feels better after the warm-up.
- Next 24–72 hours: monitor resting HR, sleep and fatigue; reschedule the threshold session once two or more signals return towards baseline.
- Weeks 1–4: agree a simple readiness routine—morning resting HR, sleep hours and a one-line wellbeing score—and practical sleep strategies around work demands.
- Weeks 1–12: plan key sessions around predictable work peaks; place reduction weeks where deadlines cluster.
- Throughout: if symptoms such as fever, chest discomfort or unusual breathlessness appear, stop training and consult a doctor.

COACH'S CORNER

Offer the athlete a choice within a range—"today we run easy, or a shortened threshold if the warm-up feels good"—rather than a flat cancellation. Athletes who feel trusted to adjust report fatigue more honestly, and the data become more reliable.

What to retest

- Resting HR, sleep and subjective fatigue daily for the next week, compared with the athlete's own baseline range.
- Performance and effort in the rescheduled threshold session: pace, heart rate and RPE compared with recent equivalents.
- Weekly readiness trends over the 8–12 weeks to see whether work stress and training peaks are better aligned.

What would change our mind

If readiness signals return to baseline within 24–48 hours and the rescheduled session goes well, the episode was likely short-term and the programme can continue unchanged. If signals remain depressed for more than a week despite reduced load, the coach should review total training load, life stress and recovery more broadly, and encourage the athlete to seek medical assessment to rule out health issues. Sustained underperformance with persistent fatigue deserves careful attention; the consensus statement on overtraining syndrome by Meeusen and colleagues emphasises that such states require exclusion of other causes by qualified professionals. Conversely, if the athlete repeatedly performs well despite poor readiness scores, the readiness model may need recalibrating for this individual.

CHECKPOINT

When your athlete's fitness and readiness disagree, what is your agreed rule for modifying the session? How will you use the next 72 hours of data to decide what happens next?

PART XI AT A GLANCE

The key idea from each chapter, for quick revision.

- | | |
|-----------|---|
| 59 | Case Study A: High VO₂max, Weak Marathon
A high VO ₂ max describes the size of the engine, not how long it can run near full power. Marathon performance depends on sustaining a large fraction of capacity for hours, often in heat. |
| 60 | Case Study B: Moderate VO₂max, Excellent Economy
A single physiological metric cannot define the whole athlete. Moderate VO ₂ max combined with excellent economy and a high threshold can produce outstanding endurance performance. |
| 61 | Case Study C: Strong Heat Response
A good heat response is evidence of adaptation to heat, not proof of general performance improvement. Record the intervention, the test conditions and the response, then test again. |
| 62 | Case Study D: Altitude Response
Measuring the exposure, the physiological response and the performance separately turns a story into evidence. Even then, the result belongs to that athlete and that protocol. |
| 63 | Case Study E: Strong Fitness, Poor Readiness
Fitness is what the athlete has built; readiness is what the athlete can express today. Treating them as the same number leads to hard sessions on the wrong days. |

PART XII — DATA SCIENCE

An index is only as honest as the mathematics underneath it. Part XII moves from physiology to the data science that turns measurements into an AKURA Endurance Index™ score. It explains why weights must be learned rather than guessed, how variables measured in different units can be placed on a common scale, how missing data should be handled without inventing certainty, why every score needs a confidence layer, and how AEI must be validated before anyone relies on it. The theme throughout is restraint: a research system earns trust by showing what it does not know as clearly as what it does.

IN THIS PART

- 64 AEI Scoring: Do Not Guess the Weights
- 65 Normalisation
- 66 Missing Data
- 67 Confidence and Uncertainty
- 68 Validation Plan

PART XII • DATA SCIENCE

Honest Numbers: Weights, Gaps, Confidence

How measurements become an AEI score without inventing certainty: learned weights, fair scaling, honest gaps and validation.

14 Women's steeplechase record entries, 2010–2026: four women, over a minute faster



Don't guess the weights

A weight is a scientific claim about importance: estimate it from data, or label it a provisional assumption.



Normalise to the question

Compare with a population for capacity; compare with the athlete's own baseline for readiness.



MNAR

Missing is not normal

Athletes may skip entries on their worst days; turning blanks into averages manufactures false reassurance.



3 tiers

High (key data within ~8 weeks), Moderate (2–6 months old), Limited (older or critical data missing).



Test on unseen data

Out-of-sample testing is non-negotiable; machine learning overfits small samples without strict testing.



6 rungs

The validation ladder

Reliability, convergent and predictive validity, calibration, subgroups, then external validation in new cohorts.

THE PROCESS



A score without a confidence label invites over-interpretation: show what the system does not know.

THE STORY

What the Record Book Remembers

PART XII — DATA SCIENCE

Open the Indian record book at the women's 3000 m steeplechase and read down the column. It starts in May 2010 in Kochi, where Sudha Singh ran 10:09.56. It ends on 27 September 2026 in Nagoya, where Parul Chaudhary ran 9:08.67 to win Asian Games bronze. Between them are fourteen entries by four women, and more than a minute of improvement. It looks like a list of times. It is really a dataset, and like any dataset it contains a story, a few surprises and some things it cannot tell you.

Reading the curve

Plot those fourteen marks against time and the line is not smooth. In 2010, when the event was brand new for Indian women, the record fell three times in six months: Sudha's 10:09.56, O. P. Jaisha's 10:03.05 at Patiala in August, and Sudha's 9:55.67 when she won the first women's steeplechase ever held at the Asian Games, in Guangzhou. Early records in a young event fall fast. There is so much room to improve that almost any serious attention moves the mark. Sudha lowered it again, to 9:47.70, in 2012.

Then comes a steep, concentrated drop. From 2014 to 2016 Lalita Babar and Sudha Singh, training in the same national group under Nikolai Snegarev, traded the record back and forth: 9:35.37, 9:34.13, 9:27.86, 9:27.09, 9:26.55, and finally 9:19.76 in the Rio Olympic heat. Six records in about two years.

And then nothing. For seven years, from August 2016 to August 2023, the line is flat. Snegarev left the AFI in 2019. A generation of steeplechasers came and went. The record did not move until Parul Chaudhary ran 9:15.31 in the 2023 World Championships final in Budapest. After that the line drops again: 9:13.39 in Doha, 9:12.46 in Gumi in 2025, and 9:08.67 in Nagoya in 2026. A data scientist looking at that shape would ask questions before drawing conclusions. What changed in 2014? What stopped in 2016? What started in 2022? The numbers do not answer; they only show where to look.

Forty-eight years

Now turn to the men's marathon. In 1978 (on 28 May, according to record lists), at the National Inter-State Championship in Jalandhar, Shivnath Singh ran 2:12:00. He was reportedly an Army man, and is reported to have raced barefoot throughout his career. His record then stood for 48 years, the longest-standing mark in Indian athletics.

On 12 April 2026, at the Rotterdam Marathon, a 28-year-old soldier from Mandi district in Himachal Pradesh ran his first marathon. Sawan Barwal finished 20th in 2:11:58. After 48 years, the record had fallen by two seconds. Even that moment came with a data glitch: Olympics.com printed the time as "2:11.58s" and the margin as "0.42 seconds", a formatting error the real numbers easily correct. Five months later, at the Asian Games in Nagoya, Barwal ran 2:11:37 and won silver, India's first men's

marathon medal at the Games since 1982. "There was only one thing in my mind: I have to go till the finishing line and how I can maintain the pace," he told The Tribune.

What do 48 years and two seconds say? That the old mark was extraordinary for its time, certainly. But also that nearly five decades of Indian marathon running, with its national camps, Army programmes and three Olympic marathoners at Rio in 2016, did not produce a faster run. No Indian man has ever run under 2:10. In 2019 alone, according to World Athletics lists compiled by Toni Reavis, there were 293 sub-2:10 performances worldwide, 123 of them by Kenyans.

Compare the men's 10,000 m. Hari Chand's 28:48.72 from Montreal in 1976 lasted 32 years. Surendra Singh's 28:02.89 in Spain in 2008 then lasted another 16. Then, within twelve months, Gulveer Singh took it to 27:41.81, 27:14.88 and 27:00.22. Two long flat stretches followed by a sudden cliff: the same shape as the women's steeplechase, and a warning against reading any single year of data as a trend.

The frozen columns

Other columns have gone quiet. Preeja Sreedharan's women's 10,000 m record of 31:50.47 dates from the 2010 Asian Games, where Kavita Raut finished less than a second behind her. L. Suriya's half-marathon record of 1:10:31 dates from Delhi in 2017. O. P. Jaisha's 2:34:43 marathon, run at the 2015 World Championships in Beijing, still stands. At the 2026 national marathon championship in New Delhi the winning woman ran 2:41:15, and, as Olympics.com noted, no Indian woman reached the Asian Games qualifying mark of 2:31:52.

A frozen record is a signal too. It can mean a truly exceptional performance, or an event that lost its coaching, its funding or its athletes. The record book cannot tell you which.

What the numbers hide

Records hide where they were made. Almost every Indian distance record since 2022 was set abroad, at fast, well-paced meets in California, Oregon, Japan, Boston, Budapest or Paris. A time from The TEN in California and a time from a hot national championship are both written in minutes and seconds, but they are not the same kind of measurement.

Records hide definitions. On 15 March 2026 Gulveer Singh ran 59:42 at the New York City Half Marathon, the first Indian under an hour. It is not the official record: the course is point to point, with start and finish about 11.6 km apart, so it is not record-eligible. Avinash Sable's 1:00:30 from 2020 remains the official mark. Gulveer's 12:59.77 for 5000 m was run indoors in Boston and ratified as the record, while his best outdoor time is 13:03.93. Same athlete, same distance, two numbers, depending on which rules you read.

Records hide errors. Hari Chand's barefoot 10,000 m from the 1976 Olympics appears as 28:48.72 in most sources and 29:48.72 in one; the slower time could never have been a record for 32 years, so it is almost certainly a typo. When the research for this book was

checked on 29 September 2026, the Wikipedia list still showed the women's steeplechase record as 9:12.46, a mark that had been beaten two days earlier. Data go stale, and nobody raises an alarm.

Records hide the athletes who never got close. A national record is one person on one day. It says nothing about the depth of the field, the runners without jobs, or the talented teenager who stopped because the family needed an income.

And records hide integrity questions. In April 2026 the Athletics Integrity Unit placed India in Category A, its "extremely high" doping-risk group. In WADA's 2024 figures, India had the highest positivity rate among major testing nations, 3.6 per cent. Sustained violations can erase results after the fact, as when Sanjivani Jadhav's 2019 Asian Championships bronze was disqualified. A number in a table is only as trustworthy as the system that produced it.

From record book to research database

Every problem in this story is a data-science problem. Curves that need context before you interpret them. Measurements taken under different conditions. Definitions that change what counts. Errors and stale entries. Missing people. Outcomes that must be checked before anyone trusts them. The record book is a small, public, carefully kept dataset, and it still has all of them. An athlete-monitoring system collecting thousands of daily readings will have them on a far larger scale. Part XII is about handling them honestly: learning weights instead of guessing, putting different units on a common scale, storing what is missing, showing confidence, and validating before believing.

Sources: <https://www.olympics.com/en/news/athletics-list-of-indian-national-record-holders> ;
<http://www.indianexpress.com/news/Jaisha-betters-steeplechase-mark--Om-Prakash-bags-shot-put-gold/657766> ;
<https://web.archive.org/web/20121107223944/http://www.hindu.com/2010/08/09/stories/2010080952941500.htm> ;
<https://www.hindustantimes.com/other-sports/national-record-a-gift-for-india-s-support-steeplechaser-sudha-singh/story-zUIqWcXotQy21aqUfvaiXK.html> ; <http://indianexpress.com/sports/rio-2016-olympics/lalita-babar-india-3000m-steeplechase-result-final-medal-2973964/> ; <https://www.olympics.com/en/news/world-athletics-championships-2023-budapest-day-9-india-results-parul-chaudhary> ; <https://www.rediff.com/sports/report/asian-games-after-double-bronze-haul-parul-now-craves-family-time/20260928.htm> ; <https://www.olympics.com/en/news/sawan-barwal-breaks-marathon-national-record-india> ; <https://www.tribuneindia.com/news/sports/who-is-sawan-barwal-himachal-runner-who-challenged-marathon-myth-to-win-asian-games-silver/> ; <https://tonireavis.com/2021/12/21/sub-210-marathoning-2021/> ; <https://www.olympics.com/en/news/new-delhi-marathon-2026-results-winners> ;
<https://www.olympics.com/en/news/gulveer-singh-india-athletics-nyc-half-marathon-2026-bronze-national-record> ;
<https://www.olympics.com/en/news/hari-chand-indian-athlete-runner-olympian-dies> ;
<https://www.aljazeera.com/sports/2026/4/20/india-in-extremely-high-doping-risk-bracket-athletics-integrity-unit> ;
<https://www.outlookindia.com/sports/others/india-highest-doping-positivity-rate-wada-2024-testing-report> ;
<https://sportstar.thehindu.com/athletics/sanjivani-jadhav-long-distance-runner-handed-two-year-suspension-for-doping/article28590718.ece>

CHAPTER 64

64. AEI Scoring: Do Not Guess the Weights

PART XII — DATA SCIENCE

Somewhere in India tonight, someone is building a slide. On it is a gleaming circle labelled "Runner Score", and below it a formula: 20 per cent VO_2max , 15 per cent running economy, 10 per cent heart-rate variability, and so on until the numbers add up neatly to 100. Investors will nod. Athletes will download the app. Coaches will compare scores in WhatsApp groups and argue about who is fitter. Almost nobody will ask the one question that matters: where did those percentages come from? Were they measured, estimated from a dataset, tested against race results or injuries? Or did they simply look reasonable at eleven o'clock at night? A number with a decimal point looks like science. Whether it is science depends entirely on how it was made.

The most dangerous part of building an athlete score is not the physiology or the software. It is the moment someone assigns attractive numbers to unvalidated concepts. A formula that gives 20% to VO_2max , 15% to running economy and 10% to heart rate variability looks scientific on a slide. It has decimal places, it adds to 100 and it can be coded in an afternoon. Yet unless those percentages were derived from data, they are arbitrary opinions dressed as measurement. This chapter explains why AEI v0.1 begins life as a data schema and research prototype, and how its weights should eventually be learned or justified.

Why guessed weights are a problem

A weight is a claim about how much a variable matters for an outcome. If VO_2max carries twice the weight of economy, the index is asserting that a unit change in one is twice as important as a unit change in the other. That assertion may be roughly true for some athletes, distances and conditions, and badly wrong for others. A marathoner with a moderate VO_2max and excellent economy, as in Case Study B, would be systematically under-rated by a formula that over-weights aerobic ceiling.

Guessed weights also hide collinearity. VO_2max , lactate threshold and recent race performance are strongly related. Adding all three with separate weights can quietly count the same underlying fitness several times, inflating the apparent precision of the score. Finally, guessed weights cannot be falsified. If nobody tested them, nobody can say whether a different set would have predicted outcomes better.

KEY IDEA

A weight in an index is a scientific claim about importance. If it has not been estimated from data or justified from strong evidence, it should be labelled as a provisional assumption, not presented as a finding.

Start with a schema, not a formula

The honest first version of AEI is a structured way of collecting and storing variables: performance results, physiological tests, training history, environment, recovery signals and context. Each field has a defined unit, a measurement method, a timestamp and a quality flag. This schema allows data to accumulate consistently across athletes and seasons. Only once enough well-described data exist can weights be estimated with any confidence.

During the prototype phase AEI can still display components separately, such as an aerobic capacity profile, an economy profile and a durability profile, without collapsing them into a single number whose weighting is invented. Coaches often find a transparent profile more useful than a composite score anyway.

Candidate modelling methods

Several statistical families can learn weights from data. Each answers a slightly different question and carries its own risks. The choice should follow the research question and the size and structure of the dataset.

Method	Typical use in AEI	Strength	Main risk
Linear regression	Predict race time from test variables	Transparent, interpretable coefficients	Overfits with many correlated inputs
Regularised regression (ridge, lasso, elastic net)	Select and shrink many candidate predictors	Handles collinearity; reduces overfitting	Shrunk coefficients harder to explain; tuning needed
Mixed-effects models	Repeated measures within athletes over time	Separates between-athlete and within-athlete effects	Needs careful specification and adequate repeats
Survival analysis	Time to injury or illness	Uses censored data; models risk over time	Outcome definitions and competing risks complicate it
Machine learning (e.g. tree ensembles)	Non-linear patterns in large datasets	Captures interactions automatically	Opaque; overfits small samples without strict testing

Regression methods and regularisation, discussed in standard texts on prediction modelling, are usually the right starting point for a modest cohort because their behaviour is well understood. Mixed-effects models suit athlete monitoring

particularly well, because each athlete contributes many observations and individuals differ in their baselines. Survival analysis is the natural home for injury outcomes, where some athletes are followed without ever being injured and must still contribute information. Machine learning can find patterns that simpler models miss, but only with strict out-of-sample testing on data the model never saw during training.

COACH'S CORNER

If a platform shows you a single fitness score, ask two questions: where did the weights come from, and on what data were they tested? If the answer is "expert judgement" alone, treat the number as a conversation starter, not a verdict. Look at the underlying components before changing training.

Out-of-sample testing is non-negotiable

Any model will fit the data it was trained on. The real test is how it performs on new athletes, new seasons or new events. Cross-validation, temporal hold-out sets and, ideally, a completely separate external cohort are the tools that separate a genuine signal from a pattern the model memorised. In sports science, where samples are often small, this discipline matters even more. A model that explains 80% of variance in 40 athletes and 20% in the next 40 has taught us mainly about the first group.

Weights can also be justified without full estimation, for example by using published effect sizes or by fixing weights in advance and then testing whether the resulting index predicts outcomes. The critical requirement is that the justification is written down, the test is prospective and the result is reported whether or not it flatters the index.

CHECKPOINT

Which variables in your current athlete dashboards are combined into a single score? Can you say how each weight was chosen, and has anyone tested whether a different weighting would predict performance or injury better?

CHAPTER 65

65. Normalisation

PART XII — DATA SCIENCE

Take two 10 km runs on the same Sunday morning. One is in Chennai at sea level, at 30 °C with heavy humidity. The other is on the roads above Ooty at 2,240 m, in cool mist. Put the two times side by side and they tell you very little about which runner is fitter. Now add a resting heart rate in beats per minute, sleep in hours and a fatigue rating from 1 to 10, and try to

add them all into one number. The biggest numbers would win simply because they are big, and a heartbeat would outweigh a night's sleep for no reason except its units. Sports data are like a crowd speaking different languages at once. Before any of it can be combined, it has to be translated into a common one, and the choice of translation changes the meaning.

VO₂max is measured in millilitres per kilogram per minute, running economy in millilitres per kilogram per kilometre, HRV in milliseconds, sleep in hours and subjective fatigue on a rating scale. These variables use different units and ranges. Adding them directly would let whichever variable has the largest numbers dominate the result. A normalisation layer is therefore required before any variables are combined. The question is not whether to normalise but which method fits the research question.

The main options

Method	Formula in words	Best suited to	Caution
Z-score	(value minus group mean) divided by group standard deviation	Comparing athletes within a well-described cohort	Sensitive to outliers; assumes a meaningful reference group
Percentile rank	Percentage of the reference group scoring at or below the value	Communicating position to athletes and coaches	Loses information about distance between athletes
Robust scaling	(value minus median) divided by interquartile range	Skewed data or data with outliers	Less familiar; still needs a reference group
Athlete-baseline deviation	(today's value minus athlete's rolling baseline) divided by athlete's own typical variation	Readiness and day-to-day monitoring	Needs enough history to establish a stable baseline

Each method answers a different question. A z-score says how unusual an athlete is relative to a group. A percentile says where the athlete sits in a ranking. Robust scaling does the same as a z-score but resists distortion by a few extreme values, such as a sensor glitch that reports a resting heart rate of 20. Athlete-baseline deviation asks something entirely different: is this athlete unusual relative to themselves?

Population comparison versus self-comparison

For readiness, deviation from the athlete's own baseline is often more useful than comparison with a population. Resting heart rate and HRV vary widely between healthy individuals, so a value that is low for one athlete may be normal for another. Plews and colleagues, writing on HRV monitoring in endurance athletes in *Sports Medicine* in 2013, argued for rolling averages and individual reference ranges rather

than single-day values judged against group norms. The same logic applies to sleep duration, subjective wellness and morning body mass.

For performance, age- and sex-specific reference data may be relevant, provided the dataset supports it. A 5 km time can reasonably be expressed relative to athletes of the same sex and age band. But the reference group must be appropriate. Norms drawn from a different climate, altitude, competitive level or measurement protocol may mislead more than they inform.

KEY IDEA

Normalise to the question. Compare athletes with a relevant population when asking about capacity; compare the athlete with their own history when asking about readiness.

Practical details that matter

The reference period for a baseline must be defined. A seven-day rolling average responds quickly but is noisy; a 28-day or 60-day window is steadier but slower to reflect genuine change. Many monitoring systems show both. The typical variation used in the denominator should also be the athlete's own, often expressed as a standard deviation or coefficient of variation across the baseline window, so that a change is judged against normal day-to-day noise.

Direction matters as well. For some variables higher is better (VO₂max), for others lower is better (resting heart rate), and for some the relationship is U-shaped (body mass for a given athlete). The normalisation layer should record direction explicitly so that combined scores do not accidentally reward a harmful change.

Finally, normalisation should be versioned. If the reference cohort grows or the baseline window changes, historical scores will shift. Storing the method and reference set alongside each score keeps the record reproducible.

COACH'S CORNER

For daily decisions, look at how far today's values sit from the athlete's own recent normal, not how they compare with teammates. Keep at least three to four weeks of morning data before trusting a baseline, and be cautious during travel, illness or major training changes when the baseline itself is moving.

A worked example

Consider an athlete whose morning resting heart rate over the past 28 days has averaged 52 beats per minute with a typical day-to-day variation of about 2 beats. Today it reads 57. Against a population, 57 is unremarkable and a z-score would barely register. Against the athlete's own baseline, it is 2.5 typical variations above normal, a

change worth noticing, particularly if sleep was short or the athlete reports feeling unwell. Now consider the same athlete's 10 km time. Here the population comparison is useful: it tells the coach where the athlete stands relative to others of the same sex and age, which helps set realistic race goals. The two variables sit in the same system but are normalised differently because they answer different questions.

Mixing methods inside one composite score is possible, but it must be deliberate and documented. A composite built partly from percentiles and partly from baseline deviations will behave in ways that are hard to interpret unless each component's scale is shown separately alongside the total.

CHECKPOINT

For each variable you monitor, do you know whether it is being compared with a population or with the athlete's own history? Is that the right comparison for the decision you are making?

CHAPTER 66

66. Missing Data

PART XII — DATA SCIENCE

Shivnath Singh ran 2:12:00 at Jalandhar in 1978, and it stood as India's marathon record for 48 years. He was an Army man, reported to have raced barefoot for his whole career, and he reportedly finished 11th in the Olympic marathon at Montreal in 1976. Yet for a runner this important, the record is full of holes. His Montreal time appears as 2:16:22 in one source and 2:15:58 in another. His coach and training base could not be verified. Even Hari Chand's famous barefoot 10,000 m from the same Games appears as 28:48.72 in most sources and 29:48.72 in one, almost certainly a typo, because the slower time could never have stood as a record for 32 years. If the most celebrated runs in Indian distance history have gaps and errors, imagine what an ordinary runner's training log looks like.

Real athletes do not produce perfect datasets. Watches fail mid-run, sleep data disappear after a firmware update, blood tests happen only a few times a year and athletes forget to enter RPE after a hard session. Any system that works with real training data must decide what to do with the gaps. The wrong answer, and the most common one in consumer products, is to fill gaps silently with "normal" values. The AKURA principle is simple: missingness must be stored explicitly, and confidence should fall when critical variables are absent.

Why missing data are not neutral

A missing value is itself information. If an athlete stops logging wellness scores during a period of heavy fatigue or personal stress, the gap is related to the very thing the system is trying to detect. Replacing it with the athlete's average would make a difficult week look ordinary. Statisticians describe three broad mechanisms, formalised by Rubin and developed in Little and Rubin's *Statistical Analysis with Missing Data*.

Type	Meaning	Running example	Implication
MCAR (missing completely at random)	Missingness unrelated to any data, observed or not	Watch battery dies on a random day	Complete-case analysis unbiased but loses power
MAR (missing at random)	Missingness explained by other observed variables	HR strap not worn on easy days, which are logged as easy	Can be handled with models or imputation using observed data
MNAR (missing not at random)	Missingness depends on the unobserved value itself	Athlete skips wellness entry when feeling worst	Bias likely; needs sensitivity analysis and explicit flags

In practice these categories overlap and cannot always be distinguished from the data alone. The safe assumption for readiness data is that some missingness is informative.

KEY IDEA

Missing is not the same as normal. A system that converts blanks into averages manufactures reassurance exactly when it may be least justified.

How AKURA should handle gaps

First, every record should distinguish between a value of zero, a value that was not measured and a value that was measured but rejected for quality reasons. These are three different states and should never share a code.

Second, the index should know which variables are critical for each output. A readiness assessment without any recovery signal is fundamentally weaker than one missing only a secondary variable. When critical inputs are absent, the confidence tier drops (see Chapter 67) or the output is withheld.

Third, where statistical imputation is used for research, it should be principled. Multiple imputation, which creates several plausible completed datasets and combines the results, reflects uncertainty in a way that single-value filling does not. It works best under MAR assumptions and should be accompanied by sensitivity analyses that ask how conclusions would change if data were MNAR.

Fourth, the pattern of missingness should itself be monitored. A sudden drop in logging compliance can be an early behavioural signal worth a conversation, regardless of what the numbers would have said.

No data, no false precision

The useful rule is: no data, no false precision. If an athlete has not completed a threshold test for eight months, the system should not report their threshold pace to the nearest second as though it were measured last week. Older values can be carried forward, but they should be labelled with their age and should contribute less confidence than fresh data. Occasional measurements such as blood markers should be displayed with their collection date, not blended silently into a daily score.

COACH'S CORNER

Make data entry easy and explain why it matters, but never punish an athlete for gaps. When logging stops, ask how the athlete is rather than assuming nothing has changed. A two-minute conversation often recovers more information than an imputation model.

Designing for fewer gaps

Good system design reduces missingness before statistics are needed. Short, mobile-friendly wellness questions completed in under a minute are more likely to be answered than long forms. Automatic syncing reduces manual entry. Reminders timed to the athlete's routine, such as after the morning run rather than at a fixed clock time, improve compliance. Some data are naturally intermittent: a blood panel every three to six months or a laboratory test twice a season. These should be planned as scheduled measurements with defined validity windows rather than treated as irregular gaps.

It also helps to decide in advance the minimum dataset required for each output. For example, a readiness assessment might require at least one objective recovery signal and one subjective report from the past 24 hours. If that minimum is not met, the system says so plainly rather than producing a number. Writing these rules down before analysis prevents quiet, after-the-fact decisions about which gaps are acceptable.

CHECKPOINT

In your current monitoring, can you tell the difference between a missing value and a normal one? What happens to the athlete's score on a day when the watch was not worn?

CHAPTER 67

67. Confidence and Uncertainty

PART XII — DATA SCIENCE

On 15 March 2026 Gulveer Singh crossed the finish line of the New York City Half Marathon in 59:42. He was third, and the first Indian ever to run the distance in under an hour. It should have been a national record. It was not. The New York course runs point to point, with start and finish about 11.6 km apart, and under the rules such a course is not record-eligible. Olympics.com kept Avinash Sable's 1:00:30 from Delhi in 2020 as the official record, while Wikipedia listed Gulveer's run anyway. Same race, same clock, two answers. The time itself is not in doubt. What differs is how much confidence each source puts in what it means. Every number an athlete produces carries this hidden property, a degree of confidence, and most systems never show it.

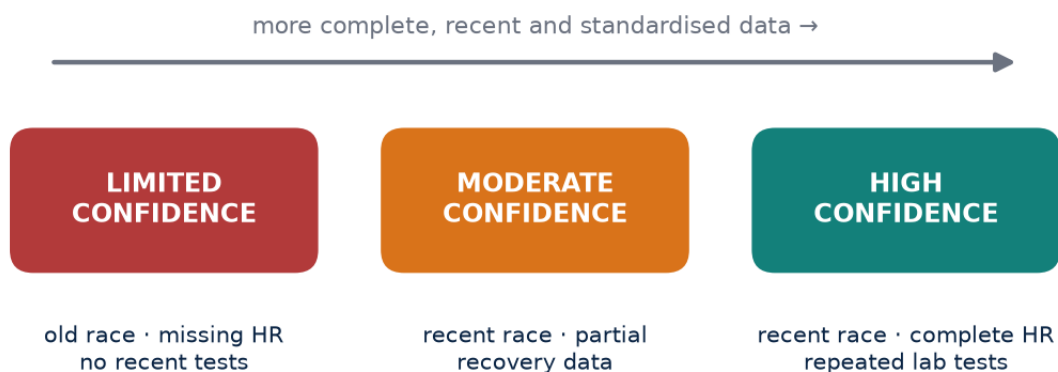


Figure 24. Every AEI output carries a confidence tier. Confidence falls when critical data are old or missing: no data, no false precision.

Every AEI output should carry a confidence layer. Two athletes can receive the same score for very different reasons. One score may rest on a race completed last month, complete heart rate data from dozens of sessions and repeated laboratory tests. Another may rest on a race from two years ago, a handful of watch-estimated values and no recovery information at all. Displaying both numbers in the same bold font, with no indication of how much evidence stands behind them, misleads the coach and the athlete. This chapter describes how AKURA should communicate uncertainty honestly.

Sources of uncertainty

Uncertainty in an athlete score comes from several places. Measurement error arises because every device and test has limited precision; a laboratory VO₂max test is more precise than a wrist-based estimate, but neither is exact. Biological variation means that even a perfect instrument would record different values on different days because the athlete genuinely changes with sleep, hydration, heat and time of day. Data age matters because fitness drifts, so a value measured months ago describes a slightly different athlete. Model uncertainty exists because the weights and structure of the index are themselves estimates (Chapter 64), and completeness matters because missing inputs reduce what can be known (Chapter 66).

Hopkins and colleagues, in their 2009 progressive statistics guidance in Medicine and Science in Sports and Exercise, encouraged sports scientists to report uncertainty with confidence limits and to interpret changes against the smallest worthwhile effect rather than relying on bare point estimates or p-values alone. The same attitude belongs in an athlete-facing interface.

Confidence tiers

The interface does not need to show statistical intervals to every user. It can show "high confidence", "moderate confidence" or "limited confidence" without pretending that the underlying physiology is known exactly. The criteria for each tier should be explicit and consistent.

Tier	Recency	Completeness	Measurement quality	Typical use
High	Key tests or races within about 8 weeks	All critical variables present; long monitoring history	Validated tests; clean sensor data	Planning training zones and race targets
Moderate	Key data 2–6 months old	Most critical variables present; some gaps	Mix of lab and field or device estimates	General guidance with periodic review
Limited	Older data or first weeks on the system	Critical variables missing	Mainly device estimates or self-report	Orientation only; prioritise retesting

The thresholds in this table are illustrative starting points for the prototype, not validated cut-offs. They should be refined as the dataset grows and as reliability studies show how quickly each variable loses relevance.

KEY IDEA

A score without a confidence label invites over-interpretation. The confidence layer tells the coach how hard to lean on the number.

Showing uncertainty without confusing people

Several design choices help. A score can be shown with a range rather than a single point, for example "AEI-Fitness 71 (likely range 67–75)". Changes can be classified as "likely real", "possibly real" or "within normal variation" relative to the athlete's typical error. The age of each input can be displayed next to it. And the system can recommend the single most useful next measurement to raise confidence, such as a time trial or a submaximal heart rate test.

The limited-confidence tier should never be hidden or treated as a failure. For a new athlete it is simply the honest starting point. As data accumulate, the tier should rise, and the athlete should see why.

COACH'S CORNER

Treat a limited-confidence score as a prompt to gather data, not as a basis for major decisions. When confidence is moderate, make conservative adjustments and re-check within a few weeks. Reserve precise pace targets for high-confidence profiles.

Research system versus motivational score

This confidence layer is one of the major differences between a research system and a motivational score. Motivational scores are designed to engage: they change often, reward activity and rarely admit ignorance. A research system is designed to be correct over time. It will sometimes say "we do not know yet", and that admission is part of its value. Athletes are generally more comfortable with honest uncertainty than product designers assume, particularly when the system also explains how to reduce it.

Confidence as a guide to action

The confidence tier should change what the system recommends, not only how the score looks. At high confidence, the system can suggest precise training paces and race targets. At moderate confidence, it can suggest ranges and recommend a check-in test within a defined period. At limited confidence, it should focus on data collection: a standardised submaximal run, a recent race or a few weeks of consistent morning monitoring. In this way, uncertainty becomes a practical plan rather than a disclaimer. Over a season, a coach should see confidence rise as the athlete's record fills in, and should notice quickly if it falls because of injury, device changes or lapses in logging.

CHECKPOINT

For the last score you used to change an athlete's training, how recent and complete were the underlying data? Would you have made the same decision if the score had been labelled "limited confidence"?

CHAPTER 68**68. Validation Plan**

PART XII — DATA SCIENCE

In 2022 the Athletics Federation of India sent four of its leading middle- and long-distance runners, Avinash Sable, Parul Chaudhary, Shankar Lal Swami and Ajay Kumar Saroj, to a 16-week camp in Colorado Springs under the American coach Scott Simmons. Same thin air, same coach, same camp. Afterwards, Sportskeeda reported, only Sable had clearly benefited. He set national records that year in Rabat and Birmingham. What should a federation conclude? That altitude works? That it does not? That it works for some athletes and not for others, or at some points in a career and not others? One camp, four athletes and no comparison group cannot settle the question, however exciting the headline result. Answering it needs a plan written before the data arrive, with outcomes defined in advance. That is what validation means.

An index is not validated because it looks sensible, because experienced coaches like it or because it correlates with race times. AEI should be validated prospectively, with a plan written before the data are collected. The plan should start with a cohort large enough to estimate reliability, then test whether the index predicts outcomes beyond simple race performance. This chapter sets out the types of evidence required and the hardest question AEI must answer.

The validation ladder

Validation is a sequence of questions, each building on the one before. If an index is not reliable, it cannot be valid; if it is not predictive, calibration hardly matters; if it works only in the development cohort, it is not yet ready for general use.

Validation type	Question asked	Example AEI test
Test–retest reliability	Does the score repeat under stable conditions?	Two assessments a week apart with no training change
Convergent validity	Does it agree with established measures of the same construct?	AEI-Fitness versus laboratory VO ₂ max and threshold
Predictive validity	Does it forecast future outcomes?	Baseline AEI predicts race time or injury over the next season
Calibration	Do predicted values match observed values across the range?	Predicted 10 km times versus actual times in each score band
Subgroup analysis	Does it perform similarly across groups?	Women and men, age bands, climates, training levels
External validation	Does it work in a new, independent cohort?	Different clubs, cities or countries using the same protocol

Reliability studies typically report the typical error of measurement and intraclass correlation, following approaches described by Hopkins in Sports Medicine in 2000 and by Atkinson and Nevill in 1998. Knowing the typical error allows every later change in AEI to be judged against measurement noise.

Calibration deserves special attention because it is often ignored. A model can rank athletes correctly while systematically over-predicting performance for everyone. Good discrimination with poor calibration can still mislead individual decisions, so both must be reported.

KEY IDEA

The easy question is whether AEI correlates with performance. The hard question is whether its extra layers improve decisions or outcomes beyond what coaches already use.

The incremental-value question

AEI almost certainly will correlate with running performance if performance is one of its inputs. That correlation proves little. The research question that justifies AKURA is incremental: do the physiological, environmental and readiness layers add predictive or practical value beyond a recent race time and a simple training log?

This can be tested in several ways. Nested models can compare a baseline model containing only race performance with an extended model adding AEI components, reporting the change in prediction error on held-out data. A stronger design compares outcomes in groups coached with and without AEI-guided decisions, ideally with randomisation where ethically and practically feasible. Outcomes should include not just performance but also injury, illness, training availability and athlete-reported experience.

COACH'S CORNER

Be willing to find out that part of the index does not help. If a component adds complexity without improving decisions, removing it is a scientific success, not a failure. Share validation results with your athletes; it builds trust.

Designing a credible study

A credible validation plan pre-registers its hypotheses, primary outcomes and analysis methods. It defines inclusion criteria, the minimum follow-up period and the handling of missing data. It reports results using established guidance for prediction models, such as the TRIPOD statement. It publishes negative as well as positive findings. And it separates the people who build the model from the data used to test it wherever possible.

For AKURA in India, a realistic path is a staged one: a reliability pilot with a few dozen athletes, a development cohort spanning multiple cities and climates, and then external validation with partner clubs and institutions. Each stage should be reported openly so that coaches can judge the strength of the evidence for themselves.

Common pitfalls

Several traps recur in the validation of athlete indices. Circularity occurs when the outcome being predicted is also an input, so the index appears to predict performance simply because it contains performance. Leakage occurs when information from the test period accidentally enters model development, for example when normalisation uses statistics from the whole dataset including future data. Small samples produce unstable estimates that shrink dramatically in new cohorts. Selective reporting, where only favourable outcomes are published, distorts the evidence base. And survivorship bias arises when athletes who drop out through injury or loss of motivation disappear from the analysis, leaving a healthier and more successful group that flatters the index. A written protocol, independent analysis and transparent reporting are the main defences.

CHECKPOINT

If AEI were removed from your coaching tomorrow, which decisions would change? What evidence would convince you that those decisions are better with it than without it?

PART XII AT A GLANCE

The key idea from each chapter, for quick revision.

- | | |
|-----------|---|
| 64 | AEI Scoring: Do Not Guess the Weights
A weight in an index is a scientific claim about importance. If it has not been estimated from data or justified from strong evidence, it should be labelled as a provisional assumption, not presented as a finding. |
| 65 | Normalisation
Normalise to the question. Compare athletes with a relevant population when asking about capacity; compare the athlete with their own history when asking about readiness. |
| 66 | Missing Data
Missing is not the same as normal. A system that converts blanks into averages manufactures reassurance exactly when it may be least justified. |
| 67 | Confidence and Uncertainty
A score without a confidence label invites over-interpretation. The confidence layer tells the coach how hard to lean on the number. |
| 68 | Validation Plan
The easy question is whether AEI correlates with performance. The hard question is whether its extra layers improve decisions or outcomes beyond what coaches already use. |

PART XIII — THE AKURA MODEL

Part XIII brings the book's physiology, measurement and data science together into a working model for coaching. It begins with the principle that the athlete, not the device, is the system being studied. It then separates two questions that are often confused: what an athlete is capable of, answered by AEI, and what an athlete should do today, answered by the AISRi readiness layer. The closed-loop coaching system shows how the two interact over time, the chapter on real-world data explains why integration is not validation, and the final chapter describes what an Indian athlete profile can and cannot mean.

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- 71 The Closed-Loop Coaching System
- 72 Garmin, Strava and Real-World Data
- 73 The Indian Athlete Profile

PART XIII • THE AKURA MODEL

Capacity and Readiness, Measured Separately

The athlete, not the device, is the system studied: AEI answers what they can do, AISRi what they should do today.

1991

Since then, Nashik's coach who stayed: "the Kenya of India"

 **n = 1**

The athlete is the laboratory

One athlete measured carefully over many cycles teaches more about them than any population average.

 **AEI**

Capacity: weeks to months

Built from races, tests and training history; guides zones, targets and season planning.

 **AISRi**

Readiness: day to day

Sleep, HR, HRV, wellness and recent load versus own baseline; guides session changes and risk.

 **10 steps**

The closed loop

From intake to AEI update, every decision and its reason must be recorded for the loop to learn.



Integration is not validation

Clean Garmin or Strava feeds can hide GPS drift, cadence lock and firmware shifts: apply quality rules.



Profile, not stereotype

Diversity within India exceeds any average: describe variation in climate, work, diet, health and access.

THE PROCESS



Measure capacity and readiness separately, then combine them in the decision, not in the score.

THE STORY

The Kenya of India

PART XIII — THE AKURA MODEL

Dawn in Nashik comes up over the Godavari, the vineyards and the low hills of the Sahyadri foothills. On the outskirts of the city is the campus of Bhonsala Military School, and for more than three decades the school has been linked to one of India's most productive distance-running groups. Nashik has no national altitude centre, no foreign coach and no sports-science laboratory. What it has had since 1991 is a coach who stayed.

The coach who stayed

Vijender Singh, a Sports Authority of India coach, has been based in Nashik since 1991, working with SAI and Bhonsala Military School. The athletes he has coached include Kavita Raut, Sanjivani Jadhav, Monika Athare, Anjana Thamke and Durga Deore. Scroll described his runners as coming from Nashik's surrounding villages, including tribal communities. Asked about them, he did not undersell the place.

Kavita Raut, Monica Atre [Monika Athare]... Sanjivani. I am telling you Nashik is the Kenya of India ... Remember, Kavita had to walk so many kilometres to reach school or fetch water.

— Vijender Singh, *The Indian Express*

He also credits altitude in part for his athletes' success, but Nashik is not a high-altitude training site in the sense that Ooty, at 2,240 m, is. The Nashik system is built from other things.

Kavita, Sanjivani, Monika

Kavita Raut was reportedly born into a tribal family in Sawarpada village, near Harsul in Nashik district. She was reportedly nicknamed the "Sawarpada Express". In 2009 she ran 1:12:50 for the half marathon, reported as a national record at the time. At the 2010 Commonwealth Games in Delhi she won 10,000 m bronze in 33:05.28, the first individual track medal by an Indian woman at the Games. A few weeks later, at the Asian Games in Guangzhou, she ran 31:51.44 for 10,000 m silver, less than a second behind Preeja Sreedharan's national record, and added 5000 m bronze. In those years she was also part of Nikolai Snegarev's national group. In 2016 she ran the Olympic marathon in Rio. She is reported to have been employed by a public-sector company, ONGC.

Sanjivani Jadhav comes from Vadalibhai, about an hour from Nashik. She is a teacher's daughter, and her grandfather, a wrestler, wanted her on the mat. Vijender Singh persuaded her parents to let her run. In 2017 she won 5000 m bronze at the Asian

Championships and 10,000 m silver at the Universiade. In 2019 the Athletics Integrity Unit gave her a two-year suspension for probenecid, a masking agent; she admitted the violation and attributed it to ayurvedic medicine, and her results from mid-2018, including a 2019 Asian Championships bronze, were disqualified. She came back to the sport. She won silver at the 2024 Asian Cross Country Championships, a third straight Indian women's title at the TCS World 10K in Bengaluru in April 2026, and the Indian elite women's race at the 2026 Mumbai Marathon.

Monika Athare, born in Nashik in 1992, ran the marathon at the 2017 World Championships in London. A cartilage injury later kept her out for a year. "I was very upset when doctors told me that I would never be able to run a marathon," she told *The Hindu*. She came back to win a half-marathon bronze in Mumbai.

The job that buys the training

Nashik is one node in a much larger national system, and most of that system is not about sport at all. It is about jobs.

The biggest employer is the Army. Most of India's elite male distance runners are soldiers: Avinash Sable of the Mahar Regiment, Gulveer Singh of The Grenadiers, Thonakal Gopi of the Artillery, Sawan Barwal, Man Singh and many more. Several joined mainly for a steady income or a sports-quota post, not a running career, and discovered serious running only after enlisting. Thonakal Gopi, the son of rice and ginger farmers in Wayanad, joined through the sports quota and became a Havildar in the Artillery; under coach Surendra Singh Bhandari of the Army Sports Institute he went on to run the Rio Olympic marathon and win the 2017 Asian Marathon Championship. The Army Sports Institute in Pune, set up on 1 July 2001 under the Army's "Mission Olympics" programme, runs a Boys Sports Company as a joint venture with SAI. The pull of the recruitment test reaches into villages. In Pauri Garhwal, a girl named Ankita Dhyani grew up running hill roads with young men training for Army physical tests. "I decided to run with them ... They never slowed down for me," she told ESPN. She went on to set a national record for the 2000 m steeplechase.

For women, the second great employer is Indian Railways. Sudha Singh worked for Central Railway in Mumbai, and it was a Railways officer who steered her into the steeplechase: "Madhu Shivdutt, my railway officer introduced me to Steeplechase," she told *Mumbai Live*. Lalita Babar, Preeja Sreedharan and, from 2015, Parul Chaudhary all drew Railways salaries. Police forces are a third route: Sunita Rani joined Punjab Police as a DSP in 2002, and Parul became a DSP in Uttar Pradesh in 2024.

The other side of the picture is Jyoti Gawate from Parbhani, a marathoner with a best of 2:45:48 who, according to a 2019 report, had no job. She took a Mumbai Police post in 2014 and quit after eight days to protect her training time; her Railways application went unanswered. She lived on prize money, trained in Parbhani without national-camp access, and was quoted as saying: "I am still jobless; what more can I do?"

Similar small, local groups exist elsewhere. At Kavinadu village near Pudukkottai in Tamil Nadu, S. Loganathan's Youth Sports Club trained G. Lakshmanan, who later won Asian Championships gold in both the 5000 m and 10,000 m, on tar roads and village streets, and is reported to have coached L. Suriya, who still holds the women's half-marathon record. Around them, government schemes add a layer of support: a Khelo India athlete receives ₹6.28 lakh a year for training, diet, kit and insurance, including a ₹10,000 monthly allowance. But the system is short of people. A ministry official told Hindustan Times in 2026 that India is "short of roughly 750 coaches".

What a model can learn from a system

Put the pieces together and India's grassroots distance system looks like this: a coach who stays for decades, athletes from villages where daily life already involves hard physical work, a group that trains together and pushes each other, and a government or Army job that turns running into a living. It has produced Commonwealth and Asian medals, national records and Olympians with very little laboratory science.

Its weaknesses are just as clear. Most of what it knows sits in coaches' heads and notebooks. When a coach retires or leaves, as Snegarev did in 2019, that knowledge leaves too. Almost none of it is recorded in a form the next coach can learn from. And a system that runs on jobs and camps works best for the athletes who get them; the ones who do not often disappear from the record.

The AKURA model does not aim to replace coaches like Vijender Singh or the job pathways that keep Indian runners running. It aims to learn from them, and to write down what they know: to record each athlete's context and each decision in a way that survives a transfer, a retirement or a new coach. The final chapters of this book describe that model, starting from the idea that has always driven Nashik: begin with the athlete in front of you.

Sources: <https://indianexpress.com/article/sports/sport-others/another-middle-distance-star-emerges-from-nashik-4739302/> ; <https://scroll.in/field/865878/how-a-former-sai-coach-is-making-nashik-the-hub-for-long-distance-running-in-india> ; <https://web.archive.org/web/20111002054457/http://hindu.com/2010/10/09/stories/2010100956762200.htm> ; <https://worldathletics.org/athletes/india/kavita-raut-14283028> ; <https://thebridge.in/athletics/india-athletics-coach-nikolai-snegarev-dead-19834> ; <https://sportstar.thehindu.com/athletics/sanjivani-jadhav-long-distance-runner-handed-two-year-suspension-for-doping/article28590718.ece> ; <https://www.deccanherald.com/sports/tcs-world-10k-triple-delight-for-sanjivani-harman-surprises-domestic-field-3981750> ; <https://www.olympics.com/en/news/mumbai-marathon-2026-race-results-times-winners-list> ; <https://www.thehindu.com/news/cities/mumbai/a-year-off-the-track-athlete-makes-comeback/article30603100.ece> ; <https://armysportsinstitute.com/> ; https://www.espn.in/athletics/story/_/id/30859627/no-turning-back-ankita-dhyani-mastery-improbable-chases-makes-unbeatable ; <https://www.mumbailive.com/en/sports/exclusive-interview-jakarta-asian-games-2018-steeplechase-silver-medalist-sudha-singh-29667> ; <https://shyamgopan.com/2019/02/04/jyoti-and-the-eight-minutes/> ; <https://www.thehindu.com/news/cities/Tiruchirapalli/athletes/article19251278.ece> ; <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2004017> ; <https://www.hindustantimes.com/sports/others/high-altitude-training-centre-to-come-up-in-shillong-101780588041378.html>

CHAPTER 69**69. The Athlete Is the Laboratory**

PART XIII — THE AKURA MODEL

Before Avinash Sable was a steeplechaser, he was a soldier. His Army postings, Olympics.com reported, took him from the cold of Siachen to the Rajasthan desert, where temperatures reached 50 °C, and on to Sikkim. He took up running seriously only in 2015, through Army cross country, and in 2017 Army coach Amrish Kumar moved him to the steeplechase. No sports scientist designed that sequence of exposures. Nobody measured what it did to his heart, his blood or his tendons. Yet by 2024 the son of farmers from Beed district held a national record of 8:09.91. Every athlete's body carries a history like this: the run to school, the climate, the job, the injuries, the coaches. Most of it is never written down. The AKURA model begins with a simple idea: start writing it down, one athlete at a time.

Wearables, blood tests and laboratory equipment are tools. The athlete remains the system being studied. This sounds obvious, yet much of modern endurance technology quietly reverses the relationship: the device becomes the authority and the athlete becomes a source of data for it. The AKURA model starts from the opposite position. Every measurement exists to understand one specific person better, and every interpretation must return to that person's experience, context and goals.

The athlete's body is the laboratory; data is the language.

— Kura

Measure, train, recover, retest, interpret

The best AKURA workflow is iterative. The coach measures the athlete's current state, prescribes training, allows recovery, retests under comparable conditions and interprets what changed. Then the cycle begins again. No single test or season provides the full picture; understanding builds through repetition.

Each step has its own discipline. Measurement must be standardised so that a later retest is comparable: same time of day, similar weather where possible, similar preceding training load, same protocol. Training must be recorded accurately enough that its effect can be linked to the result. Recovery must be long enough that the retest reflects adaptation rather than residual fatigue. Interpretation must consider measurement error before declaring an improvement or a decline.

KEY IDEA

One athlete measured carefully over many cycles can teach a coach more about that athlete than any population average. The workflow is a loop, not a line.

Individual response and the n-of-1 mindset

Group studies describe average responses. Individual athletes vary around those averages, sometimes widely. Hecksteden and colleagues, writing in the *Journal of Applied Physiology* in 2015, discussed how apparent individual differences in training response must be separated from measurement error and day-to-day variation before they are interpreted as true differences. For a coach, the lesson is twofold: individual responses are real and worth understanding, but a single before-and-after comparison is rarely enough to establish them.

Treating the athlete as a laboratory means adopting an n-of-1 mindset. Changes are introduced deliberately, preferably one at a time. Outcomes are measured repeatedly. Alternative explanations, such as a hotter test day, a new pair of shoes or a stressful work week, are recorded and considered. Over months, the coach builds a personal evidence base for that athlete: how they respond to heat, to volume increases, to altitude, to intensity blocks.

Know why each measurement exists

The coach should know why each measurement exists. Every test, device reading and questionnaire should answer a question that could change a decision. If nobody can say what action would follow from a particular result, the measurement may be adding burden without adding value.

A simple audit helps. For each variable collected, write down the question it answers, the decision it could influence and how often it genuinely needs to be measured. Many athletes are over-measured in some areas, such as daily GPS pace data, and under-measured in others, such as periodic standardised submaximal tests or honest subjective wellness reports.

COACH'S CORNER

Before adding any new test or device, write one sentence: "If this result is X, I will do Y." If you cannot complete the sentence, postpone the measurement. Schedule retests at fixed points in the training cycle so that comparisons are fair.

The athlete as a partner

The laboratory metaphor has a limit. Unlike a laboratory specimen, the athlete observes, interprets and chooses. Subjective reports of fatigue, mood and soreness have been shown in systematic review evidence to respond to training load at least as

sensitively as many objective markers. The athlete's own understanding of their body is part of the data. A good AKURA workflow explains results to athletes in plain language and invites their interpretation, so that data becomes a shared language rather than a verdict delivered from outside.

Standardising the laboratory

A laboratory is useful only when conditions are controlled. For the athlete-as-laboratory model, this means agreeing a small set of standard conditions for retests: the same course or treadmill, a similar time of day, a similar warm-up, comparable footwear, and a note of temperature and humidity. In India, where seasonal heat and humidity can change dramatically between months, recording environmental conditions is essential; a slower retest in May may reflect weather rather than lost fitness. Where perfect standardisation is impossible, the next best step is to record the differences so that they can be considered when interpreting the result.

CHECKPOINT

For one athlete you coach, list every measurement you currently collect. For each, what question does it answer and when did it last change a decision?

CHAPTER 70

70. AEI + AISRi

PART XIII — THE AKURA MODEL

In September 2026 Avinash Sable lined up for the steeplechase at the Asian Games in Nagoya. On paper he was one of the fastest men in Asia; his national record of 8:09.91, from Paris in 2024, had not been touched. But that paper was two years old. Since then he had suffered a knee injury at the Monaco Diamond League in July 2025, had surgery, and missed both the World Championships and the Commonwealth Games. His first steeplechase back, only weeks earlier, had taken 8:32.39. In Nagoya he ran 8:21.67, a season's best, and finished fourth. Was he the athlete of 8:09.91 or the athlete of 8:32.39? In a sense, both. One number described what his body had shown it could do. The other described what it could express that season. Mix up the two and every decision that follows goes wrong.

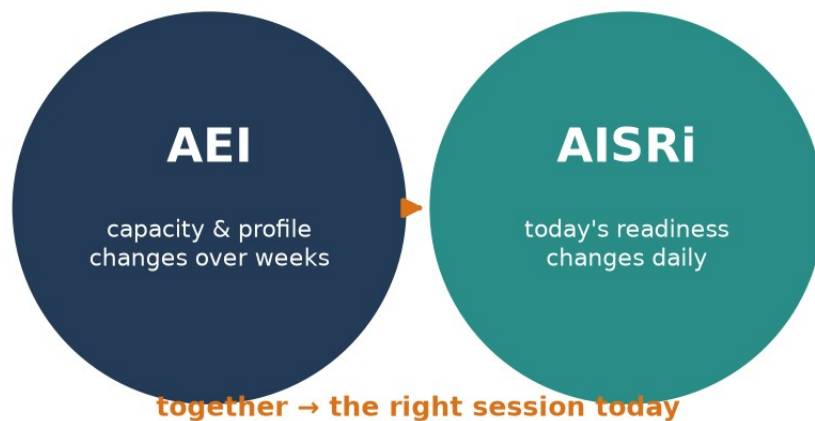


Figure 25. Two questions, two layers: AEI describes what the athlete can do; AISRi advises what the athlete should do today.

AEI and a daily readiness system should answer different questions. AEI asks: "What is the athlete's current capacity and profile?" AISRi asks: "Given today's recovery and risk signals, what training should be expressed today?" Keeping these questions separate is one of the central design decisions in the AKURA model. It prevents the common mistake of using one number for both fitness and readiness.

Two questions, two timescales

Capacity changes slowly. Aerobic fitness, economy and durability build over weeks and months and decline over similar timescales. Readiness changes quickly. Sleep, illness, travel, heat, emotional stress and the previous day's training can alter readiness within hours. A single score trying to represent both will either be too slow to catch a poor night's sleep or too jumpy to reflect genuine fitness.

AEI (capacity)	AISRi (readiness)
What can this athlete do?	What should this athlete do today?
Changes over weeks to months	Changes day to day
Built from races, tests, training history	Built from sleep, HR, HRV, wellness, recent load
Compared with population and own history	Compared mainly with the athlete's own baseline
Guides zones, targets and season planning	Guides session modification and risk management
Updated after retests and races	Updated every morning or before key sessions
Error: over-rating a detrained athlete	Error: ignoring a warning sign on a key day

Why one number fails

A strong athlete can be poorly recovered. A lower-fitness athlete can be very ready for today's easy run. If both situations are compressed into one number, the coach loses the information needed to respond correctly. Case Study E in Part XI illustrated the first pattern: high AEI-Fitness with elevated resting heart rate, poor sleep and high fatigue. The right response there was a modified session, not a reduction in the athlete's assessed capacity.

The reverse error is equally common. A beginner with high readiness should not be given a session designed for a far fitter athlete simply because they feel fresh. Readiness tells the coach how much of the athlete's capacity can safely be expressed today; it does not raise the ceiling.

KEY IDEA

Capacity is the size of the engine; readiness is how much of it can be used today. Measure them separately, then combine them in the decision, not in the score.

How the two layers interact

In practice the coach uses both. AEI sets the framework: training zones, weekly structure, realistic race targets. AISRi modulates the daily expression of that framework: proceed as planned, reduce volume, reduce intensity, swap sessions or rest. Over time, patterns in AISRi can inform AEI. Repeated poor readiness after certain types of session may indicate that the athlete's durability or recovery capacity is lower than their test results suggest, which in turn may justify revising the profile.

The two layers must also share a confidence logic. A readiness flag based on a single missing night of HRV data should carry less weight than one based on converging signals from sleep, resting heart rate and subjective wellness. Monitoring literature, including consensus work on training load by Bourdon and colleagues, emphasises combining internal and external load measures rather than relying on any single marker.

COACH'S CORNER

Plan the week from capacity; adjust each day from readiness. When readiness is low on a key session, move the session rather than abandoning it. When readiness is high but capacity is modest, enjoy the good day without inflating the workload.

A simple decision matrix

Combining the two layers in the decision rather than the score leads naturally to a four-way matrix. High capacity with high readiness supports the planned key session, performed as prescribed. High capacity with low readiness calls for a modified or

rescheduled session while leaving the long-term profile unchanged. Modest capacity with high readiness supports the planned session at the athlete's own level, perhaps with slightly more confidence to complete it fully. Modest capacity with low readiness suggests easy running, rest or recovery work, and a conversation about sleep, stress and health. The matrix is deliberately simple. Its purpose is to make the coach's reasoning explicit and recordable, so that the closed-loop system described in the next chapter can learn from it.

CHECKPOINT

Think of the last time you changed a session because an athlete seemed tired. Were you responding to a change in capacity or in readiness, and did your records reflect that distinction?

CHAPTER 71

71. The Closed-Loop Coaching System

PART XIII — THE AKURA MODEL

Before the 2023 World Championships in Budapest, Avinash Sable went back to altitude, training in Colorado Springs and St Moritz. His coach, Scott Simmons, told Hindustan Times: "His adaptation upon his return to altitude this year was almost instant." Look closely at that sentence. It contains a plan (go to altitude), an observation (how the athlete responded) and, hidden in the words "this year", a comparison with earlier years. Without a memory of how Sable had adapted before, "almost instant" would mean nothing. That is a feedback loop in miniature: act, observe, compare, adjust. Good coaches run loops like this all the time, mostly in their heads. The trouble is that heads forget, coaches change jobs and athletes move between camps. A loop that lives only in memory cannot be checked, shared or improved.

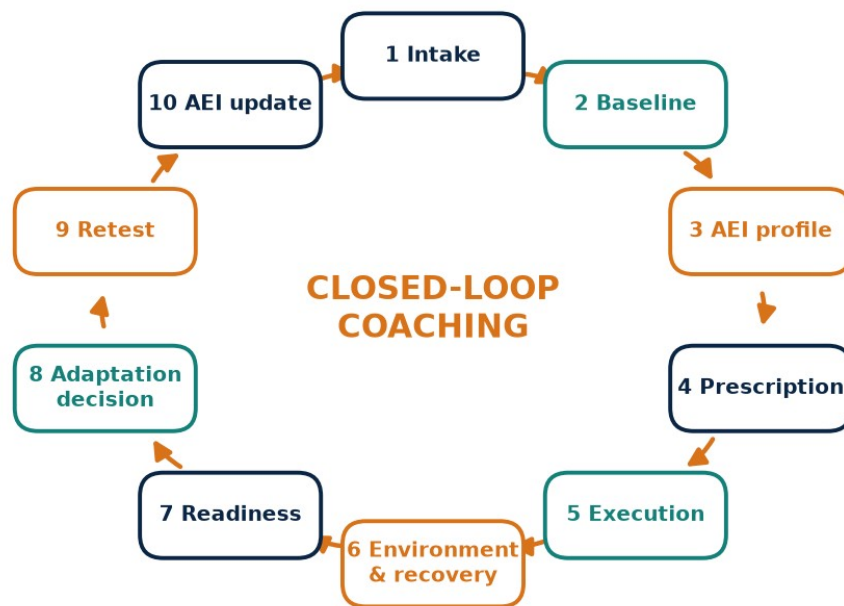


Figure 26. The ten-step AKURA closed loop: a digital coaching system that behaves like a longitudinal research study, not a static plan.

A training plan written in January and followed unchanged until the race is an open-loop system. It sends instructions but does not listen to what happens. Most real coaching is already partly closed-loop: coaches watch, adjust and learn. The AKURA model makes that loop explicit and structured, so that every stage produces data that informs the next. This is the foundation for a digital coaching platform that behaves more like a longitudinal research system than a static plan generator.

The ten steps

The complete AKURA loop has ten stages. Each one has a defined purpose, defined inputs and defined outputs.

1. Athlete intake: collect history, goals, injuries, medical screening, lifestyle, occupation, climate and access to facilities.
2. Baseline performance and physiology: record recent races and standardised field or laboratory tests to establish the starting point.
3. AEI profile: combine baseline data into a capacity profile with a stated confidence tier.
4. Training prescription: set zones, weekly structure and progression based on the profile and the athlete's goals.
5. Workout execution: the athlete completes sessions, with pace, heart rate and other data captured as available.
6. Environmental and recovery capture: record temperature, humidity, altitude, sleep, wellness and other recovery signals.

7. Readiness assessment: AISRi compares today's signals with the athlete's baseline to judge what training can be expressed.
8. Adaptation decision: the coach proceeds, modifies, moves or replaces sessions, recording the reason.
9. Retest: repeat standardised tests or races at planned intervals under comparable conditions.
10. AEI update: revise the profile and its confidence using new data, then return to prescription.

KEY IDEA

The loop is only closed if each decision and its reason are recorded. Without that record, the system cannot learn which adjustments worked.

Two loops running at different speeds

The ten steps contain two nested loops. The fast loop runs daily: workout execution, environmental and recovery capture, readiness assessment and adaptation decision (steps 5 to 8). The slow loop runs across weeks and months: prescription, retest and AEI update (steps 4, 9 and 10). Intake and baseline testing (steps 1 and 2) are repeated less often, typically at the start of a season, after injury or when goals change.

Separating the loops helps avoid two failure modes. The first is over-reacting: changing the entire programme because of a single poor morning reading. The second is under-reacting: ignoring several weeks of consistent warning signs because the plan says to continue. The fast loop handles day-to-day variation; the slow loop handles genuine changes in capacity.

Recording decisions, not just data

Most training platforms record what the athlete did. Fewer record what the coach intended and why the plan changed. For a closed-loop system to learn, step 8 must be documented: the planned session, the actual session, the readiness signals at the time and the reason for any change. Over months, these records allow the coach, and eventually the research team, to ask whether modifications based on readiness improved outcomes compared with sticking to the plan.

This record also protects the athlete. If a pattern of repeated modifications appears around certain sessions, the coach can identify a mismatch between prescription and capacity before it becomes an injury or a period of prolonged fatigue.

COACH'S CORNER

Add one field to your training log: "planned versus done, and why". Review it monthly. Patterns in that single field often reveal more about an athlete's needs than any new device.

Keeping the human in the loop

The closed loop is not a case for automation replacing coaches. Algorithms can flag signals, estimate confidence and suggest options, but the adaptation decision involves context that data rarely captures fully: exams, family responsibilities, motivation, a niggle the athlete mentioned in passing. The AKURA model treats software as a structured assistant to the coach and athlete, not as the decision-maker.

An example cycle

Imagine a club marathoner beginning a 16-week block. Intake records a desk job, a two-year training history and a previous calf strain. Baseline testing includes a 5 km time trial and a submaximal heart rate test on a flat loop. The AEI profile shows good aerobic capacity but limited durability, with moderate confidence. Prescription emphasises gradual long-run progression and strength work. Over the following weeks, daily readiness occasionally flags poor sleep during a busy work period, and two long runs are shortened. At week eight, a retest shows improved submaximal heart rate at the same pace. The AEI update raises durability slightly and confidence to high, and prescription adjusts for the second half of the block. Every step is recorded, so the next cycle begins from evidence rather than memory.

CHECKPOINT

Which of the ten steps are currently missing or informal in your coaching? Which one, if made systematic, would most improve your decisions?

CHAPTER 72

72. Garmin, Strava and Real-World Data

PART XIII — THE AKURA MODEL

Run along Mumbai's Marine Drive before dawn with a GPS watch, then look at the map afterwards. The line mostly hugs the promenade, but near the tall buildings at Nariman Point it may wander out over the sea, and a steady 5:00 per kilometre may show a sudden split of 3:40, as though the runner had briefly become an Olympian. Upload the file and it turns into a neat, shareable map with a number beside it. Upload thousands of such files to a research database and those small wanderings become a big problem. The same watch that gives a runner priceless daily records can also generate confident-looking errors: a heart rate locked to cadence, a hill that never existed, a treadmill run that was really 800 m short. Automatic data are wonderful. They are not automatically true.

A modern endurance system can combine structured workouts with GPS, heart rate, elevation, pace, cadence and training history. Consumer platforms such as Garmin Connect and Strava, along with many coaching platforms that connect to them, make it possible to collect this information automatically from thousands of sessions. For AKURA, integrations are valuable because they reduce manual entry and increase the amount of real-world data available. But integration is not validation. A clean API feed can still contain poor sensor data.

The promise and limits of automatic data

Automatic data collection solves one of the biggest problems in athlete monitoring: incomplete manual logs. Every run is captured with a timestamp, a route and a set of physiological signals. For an Indian research system hoping to follow many athletes across diverse climates and terrains, this is transformative.

The limits are equally real. Devices differ in sensor quality, sampling rate and processing algorithms. Firmware updates can change how a variable is calculated without notice. Estimated values, such as device-derived VO_2max or recovery time, are produced by proprietary models whose validity varies between individuals and conditions. And every sensor has characteristic failure modes that produce plausible-looking but wrong numbers.

KEY IDEA

A data feed that arrives cleanly formatted is not therefore accurate. Data quality rules must sit between the device and the index.

Data-quality rules

The table below lists common artefacts in wearable running data and how AKURA should handle them. The rules apply regardless of device brand; all manufacturers face similar physical limitations.

Artefact	What happens	Detection rule	Handling
GPS drift	Position errors inflate distance or create pace spikes, especially near tall buildings, tree cover or tracks	Implausible pace or acceleration jumps; track laps not matching 400 m	Smooth or flag segment; prefer footpod or known distance
Optical HR cadence lock	Wrist sensor locks onto step rate instead of heartbeat	HR suddenly matches cadence for a sustained period	Exclude segment from HR analysis; recommend chest strap
Missing HR	Sensor dropout or strap not worn	Gaps or zero values in HR stream	Store as missing, not zero; lower session confidence
Auto-pause artefacts	Device pauses at traffic lights or slow uphill sections	Moving time far below elapsed time; gaps in stream	Use moving and elapsed time appropriately; flag intervals
Elevation noise	Barometric or GPS elevation drifts with weather or signal	Implausible total climb for known flat route	Correct with terrain data or flag
Treadmill mis-calibration	Wrist-estimated distance differs from belt	Large mismatch with treadmill display	Use manual entry where available; flag pace data
Device or firmware change	Calculated metrics shift without training change	Step change in a metric at a known update date	Mark discontinuity; recalculate baseline

Heart rate data from wrist-based optical sensors deserve particular care during intervals, in cold conditions, with darker skin tones or tattoos and with loosely worn straps. These are recognised potential limitations of optical sensing, although the evidence on skin tone is mixed, and they are a reason to validate data quality rather than a reason to exclude any athlete group.

Building a quality layer

A practical quality layer does three things. It checks each incoming session against physical plausibility rules, such as maximum realistic pace and heart rate ranges for that athlete. It assigns a quality flag to each data stream rather than a single pass or fail for the whole session. And it feeds those flags into the confidence layer described in Chapter 67, so that a score built from flagged data is labelled accordingly.

Where possible, the quality layer should also learn the athlete. A heart rate of 205 may be implausible for one athlete and routine for another. Athlete-specific plausibility ranges, built from clean historical data, reduce false alarms without letting genuine errors through.

COACH'S CORNER

Use a chest strap for sessions where heart rate matters, such as threshold tests and intervals. Run key time trials on a measured course or track rather than trusting GPS distance. When a number surprises you, look at the raw graph before changing the plan.

Estimated metrics and privacy

Many devices report derived values such as estimated VO_2max , training status, recovery time or race predictions. These estimates can be helpful for engagement and trend-watching, but they are not direct measurements. AKURA should store them as device estimates, clearly separated from laboratory or field-test values, and should not use them as primary inputs to AEI unless their agreement with measured values has been checked for the athlete concerned.

Integrations also carry responsibilities. Athletes should know which data are imported, how they are stored, who can see them and how to withdraw consent. Location data in particular can reveal home addresses and routines, so research datasets should remove or coarsen GPS traces where they are not needed for the analysis.

CHECKPOINT

Look at the last three sessions of an athlete's data. Can you spot any of the artefacts in the table? How would each have affected a score if left uncorrected?

CHAPTER 73**73. The Indian Athlete Profile**

PART XIII — THE AKURA MODEL

Look at where some of India's best distance runners of the last decade actually came from. Avinash Sable ran 6 km to school in Beed district on the Deccan plateau. Thonakal Gopi grew up in Wayanad, about 1,000 m up in the Western Ghats, the son of rice and ginger farmers. Seema herded cattle and ran barefoot in Chamba, Himachal Pradesh. G. Lakshmanan learned on tar roads and village streets near Pudukkottai in Tamil Nadu. Parul Chaudhary ran on paths between sugarcane fields near Meerut. Ankita Dhyani's village in Pauri Garhwal sits at about 1,400 m. Plateau, hill and plain; dry heat, humidity and altitude in every combination. There is no single "Indian runner" in that list: no single body, climate or story.

Anyone building a model of Indian athletes has to start by admitting how varied they are.

Indian endurance athletes are not one physiological population. A distance runner training in the humid heat of Chennai, a trail athlete living at altitude in Ladakh, a masters marathoner in Bengaluru who trains before an office commute, and a young athlete in a sports hostel with limited access to laboratory testing all face different conditions and respond to training in their own ways. Climate, altitude, age, sex, nutrition, occupation, training history and access to facilities vary widely. Any system that treats "the Indian athlete" as a single type risks replacing one set of borrowed assumptions with another.

Why local longitudinal data matter

Much of the published endurance science, and many of the reference values built into consumer devices, come from populations studied in different climates, with different training cultures and different diets. Some of those findings transfer well; others may not. The honest answer is often that we do not yet know.

The value of an Indian research system is precisely that it can collect local longitudinal data instead of importing assumptions from another population. Repeated measurements of the same athletes across seasons, including pre-monsoon heat, monsoon humidity and cooler winter months, can show how environment and training interact in real Indian conditions. This is information no imported norm can provide.

KEY IDEA

Diversity within India is larger than any single average can capture. The goal is to describe variation carefully, not to create a new stereotype.

Profile variables

An AKURA athlete profile should record the context that shapes physiology and training, so that analyses can account for it rather than assume it away.

Domain	Variables	Why it matters
Demographics	Age, sex, body mass, stature	Reference ranges and training response vary
Environment	Home climate, typical training temperature and humidity, altitude of residence	Heat and altitude affect pace, HR and recovery
Training history	Years of training, weekly volume, event focus, coaching access	Explains current capacity and adaptation rate
Lifestyle and occupation	Work hours, commute, shift work, sleep pattern	Affects recovery and available training time
Nutrition	Dietary pattern (e.g. vegetarian), energy availability, iron and vitamin D status where tested	Influences adaptation, health and injury risk
Health	Injury history, menstrual health where relevant, illness, medications	Informs safe prescription and risk flags
Facilities and data access	Track, trails, gym, laboratory access, device type	Determines which tests are feasible and data quality

Recording these variables does not mean every athlete must provide all of them. Sensitive items should be optional, collected with informed consent and stored securely. The aim is to allow analyses to adjust for context and to ensure that conclusions are not driven by hidden differences in circumstances.

Avoiding stereotypes

Research on populations must never slide into assumptions about individuals. Observed differences between groups often reflect environment, opportunity, nutrition and training history far more than fixed biological traits. The AKURA dataset should be analysed with this in mind: variation within any group will usually be larger than differences between groups, and every individual athlete should be assessed on their own data.

This is also why subgroup analysis in validation (Chapter 68) matters. If AEI performs well for well-resourced urban runners but poorly for athletes from rural or less-equipped settings, the index must be improved before it is used to make decisions about those athletes.

COACH'S CORNER

Record where and when your athletes train, not just what they did. A slow tempo run on a hot, humid afternoon may reflect excellent effort. Compare each athlete with their own history in similar conditions before drawing conclusions.

An evidence base for India, open to the world

AKURA's long-term dataset should become an evidence base for Indian endurance coaching while remaining open to international comparison. Using common definitions, standard protocols and transparent reporting allows Indian findings to be compared with those from other countries, strengthening both. Over time, such a dataset could help Indian coaches move from borrowed averages to locally grounded, individually interpreted evidence.

CHECKPOINT

Which context variables do you currently record for your athletes? Which missing ones might explain differences in performance you have attributed to talent or effort?

PART XIII AT A GLANCE

The key idea from each chapter, for quick revision.

- | | |
|-----------|--|
| 69 | The Athlete Is the Laboratory
One athlete measured carefully over many cycles can teach a coach more about that athlete than any population average. The workflow is a loop, not a line. |
| 70 | AEI + AISRI
Capacity is the size of the engine; readiness is how much of it can be used today. Measure them separately, then combine them in the decision, not in the score. |
| 71 | The Closed-Loop Coaching System
The loop is only closed if each decision and its reason are recorded. Without that record, the system cannot learn which adjustments worked. |
| 72 | Garmin, Strava and Real-World Data
A data feed that arrives cleanly formatted is not therefore accurate. Data quality rules must sit between the device and the index. |
| 73 | The Indian Athlete Profile
Diversity within India is larger than any single average can capture. The goal is to describe variation carefully, not to create a new stereotype. |

PART XIV — RESEARCH PROTOCOLS

Ideas become knowledge only when they are tested in a way that another coach could repeat. This Part converts the questions raised throughout the book into four practical pilot protocols: a heat-acclimation study suited to Chennai's climate, an altitude study built around Ooty and Kodaikanal, a marathon durability protocol, and a crossover design comparing heat, altitude and control blocks in the same athletes. Each chapter is written as a mini study protocol with aims, hypotheses, participants, design, measurements, outcomes, safety rules, analysis and ethics. None of these designs is final. They are starting templates, meant to be refined with qualified scientists, clinicians and an institutional ethics committee before a single athlete is tested.

IN THIS PART

- 74 Heat Study Protocol
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- 77 Heat Versus Altitude Crossover

PART XIV • RESEARCH PROTOCOLS

Four Pilot Studies a Coach Can Repeat

Heat, altitude, durability and crossover protocols: starting templates to refine with scientists, clinicians and an ethics committee.

1 second

Gap between Gopi (2:15:25) and Kheta Ram in the Rio 2016 marathon

 **8–10**

Heat study (Chennai)

Baseline in week 2, 8–10 controlled heat sessions in weeks 3–4, post-test in week 6, optional decay check.

 **Wk 4–6**

Altitude study (Ooty)

Chennai baseline, iron review, camp at Ooty or Kodaikanal, post-tests and follow-up to week 12.

 **3 runs**

Marathon durability

Three standard long runs at fixed pace and fuelling, then compare long-run drift with race fade.

 **26 wk**

Heat vs altitude crossover

Heat, altitude and control blocks in randomised order, with washouts, in the same athletes.



Log the environment

If baseline and post-tests differ in heat and humidity, a change in HR may simply be the weather.

 **Ethics**

Before any athlete

Refine each design with qualified scientists, clinicians and an institutional ethics committee first.

THE PROCESS



The value is discovering who responds to heat, who to altitude, and who to neither.

THE STORY

Three Soldiers and One Second

PART XIV — RESEARCH PROTOCOLS

It is August 2016, and three men in Indian vests are waiting on an Olympic start line in Rio de Janeiro. All three are soldiers of the Indian Army. None of them is famous at home. Ahead of them lie 42.195 kilometres of city streets, and behind them lie three of the most different childhoods Indian distance running could produce: a hill farm in Kerala, the sand country of western Rajasthan, and the mountain roads of Uttarakhand.

The first is Thonakal Gopi. He comes from Sulthan Bathery in Wayanad, a town that sits about 1,000 metres above sea level among forest, spice gardens and paddy. His parents were farmers who grew rice and ginger. He joined the Army through the sports quota and rose to Havildar in the Artillery regiment, and the Army lists him at the Army Sports Institute in Pune. He did not enlist to become an Olympian. Like many Indian distance runners, he found that the Army offered something rare: a steady job that allowed him to run.

The second is Kheta Ram. He is reported to have grown up in Barmer district, in the desert of western Rajasthan, and to have run four kilometres to school. He serves in the Jat Regiment. Think of a Barmer summer afternoon, the horizon shimmering, the sand too hot to stand on barefoot, and then think of a boy running to class across it every morning.

The third is Nitendra Singh Rawat, reported to come from Garur in Bageshwar district, Uttarakhand, where roads are cut into hillsides and every run is either up or down. In January 2016, at the Mumbai Marathon, Rawat ran 2:15:48. That time broke Ram Singh Yadav's Indian course record of 2:16:59 and booked his place in Rio. According to the reporting, all three of the Rio marathoners qualified at that same Mumbai race.

The same mountain

What makes the three men interesting to a scientist is not only how different they were. It is how much they had in common. They served in the same Army. They trained under the same coach, Surendra Singh Bhandari of the Army Sports Institute (his first name appears as Surinder in some reports). And before Rio they shared the same mountain.

ESPN described Gopi's preparation as a "ten-month training stint at a high-altitude Ooty facility". Ooty sits at 2,240 metres in the Nilgiris, high enough for mist to roll across the tea gardens in the early morning and for a sea-level runner to feel the air go thin on the first hill out of town. Bhandari's altitude block there covered all three marathoners. Rawat had already credited the Ooty camp directly after Mumbai.

I was personally confident after the training we had been put through in the national camp at Ooty by our coach (Surinder Singh) Bhandari.

— Nitendra Singh Rawat (Sportskeeda/IANS, 2016)

Picture that camp as any coach who has taken a group to the hills will recognise it: cold mornings, steep roads, runners in tracksuits steaming in the mist, long days of eating and sleeping and running again. Three men, one coach, one altitude, one goal.

The race

Then Rio. Gopi finished 25th in 2:15:25, a personal best. Kheta Ram finished 26th in 2:15:26, also a personal best, one second behind his teammate. Read that again. After more than two hours of running, after months at altitude, two men from a Kerala hill farm and a Rajasthan desert crossed the line one second apart.

Rawat's day went differently. He finished 84th in 2:22:52. The reporting afterwards explained why: he had run with a hamstring injury. Seven months after setting an Indian course record in Mumbai, he was seven minutes slower on the biggest stage of his life.

To place these times in history: in 2016 the Indian men's marathon record still belonged to Shivnath Singh, who had run 2:12:00 at Jalandhar in 1978. It would stand for about 48 years, until Sawan Barwal ran 2:11:58 in Rotterdam in April 2026. The Rio trio were chasing ghosts from before they were born.

What happened next

The three careers kept moving. Gopi won the 2017 Asian Marathon Championship in Dongguan in 2:15:48, the first Indian to take that title. In January 2018 he won the Indian elite race at the Mumbai Marathon in 2:16:51. Second, in 2:16:54, was Nitendra Singh Rawat, who had recovered after Rio and trained with the national camp in Bengaluru. In 2019 Rawat ran 2:15:52 in Mumbai, four seconds outside his own course record, and in 2022 he won the national marathon title in 2:16:04. Gopi kept going longer still: at the Valencia Marathon in December 2025 he ran 2:12:23, a personal best just 23 seconds outside the old national record, after four more months of training at Ooty.

The experiment nobody wrote down

Now look at Rio the way a research scientist would. Three trained athletes, all from the same organisation. One coach. One altitude camp of the same length, as far as the reports show. One qualifying race. One Olympic course, on one day, in one set of conditions. That is about as close to a natural experiment as elite sport ever gets. Two athletes responded almost identically. The third was compromised by an injury.

What did the camp actually do to them? Did their haemoglobin mass rise, and by how much? At what altitude did they sleep, and at what altitude did they run their hard sessions? How many kilometres a week did each man run in the final ten weeks? Did heart rate at a fixed pace fall during the camp, and did it fall equally for all three? When did Rawat's hamstring first complain, and what was his training load in the fortnight before? How long after coming down from Ooty did each man race best?

The public record answers none of these questions. The data may exist in a coach's notebook or an Army file. But a search for this book found no peer-reviewed study of Indian athletes' blood responses at Ooty at all. Generations of India's best runners have gone up the ghat road, and almost nothing about what their bodies did up there has been published in a form another coach could read, check and use.

This is not a criticism of the coach or the athletes. They did their jobs superbly: two personal bests at an Olympic Games is a real achievement. It is a description of a system in which experience accumulates in people rather than in records. When a coach retires, the knowledge often retires with him.

What a protocol would have caught

A simple research protocol, built with an ethics committee and a doctor, would not have changed the race. It would have changed what came after. It would have recorded a baseline before the camp: blood measures, a submaximal run at a fixed pace, a time trial. It would have logged the dose: sleeping altitude, training altitude, days at camp. It would have tracked load, sleep and resting heart rate every day. It would have logged every niggle, including the first one in a hamstring. And it would have fixed the retest dates in advance, so that nobody could pick the best result after the fact.

With three athletes you cannot prove anything about altitude. You can describe how each individual responded, and that description would have been worth a great deal. When Gopi returned to Ooty for four months before Valencia in 2025, a coach could have compared his response nine years later with his response before Rio, instead of relying on memory. Afterwards Gopi said simply: "I think the high altitude at Ooty helped my performance." He may be right. A protocol would tell us how, how much, and whether the same is true for the next soldier who goes up the mountain.

The four chapters that follow are written so that the next camp, the next heat block and the next marathon produce knowledge as well as results. We begin where every Indian runner begins, whether they choose to or not: with the heat.

Sources: https://www.espn.com/athletics/story/_id/17475379/india-marathon-runner-olympics-thonackal-gopi-talks-espn-exclusive-interview ; <https://www.firstpost.com/sports/rio-olympics-2016-thanackal-gopi-and-kheta-ram-clock-personal-best-to-finish-25th-26th-in-marathon-2969066.html> ; <https://timesofindia.indiatimes.com/sports/rio-2016-olympics/india-in-olympics-2016/athletics/rio-olympics-indias-thanackal-gopi-kheta-ram-clock-personal-best-in-marathon/articleshow/53800667.cms> ; <https://sportskeeda.com/running/nitendra-rawat-credits-coach-success-mumbai-marathon> ; <https://scroll.in/field/865807/mumbai-marathon-on-the-road-to-redemption-nitendra-singh-rawat-puts-up-a-stellar-performance> ; https://www.espn.com/espn/story/_id/25815512/i-ensure-the-best-marathon-runner-india-says-resurgent-nitendra-rawat-mumbai-win ; <https://www.olympics.com/en/news/asian-marathon-championships-2024->

india-results ; <https://shyamgopan.com/2025/12/11/gopi-23-secs-short-of-national-record-sets-new-pb-at-valencia-marathon/> ; <https://www.olympics.com/en/news/sawan-barwal-breaks-marathon-national-record-india> ; <https://www.thehindu.com/news/national/tamil-nadu/udhagamandalam-boasts-south-indias-only-high-altitude-sports-training-centre/article68333440.ece> ; https://en.wikipedia.org/wiki/Kheta_Ram ; https://en.wikipedia.org/wiki/Nitendra_Singh_Rawat

CHAPTER 74

74. Heat Study Protocol

PART XIV — RESEARCH PROTOCOLS

It is just after five in the morning on Marina Beach, and the air is already heavy and damp. A club runner jogs past the fishing boats pulled up on the sand, her shirt dark with sweat before the first kilometre is done. By the time the sun clears the Bay of Bengal, the heart rate on her watch has drifted well above where it sat in December, at exactly the same pace. Her coach shrugs: Chennai makes you tough, everybody knows that. Perhaps it does. But nobody in the group can say by how much, how quickly, or whether the runner who trains at 5 a.m. adapts differently from the one who trains after work in an air-conditioned gym. The research gathered for this book found no peer-reviewed heat-stress study of an Indian marathon at all. The heat in Chennai is free and arrives every morning. The knowledge about what it does to Indian runners does not.

Chennai offers something many high-performance centres pay to recreate: sustained heat and humidity for most of the year. That natural environment is both an opportunity and a risk. If AKURA wants to know whether a structured heat-acclimation block changes an athlete's physiology and performance, it must stop relying on the general impression that "training in Chennai already makes us heat-adapted" and instead run a controlled, documented, repeatable study. This chapter outlines a pilot design. It is deliberately modest: the goal is to generate clean data from a small group, learn what is practical, and prepare for a larger study.

Aim and hypotheses

The aim is to quantify the physiological and performance response of trained endurance runners to a structured heat-acclimation block, and to describe how much that response varies between individuals. Three working hypotheses guide the design. First, heart rate at a fixed submaximal pace in standardised warm conditions will fall after the block. Second, thermal comfort and perceived exertion at that pace will improve. Third, the size of the response will differ meaningfully between athletes, and part of that difference may be explained by prior heat exposure. The study does not assume that every athlete will improve, and a null or mixed result is an acceptable outcome.

Participants, inclusion and exclusion

Recruit a small pilot group (for example 12 to 20 runners) with stable training for at least several weeks, no current injury, and a recent race or time-trial result that anchors their performance level. Inclusion: adult athletes, regular training of a defined minimum weekly volume, willingness to complete all visits, and medical clearance. Exclusion: history of exertional heat illness without specialist clearance, known cardiovascular, renal or metabolic conditions, current febrile or gastrointestinal illness, medications that impair thermoregulation, and pregnancy. Record sex, age, training history, home climate and typical training time of day, because a runner who trains at 5 a.m. on the Marina and one who trains in an air-conditioned gym arrive with different heat histories.

Design and timeline

Week	Phase	Main activity
1	Familiarisation	Consent, medical screening, practise test protocol
2	Baseline	Standardised submaximal run and time trial in logged conditions
3–4	Heat block	Controlled heat sessions (e.g. 8–10 sessions) with normal training
5	Short recovery	Reduced load, no extra heat sessions
6	Post-test	Repeat baseline tests under matched conditions
8–9	Decay check (optional)	Repeat submaximal test to observe retention

A control or crossover condition strengthens causal inference. In the simplest version, a matched control group continues normal training without the extra heat sessions. In a stronger version, each athlete completes both a heat block and a control block in randomised order with a washout between them. Heat sessions can be outdoor runs in logged conditions, indoor sessions in a heated room, or post-exercise passive heat exposure, but the method must be fixed in advance and applied consistently.

Measurements and outcomes

Before every test record temperature, relative humidity and, ideally, wet-bulb globe temperature (WBGT), along with time of day, wind, surface and footwear. Primary outcomes can include heart rate at a fixed pace, core-temperature response where it can be measured safely and with appropriate equipment, body-mass change as a sweat-loss estimate, thermal comfort, and performance in a standardised time trial. Secondary outcomes can include plasma-volume estimates from haemoglobin and haematocrit, sweat rate, sleep and resting heart rate, and blood markers where appropriate and clinically supervised.

KEY IDEA

A heat study is only as good as its environmental log. If baseline and post-tests are run in different heat and humidity, the change in heart rate may simply reflect the weather, not the athlete.

Safety and stop rules

Each heat session requires a supervisor trained in recognising heat illness, access to shade, cooling methods and fluids, and a clear emergency plan with the nearest hospital identified. Stop a session immediately if the athlete shows confusion, unusual behaviour, loss of coordination, has hot skin (dry or still sweating), collapses, vomits, or reports a severe headache or dizziness; begin cooling and seek emergency medical care. Predefine upper limits for environmental conditions and, where core temperature is monitored, a termination threshold set by the medical lead. Athletes who are unwell, underslept or dehydrated on arrival do not complete the session that day.

Analysis and ethics

Report individual responses alongside group means, with confidence intervals, and compare each change against the test's typical error and a predefined smallest worthwhile change. Adjust for environmental differences where possible and state clearly what could not be controlled. The protocol must receive approval from an institutional ethics committee before recruitment, every participant must give written informed consent after a plain-language explanation of risks, and a qualified physician must provide medical oversight throughout. Participants may withdraw at any time without consequence.

Practical notes for Chennai

In Chennai the difficulty is often not creating heat but controlling it. Outdoor conditions can change sharply between a 6 a.m. and an 8 a.m. start, and sea breeze, cloud cover and pre-monsoon humidity can make two apparently similar mornings very different. For that reason the baseline and post-tests are best run indoors on a treadmill in a room whose temperature and humidity can be held steady, or outdoors only when WBGT falls within a pre-agreed band. Keep the same treadmill, footwear, time of day and pre-test meal. Ask athletes to avoid hard training for 48 hours before each test and to record their sleep for the preceding two nights. Small details like these decide whether the final data can be trusted.

COACH'S CORNER

Even without a formal study, you can borrow this structure. Test a fixed-pace run in logged conditions, add a planned heat block, then retest in similar conditions. Keep the notes; a season of carefully recorded mini-experiments is worth more than a lifetime of impressions.

CHECKPOINT

Can you name the exact conditions of your athlete's last baseline test? If you repeated it today, how would you know whether a change came from adaptation or from the weather?

CHAPTER 75**75. Altitude Study Protocol**

PART XIV — RESEARCH PROTOCOLS

Morning mist hangs over the Nilgiris, and the road out of Ooty climbs through tea gardens into cold, thin air. Indian distance runners have been coming here for generations. Before the Rio Olympics, Army marathoner T. Gopi spent what ESPN described as a ten-month training stint at a high-altitude Ooty facility. Before the 2025 Valencia Marathon he returned for four months, and afterwards said, "I think the high altitude at Ooty helped my performance." Sawan Barwal's Project 2:09 group normally spends about four winter months at Ooty and Wellington; he broke the national marathon record in 2026. In February 2024 Tamil Nadu opened a dedicated high-altitude training centre at Brecks Ground. The mountain is busy. Yet a search for peer-reviewed studies of Indian athletes' blood responses at Ooty comes back empty. Thousands of runners have climbed the ghat road, and almost nothing about what their bodies did up there has been written down in a form another coach could check and use.

The Nilgiris and the Palani hills give South Indian athletes access to moderate altitude within a day's travel of Chennai. Ooty sits at about 2,240 metres and Kodaikanal at a little over 2,100 metres, which places both in the range often used for altitude training camps. Many Indian coaches already send athletes there. What is usually missing is measurement: a clear record of dose, a baseline, a controlled training load and a proper retest. This protocol treats an altitude camp as an experiment rather than a pilgrimage.

Aim and hypotheses

The primary question is the size and timing of each athlete's response, not whether altitude "works". The study should not begin with the assumption that performance must improve. Working hypotheses: haemoglobin mass, where it can be measured with a validated method, will show a small average increase after sufficient exposure; individual responses will vary widely; and changes in blood measures will not map perfectly onto changes in race performance. If haemoglobin mass changes but race performance does not, that is still a scientifically meaningful result.

Participants, inclusion and exclusion

Recruit trained endurance athletes who can commit to the full camp and follow-up. Inclusion: stable training, medical clearance, and baseline iron status reviewed by a clinician. Exclusion: iron deficiency that has not been treated, anaemia, known cardiopulmonary disease, previous severe altitude illness, sickle-cell trait or disease unless cleared by a specialist, recent illness, and pregnancy. Record prior altitude exposure, home altitude and any iron supplementation already in use.

Design and timeline

Week	Phase	Main activity
1–2	Baseline (Chennai)	Blood measures, ferritin, performance test, submaximal test
3	Pre-camp	Iron review with clinician; set sleeping and training altitude
4–6	Camp (Ooty or Kodaikanal)	Controlled training load; daily sleep, SpO ₂ , RHR, wellness
7	Return	Monitor first-week responses at sea level
8–9	Post-test	Repeat blood measures and performance tests
10–12	Follow-up	Observe timing of best performance and decay

Define sleeping and training altitude precisely, because "we went to Ooty" can mean very different exposures depending on where the athlete slept and where the sessions took place. Log hours per day at altitude. Control training load so that any change is not simply a result of a heavier or lighter training block. A sea-level control group training in Chennai under a similar programme, where feasible, makes interpretation much stronger.

Measurements and outcomes

Primary outcome: change in haemoglobin mass by an appropriate specialist method, or, where that is unavailable, carefully standardised haemoglobin concentration and haematocrit interpreted with caution because concentration is affected by plasma volume. Secondary outcomes: time-trial or race performance, heart rate at fixed pace, SpO₂ trend during the camp, sleep quality, resting heart rate and heart-rate variability

trend, body mass, ferritin, and athlete-reported wellness. Every blood draw should follow a standard posture, time of day and hydration protocol.

KEY IDEA

Haemoglobin concentration and haemoglobin mass are not the same thing. A concentration can rise simply because plasma volume fell. Only a total-mass method tells you whether the athlete actually built more red-cell capacity.

Safety and stop rules

Altitude exposure should be medically supervised when indicated. Brief all athletes on warning signs of altitude illness: worsening headache, nausea or vomiting, unusual breathlessness at rest, a persistent cough, confusion, loss of coordination or extreme fatigue. Any of these requires the athlete to stop training, be assessed by a clinician, and descend if symptoms are serious or worsening. Reduce training intensity during the first days of the camp. Monitor iron status and sleep, and stop the study for any athlete whose symptoms or blood results raise clinical concern.

Analysis and ethics

Report individual response trajectories as well as means, with confidence intervals, and compare changes against the measurement's typical error. Pre-register the timing of retests so that the "best" post-camp result is not selected after the fact. The protocol requires institutional ethics committee approval, written informed consent, and a named physician responsible for blood testing, iron decisions and altitude illness. Blood abnormalities must be interpreted by qualified clinicians, not by the coach or the algorithm.

Practical notes for Ooty and Kodaikanal

Hill-station camps bring their own complications. Road travel from Chennai is long, and the first days are often disturbed by travel fatigue, cold nights and changed food. Terrain in both towns is hilly, so paces are not directly comparable with flat roads in the plains; use heart rate and RPE to guide intensity and record elevation gain for every session. Tourist seasons can affect accommodation, noise and sleep. Plan retesting at sea level in Chennai under the same conditions as baseline, and schedule it in advance rather than choosing the day the athlete "feels good". Finally, remember that nutrition, particularly energy and iron intake, must be adequate at altitude; an athlete who under-eats at camp may show no haematological response for reasons that have nothing to do with altitude itself.

COACH'S CORNER

Before the next camp to Ooty or Kodaikanal, write down five things: sleeping altitude, training altitude, days at camp, planned training load and the exact retest dates. If you cannot write them down, you are not yet running an experiment.

CHECKPOINT

What result would convince you that a particular athlete does not benefit much from altitude? Are you prepared to accept that result if the data show it?

CHAPTER 76**76. Marathon Durability Protocol**

PART XIV — RESEARCH PROTOCOLS

Mumbai, January 2018. After 42.195 kilometres of city roads, T. Gopi wins the Indian elite men's race in 2:16:51. Nitendra Singh Rawat, who had set the Indian course record two years earlier, is second in 2:16:54. After more than two hours of running, the two soldiers are separated by three seconds on the clock. Eighteen months before that, at the Rio Olympics, Gopi and his Army teammate Kheta Ram had finished 25th and 26th, one second apart. Somewhere in the closing stages of each race, bodies that looked evenly matched at the start began to separate by tiny amounts. Some athletes hold their pace as fatigue piles up; others slowly unravel, and nothing in a fresh laboratory test warned that they would. That quality, the ability to keep running well when already tired, rarely appears on a test report. It decides marathons.

Two athletes can share the same $VO_2\text{max}$, the same threshold pace and the same half-marathon time and still finish a marathon ten minutes apart. The difference often lies in durability: how well physiology and performance hold up as duration and fatigue accumulate. Durability has become an important topic in endurance science because traditional laboratory tests are usually performed on fresh athletes. This chapter proposes a practical protocol to measure how pace, heart rate and perceived effort relationships change with duration in real training.

Aim and hypotheses

The aim is to develop and test a repeatable method for describing durability in marathon runners. Hypotheses: first, the relationship between pace and heart rate will drift in a measurable way after a certain duration; second, the pattern of drift will differ between athletes; third, that drift will relate to how much an athlete slows in the final part of a marathon. A durability index can be built from the rate of deterioration after a defined duration, but the formula should be validated before being used as a score.

Participants, inclusion and exclusion

Recruit marathon runners with at least one completed marathon and a planned race within the study window. Inclusion: stable training, capacity to complete long runs of at least 90 to 150 minutes, medical clearance, and a heart-rate device that records reliably. Exclusion: current injury, recent illness, or any condition making prolonged exercise in heat unsafe. In Chennai, schedule long runs early in the morning and log conditions carefully, because heat is the most common confounder of any durability measure in Indian cities.

Design and timeline

Week	Phase	Main activity
1	Baseline	Fresh submaximal test at fixed pace; time trial
2–3	Standard long run 1	Fixed pace, fixed fuelling, full data capture
6–7	Standard long run 2	Same route, time and protocol
10–11	Standard long run 3	Repeat to measure reliability and trend
Race	Marathon	Race data with splits, fuelling and conditions
Race + 2	Review	Compare long-run drift with race fade

Use standardised long runs or controlled race segments. The same route, start time, target pace, footwear and fuelling plan should be used each time. A useful structure is a steady long run with short fixed-pace "probe" segments at the start and near the end, so that heart rate and RPE at identical pace can be compared fresh and fatigued.

Measurements and outcomes

Record pace, heart rate, RPE at set intervals, fuelling (grams of carbohydrate and timing), fluid intake, body mass before and after, temperature, humidity and WBGT, and a brief post-run note on muscle soreness, gut comfort and cramps. Primary outcome: change in heart rate and RPE at the fixed probe pace between early and late segments. Secondary outcomes: pace decline in the race's final 10 kilometres, cardiac drift, and athlete-reported limiters.

KEY IDEA

The most useful result may be identifying whether the athlete loses performance through metabolic, thermal, pacing or musculoskeletal pathways. Each pathway calls for a different training response.

Interpreting the pathways

A metabolic pattern may show progressive heart-rate and RPE rise with poor fuelling or low glycogen availability. A thermal pattern may show sharper drift on hotter days,

larger body-mass loss and improvement when conditions are cooler. A pacing pattern may show a fast early section followed by inevitable decline. A musculoskeletal pattern may show relatively stable heart rate but rising soreness, form breakdown and pace loss. These are working categories, not diagnoses, and many athletes show a mix.

Safety, analysis and ethics

Stop a long run if heat-illness warning signs appear, if the athlete reports chest pain, fainting or unusual breathlessness, or if an injury changes running mechanics. Carry fluids and have a support vehicle or check-in points on longer routes. Analyse within-athlete changes against typical error, and check reliability across the three long runs before calculating any index. As with every AKURA protocol, obtain institutional ethics committee approval, written informed consent, and medical oversight.

Building and validating a durability index

A simple prototype index might express the percentage change in heart rate at the probe pace between the early and late segments, adjusted for temperature. Before such a number appears on any dashboard it must pass three tests. First, reliability: does it give similar values when the same athlete repeats the same long run in similar conditions? Second, sensitivity: does it change when durability is expected to change, for example after a block of specific long-run training? Third, validity: does it relate to what matters, such as the degree of slowing in the final 10 kilometres of a marathon? Only when all three questions have acceptable answers should the index be reported as a score rather than a research variable.

COACH'S CORNER

Add two short fixed-pace segments to your athlete's next long run, one early and one late. Record heart rate and RPE for both. Over a season, that simple habit will teach you more about marathon fade than any single laboratory test.

CHECKPOINT

When your athlete slowed in their last marathon, which pathway was most likely responsible? What data would you need to be sure?

CHAPTER 77

77. Heat Versus Altitude Crossover

PART XIV — RESEARCH PROTOCOLS

Chennai Central, late evening. A coach stands on the platform with a list of eight runners and a budget that will cover one block of specialised training

before the season's target race. Option one: put the squad on the overnight train towards the Nilgiris and spend three weeks at Ooty, 2,240 metres above sea level. Option two: stay at home, where the heat and humidity of a Chennai summer arrive every morning without a ticket. India's map of choices is growing. Hindustan Times reported in 2026 that SAI's campuses at Ooty and Shilaroo are the country's only two high-altitude training centres, with a new centre planned for Shillong; separately, SAI has approved a Khelo India centre at Leh. Heat, meanwhile, is available almost everywhere. What nobody can give that coach is a direct comparison: for these particular eight athletes, which block would do more, and for which goal?

Throughout this book we have treated heat and altitude as separate stimuli with some overlapping effects. Both can influence plasma volume, heart-rate responses and perceived effort; altitude may also increase haemoglobin mass over sufficient exposure, while heat acclimation primarily improves thermoregulatory and cardiovascular responses to heat. For an Indian coach who can choose between Chennai's heat and the hills of Ooty or Kodaikanal, the practical question is obvious: which block, for which athlete, for which goal? A crossover study is the most informative way to ask it.

Aim and hypotheses

The aim is to compare heat, altitude and control blocks in the same athletes, so that each athlete acts as their own comparison. Hypotheses: heat blocks will produce larger changes in thermal response and heart rate in the heat; altitude blocks will produce larger changes in haemoglobin mass; both may produce changes in submaximal heart rate; and individual response patterns will differ. Outcomes should distinguish plasma volume, Hb mass, submaximal heart rate, thermal response, performance and recovery. This allows the research team to ask which adaptation overlaps and which is unique.

Participants, inclusion and exclusion

Recruit trained athletes able to commit for a long study period, ideally six months or more. Inclusion and exclusion criteria combine those of Chapters 74 and 75: medical clearance, clinician-reviewed iron status, no history of serious heat or altitude illness without specialist clearance, no relevant cardiovascular or respiratory conditions, and no current injury. Because the study is long, plan for drop-outs and seasonal race schedules.

Design and timeline

Period	Block	Notes
Weeks 1–2	Baseline testing	Blood, heat-response test, performance test
Weeks 3–6	Block A (randomised)	Heat, altitude or control
Weeks 7–12	Washout	Normal training; retest at end
Weeks 13–16	Block B (randomised)	Second condition
Weeks 17–22	Washout	Normal training; retest at end
Weeks 23–26	Block C (randomised)	Third condition, then final tests

Randomisation and washout are desirable where practical. Randomising the order protects against season, fitness trend and race calendar being mistaken for block effects. The washout must be long enough for the main adaptations to return towards baseline; haemoglobin mass changes, in particular, persist for weeks. Where a full three-way crossover is impractical, a two-condition version (heat versus control, or altitude versus control) is a sensible first step.

Measurements and outcomes

Use identical tests at every time point: haemoglobin mass by a specialist method where available, plasma-volume estimates, heart rate at fixed pace in both temperate and warm conditions, core or skin temperature where safely measured, sweat rate, time-trial performance, resting heart rate, heart-rate variability trend and sleep. Log training load throughout so that each block is matched in volume and intensity as closely as possible.

KEY IDEA

The value of a crossover is not only in the average result. It is in discovering which athletes respond more to heat, which to altitude, and which to neither.

Safety and stop rules

The study must be designed around safety and qualified medical oversight, particularly for hypoxic exposure and blood testing. All stop rules from the heat and altitude protocols apply. Any athlete showing heat-illness or altitude-illness warning signs stops immediately and is assessed; any abnormal blood result is referred to a clinician. Limit the total number of blood draws to what is needed and explain each one in the consent form.

Analysis and ethics

Use mixed-effects models or similar methods appropriate for repeated measures, report individual responses with confidence intervals, and account for order and carry-

over effects. Be explicit that a small pilot crossover will generate hypotheses rather than definitive answers. Institutional ethics committee approval, informed consent and a named medical lead are mandatory, and participants should receive their own results in a clear, plain-language summary.

Why the comparison matters for India

Travel to hill stations costs money, time and time away from work or study, while heat is available at home for much of the year in Chennai and many other Indian cities. If a well-designed crossover showed that, for some athletes and some goals, a controlled heat block produced useful changes at a fraction of the cost, that would matter greatly for Indian endurance development. Equally, if altitude produced unique benefits that heat could not replicate for particular athletes, that would justify investment in camps. The study is not designed to declare a winner. It is designed to help coaches match the right stimulus to the right athlete at the right time, with evidence rather than habit.

COACH'S CORNER

If you cannot run a crossover, you can still think like one. Record each athlete's response to every heat block and every camp over the years. Across a career, those records form a personal crossover study.

CHECKPOINT

For your squad, do you currently know which athletes respond best to heat and which to altitude? If not, what would be the first block you could measure properly?

PART XIV AT A GLANCE

The key idea from each chapter, for quick revision.

- | | |
|-----------|--|
| 74 | Heat Study Protocol
A heat study is only as good as its environmental log. If baseline and post-tests are run in different heat and humidity, the change in heart rate may simply reflect the weather, not the athlete. |
| 75 | Altitude Study Protocol
Haemoglobin concentration and haemoglobin mass are not the same thing. A concentration can rise simply because plasma volume fell. Only a total-mass method tells you whether the athlete actually built more red-cell capacity. |
| 76 | Marathon Durability Protocol
The most useful result may be identifying whether the athlete loses performance through metabolic, thermal, pacing or musculoskeletal pathways. Each pathway calls for a different training response. |

77

Heat Versus Altitude Crossover

The value of a crossover is not only in the average result. It is in discovering which athletes respond more to heat, which to altitude, and which to neither.

PART XV — IMPLEMENTATION

Research frameworks fail when they cannot be used on an ordinary Tuesday morning. This Part turns AEI from a set of ideas into tools a coach can open, read and act on. It describes the AEI Athlete Card, a standard record that keeps raw data separate from derived scores; the AEI Dashboard, which answers what an athlete can do, how ready they are today and what has changed; the coach's interpretation layer, which ensures human context is never hidden by an algorithm; and the safety rules that define where automation must stop. The principle is simple: every number must be traceable, and every recommendation must be open to human review.

IN THIS PART

- 78** The AEI Athlete Card
- 79** The AEI Dashboard
- 80** The Coach's Interpretation Layer
- 81** Safety Rules

PART XV • IMPLEMENTATION

Traceable Numbers, Reviewable Decisions

The AEI Athlete Card, Dashboard, coach interpretation layer and safety rules turn ideas into tools for an ordinary Tuesday.

44 S Ankita Dhyani's steeplechase change, Tokyo 2025 to Nagoya 2026

 **12**

The AEI Athlete Card

Twelve sections from identity to clinical notes: raw data first, scores second, everything dated.



A record that travels

With consent, one card follows the athlete from hostel to academy to national camp.

 **3 Qs**

The AEI Dashboard

What can they do? How ready today? What changed? And how sure are we?



Override with a reason

A coach's override with a reason captures human knowledge the model does not yet have.



Know when not to automate

No readiness score overrides symptoms: chest pain, fainting or heat-illness signs stop the session.



Defer to clinicians

Blood results, iron, supplements and medication decisions belong to clinicians, never to the AEI system.

THE PROCESS



A dashboard is not a verdict; it prompts a coach–athlete conversation grounded in evidence.

THE STORY

The Girl Who Ran With the Recruits

PART XV — IMPLEMENTATION

On the hill roads of Pauri Garhwal, the mornings belong to young men. You see them in any Uttarakhand hill district: groups of teenagers in cheap trainers running up switchbacks above terraced fields, training for the physical tests that stand between them and a job in the Army. They run hard, because the test is hard and the job matters.

In Maroda, a village at about 1,400 metres, one of a farmer's four children watched them and made a decision. Her name was Ankita Dhyani, and she told ESPN what happened next: "I decided to run with them ... They never slowed down for me."

Nobody slowed down, so she sped up. The route from that hill road to an Olympic Games ran through a sports hostel in Rudraprayag in 2017, then to Bhopal. In late 2018 she joined the Elite Distance Running Programme there, coached by the Dutch former marathoner Hugo van den Broek, whose group travelled to Iten in Kenya once a year. From 2019 she was at the National Centre of Excellence in Bhopal under coach Toppo, who pointed out to ESPN that the hills had already done some of the work: "because she was running in Uttarakhand, she had already been doing a lot of high-altitude training."

The results followed, not in a straight line but in a series of steps. She ran the 1500 m and 5000 m at the 2021 World U20 Championships in Nairobi. She ran the heats of both the 5000 m and the steeplechase at the Paris Olympics in 2024. In Jerusalem in August 2025 she set a national record of 6:13.92 for the 2000 m steeplechase. A month later, at the World Championships in Tokyo, she ran 10:03.22 in her heat. Then, on 27 September 2026 in Nagoya, she ran 9:19.24 for fourth place at the Asian Games, a personal best, in the same race in which Parul Chaudhary took bronze with a national record of 9:08.67.

Reta, Chamba

Further west along the Himalaya, in Chamba district of Himachal Pradesh, another girl was running barefoot. Seema was born in Reta village in 2001, the youngest of six children. Her father died when she was twelve. She grew up herding cattle and running barefoot.

A SAI hostel gave her structure. She won junior national gold at Ranchi in 2015. Nine years later, at the Asian Cross Country Championships in Hong Kong in 2024, she won gold in 37:20, with Sanjivani Jadhav given the same time for silver. Then came a run of marks that show an athlete climbing steadily: 1:11:23 to win the Indian elite race at the Delhi Half Marathon in October 2025, 52 seconds outside L. Suriya's 2017 national record of 1:10:31; a national record of 1:26:04 for 25 km in Kolkata that December; 32:02.43 for 10,000 m in March 2026.

In Nagoya, on 24 September 2026, she won 10,000 m bronze at the Asian Games. She became only the third Indian woman to win an Asian Games medal in the event, and the first in 16 years. Afterwards she described it plainly.

The race was very tactical. Everyone ran for the medal.

— Seema (PTI, via Outlook India, 2026)

Amroha to Los Angeles

The third story begins in Amroha, Uttar Pradesh, where K. M. Deeksha was born in 1999. She competes for Madhya Pradesh and, as of 2024, had trained for about five years under coach S. K. Prasad at the Madhya Pradesh Athletics Academy in Bhopal.

On 11 May 2024, at the Sound Running Track Fest in Los Angeles, she finished third in the 1500 m in 4:04.78. It was a national record. To understand what that meant, follow the record back. Sunita Rani ran 4:06.03 at the 2002 Asian Games in Busan, and that mark stood until Harmilan Bains ran 4:05.39 in Warangal in 2021. Deeksha's run took the record under 4:05 for the first time. Nineteen years for the first improvement; three years for the second. The line of Indian middle-distance running had started to move.

The others on the road

They are not alone. Parul Chaudhary, who started with a barefoot school race in a Meerut village, now holds national records at 3000 m, 5000 m and the steeplechase. After her double bronze in Nagoya she said: "It feels really good. Last time in Hangzhou, I also took two medals, but this time the colour has changed..." Priti Lamba, steeplechase bronze medallist at the Hangzhou Asian Games, once broke her ankle in two places at a water jump at the Patiala national camp, and still runs with a rod in it. Harmilan Bains had knee surgery in 2022 and missed both the Commonwealth Games and the World Championships.

Each of these careers is also a trail of data: race times, injuries, camps, coaches, flights, hostels. Most of it lives in scattered places, and some of it is already lost.

What implementation means for them

Look at Ankita Dhyani's path again, this time as a data problem. A hill village. A hostel in Rudraprayag. An elite programme in Bhopal with an annual camp in Kenya. A national centre. Junior championships in Nairobi, an Olympics in Paris, a Grand Slam meet in Jerusalem, World Championships in Tokyo, Asian Games in Nagoya. At every transition a new coach, doctor or physio met an athlete whose history arrived, at best, in someone's memory and a phone gallery.

An athlete card of the kind described in Chapter 78 is a simple idea: one record that travels with the athlete, with her consent, from hostel to academy to national camp. It keeps raw measurements separate from derived scores, marks what is missing as missing, and dates everything. When a new coach meets her, the card tells them what

altitude blocks she has done, how she responded, and which injuries have come and gone.

A dashboard of the kind described in Chapter 79 asks a sharper question. Between Tokyo in September 2025 (10:03.22) and Nagoya a year later (9:19.24), something changed by 44 seconds. The public record does not say what. It might have been conditions, health, training, travel or tactics; nobody outside her team can know, and nobody should guess. A good dashboard would not guess either. It would show readiness, environment, load and recent interventions side by side, so that the people responsible for her could see the pattern, and learn from it.

And safety, the subject of Chapter 81, is not abstract for this generation. A 2025 study of 104 female athletes in Tamil Nadu, from volleyball and ball-badminton, found that about 53.8% were anaemic, against a background in which a large survey found anaemia in 44% of Indian adolescent girls and 41% of adult women. Those were not runners, and the numbers cannot be transferred directly. But any system that collects blood values from young Indian women athletes must put a clinician, not a coach and not an algorithm, in charge of interpreting them. In April 2026 the Athletics Integrity Unit also moved India into its highest-risk category, which means more out-of-competition testing, including blood tests, for distance runners. The athlete's body is becoming a data record whether she likes it or not. The question is whether that record is kept securely, read honestly and used for her benefit.

The girl who ran with the recruits did not need a dashboard to get started. She needed hills, stubbornness and people who did not slow down. What she and the runners who follow her deserve now is a system that remembers what worked, notices early when something is wrong, and knows when to stop and call a doctor. The next four chapters describe how to build one.

Sources: https://www.espn.in/athletics/story/_/id/30859627/no-turning-back-ankita-dhyani-mastery-improbable-chases-makes-unbeatable ; <https://worldathletics.org/athletes/india/ankita-dhyani-14818406> ; <https://worldathletics.org/news/feature/indian-endurance-project-kenya-netherlands-distance-running-broek-pal-murli> ; <https://www.olympics.com/en/news/grand-slam-jerusalem-2025-ankita-women-2000-m-steeplechase-report> ; <https://thebridge.in/athletics/world-chsips-2025-parul-chaudhary-ankita-dhyani-women-3000m-steeplechase-54592> ; <https://www.deccanchronicle.com/sports/asian-games-2026-indias-parul-chaudhary-wins-bronze-in-womens-3000m-steeplechase-1991012> ; <https://www.outlookindia.com/sports/others/seema-kumari-asian-games-2026-bronze-medal-mother-health-athletics-india> ; <https://www.tribuneindia.com/news/himachal/chamba-girl-strikes-gold-in-hong-kong/> ; <https://worldathletics.org/athletes/india/-seema-14734780> ; <https://www.shethepeople.tv/shesport/athlete-k-m-deeksha-national-record-for-1500m-race-4564973> ; <https://www.olympics.com/en/news/who-is-harmilan-bains-india-athlete-medals-records> ; <https://www.rediff.com/sports/report/asian-games-after-double-bronze-haul-parul-now-craves-family-time/20260928.htm> ; <https://indianexpress.com/article/sports/sport-others/priti-lamba-overcomes-injury-setback-to-triumph-at-federation-cup-athletics-8614902/> ; <https://pubmed.ncbi.nlm.nih.gov/40443248/> ; <https://pubmed.ncbi.nlm.nih.gov/39779946/> ; <https://www.aljazeera.com/sports/2026/4/20/india-in-extremely-high-doping-risk-bracket-athletics-integrity-unit>

CHAPTER 78

78. The AEI Athlete Card

PART XV — IMPLEMENTATION

Try to build an athlete card for Shivnath Singh. The headline number is easy: 2:12:00, run at Jalandhar in 1978, a national marathon record that stood for about 48 years until Sawan Barwal broke it in Rotterdam in April 2026. Then the fields start to go blank. His coach: unverified. His training base: unverified. He is reported to have finished 11th in the marathon at the 1976 Montreal Olympics, but even that result comes down to us with two different times, depending on which source you read. He is reported to have run barefoot throughout his career. Of weekly mileage, heart rates, blood values or the weather on the day of the record, nothing survives in the public record. One of the greatest performances in the history of Indian running is, as data, almost empty. The athlete card exists so that the next record-breaker leaves behind more than a single number.

Every athlete in the AKURA system should have one standardised card. It is the single place a coach, scientist or clinician can go to see who the athlete is, what they have done, how they have been tested, and how confident we are in each piece of information. Without a standard card, data scatter across watches, spreadsheets, message threads and memory. With one, the athlete becomes a coherent research record that grows over a career.

What the card contains

The card should contain demographics relevant to sport, training history, recent race results, physiological tests, environmental exposure, injury and recovery flags, equipment and data confidence. The table below shows a proposed structure. It is a starting point and will change as the system learns what coaches actually use.

Section	Fields
Identity	Athlete ID, age band, sex, event focus, home city, coach
Training history	Years training, typical weekly volume, recent block summary
Performance anchor	Recent 5K/10K/HM/marathon results with date and conditions
Physiological tests	Threshold marker, VO ₂ -related measure, economy, Hb/Hb mass
Cardiovascular	Resting HR, HR at fixed pace, HRV trend
Environment	Heat exposure log, altitude exposure log, home climate
Durability	Standard long-run drift, marathon fade
Recovery flags	Sleep trend, soreness, illness, injury notes
Equipment	Watch, HR strap, footwear, device settings
Interventions	Heat blocks, camps, nutrition changes, with dates
Data confidence	Protocol version, missing data, quality flags
Clinical notes	Clinician-entered only; restricted access

Raw data first, scores second

The card should separate raw data from derived scores. Coaches must be able to inspect the raw measurement behind every conclusion. If the card shows an AEI Oxygen domain value, one tap should reveal the actual haemoglobin result, the date, the laboratory, the method and the protocol notes. If a durability rating appears, the coach should see the long runs from which it was calculated. This is not bureaucracy; it is the foundation of trust.

KEY IDEA

A score without traceability is difficult to trust. Every number on the AEI card must lead back to a measurement, a date and a method.

Confidence and privacy

Each field should carry a confidence marker: high when the data are recent, standardised and from a reliable device or laboratory; lower when data are old, self-reported or collected under unknown conditions. Missing data should be shown as missing, not quietly filled in. Because the card holds sensitive health information, access should be role-based. Clinical notes and blood results should be visible only to those who need them, athletes should be able to see their own card, and consent for research use should be recorded explicitly. Indian data-protection requirements should be followed, and athletes should know how their data are stored and who can see them.

Keeping the card alive

A card that is filled in once and forgotten is useless. Updating should be part of routine: race results added within a few days, test results on the day, environmental logs with every heat block or camp, and a quarterly review in which coach and athlete read the card together. Over time the card becomes the athlete's physiological biography, one that can show, for example, how a runner from Madurai responded to three successive Kodaikanal camps, or how heat tolerance changed across four Chennai summers.

Using the card in practice

Imagine a coach preparing a runner for a winter marathon. Before planning, the coach opens the card and sees the most recent half-marathon anchor with its date and conditions, the heart rate at fixed pace from the last three tests, a haemoglobin result from before the previous Kodaikanal camp, the heat-session log from the summer, and a note that the athlete had a calf strain in August. Each item shows its confidence. In a few minutes the coach can see what is known, what is uncertain and what needs testing before the key block begins. That is the purpose of the card: not to decide for the coach, but to make sure the coach decides with the full picture.

A card also makes handovers safer. When an athlete moves from a school coach to a university programme, or from a club in Coimbatore to a training group in Chennai, the new coach inherits a documented history instead of starting from guesswork. With the athlete's consent, the card can travel with them.

COACH'S CORNER

Start with a paper or spreadsheet version if needed. Fill in only the sections you can support with real data, mark everything else as missing, and review it with the athlete at the end of each training block.

CHECKPOINT

If a colleague asked you tomorrow for the raw measurement behind your athlete's most important training decision, could you find it in less than a minute?

CHAPTER 79

79. The AEI Dashboard

PART XV — IMPLEMENTATION

It is 5:40 a.m. in a training hostel, and the coach's phone has already buzzed thirty times. Screenshots of watch data. A voice note about a sore calf. A photo of a thermometer on a verandah. Somebody's sleep score,

somebody's resting heart rate, somebody's GPS track that cuts straight through a building. Each message is a fragment of truth, and together they are almost useless, because nobody can read thirty fragments before the 6 a.m. session begins. The modern Indian coach is not short of data. Watches are cheap, apps are free, and a squad of fifteen runners can generate thousands of numbers in a single week. What the coach is short of is attention. The real skill is no longer collecting information. It is knowing, in the few minutes before the athletes arrive at the track, which three facts matter today.

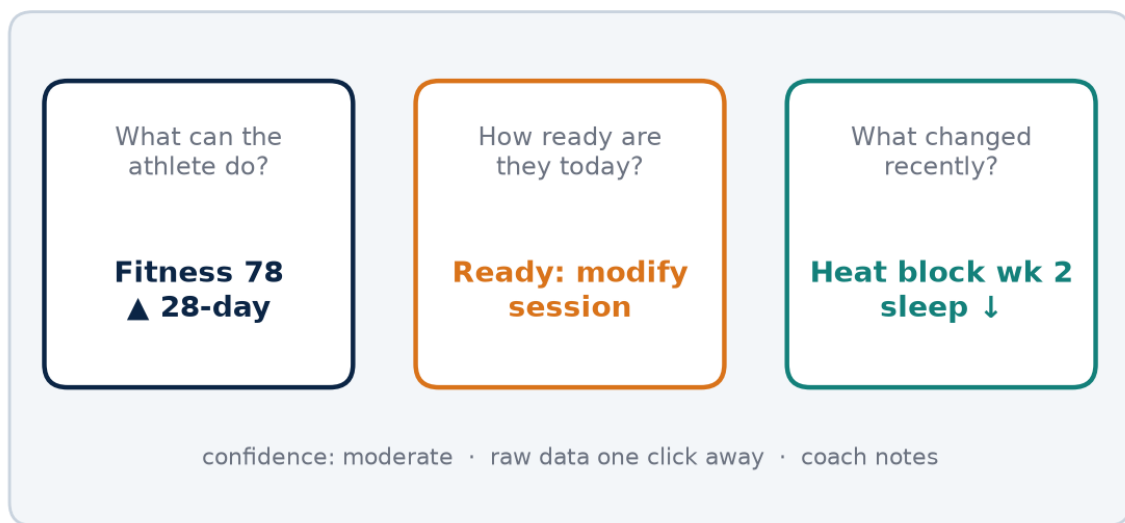


Figure 27. Dashboard concept (mock-up): three questions answered at a glance, with confidence and raw data always visible.

A card holds everything; a dashboard shows what matters now. Coaches work under time pressure, often with many athletes and limited staff. A dashboard that tries to show every metric at once becomes noise. The AEI dashboard is designed around restraint: a small number of panels that together answer the coach's most urgent questions, with the raw card always one click away.

Three questions

The dashboard should answer three questions quickly. What can the athlete do? How ready are they today? What changed recently? Everything else is secondary. The first question is answered by the performance anchor and fitness trend. The second is answered by readiness indicators such as sleep, resting heart rate, heart-rate variability trend and subjective wellness, which in the wider AKURA ecosystem feed the AISRi daily readiness layer. The third is answered by recent interventions, environmental exposure and training-load changes.

Proposed panels

A useful dashboard should show current performance anchor, fitness trend, readiness trend, key physiological domains, environmental exposure, training load, durability, confidence and recent interventions.

Panel	What it shows	Question answered
Performance anchor	Most recent standardised race or time trial	What can they do?
Fitness trend	HR at fixed pace, threshold trend over weeks	What can they do?
Readiness trend	Sleep, RHR, HRV trend, wellness (AISRI)	How ready today?
Physiological domains	AEI domain values with confidence	What can they do?
Environment	Recent heat and altitude exposure, WBGT	What changed?
Training load	Acute and chronic load, recent changes	What changed?
Durability	Long-run drift trend	What can they do?
Confidence	Data quality and missing-data flags	How sure are we?
Interventions	Heat blocks, camps, nutrition, equipment	What changed?

Design principles

Show trends rather than single values, because a single low heart-rate-variability reading means much less than a two-week decline. Display confidence alongside every value, using clear visual cues for low-quality or missing data. Separate fitness from readiness visually, so that a tired athlete is not mistaken for an unfit one. Keep colour meanings consistent and avoid alarming reds for normal day-to-day variation. Allow the coach to annotate the dashboard directly, and make sure those annotations are saved.

KEY IDEA

A dashboard is not a verdict. It is a prompt for a conversation between coach and athlete, grounded in evidence.

Avoiding dashboard traps

Dashboards can create false precision. A number displayed to one decimal place feels more certain than it is. They can also encourage over-monitoring, where athletes become anxious about daily fluctuations. The AEI dashboard should round sensibly, show ranges where appropriate, and remind users that daily readiness signals are context-dependent. Athletes should see a simplified version focused on their own progress and practical guidance, not a wall of physiological variables.

An example morning

At 6 a.m. a coach in Chennai opens the dashboard for a squad of fifteen runners. Most readiness panels show normal variation. One athlete's resting heart rate has been elevated for three days, sleep has fallen, and the wellness note mentions a sore throat. Another athlete's fitness trend shows heart rate at fixed pace improving steadily since a heat block ended two weeks earlier. A third has a low-confidence flag because their chest strap was not worn. Within a minute the coach knows whom to call, whose session to modify, and whose data to ignore until it is fixed. That is what a good dashboard does: it directs attention, and then it gets out of the way.

Building gradually

A dashboard should grow with the data. A club with only race results and watch heart rate can begin with the performance anchor, fitness trend and training-load panels. Readiness and environment panels can be added once check-ins and environmental logs become routine, and physiological domains once standardised testing is in place. Adding panels before the data exist only creates empty boxes and false impressions of completeness.

Finally, the dashboard must be honest about what it does not know. A blank panel labelled "no recent data" is more useful than a confident-looking estimate built on stale measurements. Coaches quickly learn to distrust tools that look precise but prove wrong, and that distrust spreads to the whole system.

COACH'S CORNER

Look at your current monitoring tools and ask whether they answer the three questions within ten seconds. If they do not, remove panels until they do.

CHECKPOINT

Which single panel on your current dashboard do you look at most, and which have you never used to make a decision?

CHAPTER 80

80. The Coach's Interpretation Layer

PART XV — IMPLEMENTATION

Look at the results of the Rio 2016 Olympic marathon and you will find Nitendra Singh Rawat in 84th place, in 2:22:52. An algorithm reading that line might conclude that he was unfit, badly paced or poorly prepared. Seven months earlier he had run 2:15:48 in Mumbai, an Indian course

record that qualified him for Rio. The result sheet cannot tell you what the reporting did: that he ran in Rio with a hamstring injury. After recovering and training with the national camp in Bengaluru, he ran 2:16:54 in Mumbai in 2018, and in 2019 he ran 2:15:52 there, four seconds outside his own course record. The Rio result is accurate. It is simply incomplete. Every coaching system faces this problem every day: numbers arrive without their stories, and a model that sees only numbers will sometimes be confidently wrong.

No algorithm, however carefully validated, sees the whole athlete. It sees what it has been given. A runner's heart rate may be high because of fitness loss, but also because of a late shift at work, a family crisis, a fever starting, a new medication, a hot bus journey, or a watch strap worn loosely. The coach's interpretation layer exists so that this context is never lost and never overridden silently by a number.

Context the algorithm cannot see

No algorithm should hide context. Travel, family stress, work shifts, illness, medication, menstrual-cycle factors where relevant, equipment changes and unusual weather can alter performance. In the Indian setting, add long commutes, exam seasons for student athletes, festival periods with changed sleep and diet, monsoon disruptions and power cuts that affect sleep in hot weather. Each of these can shift the physiological signals on which AEI and AISRi depend.

Annotation and override

The coach should be able to annotate events and override a recommendation with a reason. If the system suggests a hard session but the coach knows the athlete travelled overnight by train from Coimbatore, the coach should be able to change the plan and record why. The override is not a failure of the system. It is information. Those overrides become research data themselves, because they record exactly where the model and the expert disagreed.

KEY IDEA

An override with a reason is one of the most valuable data points a coaching system can collect. It captures human knowledge the model does not yet have.

Learning from the coach

Over time, the system can learn which signals the coach repeatedly uses correctly. If a coach consistently reduces load when a particular athlete's sleep drops below a personal threshold, and those decisions are followed by better sessions, that pattern can inform future recommendations. Equally, if some overrides are followed by poor outcomes, the system can highlight this gently. The aim is mutual learning: the

algorithm becomes more context-aware, and the coach becomes more aware of their own patterns and biases.

Guarding against bias

Human interpretation brings its own risks. Coaches can hold assumptions about athletes based on background, gender, region or appearance. The interpretation layer should encourage evidence-based notes, not labels. Write "athlete reported poor sleep for three nights due to exams", not "this athlete is always lazy before races". Keep annotations factual, respectful and useful, remembering that athletes may be able to read them.

Annotations in practice

A good annotation is short, specific and timed. It states what happened, when, and what was changed. For example: "Returned from wedding travel, two nights of four hours' sleep; replaced intervals with 40 minutes easy." Or: "Hot humid morning, WBGT high; moved tempo to treadmill in the evening." Standard tags such as travel, illness, work, exams, heat, equipment and menstrual-cycle note (entered only with the athlete's consent) make these notes searchable later. When the research team later analyses why a fitness trend dipped or a test result looked unusual, these annotations often provide the explanation that no sensor could.

Athlete voice

The interpretation layer is not only for the coach. Athletes should be able to add their own notes about how sessions felt, what else was happening in their lives, and what they think explains a good or poor result. Athletes often notice patterns before the data show them. Including their voice improves the data, builds trust and helps athletes develop the self-awareness that is the long-term goal of the whole AKURA approach.

When the coach and system disagree

Disagreement should be expected and welcomed. When the system and the coach point in different directions, the useful response is neither to obey the algorithm blindly nor to ignore it, but to ask why they differ. Perhaps the model lacks context; perhaps the coach is reacting to a single vivid memory. Recording the disagreement, the decision and the outcome turns each case into evidence. Over hundreds of such cases, patterns emerge that neither the coach nor the model could have seen alone, and that shared record becomes one of the most distinctive assets of the AKURA system.

COACH'S CORNER

Build the habit of a one-line note whenever you change a planned session: what you changed and why. After three months, read the notes back. You will learn a great deal about your own coaching.

CHECKPOINT

When did you last override a plan based on context the data could not see? Did you record it, and what happened afterwards?

CHAPTER 81**81. Safety Rules**

PART XV — IMPLEMENTATION

Rio de Janeiro, August 2016. O. P. Jaisha, who a year earlier had set the Indian women's marathon record of 2:34:43 in Beijing, collapses after crossing the finish line of the Olympic marathon. Afterwards she says that Indian officials gave her no refreshments along the course. Scroll quoted her: "All the [other] countries had their stalls at every two kilometres, but our country's stall was empty." The Athletics Federation of India disputed her account, and her teammate Kavita Raut said she had had enough water. Years later, what happened that morning is still argued over. That is exactly the point. When a safety plan exists only in people's heads, nobody can check afterwards what was planned, what was provided and where the chain broke. A written plan, briefed in advance and reviewed without blame afterwards, protects athletes and officials alike.

Every measurement system eventually reaches a point where the right answer is not a number but a person. AKURA's safety rules define that point clearly. They apply to every protocol in this book, every dashboard, and every automated recommendation. They are not optional extras; they are the conditions under which the rest of the system is allowed to operate. Nothing in this book replaces the judgement of a qualified doctor.

The core rules

1. No readiness score should override symptoms requiring medical attention. If an athlete reports chest pain, fainting, unusual breathlessness, palpitations or signs of illness, the session stops and the athlete is assessed, whatever the dashboard says.
2. Heat training should be stopped when heat-illness warning signs appear. Every heat session needs supervision, cooling, fluids and an emergency plan.
3. Altitude exposure should be medically supervised when indicated, especially for athletes with medical conditions, low iron status or previous altitude illness.
4. Blood abnormalities should be interpreted by qualified clinicians. Coaches and algorithms may flag a result but must not diagnose or prescribe treatment.

5. No supplement, iron or medication decision should be made by the AEI system. These belong to clinicians.
6. Athletes must be able to stop any test or session at any time without penalty.
7. Any result the system cannot confidently interpret must be flagged for human review.

Heat-illness warning signs

Stop exercise immediately, begin cooling and seek emergency medical care if an athlete shows any of the following:

- Confusion, disorientation, irritability or unusual behaviour
- Hot skin, whether dry or still sweating (many exertional heatstroke casualties are still sweating)
- Collapse or loss of consciousness
- Vomiting or persistent nausea
- Severe headache, dizziness or loss of coordination

Exertional heat stroke is a medical emergency. Rapid whole-body cooling, ideally cold-water immersion started on site as soon as heatstroke is suspected, and emergency care save lives. In Chennai and other hot Indian cities, every training group should know where the nearest emergency department is and how to reach it quickly.

Altitude warning signs

Stop training and arrange clinical assessment if an athlete at Ooty, Kodaikanal or higher altitude shows:

- Worsening headache that does not settle with rest
- Nausea, vomiting or loss of appetite with fatigue
- Unusual breathlessness at rest or a persistent cough
- Confusion, unsteadiness or extreme drowsiness
- Chest tightness or coughing up frothy or blood-stained sputum

Serious or worsening symptoms require descent and urgent medical care. Moderate altitudes carry lower risk than high mountains, but individual susceptibility varies and warning signs should never be ignored.

KEY IDEA

A research system must know when not to automate. The safest algorithm is the one that can say: "This result needs human review."

Defer to clinicians

This book discusses physiology to improve coaching decisions, not to diagnose disease. Where symptoms, blood results, heart rhythm irregularities, disordered eating, repeated injuries or menstrual disturbances appear, the coach's role is to refer to an appropriately qualified clinician. The AEI system should make referral easy and never make it feel like failure.

Building a safety culture

Rules on paper are not enough. Safety depends on a culture in which athletes feel able to report symptoms without being seen as weak, and in which coaches reward honest reporting. In many Indian sporting environments, athletes may hide illness or discomfort to avoid losing selection or disappointing a coach. The AKURA system should make it normal, even expected, to say "I don't feel right today". Assistant coaches and senior athletes should be trained in first response, including how to cool an athlete rapidly and how to call for emergency help. After any incident or near miss, the team should review what happened and what can be improved, without blame.

Data safety and privacy

Safety also includes the athlete's information. Blood results, menstrual-cycle notes, injury histories and mental-health comments are sensitive. They should be stored securely, shared only with those who need them, and never used against an athlete in selection or contract discussions without clear consent. Athletes should be able to see, correct and, where appropriate, withdraw their data.

COACH'S CORNER

Print the heat and altitude warning signs and keep them in your kit bag, at the track and on the team group. Brief every new athlete and every assistant coach before their first hot session or first camp.

CHECKPOINT

If an athlete collapsed during a hot session tomorrow, does everyone in your group know exactly what to do in the first five minutes?

PART XV AT A GLANCE

The key idea from each chapter, for quick revision.

78

The AEI Athlete Card

A score without traceability is difficult to trust. Every number on the AEI card must lead back to a measurement, a date and a method.

-
- 79** **The AEI Dashboard**
A dashboard is not a verdict. It is a prompt for a conversation between coach and athlete, grounded in evidence.
-
- 80** **The Coach's Interpretation Layer**
An override with a reason is one of the most valuable data points a coaching system can collect. It captures human knowledge the model does not yet have.
-
- 81** **Safety Rules**
A research system must know when not to automate. The safest algorithm is the one that can say: "This result needs human review."
-

PART XVI — PHILOSOPHY

This Part steps back from protocols and dashboards to ask what the whole enterprise is for. Endurance communities everywhere carry myths, and many contain a real observation wrapped in an inaccurate explanation. AKURA's task is not to mock those beliefs but to test them, keeping what the evidence supports and letting go of what it does not. These closing chapters set out the attitude behind the method: humility before data, respect for the athlete, and a cycle of training, measuring, understanding and adapting that continues across a career. They end with the real goal of AEI, which is not a famous number but athletes who understand themselves better.

IN THIS PART

- 82** From Myth to Measurement
- 83** Train. Measure. Understand. Adapt.
- 84** The Real Goal of AEI

PART XVI • PHILOSOPHY

Train. Measure. Understand. Adapt.

Many myths wrap a real observation in an inaccurate explanation; AKURA tests them with humility and respect for the athlete.

28:48.72

Hari Chand's 1976 10,000 m national record — it stood 32 years

**The useful kernel**

Keep the real observation inside a myth, and test the explanation wrapped around it.

**Myth to hypothesis**

A myth becomes useful the moment it is a hypothesis that could be proved wrong.

**Humility in both directions**

Do not mock old beliefs, and do not overclaim for data: let evidence decide what stays.

 4 words**A loop, not a line**

Training without measuring is guessing; measuring without understanding is noise; understanding without adapting changes nothing.

**32 · 48****Progress in decades**

Records that stood 32 and 48 years show endurance progress is measured in decades, not seasons.

**The real goal**

Not a famous number, but athletes and coaches who understand the body and decide better.

THE PROCESS

Train

Measure

Understand

Adapt

Belief can create a hypothesis, but the athlete's truth must be told by data.

THE STORY

The Barefoot Record

PART XVI — PHILOSOPHY

Montreal, July 1976. The Olympic 10,000 metres heats are under way, and among the vests and spikes of the world's best distance runners is a pair of bare feet. They belong to Hari Chand, a 23-year-old policeman from Hoshiarpur district in Punjab. He runs the whole race barefoot. He finishes eighth in his heat and does not reach the final.

On the scoreboard, it looks like an ordinary early exit. In the Indian record books, it becomes something else. The time Hari Chand ran that day was a national record, and it stood for 32 years. Sources disagree on the exact figure: Wikipedia and other accounts give 28:48.72, while Olympics.com prints 29:48.72, almost certainly a typing error, since a slower time could hardly have survived as the record for three decades. Even the most famous run of his life comes down to us with a question mark attached.

A policeman who ran

Hari Chand was born on 1 April 1953. The Times of India gives his village as Ghorewaha in Hoshiarpur district. He served with the Central Reserve Police Force, where Olympics.com describes him as an inspector, and he is reported to have retired as a Commandant. He ran the 5000 m and the 10,000 m, and later the marathon.

Imagine the Punjab of his youth as any traveller would see it: flat, green wheat country stretching to the Shivalik foothills, winter fog lying on the fields, summers that bake the canal roads. That is a picture of the place, not of the man. What we know about Hari Chand himself comes almost entirely from results.

And the results are remarkable. At the 1975 Asian Championships in Seoul, he won gold in the 10,000 m and bronze in the 5000 m, and received the Arjuna Award for them. The runner-up in that 10,000 m, according to Wikipedia, was Shivnath Singh, the Army runner who would soon set a marathon record of his own. After Montreal came 1978, the year Indian distance running seemed to catch fire. In May, Shivnath Singh ran 2:12:00 for the marathon at Jalandhar. At the Asian Games in Bangkok that year, Hari Chand won both the 5000 m and the 10,000 m. Double gold.

He went to the Moscow Olympics in 1980 and went out in the heats of the 10,000 m; Wikipedia records that he also ran the marathon there, finishing 22nd. He missed the 1982 Asian Games in Delhi because of a viral fever. Then the record books go quiet.

He died on 13 June 2022, aged 69. The Times of India's obituary headline said that he had died unsung.

What the record books do not say

Try to reconstruct Hari Chand's training, and the pages turn blank almost immediately. Who coached him? Unverified. Where did he train? Unverified. How many kilometres

did he run a week, on what surfaces, in what heat, with what food? Nothing survives in the public record. His coach and training base are among the most basic facts a sport could keep, and for one of its greatest runners, India did not keep them.

The same is true of Shivnath Singh, reported to have run barefoot throughout his career. His 2:12:00 from 1978 was the longest-standing record in Indian athletics. It lasted about 48 years, until Sawan Barwal ran 2:11:58 in Rotterdam in April 2026 and then 2:11:37 for Asian Games silver in September. Shivnath Singh's coach and training base are also unverified. Even his 11th-place time at the Montreal Olympic marathon appears in two different versions.

The long view

Line up the Indian distance records and a strange pattern appears. The 10,000 m record set by Hari Chand in 1976 stood until Surendra Singh ran 28:02.89 in Vigo, Spain, in 2008. Gopal Saini's steeplechase record from 1981 stood for 37 years, until Avinash Sable began breaking it in 2018. Bahadur Prasad's 5000 m record from 1992 stood about 30 years. Shivnath Singh's marathon stood 48.

Then, in a rush, they fell. Gulveer Singh took the 10,000 m record to 27:41.81 in 2024 and then to 27:00.22 in California in 2025. Sable broke the steeplechase record ten times. The marathon record went twice in one year.

Why the long silence, and why the sudden rush? The honest answer is that nobody can say for certain. What the evidence does show is a pattern: almost every record since 2022 was set at an overseas meet with fast pacing, in an era of foreign coaching, altitude camps abroad and racing against the world's best. That is an observation. It is not yet an explanation, and it certainly does not prove that the athletes of the 1970s were less talented. They ran what they ran with what they had.

How myths grow in empty space

When measurements are missing, stories rush in to fill the space. Two of the great Indian distance records of the 1970s belong to men reported to have run barefoot. It is very tempting to build a theory on that. Perhaps bare feet made them tough, and modern shoes have made runners soft. Or perhaps the opposite: imagine what they could have done in proper shoes. Both stories are satisfying. Both are repeated. Neither can be tested, because we do not know how these men trained, how their bodies responded, or what the conditions were on the days they ran.

Every endurance culture produces myths like this. They usually start with a real observation, such as a barefoot runner who was genuinely fast or a hill village that genuinely produced champions, and then attach an explanation that nobody has checked. Over time the explanation hardens into common sense. Coaches repeat it to athletes. Athletes repeat it to their own students. Fifty years later it is simply what everybody knows.

Sawan Barwal, whose Asian Games silver in 2026 was India's first men's marathon medal at the Games since 1982, described his racing to The Tribune in words with no myth in them at all: "There was only one thing in my mind: I have to go till the finishing line and how I can maintain the pace."

Consider the contrast with today. A club runner in Chennai with a cheap sports watch now collects more data in a single Sunday long run than survives about Hari Chand's entire career: pace for every kilometre, heart rate every second, temperature, elevation, cadence. The problem has flipped. The 1970s had champions and almost no data. The 2020s have oceans of data and, for most runners, very little understanding of what it means. Neither situation, on its own, turns observation into knowledge. That takes a method.

Legacy and patience

Hari Chand's legacy is not only two gold medals in Bangkok and a record that outlived most of the careers that chased it. It is also a question he leaves behind: how many great Indian athletes have been reduced, in memory, to a single time and a single story, because nobody wrote down the rest?

The answer to that question is not to mock the old stories. It is to take their kernel of truth seriously enough to test it, and to make sure that the next Hari Chand leaves behind not only a record but a record of how it was made. Patience is part of this. Records that stood for 32 and 48 years remind us that progress in endurance running is measured in decades, not seasons. A research system has to think on the same scale.

That is where this Part of the book begins: with the move from myth to measurement.

Sources: <https://www.olympics.com/en/news/hari-chand-indian-athlete-runner-olympian-dies> ; <https://timesofindia.indiatimes.com/sports/more-sports/athletics/olympian-hari-chand-who-created-10000m-national-record-barefoot-in-montreal-dies-unsung/articleshow/92183859.cms> ; https://en.wikipedia.org/wiki/Hari_Chand ; https://en.wikipedia.org/wiki/Shivnath_Singh ; <https://www.olympics.com/en/news/sawan-barwal-breaks-marathon-national-record-india> ; <https://www.tribuneindia.com/news/sports/who-is-sawan-barwal-himachal-runner-who-challenged-marathon-myth-to-win-asian-games-silver/> ; <https://www.hindustantimes.com/sports/others/after-asian-games-10-000m-silver-kartik-targets-national-record-101697292334456.html> ; <https://www.olympics.com/en/news/who-is-avinash-sable-india-athlete-steeplechase> ; <https://www.olympics.com/en/news/gulveer-singh-india-10000m-national-record-the-ten-2025-athletics-california> ; <https://www.olympics.com/en/news/athletics-list-of-indian-national-record-holders>

CHAPTER 82

82. From Myth to Measurement

PART XVI — PHILOSOPHY

"Kavita Raut, Monica Atre [Monika Athare]... Sanjivani. I am telling you Nashik is the Kenya of India ... Remember, Kavita had to walk so many kilometres to reach school or fetch water." The speaker is Vijender Singh,

the coach who has worked in Nashik since 1991, talking to The Indian Express. It is a wonderful line, and it contains a real observation: his group, drawn from Nashik and its surrounding villages, has produced a Commonwealth Games medallist, a World Championships marathoner and Asian medallists. It also contains the seed of a myth. Is the secret the long walks to school? The terrain? The coach himself? Or simply which children stayed in the sport long enough to be noticed? The same shape of explanation has been applied to Kenya, to Ethiopia and to the hills of Uttarakhand. A coach who loves his athletes is entitled to his metaphor. A scientist has to ask what would need to be measured to know whether it is true.

Every endurance community has myths. In India we hear that runners from hill regions are naturally stronger, that Chennai's heat alone makes athletes tough, that a camp in Ooty automatically adds a minute to a marathon, that a certain diet or a certain shoe is the secret. Internationally, the myths are often about East African runners, about genes, about altitude, about poverty and hunger. Some contain a useful observation wrapped in an inaccurate explanation. This book has tried to treat every such belief with curiosity rather than contempt.

The useful kernel

A myth usually survives because it contains something true. Athletes who live and train at moderate altitude do experience real physiological challenges. Athletes who grow up running long distances as children may accumulate substantial aerobic training. Hot climates do produce heat adaptation. The problem arises when a partial truth becomes a complete explanation, or when a population-level observation is turned into a statement about every individual. That is where stereotypes form, and stereotypes are both scientifically weak and unfair to athletes.

Separating observation from conclusion

The job of AKURA is not to ridicule the myth. It is to separate the useful observation from the unsupported conclusion. That requires a simple discipline: state the belief as a testable hypothesis, identify what would be measured, predict what result would support or contradict it, and then collect the data. "Altitude makes runners faster" becomes "a three-week camp at a defined sleeping altitude will increase haemoglobin mass and time-trial performance in this group, with a measurable individual response". The second statement can be wrong, and that is exactly why it is useful.

MYTH VS EVIDENCE

Myth: Athletes from a particular region or ethnic group are naturally built for endurance, so training matters less.

Evidence: Elite endurance performance is multifactorial. Training history, environment, economy, motivation, opportunity, coaching and support systems all matter, and individual variation within any group is large. No population label predicts an individual athlete's potential.

Belief and data

Belief is not the enemy of science. Coaches need conviction to commit to a plan, and athletes need belief to endure hard training. Belief is often where a good question starts. But belief cannot be the end point. The athlete deserves to know what is actually happening in their own body, and that can only come from careful measurement interpreted honestly.

நம்பிக்கையால் hypothesis உருவாகலாம். ஆனால் athlete-ன் உண்மையை data தான் சொல்ல வேண்டும்.

— Kura

In English, this means: "Belief can create a hypothesis. But only data can tell the athlete's truth." The line is written in the mixed Tamil and English that many Chennai coaches and athletes actually speak, and it captures the spirit of this book. Faith, tradition and intuition are welcome at the beginning of the research process. They generate the questions. The answer, however, must come from what the athlete's body shows when it is measured properly.

Humility in both directions

Measurement also has limits. Devices have error, tests have poor reliability when not standardised, and statistics can mislead when sample sizes are small. Moving from myth to measurement does not mean replacing one certainty with another. It means holding every conclusion with an appropriate level of confidence, updating it as new evidence arrives, and being honest when the answer is "we don't know yet".

KEY IDEA

A myth becomes useful the moment it is turned into a hypothesis that could be proved wrong.

Myths in the Indian context

Indian endurance culture has its own inherited beliefs: that drinking water during runs makes athletes weak, that pain must always be pushed through, that a particular village or state produces "natural" runners, that women should train less, or that recovery is laziness. Some of these beliefs once contained a grain of practical sense in a particular setting; others were never supported at all. Treating them as hypotheses allows coaches to keep the respect for hard work and tradition while discarding practices that harm athletes. That process should be gentle and respectful, because the people who hold these beliefs often care deeply about their athletes.

Ultimately, the move from myth to measurement is a move from arguing about who is right to asking what is true for this athlete. That shift lowers the temperature of debates between coaches, and it keeps the conversation centred on the person who matters most.

COACH'S CORNER

Write down three beliefs you hold about training that you have never tested. For each, write what you would measure and what result would change your mind. Choose one to test this season.

CHECKPOINT

Which training belief in your group is most widely accepted and least often questioned? What would it take to test it fairly?

CHAPTER 83

83. Train. Measure. Understand. Adapt.

PART XVI — PHILOSOPHY

In July 2025, at the Monaco Diamond League, Avinash Sable injured the ACL and meniscus of his right knee. He had surgery. He missed the World Championships in Tokyo that year and the Glasgow Commonwealth Games in 2026. Speaking about his injuries, he offered one of the most honest sentences an elite athlete can say in public: "I think I might have rushed my return a bit, which caused the injury to recur." In September 2026 he ran his first steeplechase back, 8:32.39, at the Indian Athletics Series final. Weeks later in Nagoya he ran 8:21.67, his best of the season, to finish fourth at the Asian Games. Train, test, learn, adjust, train again. The road back was a loop rather than a straight line, and one of its most valuable lessons came from a step that went wrong.

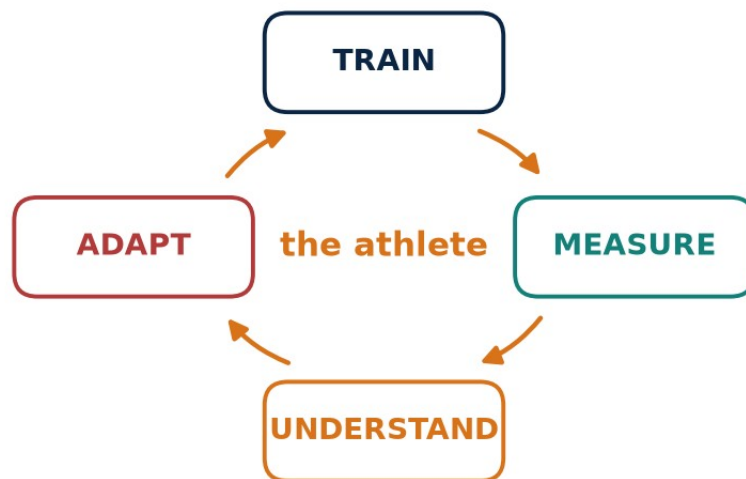


Figure 28. *Train. Measure. Understand. Adapt. The cycle repeats for the athlete's entire career.*

Training without measurement can still work. Coaches have produced great athletes with a stopwatch, a notebook and years of experience. But measurement makes the learning process clearer. It shortens the time needed to discover what works for an individual, and it protects athletes from repeating mistakes that nobody noticed. Measurement without interpretation is just data. Interpretation without humility becomes dogma. AKURA's approach can be summarised in four words.

The four words

1. **Train.** Do the work, consistently and intelligently. Training is the stimulus that drives adaptation. No amount of measurement replaces months and years of well-structured running, strength work and recovery. The training plan should be clear enough that it can be described, recorded and repeated.
2. **Measure.** Capture what happened and how the body responded, using standardised protocols. That includes performance, heart rate, perceived effort, environment, sleep and, where appropriate, laboratory tests. Measure carefully, log context, and note the confidence of each piece of data.
3. **Understand.** Interpret the measurements in the context of the athlete's life, history and goals. Ask what changed, why it might have changed, and whether the change is larger than measurement error. Separate fitness from readiness, adaptation from noise, and correlation from cause.
4. **Adapt.** Change the plan based on what was learned. Adaptation applies to the athlete's body and to the coach's thinking. If the data disagree with the hypothesis, revise the hypothesis. Then return to training, and the cycle begins again.

A cycle, not a line

The cycle repeats for the athlete's entire career. Each loop adds to the athlete's record and sharpens the coach's understanding. Early loops may be crude, with few measurements and much uncertainty. Later loops can be more precise, because the athlete's baseline, typical responses and personal limiters are known. This is the practical meaning of "the athlete is the laboratory": every block is an experiment, and the results accumulate.

KEY IDEA

Train, measure, understand and adapt is a loop. Skipping any step breaks the learning: training without measuring leaves you guessing, measuring without understanding leaves you with noise, and understanding without adapting changes nothing.

Speed of the loop

Different questions run on different cycle lengths. Daily readiness decisions, the domain of AISRi, run on a loop of hours. Training-block decisions run over weeks. Questions about heat or altitude responses run over months. Questions about long-term development run over years. A good system keeps all of these loops turning without confusing them, so that a single bad night's sleep is not mistaken for a failed training philosophy.

Where things go wrong

The loop fails in predictable ways. Some coaches train hard but measure nothing, so they cannot learn from the season. Some collect vast amounts of wearable data but never interpret it. Some interpret confidently but never change their minds. And some adapt constantly, changing plans every week in response to noise. The discipline is to move through all four steps with patience, allowing enough time for training to take effect before judging it.

An example loop

Consider a club runner in Chennai preparing for a half marathon. The coach plans a four-week block of steady runs and one weekly threshold session (Train). Each week they record heart rate at a fixed easy pace on the same route before 6:30 a.m., along with WBGT, sleep and RPE (Measure). After four weeks, heart rate at the fixed pace has fallen by more than the typical day-to-day variation, while threshold sessions feel easier; the coach notes that two of the hottest mornings produced outlying values and excludes neither but annotates both (Understand). The next block increases threshold volume slightly and adds a durability probe to the long run (Adapt). Nothing about this is complicated, but it turns a season of running into a season of learning.

The loop for the coach

The same four words apply to coaches themselves. A coach trains through practice, measures through records and reflection, understands through study, mentoring and conversation with scientists and clinicians, and adapts by changing methods when evidence demands it. A coach who stops learning eventually coaches the same year over and over again. The AKURA cycle is meant to keep both athlete and coach growing, season after season, so that experience becomes expertise rather than habit.

COACH'S CORNER

At the end of each training block, hold a fifteen-minute review with the athlete using the four words as headings: what did we train, what did we measure, what do we understand now, and what will we change?

CHECKPOINT

Which of the four steps is weakest in your current coaching practice? What one habit would strengthen it?

CHAPTER 84

84. The Real Goal of AEI

PART XVI — PHILOSOPHY

In 2011, a schoolgirl from Iklauta village near Meerut lined up for an 800 m race without ever having trained. "Papa asked me if I wanted to run and I agreed. I had never practised before that school race," Parul Chaudhary told The Indian Express years later. She ran barefoot that day, and afterwards trained on the paths between sugarcane fields. Fifteen years later, on 27 September 2026 in Nagoya, she ran the steeplechase in 9:08.67 for Asian Games bronze and a national record, and followed it with bronze in the 5000 m. Between those two moments lie several coaches, a railway job, a police job, a training base in Colorado Springs and seven national records in five seasons. No single number from that journey explains it. What explains it is a long accumulation of learning about one particular body, by the athlete and by the people around her.

It is tempting for any new index to chase attention: a single number that can be compared, ranked and shared. That is not the purpose of the AKURA Endurance Index. The real goal is not to produce a famous number. The goal is to help athletes become more understandable to themselves and their coaches. An athlete who knows why a race went well or badly, which environments suit them, how they respond to heat or

altitude, and what their early warning signs of overload look like is better equipped for a long career than one who simply knows a score.

What success would look like

If AEI eventually becomes a validated index, its success should be measured by whether it improves training decisions, reduces avoidable overload, identifies useful interventions and helps athletes perform more consistently. These are practical outcomes that can themselves be studied. Do coaches using AEI make decisions they later judge to be better? Do athletes experience fewer periods of unplanned fatigue or injury? Do interventions such as heat blocks or altitude camps become better targeted? Do race performances become more predictable? If the answer to these questions is no, the index has not yet earned its place, however elegant its mathematics.

The number and the system

The number is the interface. The learning system is the product. A score is only a convenient way to summarise a large body of information. The real value lies underneath: the athlete card with traceable raw data, the standardised protocols, the environmental logs, the coach's annotations and overrides, and the accumulating record of how each athlete responds. If the score were removed tomorrow, a well-built AKURA system should still help coaches coach better.

KEY IDEA

AEI succeeds not when its number becomes famous but when athletes and coaches understand the body better and make better decisions because of it.

A national opportunity

India has vast numbers of endurance athletes training across extraordinary diversity of climate, altitude, culture and circumstance: coastal heat in Chennai and Mumbai, hill stations in the Nilgiris and the Himalaya, dry heat in the north-west, humid monsoon seasons almost everywhere. Very little of this natural variation has been studied systematically. A carefully built, ethically run, openly reported research system could turn this diversity into knowledge that benefits Indian athletes first and contributes to global endurance science as well.

Staying honest

The final responsibility of AEI is honesty. It must report uncertainty, admit when it is wrong, allow the data to disagree with its designers, and keep the athlete's health and dignity above any performance goal. It should be open to independent validation and willing to be revised or even replaced if better methods emerge. That humility is not weakness; it is what makes a research system trustworthy.

What AEI must never become

It is equally important to state what AEI should not be. It should not become a selection tool that excludes young athletes on the basis of a single test. It should not be used to label athletes by region, background or appearance. It should not replace a coach's relationship with an athlete, or a clinician's judgement about health. And it should not be used to pressure athletes into training through illness or injury because "the numbers look fine". Any use of the index that reduces an athlete's dignity, safety or opportunity is a misuse, regardless of the data behind it.

An invitation

This edition is a prototype, and prototypes are meant to be tested, broken and rebuilt. Coaches who try the protocols, athletes who keep careful logs, scientists who challenge the assumptions and clinicians who point out safety gaps all become part of the research. The long-term value of AEI will depend less on the ideas written here than on the quality of the data and the honesty of the people who collect and interpret it. If that community forms, the index will improve with every season, and Indian endurance athletes will have a research system built around their own bodies, climates and ambitions.

COACH'S CORNER

When you share any AEI or readiness number with an athlete, always pair it with an explanation of what it means for their training this week. A number without a conversation teaches nothing.

CHECKPOINT

If the AEI score disappeared tomorrow, what would your athletes still know about themselves because of the work you have done together?

PART XVI AT A GLANCE

The key idea from each chapter, for quick revision.

82

From Myth to Measurement

A myth becomes useful the moment it is turned into a hypothesis that could be proved wrong.

83

Train. Measure. Understand. Adapt.

Train, measure, understand and adapt is a loop. Skipping any step breaks the learning: training without measuring leaves you guessing, measuring without understanding leaves you with noise, and understanding without adapting changes nothing.

84

The Real Goal of AEI

AEI succeeds not when its number becomes famous but when athletes and coaches understand the body better and make better decisions because of it.

PART XVII — THE AKURA TOOLKIT

This Part turns the ideas of the book into tools a coach can use on Monday morning. It gives AKURA pace and race-equivalence charts, explains how to adjust them for Chennai heat and hill-station altitude, and sets out three roadmaps: a 24-week marathon roadmap, a long-term athlete development roadmap, and the research roadmap that will take the AKURA Endurance Index from prototype to validated tool. Every chart is a starting point, to be checked against the athlete's own heart rate, effort and response.

IN THIS PART

- 85** The AKURA Pace Charts
- 86** Adjusting Paces for Heat and Altitude
- 87** The 24-Week Marathon Roadmap
- 88** The Athlete Development Roadmap
- 89** The AEI Research Roadmap

PART XVII • THE AKURA TOOLKIT

From One Race to Every Training Pace

Pace and race-equivalence charts, heat and altitude adjustments, and three roadmaps — always checked against HR and effort.

3:08/km

Thonakal Gopi's average pace for 2:12:23, Valencia 2025

20:00

Equivalent performances

A 20:00 5K equates to about 41:42 for 10K, 1:32:00 half marathon and 3:11:49 marathon.

4:15

Training paces

From a 20:00 5K: aerobic 5:19–6:05, marathon 4:33, threshold 4:15, power 3:52 min/km.

WBGT

Adjust for heat

WBGT 18–23: easy runs often 2–4% slower; 23–28: 5–8% slower; above 28: short, very easy or indoors.

Days 1–4

Adjust for altitude

Days 1–4 by effort only, no threshold or speed; from day 11, consider descending for key fast sessions.

24 wk

Marathon roadmap

Foundation, aerobic build, specific strength, race-specific, taper: each phase ends with a test.

5 + 5

Two more roadmaps

Athlete development: foundation to high performance. AEI research: data foundation to external validation.

THE PROCESS

Foundation

Aerobic build

Specific strength

Race-specific

Taper, race

Charts are a starting point: in heat, hold the effort and let the pace fall.

THE STORY

Three Minutes Eight Seconds

PART XVII — THE AKURA TOOLKIT

Valencia, 7 December 2025. The Spanish city's marathon course is famous for being flat and fast, and on this morning Thonakal Gopi is running the race of his life. He crosses the line in 2:12:23. It is a personal best. It is also 23 seconds outside the national record of 2:12:00 that Shivnath Singh set at Jalandhar in 1978, a record that has outlived almost every runner who ever chased it.

Take a calculator and divide 2:12:23 by 42.195. The answer is about 3:08 per kilometre. Not one fast kilometre: forty-two of them, one after another, each covered in about three minutes and eight seconds. Most club runners in India would struggle to hold that pace for one lap of a track. Gopi held it for more than two hours.

This is how coaches think about races. Not as a single time, but as a pace, a rhythm, a number per kilometre that can be trained, tested and planned. It is exactly how this Part of the book thinks too.

From Wayanad

Gopi comes from Sulthan Bathery in Wayanad, Kerala, a hill town about 1,000 metres above sea level. Picture the district in the monsoon: rain drumming on the pepper vines, red earth roads running between paddy fields and forest. His parents were farmers who grew rice and ginger. He joined the Army through the sports quota and became a Havildar in the Artillery regiment, attached to the Army Sports Institute in Pune.

His events were the 10,000 m and the marathon. Under Army coach Surendra Singh Bhandari he spent what ESPN described as a ten-month training stint at a high-altitude facility in Ooty before the 2016 Rio Olympics. In Rio he finished 25th in 2:15:25, then a personal best, one second ahead of his Army teammate Kheta Ram.

Champion of Asia

The following year, at the Asian Marathon Championship in Dongguan, China, Gopi won gold in 2:15:48. No Indian had won the title before him. The next Indian to win it was Man Singh, in Hong Kong in 2024.

The wins kept coming at home. In January 2018 he won the Indian elite race at the Mumbai Marathon in 2:16:51, with Nitendra Singh Rawat second in 2:16:54. In Seoul in 2019 he ran 2:13:39, a new personal best. At Mumbai in 2023 he was again the first Indian man, in 2:16:41.

Then came Valencia. For that race he went back to the Nilgiris, spending four months training at Ooty, while Bengaluru had been his base for the previous six years. Afterwards he said simply: "I think the high altitude at Ooty helped my performance."

Four months after Valencia, in April 2026, he ran the Rotterdam Marathon and finished 23rd in 2:13:16, well inside the Athletics Federation of India's Asian Games standard of 2:15:04. In the same race, Sawan Barwal ran 2:11:58 on his marathon debut and finally broke Shivnath Singh's record, by two seconds. The old mark that Gopi had missed by 23 seconds in December fell to a younger man in April. That is how records usually fall: not to one hero but to a generation pushing on the same door.

A career as a pace chart

Now do what this Part asks every coach to do. Turn Gopi's results into paces.

Rio 2016, 2:15:25: about 3:13 per kilometre. Dongguan 2017, 2:15:48: also about 3:13. Seoul 2019, 2:13:39: about 3:10. Valencia 2025, 2:12:23: about 3:08. Rotterdam 2026, 2:13:16: just under 3:10.

Across nine years, the average pace moved by about four and a half seconds per kilometre. That sounds like nothing; across 42 kilometres it is about three minutes. For an elite marathoner, that is the difference between being a good national runner and knocking on the door of a record nearly 48 years old. The improvement did not arrive in a single leap. It came in small steps, separated by years, with results that went up and down along the way. A coach who looked only at Rotterdam, a slower run than Valencia, might have called it a decline. A coach who looked at the whole line would see a runner still holding paces that, in 2016, had been beyond him.

Chart 1 in the next chapter uses a well-known formula, Riegel's endurance equation, to translate a 5K time into equivalent performances at longer distances. Run the arithmetic backwards from Gopi's 2:12:23 and the model says it corresponds to a 5K of roughly 13:48. That is not Gopi's actual 5K time; it is simply what the formula implies. It is also well off the top of the chart, which begins at 15:00, because the chart is built for club runners and developing athletes, not for Asian champions. Even for elites, though, the logic is the same. A marathon pace is a fraction of what the athlete can do over shorter distances, and how large that fraction is depends on durability, economy, fuelling and heat.

A career as a roadmap

Gopi's preparations also read like roadmaps. Ten months at Ooty before Rio. Four months at Ooty before Valencia, followed four months later by Rotterdam. Sawan Barwal's group, according to The Bridge, normally follows a similar rhythm: about four months at Ooty and Wellington from November to February, then pre-competition work in Bengaluru. Each is a sequence of phases: a long base, a specific block, a taper, a race and then a review before the next cycle begins.

The 24-week marathon roadmap in Chapter 87 is a shorter, simpler version of the same idea, built for runners who have jobs, families and one target race a year. The athlete development roadmap in Chapter 88 is the long version, the one that describes how a boy from a Wayanad farm becomes an Army runner, then an Olympian, then champion of Asia. And the research roadmap in Chapter 89 is about making sure that the next time an Indian runner spends four months at Ooty and runs a personal best, we know more precisely what the mountain contributed.

What the charts are for

A pace chart will not make anyone a champion. Gopi's 3:08 per kilometre rests on years of Army training, altitude blocks, coaches, races won and races lost. But charts do something important: they make training honest. They turn a vague intention such as "run easy" or "do some tempo" into a number that can be checked against heart rate, effort and the weather, and then corrected.

Charts also need adjusting, and in India that usually means heat. A pace that is comfortable on a cool December morning in the Nilgiris can be dangerous on a humid May morning in Chennai. Chapter 86 explains how to let the pace fall while the effort stays the same, and why a heart-rate ceiling and a stop rule matter more than any number on a chart. Altitude changes the arithmetic too: the first days in Ooty are slower, and a runner who chases sea-level splits up there is fighting the mountain rather than learning from it. An elite marathoner and a first-time half-marathoner face the same physics. The difference is only in the numbers. The principle, that the body sets the pace and the chart is only the starting point, applies to both.

Somewhere in Kerala tonight, a young runner will look at a race result and wonder what it means. The tools in this Part are built to answer that question, on an ordinary Monday morning, for an ordinary athlete with extraordinary hopes. We start with the charts.

Sources: <https://shyamgopan.com/2025/12/11/gopi-23-secs-short-of-national-record-sets-new-pb-at-valencia-marathon/> ; <https://www.indiasportshub.com/articles/gopi-thonakals-valencia-marathon-run-rekindles-indian-marathon-hope> ; https://www.espn.com/athletics/story/_/id/17475379/india-marathon-runner-olympics-thonackal-gopi-talks-espn-exclusive-interview ; <https://www.facebook.com/Indianarmy.adgpi/posts/havildar-gopi-thonakal-of-indianarmy-army-sports-institute-asi-pune-won-the-16th/736997223163576/> ; <https://www.firstpost.com/sports/rio-olympics-2016-thonackal-gopi-and-kheta-ram-clock-personal-best-to-finish-25th-26th-in-marathon-2969066.html> ; <https://www.olympics.com/en/news/asian-marathon-championships-2024-india-results> ; <https://scroll.in/field/865807/mumbai-marathon-on-the-road-to-redemption-nitendra-singh-rawat-puts-up-a-stellar-performance> ; <https://www.olympics.com/en/news/mumbai-marathon-2023-results-winners> ; <https://www.olympics.com/en/news/sawan-barwal-breaks-marathon-national-record-india> ; <https://thebridge.in/athletics/ajith-markose-decode-sawan-barwal-48-year-old-record-breaking-run-56362>

CHAPTER 85

85. The AKURA Pace Charts

PART XVII — THE AKURA TOOLKIT

Rotterdam, 12 April 2026. Sawan Barwal, a soldier from Mandi district in Himachal Pradesh, is running his first marathon. He finishes 20th in 2:11:58, and a national record that had stood since Shivnath Singh ran 2:12:00 at Jalandhar in 1978 is gone by two seconds. Divide his time by 42.195 and you get the number every coach instinctively reaches for: about 3:08 per kilometre, held for more than two hours. In September, at the Asian Games in Nagoya, he lowers the record to 2:11:37 and wins silver. "There was only one thing in my mind: I have to go till the finishing line and how I can maintain the pace," he told The Tribune. Maintain the pace. For a national record holder and for a beginner in Anna Nagar alike, training starts with the same question: what pace, and why that one?

Every coach eventually needs a simple table: "My athlete just ran this race. What paces should they train at?" Pace charts such as Jack Daniels' VDOT tables have helped a generation of runners answer that question. AKURA's charts serve the same practical purpose, but they are built from a transparent, openly stated model so that any coach can check the arithmetic, and they are designed to be adjusted by the environment and readiness layers described in earlier chapters.

How the charts are calculated

The charts start from one recent, honest race result, ideally a 5K run on a flat course in cool conditions. From that anchor, equivalent performances at other distances are estimated with Riegel's well-known endurance formula (Riegel PS, Athletic records and human endurance, American Scientist, 1981): $\text{predicted time} = \text{known time} \times (\text{new distance} \div \text{known distance})^{1.06}$. The same relationship is used to estimate the pace an athlete could hold for a given duration, which gives the training paces.

- **Threshold pace** is the estimated pace the athlete could race for about 60 minutes. It should feel controlled-hard.
- **Power pace** is the estimated pace for an all-out effort of about 11 minutes, used for VO₂max-type intervals of 2–5 minutes.
- **Speed** is given as a 400 m repetition time, based on the pace for an all-out effort of about 5 minutes, used with full recovery.
- **Marathon pace** is the Riegel marathon prediction divided by 42.195 km.
- **Recovery, Aerobic and Endurance** paces are set as percentages of threshold speed: Recovery slower than 70%, Aerobic 70–80%, and Endurance 80–88%.

KEY IDEA

AKURA pace charts turn one recent race into a full set of training paces using a published, transparent formula. They are a starting point: heart rate, effort, heat and readiness decide the final pace on the day.

Chart 1: Equivalent race performances

Find the athlete's current 5K time in the first column. The other columns show what the model predicts at longer distances for a runner whose endurance is well developed for that distance.

5K	10K	Half marathon	Marathon
15:00	31:16	1:09:00	2:23:52
16:00	33:22	1:13:36	2:33:27
17:00	35:27	1:18:12	2:43:03
18:00	37:32	1:22:48	2:52:38
19:00	39:37	1:27:24	3:02:14
20:00	41:42	1:32:00	3:11:49
21:00	43:47	1:36:36	3:21:25
22:00	45:52	1:41:12	3:31:00
23:00	47:57	1:45:48	3:40:36
24:00	50:02	1:50:24	3:50:11
25:00	52:07	1:55:00	3:59:47
26:00	54:12	1:59:36	4:09:22
27:00	56:18	2:04:12	4:18:58
28:00	58:23	2:08:48	4:28:33
29:00	1:00:28	2:13:24	4:38:09
30:00	1:02:33	2:18:00	4:47:44
32:00	1:06:43	2:27:12	5:06:55
34:00	1:10:53	2:36:24	5:26:06
36:00	1:15:03	2:45:36	5:45:17
38:00	1:19:14	2:54:48	6:04:28
40:00	1:23:24	3:04:00	6:23:39

The marathon column assumes marathon-specific training: regular long runs and enough weekly volume. Runners on lower mileage, beginners and athletes racing in

heat usually run slower than predicted, often by 5–10% or more. This gap reflects the durability limits described in Chapter 36; the difference between an athlete's 5K prediction and actual marathon result is useful data, not a failure.

Chart 2: AKURA training paces (min/km)

5K	Recovery (slower than)	Aerobic	Endurance	Marathon	Threshold	Power	Speed 400 m
15:00	4:38	4:03–4:38	3:41–4:03	3:25	3:15	2:57	1:08
16:00	4:56	4:19–4:56	3:55–4:19	3:38	3:27	3:08	1:12
17:00	5:13	4:34–5:13	4:09–4:34	3:52	3:39	3:19	1:16
18:00	5:30	4:49–5:30	4:23–4:49	4:05	3:51	3:30	1:20
19:00	5:48	5:04–5:48	4:37–5:04	4:19	4:03	3:41	1:25
20:00	6:05	5:19–6:05	4:50–5:19	4:33	4:15	3:52	1:29
21:00	6:22	5:34–6:22	5:04–5:34	4:46	4:27	4:03	1:33
22:00	6:39	5:49–6:39	5:18–5:49	5:00	4:39	4:14	1:37
23:00	6:56	6:04–6:56	5:31–6:04	5:14	4:51	4:25	1:41
24:00	7:13	6:19–7:13	5:45–6:19	5:27	5:03	4:36	1:45
25:00	7:30	6:34–7:30	5:58–6:34	5:41	5:15	4:46	1:50
26:00	7:47	6:49–7:47	6:12–6:49	5:55	5:27	4:57	1:54
27:00	8:04	7:04–8:04	6:25–7:04	6:08	5:39	5:08	1:58
28:00	8:21	7:19–8:21	6:39–7:19	6:22	5:51	5:19	2:02
29:00	8:38	7:33–8:38	6:52–7:33	6:36	6:03	5:29	2:06
30:00	8:55	7:48–8:55	7:05–7:48	6:49	6:14	5:40	2:10
32:00	9:28	8:17–9:28	7:32–8:17	7:16	6:38	6:01	2:18
34:00	10:02	8:47–10:02	7:59–8:47	7:44	7:01	6:23	2:26
36:00	10:35	9:16–10:35	8:25–9:16	8:11	7:25	6:44	2:35
38:00	11:08	9:45–11:08	8:52–9:45	8:38	7:48	7:05	2:43
40:00	11:42	10:14–11:42	9:18–10:14	9:06	8:11	7:26	2:51

COACH'S CORNER

Use the chart to set the first week of training, then let the athlete's response correct it. If heart rate at "Aerobic" pace drifts well above the athlete's usual easy range, or the threshold session feels like a race, the chart is too fast for this athlete today. Re-anchor the chart after every honest race or time trial, roughly every 6–8 weeks.

Using the charts safely

A chart is only as good as its anchor. An old race, a downhill course, a race run on a cool December morning or a time trial run while unwell will all give misleading paces. Where the athlete has no recent race, use a 5K time trial on a flat measured course, run early in the morning, after two easy days.

The easy paces are deliberately wide. On hot days, after poor sleep or in the first days at altitude, athletes should run towards the slow end of the range or slower. Easy running is defined by effort and heart rate first; the pace column is a guide, not a target.

FIELD PROTOCOL

1. Setting an athlete's paces
2. Run a 5K time trial on a flat, measured course in the coolest part of the day
3. Record time, temperature, humidity and average heart rate
4. Find the nearest 5K time in Chart 2 and write the paces into the athlete's AEI card
5. For the first two weeks, check heart rate and RPE in every session against the prescribed domain
6. Adjust the paces for heat and altitude using Chapter 86
7. Retest after 6–8 weeks or after a race

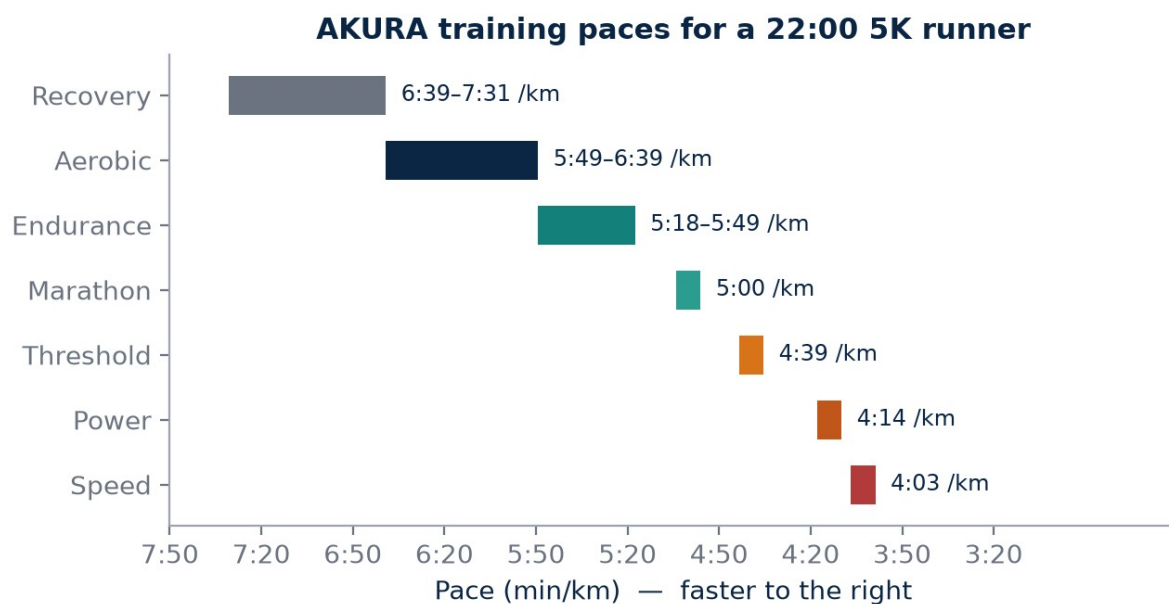


Figure 29. Example AKURA training paces for an athlete with a 22:00 5K. Easy domains are deliberately wide; the harder domains are narrow and used in smaller doses.

MYTH VS EVIDENCE

Myth: Faster training paces make faster racers.

Evidence: Training much faster than the chart usually adds fatigue without adding adaptation. Most improvement comes from consistent Aerobic and Endurance work plus well-judged Threshold sessions.

CHECKPOINT

Take one of your athletes. What 5K time would you use as their anchor, how recent is it, and in what conditions was it run? How confident are you in the paces it produces?

CHAPTER 86**86. Adjusting Paces for Heat and Altitude**

PART XVII — THE AKURA TOOLKIT

Bengaluru, April 2026. The TCS World 10K is run in hot, humid conditions, and the Indian women's race is won by Sanjivani Jadhav in 35:01, her third straight title. Her own event record, set in 2018, is 33:38. Nobody can say from the result sheet how much of that 83-second gap belongs to the weather and how much to fitness, tactics or the eight years between the two races. That uncertainty is exactly the problem a coach faces every summer in Chennai. A runner who covered an aerobic loop at 5:30 per kilometre in January covers the same loop at 6:00 in May and comes home convinced she is losing fitness. Often she is not. She is paying the price of running in heat, and that price is paid in pace. The question is how to set training paces that respect it.

A pace chart assumes a cool, still, sea-level day. Chennai in May is not that day, and neither is the first week in Ooty. This chapter gives AKURA's working adjustments. They are practical coaching guidance drawn from the physiology in Parts III and IV, not validated equations, and they are always overruled by heart rate, perceived effort and warning signs.

Heat and humidity

Marathon results slow progressively as environmental heat stress rises, and slower runners, who are exposed for longer, tend to be affected more. In humid heat, sweat evaporates poorly, so heart rate climbs at the same pace. The practical response is to hold effort, not pace.

WBGT (°C)	Easy and endurance runs	Threshold and interval sessions	Races and long runs
Below 18	Use chart paces	Use chart paces	Use chart paces
18–23	Run by heart rate; often 2–4% slower	Slow by about 2–4%, or shorten reps	Expect a small slowdown; drink to plan
23–28	Run by heart rate; often 5–8% slower	Move to cooler hours, cut volume, slow by 5% or more	Reduce goals; heat-acclimated athletes only
Above 28	Move indoors or rest; outdoors only short, very easy and supervised	Postpone	Follow medical and event guidance; consider not racing

On a humid Chennai evening, a 5:30/km aerobic run might become 5:45–6:00/km at the same heart rate. That is not lost fitness; it is the price of thermoregulation. The AEI records the environment so that the slower pace is interpreted correctly.

KEY IDEA

In heat, hold the effort and let the pace fall. A heart-rate ceiling and a stop rule protect the athlete better than any pace target.

Altitude

At the moderate altitudes of Ooty and Kodaikanal, oxygen pressure is roughly a quarter lower than at sea level. Easy running feels harder and fast intervals are noticeably slower in the first days.

Camp phase	Easy running	Threshold	Power and speed
Days 1–4	By effort only; expect clearly slower paces	Avoid	Avoid
Days 5–10	By effort; paces begin to return	By effort; typically a few per cent slower than sea-level pace	Short reps with long recovery; do not chase sea-level times
Days 11 onwards	Close to normal effort	By heart rate and RPE	Consider descending to lower ground for key fast sessions
Return to sea level	Normal	Normal after a few days	Retest before racing

COACH'S CORNER

Write two numbers on every hot-weather or altitude session plan: the heart-rate ceiling and the stop rule. For example: "Easy 50 minutes, heart rate below 150; stop and cool down if dizzy, confused, nauseous or if sweating stops." The pace is recorded afterwards, not chased during the run.

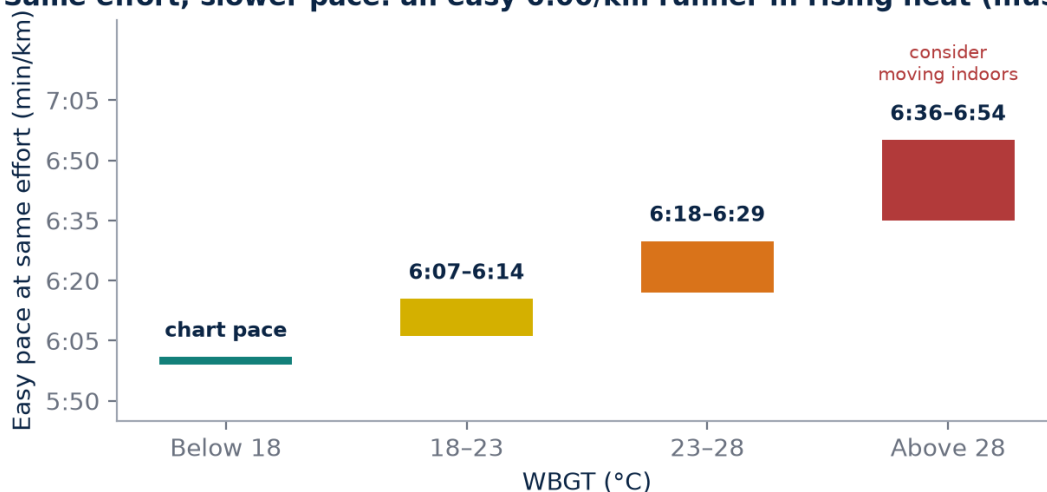
Same effort, slower pace: an easy 6:00/km runner in rising heat (illustrative)

Figure 30. Illustrative only: how the same easy effort translates into slower paces as WBGT rises. Use the athlete's own heart rate and response to set the real adjustment.

CHECKPOINT

What heart-rate ceiling would you give your athletes for easy runs in May, and what is your stop rule?

CHAPTER 87**87. The 24-Week Marathon Roadmap**

PART XVII — THE AKURA TOOLKIT

The record Sawan Barwal broke in Rotterdam was built on a calendar. According to The Bridge, his group in Reliance Foundation's Project 2:09, coached by Ajith Markose, normally spends about four months, from November to February, at Ooty and Wellington in the Nilgiris, then moves to Bengaluru for pre-competition work. "For Sawan, most of the time he was staying in high altitude," Markose told The Bridge. The Tribune described a load of about 200 kilometres a week, with 40–42 km long runs on alternate weeks. Those numbers belong to a professional athlete

training full-time, and no club runner should copy them. What any runner can copy is the structure: a long base in one place, a sharper block in another, a deliberate sequence of phases, each with its own purpose, all pointing at a single race date circled months in advance.

A marathon is built over months, not weeks. The AKURA marathon roadmap divides 24 weeks into five phases, each with a clear purpose, a key session type and a planned retest. The roadmap is a map, not a timetable: each phase ends only when the athlete has absorbed it, and the AEI trend decides whether to move on, repeat or reduce.

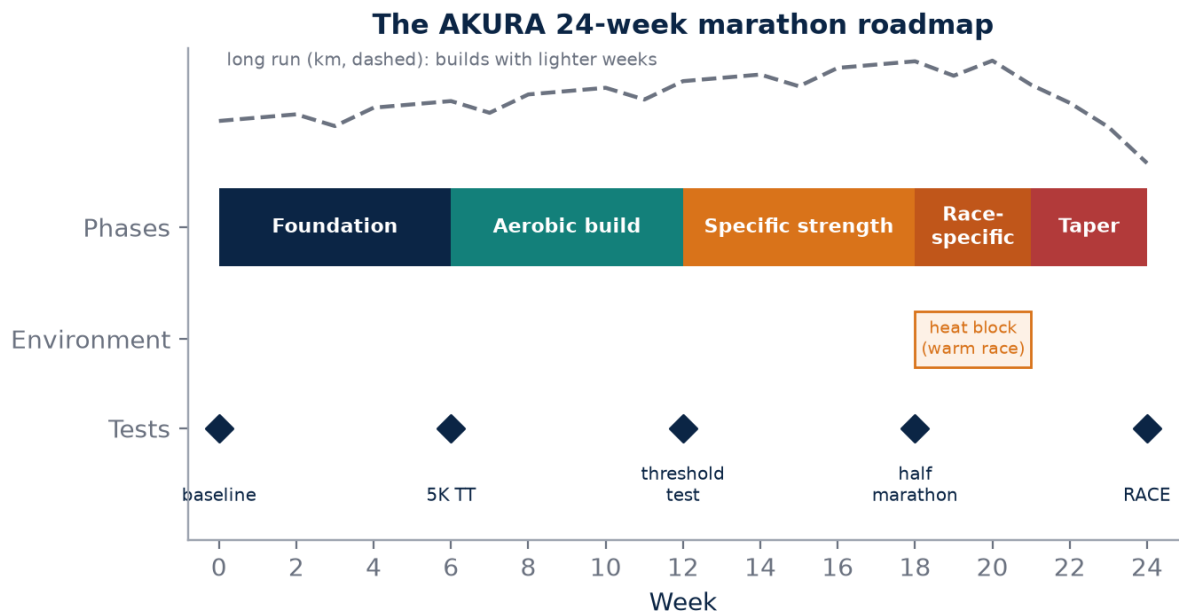


Figure 31. The AKURA 24-week marathon roadmap: five phases, four testing points, a planned heat block for a warm-weather race and a progressive long run.

The five phases

Weeks	Phase	Main purpose	Key sessions	Test at end
1–6	Foundation	Consistency, aerobic volume, strength habit	Aerobic runs, strides, strength twice a week	5K time trial; easy pace at fixed HR
7–12	Aerobic build	Raise volume and long-run length	Long run, Endurance runs, one Threshold session	Threshold test; long-run decoupling
13–18	Specific strength	Sustain higher fractions of capacity	Threshold, marathon-pace blocks in long runs	Half-marathon or 15K race
19–21	Race-specific	Rehearse race pace, fuelling and heat plan	Long runs with marathon-pace finish; heat sessions if race is warm	Final long run review
22–24	Taper and race	Arrive fresh, keep sharpness	Reduced volume, short marathon-pace efforts	Race

Progression rules

- Increase the long run gradually, typically by 2–3 km every one to two weeks, with a lighter week every third or fourth week.
- Add only one new stress at a time: volume, intensity, heat or hills.
- If AEI-Readiness is repeatedly low, or pace-HR decoupling on long runs worsens, hold the current week rather than progressing.
- Practise race-day fuelling in every long run from week 13 onwards.
- For a warm race, schedule a controlled 10–14 day heat-acclimation block in weeks 19–21, as described in Chapter 20, not in the final week.

KEY IDEA

The roadmap moves forward only when the athlete has absorbed the current phase. Testing points turn the plan into a feedback loop.

COACH'S CORNER

Put the four testing weeks into the calendar on day one and treat them as fixed appointments. The results are what allow you to adjust the plan with confidence rather than guessing.

An example: a Chennai club marathoner

Consider an illustrative athlete with a 25:00 5K preparing for a January marathon in Chennai. Chart 2 suggests Aerobic runs around 6:34–7:30/km, Threshold around 5:15/km and marathon pace around 5:41/km. Because the Foundation and Aerobic phases fall in the monsoon and early winter, most sessions are run early in the morning and by heart rate. The heat block is unnecessary for a cool January start but essential if the athlete is racing a warm event. The half-marathon test in week 18 is used to re-anchor paces for the final phase.

CHECKPOINT

Where would your next target race fall in this roadmap if you started today? Which phase would you have to shorten, and what risk does that create?

CHAPTER 88

88. The Athlete Development Roadmap

PART XVII — THE AKURA TOOLKIT

"I truly learned about competitive running only after joining the army," Gulveer Singh has said. He grew up in a farming family in Aligarh district, Uttar Pradesh, and joined the Army through the sports quota in 2018. Seven years later he ran 27:00.22 for 10,000 m, a national record, and in 2026 he won three Asian Games medals. Avinash Sable, from a farming family in Beed district, Maharashtra, joined the Army after Class 12 and took up running through Army cross country in 2015. An Army coach moved him to the steeplechase in 2017, and in Paris in 2024 he became the first Indian to reach the Olympic final in the event. Both stories unfold in stages: finding the sport, training consistently, receiving structured coaching, then racing internationally with specialist support. The order of those stages mattered as much as the speed at which they were climbed.

A world-class endurance athlete is built over many years. The AKURA development roadmap describes five stages, from a school or beginner runner to a performance athlete. Each stage has its own training priorities and, just as importantly, its own level of measurement. Measuring too much too early can be as harmful as measuring too little too late.

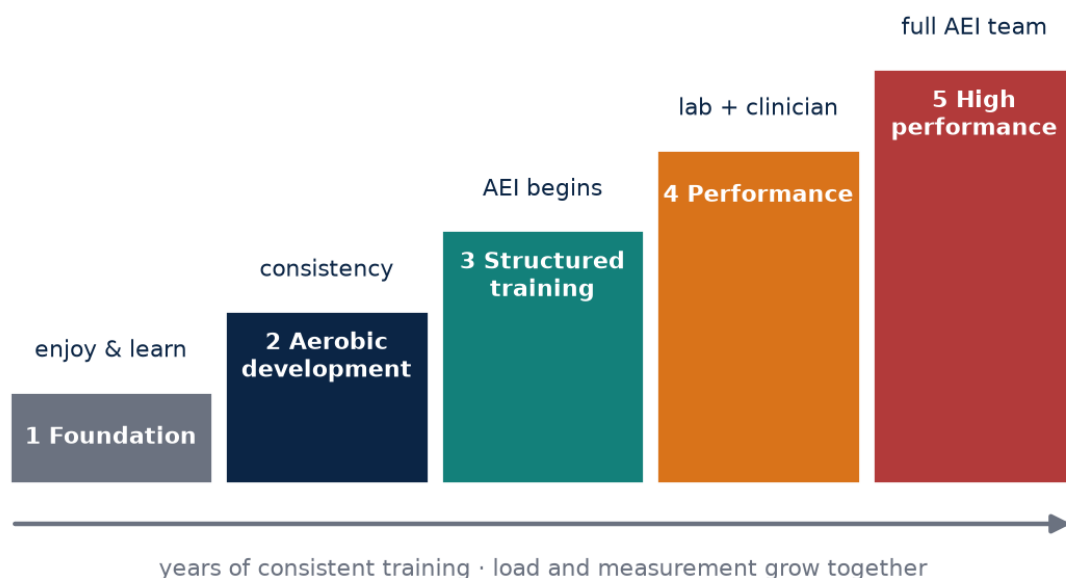


Figure 32. The AKURA athlete development roadmap: five stages in which training load and measurement depth grow together.

Stage	Typical athlete	Training priority	Measurement level
1 Foundation	School runner or adult beginner	Enjoyment, running skill, all-round athleticism	Race times, simple logs, wellbeing
2 Aerobic development	Regular club runner	Consistent aerobic volume, strength basics	Pace-HR, sleep, RPE, 5K tests
3 Structured training	Competitive club or college runner	Threshold work, long runs, planned progression	AEI-Fitness and Readiness, heat logging
4 Performance	State or national-level athlete	Specific preparation, heat and altitude blocks	Lab tests, blood markers with clinician
5 High performance	International-level athlete	Individualised, multidisciplinary support	Full AEI profile with medical team

Moving between stages

Athletes move up when they can absorb the current stage consistently for months, not when they run one good race. Age matters too: young athletes should build broad athletic ability and enjoyment before specialising in high volume. For adults who start late, the stages still apply, but progress is limited more by tissue tolerance and life load than by physiology.

KEY IDEA

Training load and measurement should grow together. Each stage asks a different question, and the AEI adds detail only when the athlete is ready to use it.

COACH'S CORNER

Write each athlete's stage on their AEI card. It prevents two common mistakes: giving a beginner a laboratory test battery they do not need, and coaching a national-level athlete with nothing more than a stopwatch.

CHECKPOINT

Which stage is each of your athletes at? What single step would move each of them to the next stage?

CHAPTER 89

89. The AEI Research Roadmap

PART XVII — THE AKURA TOOLKIT

In 1991, a team of physiologists published a study in the Japanese Journal of Physiology on 67 Bengali female athletes from different sports, tested on a cycle ergometer. The long-distance runners had the highest relative VO_2max in the group: 43.0 ml/kg/min. Four years later, Saltin and colleagues reported Kenyan senior runners averaging about 79.9 ml/kg/min at sea level. The comparison is unfair in many ways: different decades, different athletes, different equipment, different levels of competition. That is precisely the point. Published physiological data on Indian endurance runners are scarce, scattered and hard to compare with anything. In April 2025 India inaugurated its National Centre of Sports Science and Research in New Delhi. The laboratories now exist. What is still missing is a shared method: consistent data, reliable tests, and models that are checked against reality before anyone is asked to trust them.

The AKURA Endurance Index is presented in this book as Prototype Edition 0.1. The research roadmap sets out how it can become a validated tool. Each phase has a clear output and a clear test, and the index should not be used for high-stakes decisions until the later phases are complete.

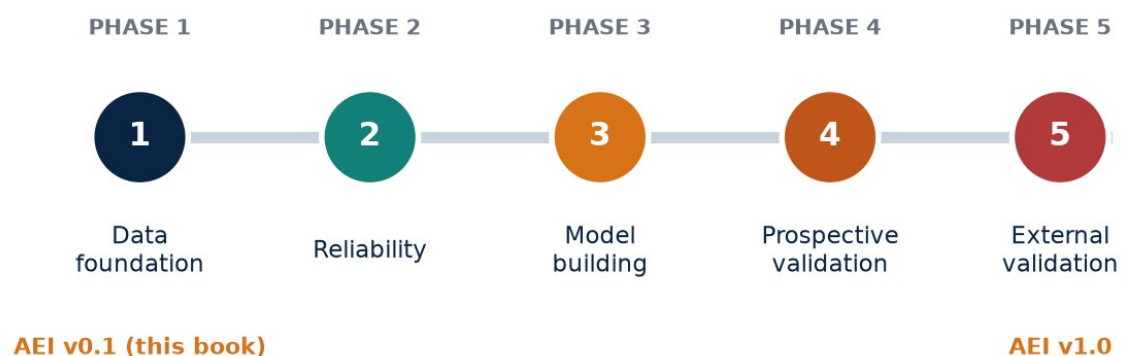


Figure 33. The AEI research roadmap, from prototype data schema to externally validated index. Timings are targets and depend on data quality and collaboration.

Phase	Goal	Main output	Success test
1 Data foundation	Collect clean, consistent data	AEI card, data fields, quality rules	Complete records for a pilot group
2 Reliability	Show that measurements repeat	Typical error for each test	Known smallest worthwhile change
3 Model building	Learn domain weights from data	First calibrated AEI model	Beats race time alone on held-out data
4 Prospective validation	Test predictions in new seasons	Validation report	Good calibration and useful decisions
5 External validation	Test in other cities and groups	AEI version 1.0	Works beyond the original group

What will count as success

The AEI will be judged by the tests set out in Chapter 68. Does it predict outcomes better than race performance alone? Does it help coaches make better decisions, reduce avoidable overload and identify useful interventions? If a domain adds nothing, it should be removed. If the index works in Chennai but not in Delhi or Kodaikanal, the model must be revised.

KEY IDEA

AEI becomes a validated index only by passing each research phase in order. Until then, report the domains separately, with their confidence.

COACH'S CORNER

Coaches can contribute from day one. Every standardised test, logged session, environmental note and reasoned override on an AEI card becomes part of the evidence base that will validate or correct the index.

CHECKPOINT

What data could your group collect this season, with the tools you already have, that would help move the AEI from Phase 1 to Phase 2?

PART XVII AT A GLANCE

The key idea from each chapter, for quick revision.

- | | |
|-----------|--|
| 85 | The AKURA Pace Charts
AKURA pace charts turn one recent race into a full set of training paces using a published, transparent formula. They are a starting point: heart rate, effort, heat and readiness decide the final pace on the day. |
| 86 | Adjusting Paces for Heat and Altitude
In heat, hold the effort and let the pace fall. A heart-rate ceiling and a stop rule protect the athlete better than any pace target. |
| 87 | The 24-Week Marathon Roadmap
The roadmap moves forward only when the athlete has absorbed the current phase. Testing points turn the plan into a feedback loop. |
| 88 | The Athlete Development Roadmap
Training load and measurement should grow together. Each stage asks a different question, and the AEI adds detail only when the athlete is ready to use it. |
| 89 | The AEI Research Roadmap
AEI becomes a validated index only by passing each research phase in order. Until then, report the domains separately, with their confidence. |

PART XVIII — INDIA'S ELITE RUNNERS: AN AEI ANALYSIS

This Part turns the AKURA lens on India's own best distance runners. It maps where they come from, profiles the men and women who have carried the national records from 1500 m to the marathon, tracks how those records have moved or stood still, and measures how far they sit from world level. It then asks what the ten AEI domains can and cannot say about them from public information alone, and ends with a practical roadmap. Every fact comes from verified public sources; where the evidence runs out, the text says so and offers hypotheses rather than conclusions.

IN THIS PART

- 90 The Indian Distance Map
- 91 The Men: From 1500 m to the Marathon
- 92 The Women: Steeple to Marathon
- 93 Records Over Time
- 94 How Far from the World?
- 95 Through the AEI Lens
- 96 A Roadmap for the Next Indian Elite

PART XVIII • INDIA'S ELITE RUNNERS

India's Distance Runners Through the AEI Lens

Where India's elites come from, how records moved, how far they sit from world level, and what to measure next.

3–19% Gap between Indian national records and world records, by event

Army

The men's main pathway

For Indian men the Army is the main pathway: salary, sports-quota entry, ASI Pune, camp access.

Jobs

Women: the job as contract

Railways and police jobs act as the professional contract that allows women to train at elite level.

10 NRs

Records move in bursts

Sable broke Gopal Saini's 37-year-old 8:30.88 steeplechase record ten times, down to 8:09.91.

Late

Late developers

Several record holders reached elite level late; talent identification closing at 17 would have missed them.

19.1%

The marathon gap

Men's track events sit about 3–5% off world records; marathons and women's long events about 9–19%.

Last 10 km

Durability: the key limiter

For Indian marathoners, durability is the most likely limiter, and cheap to measure: late-race pace and HR decay.

THE PROCESS



India needs better links between the pathways it already has, not a new pathway.

THE STORY

The Soldiers' Road

PART XVIII — INDIA'S ELITE RUNNERS: AN AEI ANALYSIS

It is a little after dawn at an Army recruitment rally somewhere in the Deccan. The ground is dry and the air still cool. Several hundred young men in vests and cheap canvas shoes stand in lines, waiting for the 1.6 km run that will decide whether they get a uniform. Most of them are farmers' sons. Almost none of them have ever heard of a lactate threshold, a steeplechase barrier or a place called Colorado Springs. They are here for a salary, a pension, a bit of respect back in the village.

Somewhere in a line like this, in the years that follow, India's distance-running future is being quietly sorted.

That scene is imagined. What follows is not.

A boy from Beed

Avinash Sable was born on 13 September 1994 in Mandwa village, in the Beed district of Maharashtra. His parents were farmers. As a boy he is reported to have run about 6 km to school. After Class 12 he joined the Army, in 5 Mahar Regiment.

He did not join to become a runner. His postings took him to the Siachen glacier, to Sikkim, and to the Rajasthan desert, where temperatures have been reported at up to 50 °C. He came to competitive running only in 2015, through Army cross country. Around 2017, an Army coach named Amrish Kumar saw him and moved him into the 3000 m steeplechase, an event he had never run.

Within two years he was in a World Championships final. He went on to break the national record ten times, win Asian Games gold in 2023 and a Commonwealth Games silver in 2022, and become the first Indian to reach an Olympic steeplechase final, in Paris in 2024.

Pause on the sequence. Farm. Uniform. Glacier. Desert. Cross-country race. A coach who happens to be watching. Nobody designed that path for Avinash Sable. The Army simply put him in the places where it could happen.

"Only after joining the army"

Four years younger, and far to the north, Gulveer Singh was born on 1 June 1998 in Sirsa village, Aligarh district, Uttar Pradesh, into a farming family. He joined the Army through the sports quota in 2018, in The Grenadiers. He was posted in Arunachal Pradesh, where his duties included patrolling at altitude. He moved from the Army Sports Institute to the national camp. He is now a Junior Commissioned Officer, and he holds the Indian records at 3000 m, 5000 m and 10,000 m.

He has been unusually direct about where his running began.

I truly learned about competitive running only after joining the army.

— Gulveer Singh (Olympics.com, quoting an Inspire Institute of Sport interview)

Read that twice. One of the most successful distance runners India has ever produced says that his sport, in any serious sense, began inside a military institution.

The institute in Pune

The institution has an address. On 1 July 2001 the Army set up the Army Sports Institute in Pune under a programme it called "Mission Olympics". Its Boys Sports Company runs as a joint venture with the Sports Authority of India.

The name turns up again and again in the biographies of India's best distance men. Thonakal Gopi, a Havildar in the Artillery, has been listed by the Army as an ASI Pune athlete. Kheta Ram's pathway ran through it, according to reference sources. Gulveer came through it. Sawan Barwal, who joined the Army in 2019 and now holds the national marathon record, is reported to have trained there under an Army coach earlier in his career.

Look at who these men were before the uniform. Gopi grew up in Sulthan Bathery, in Wayanad in Kerala, about 1,000 m above sea level; his parents farmed rice and ginger, and he joined the Army through the sports quota. Kheta Ram is reported to have grown up in the Barmer desert of Rajasthan and to have run 4 km to school. Nitendra Singh Rawat is from Garur in the hills of Uttarakhand. G. Lakshmanan comes from Sokkurani in Pudukkottai district, Tamil Nadu, "born in a poor family", as The Hindu put it; he first trained with S. Loganathan of the Kavinadu Youth Sports Club, on tar roads and village streets, before the Army took him further. In 2017 he won both the 5000 m and the 10,000 m at the Asian Championships.

Desert, hill, backwater, paddy field. Four corners of the country, one uniform.

Three soldiers in Rio

In August 2016, all three of India's men in the Olympic marathon were Army runners: Gopi, Kheta Ram and Rawat. All three had qualified at the Mumbai Marathon that January. Before the Games they had trained at altitude in Ooty under Surendra Singh Bhandari, an Army Sports Institute coach. Rawat had already credited the Ooty camp for his Mumbai run in January.

I was personally confident after the training we had been put through in the national camp at Ooty by our coach (Surinder Singh) Bhandari.

— Nitendra Singh Rawat (Sportskeeda, 2016)

Gopi and Kheta Ram finished 25th and 26th in Rio, one second apart, both in personal bests. It was not a medal. But it was a system result: three men from three states, recruited, housed, paid and coached by the same institution, sent to the same mountain, arriving at the same start line.

The pattern stretches back further than Pune. Shivnath Singh, whose 1978 marathon record stood for 48 years, was an Army man. Hari Chand, who ran the 10,000 m barefoot at the 1976 Olympics and won Asian Games double gold in 1978, joined the Central Reserve Police Force in 1972 and served as an inspector. For half a century, the fastest Indian men have mostly worn a uniform.

The women's road runs on rails

The women's road looks different. None of the women profiled in this Part came through the Army. Their equivalent has been the Railways, and later the police.

Sudha Singh, born in Raebareli, Uttar Pradesh, in 1986, worked for Central Railway in Mumbai. It was a railway officer who changed her event.

Madhu Shivdutt, my railway officer introduced me to Steeplechase.

— Sudha Singh (Mumbai Live)

She won the first women's steeplechase ever held at the Asian Games, in 2010. Lalita Babar, a farmer's daughter from Satara, ran for the Railways and reached the Rio Olympic final. Preeja Sreedharan, from Idukki in Kerala, whose father, a manual labourer, reportedly died when she was eight, became a superintendent with Southern Railway and set a 10,000 m national record in 2010 that still stands. Parul Chaudhary took a Western Railway ticket-examiner job in 2015, before the national records and Asian Games medals, and joined the Uttar Pradesh Police as a Deputy Superintendent in 2024. Sunita Rani, the 2002 Asian Games 1500 m champion, joined the Punjab Police through the sports quota.

For these women, the job was the professional contract. It paid the rent, released them to national camps and gave their families a reason to let them keep running.

The runner without a road

To see what the system gives, look at someone it missed.

Jyoti Gawate, born in 1987 in Parbhani, Maharashtra, became one of India's best women marathoners, with a 2:45:48 in Mumbai in 2019. Until 2015 she was reported to live with her family in a wooden hut. In 2014 she took a job with the Mumbai Police but quit after eight days to protect her training time. Her application to the Railways went unanswered. She trained in Parbhani with her long-time coach, Ravi Raskatla, without access to the national camp, and lived on prize money.

I am still jobless; what more can I do?

— Jyoti Gawate (Mid-Day, via shyamgopan.com)

Jyoti's story is the other side of the soldiers' road. The talent was there, and so was the work. The route was not.

The system as the main character

Put these lives side by side and one thing becomes obvious. The most important figure in Indian distance running is not any single athlete. It is an institution: the Army, the Railways, the police, a Sports Authority of India hostel, a long-serving coach in one town. The villages supply hardy, determined young people in every corner of the country. The institutions decide which of them become international athletes.

That raises harder questions. Where exactly do these pathways reach, and where do they stop? Which states feed which institutions? What does each one actually provide: a salary, a coach, altitude, a race abroad? And how many Jyoti Gawates are running village roads right now, with nobody watching?

To answer those questions, we need a map.

Sources: <https://www.olympics.com/en/news/who-is-avinash-sable-india-athlete-steeplechase> ; <https://www.olympics.com/en/news/who-is-gulveer-singh-india-athlete-long-distance-runner> ; <https://armysportsinstitute.com/> ; https://www.espn.com/athletics/story/_/id/17475379/india-marathon-runner-olympics-thonackal-gopi-talks-espn-exclusive-interview ; <https://www.facebook.com/Indianarmy.adgpi/posts/havildar-gopi-thonakal-of-indianarmy-army-sports-institute-asi-pune-won-the-16th/736997223163576/> ; <https://sportskeeda.com/running/nitendra-rawat-credits-coach-success-mumbai-marathon> ; <https://www.firstpost.com/sports/rio-olympics-2016-thonackal-gopi-and-kheta-ram-clock-personal-best-to-finish-25th-26th-in-marathon-2969066.html> ; <https://www.thehindu.com/news/cities/Tiruchirapalli/athletes/article19251278.ece> ; <https://www.olympics.com/en/news/hari-chand-indian-athlete-runner-olympian-dies> ; <https://www.mumbailive.com/en/sports/exclusive-interview-jakarta-asian-games-2018-steeplechase-silver-medalist-sudha-singh-29667> ; <https://indianexpress.com/article/sports/sport-others/indian-long-distance-runner-to-retire-from-international-events/> ; <https://indianexpress.com/article/sports/asian-games/running-sugarcane-field-parul-chaudhary-asian-games-2023-hangzhou-8943068/> ; <https://www.bhaskar.com/news/this-lady-ips-became-an-athlete-found-padma-shri-5862935.html> ; <https://shyamgopan.com/2019/02/04/jyoti-and-the-eight-minutes/> ; <https://www.shethepeople.tv/news/meet-jyoti-gawate-the-fastest-indian-woman-at-mumbai-marathon/>

CHAPTER 90

90. The Indian Distance Map

PART XVIII — INDIA'S ELITE RUNNERS: AN AEI ANALYSIS

Take the night train north out of Pune and, by morning, you could be anywhere on the map of Indian distance running. The Army Sports Institute sits behind you in the cantonment. Ahead lie the villages of Beed and Satara, where children have been reported running kilometres to

school; the hostels of Nashik; the sugarcane country around Meerut and Aligarh; the hill districts of Uttarakhand and Himachal; and, far to the south, the Nilgiri mist over Ooty, where half of India's elite seem to have spent a winter. Put a pin in the home village of every Indian who has held a distance record or won an Asian medal in the last twenty years, and a pattern appears. It is not a single region. It is a network of pathways: a uniform, a railway job, a coach who stayed in one town for thirty years, and a mountain camp.

This chapter maps where India's elite distance runners come from and, more importantly, which institutions carried them from village roads to international start lines. The athletes counted here are the ones profiled in Chapters 91 and 92. The map is built only from verified biographies; where a birthplace or employer could not be confirmed, the athlete is placed in the "other" group rather than guessed.

Pathways, not places

Indian distance running is often described by region: "the Kenya of India" has been claimed for Nashik by its own coach, and hill districts are frequently credited with producing runners. The public evidence suggests a more useful framing. Geography supplies the raw material, usually a rural, physically demanding childhood; institutions supply the conversion. Almost every elite profiled in this Part had both.

Region / pathway	Athletes (examples)	What the system provides
Army / Services	Sable, Gulveer, Sawan, Gopi, Kheta Ram, Rawat, Man Singh, Lakshmanan, Jinson, Shivnath Singh	Salary, sports-quota entry, ASI Pune, service coaches, camp access
Railways	Parul Chaudhary, Sudha Singh, Lalita Babar, Preeja Sreedharan	Sports-quota job, time to train, national-camp release
Police / CRPF	Hari Chand, Sunita Rani, Parul (from 2024)	Stable career and rank; later-career security
Nashik (Vijender Singh)	Kavita Raut, Sanjivani Jadhav, Monika Athare	A single long-serving coach and a local training group
SAI and state academies	Seema (SAI hostel), K. M. Deeksha (MP academy, Bhopal), Ankita Dhyani (Rudraprayag hostel, NCOE Bhopal)	Hostel, coaching, competition entry from the teens
Village clubs and independents	L. Suriya (Kavinadu club, reported), Jyoti Gawate, Priti Lamba	Local coach or family support; little institutional backing

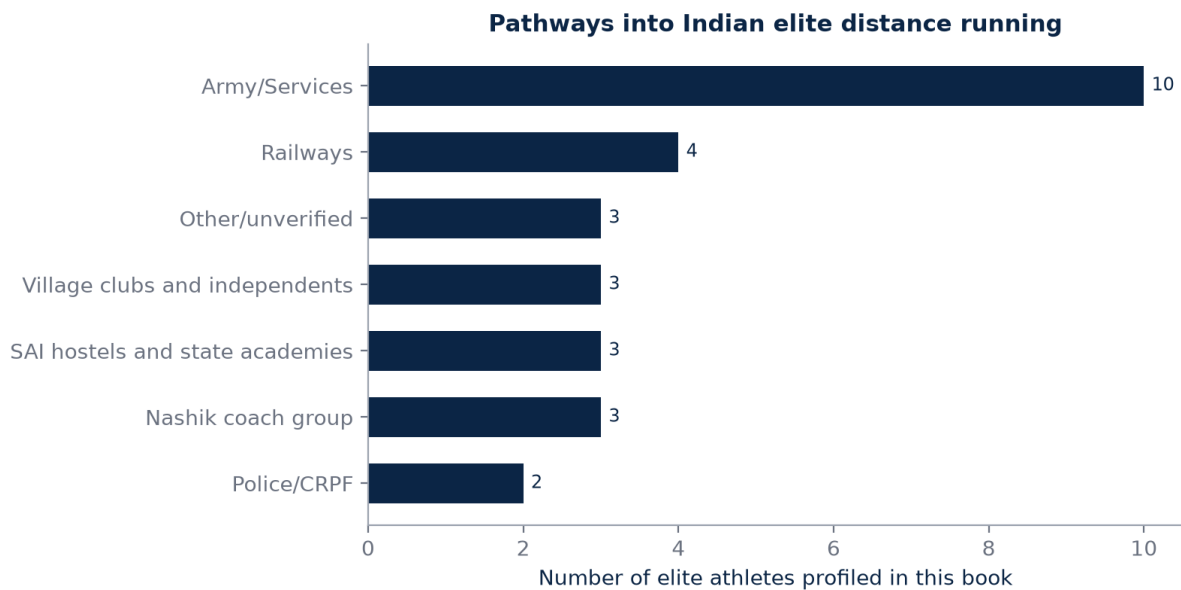


Figure 34. Pathways of the 28 Indian elite distance runners named in the tables of Chapters 90 to 92, grouped by the institution that carried them to international level. Counts are from verified biographies only.

The Army engine

Of the twelve men profiled in the next chapter, ten are, or were, Army soldiers. They include the national record holders in the steeplechase (Avinash Sable), 3000 m, 5000 m and 10,000 m (Gulveer Singh), marathon (Sawan Barwal) and 1500 m (Jinson Johnson). The Army Sports Institute in Pune, set up in July 2001 under the Army's "Mission Olympics" programme, recurs in these biographies again and again.

Two features of this pathway stand out. First, several athletes joined mainly for a secure job and only discovered competitive running after enlisting. Sable came to running through Army cross country in 2015; Gulveer has said that he truly learned competitive running only after joining. Second, the Army recruits from exactly the rural, farming and hill families that also produce hardy young runners, so it acts as a nationwide scouting net even when it is not trying to be one.

KEY IDEA

For Indian men, the Army is not one pathway among many; it is the main pathway. Any national talent strategy that ignores the services, or duplicates what they already do, will waste scarce resources.

Where the women come from

The women's map looks different. No woman profiled here came through the Army. Their equivalent is the Railways, which employed Sudha Singh, Lalita Babar, Preeja Sreedharan and Parul Chaudhary, and the police, which employed Sunita Rani and now Parul. A job is, in effect, the professional contract. Jyoti Gawate, a marathoner from

Parbhani who, according to 2019 reports, had no job and lived on prize money without national-camp access, shows what happens without one.

Regional clusters matter more for women than for men. Vijender Singh, a SAI coach based in Nashik since 1991, has worked with Kavita Raut, Sanjivani Jadhav and Monika Athare, among others. Satara (Lalita Babar), Beed (Sable) and Buldhana (Poonam Sonune) make rural Maharashtra the densest single region on the women's and men's maps combined.

The hill districts and the south

Uttarakhand has produced Nitendra Singh Rawat (Garur), Man Singh (Pithoragarh) and Ankita Dhyani, whose village of Maroda sits at about 1,400 m. Himachal Pradesh has produced Sawan Barwal (Mandi) and Seema (Chamba). Western Uttar Pradesh, which is not hilly at all, has produced Gulveer Singh, Parul Chaudhary, Kartik Kumar and K. M. Deeksha. That alone should caution anyone who thinks altitude of birth explains Indian distance talent.

In the south, Kerala supplied Gopi, Jinson Johnson, Preeja Sreedharan and O. P. Jaisha. Tamil Nadu supplied G. Lakshmanan and L. Suriya, both reported to have trained at a youth club in Kavinadu village near Pudukkottai. Tamil Nadu also hosts the most important domestic altitude venue: Ooty, at 2,240 m, where a state high-altitude training centre opened in February 2024 and which a ministry official has described as the preferred choice for middle- and long-distance runners.

Home region	Profiled athletes	Common next step
Maharashtra	Sable, Lalita, Kavita, Sanjivani, Monika, Jyoti	Army, Railways, Nashik group
Uttar Pradesh	Gulveer, Parul, Sudha, Kartik, Deeksha	Army, Railways, state academy
Uttarakhand / Himachal	Rawat, Man Singh, Ankita, Sawan, Seema	Army, SAI hostel
Kerala / Tamil Nadu	Gopi, Jinson, Preeja, Jaisha, Lakshmanan, Suriya	Army, Railways, village club
Punjab / Haryana / Rajasthan / Bihar	Hari Chand, Sunita, Harmilan, Priti, Kheta Ram, Shivnath	CRPF, police, Army, private coach

COACH'S CORNER

When you scout, look at pathways as well as times. A 16-year-old with a modest 3000 m time who is two years from a services or Railways trial may have a better route to the top than a faster athlete with no job prospect. Help athletes and families understand the sports-quota routes early, and keep training logs that a recruiting coach can read.

What the map does not show

The map shows who reached the top, not who dropped out. We have no public data on how many talented teenagers from Beed or Chamba never found a coach, a hostel place or a job. Coorg, Dharamshala and Kodaikanal are sometimes mentioned as running regions, but we could not verify any documented elite pathway from them. The map is a record of survivors, and that bias should be kept in mind when drawing lessons from it.

CHECKPOINT

Which pathways exist within 100 km of where you coach: an Army recruiting centre, a Railways sports-quota office, a SAI hostel, a state academy? Which of your athletes could realistically use each one, and when?

CHAPTER 91

91. The Men: From 1500 m to the Marathon

PART XVIII — INDIA'S ELITE RUNNERS: AN AEI ANALYSIS

In March 2025, at a track meet called The TEN in San Juan Capistrano, California, an Indian Army soldier finished sixth in a 10,000 m race. The placing barely registered in world athletics news. The time did: 27:00.22. Gulveer Singh, born in 1998 into a farming family in Sirsa village, Aligarh district, had just run the 10,000 m more than a minute faster than the national record that had stood from 2008 until 2024. The Sports Authority of India described him as the third-fastest Asian of all time. Two years earlier the national record had been 28:02.89, set in 2008. Something had changed, and it was not the Aligarh soil.

This chapter profiles India's leading men from 1500 m to the marathon, past and present, and looks for the patterns that link them. All personal bests and dates are from World Athletics profiles, Olympics.com and major news reports. Where a coach or base could not be confirmed, the table says so.

The profiles

Athlete	Event	PB (year)	Base / coach	Notes
Shivnath Singh	Marathon	2:12:00 (1978)	Army; coach unverified	NR for 48 years; reported barefoot
Hari Chand	10,000 m	28:48.72 (1976)	CRPF; coach unverified	Asian Games 5000/10,000 m gold 1978
Jinson Johnson	1500 m	3:35.24 (2019)	Army; Colorado Springs, Simmons	NR still stands; 2018 Asian Games gold
Ajay Kumar Saroj	1500 m	3:37.83 (2024)	Coach unverified	Asian Games silver 2023
Avinash Sable	Steeplechase	8:09.91 (2024)	Army, Amrish Kumar; Simmons from 2022	10 NRs; Paris 2024 finalist
Gulveer Singh	5000 / 10,000 m	12:59.77 sh. tr. (2025); 27:00.22 (2025)	Army; Colorado Springs, Simmons	NRs 3000–10,000 m; 59:42 half
G. Lakshmanan	5000 / 10,000 m	13:35.69 (2017, reported)	Army; S. Loganathan, Surender Singh	Asian Champs double gold 2017
Sawan Barwal	Marathon	2:11:37 (2026)	Army; Markose, later Simmons; Ooty, Colorado Springs	NR; Asian Games silver 2026
Thonakal Gopi	Marathon	2:12:23 (2025)	Army; Bhandari; Ooty, Bengaluru	Asian Marathon champion 2017
Man Singh	Marathon	2:13:25 (2025)	Army; coach unverified	Asian Marathon champion 2024
Kheta Ram	Marathon	2:15:26 (2016)	Army; Bhandari, Ooty	Rio 2016, 26th
Nitendra Singh Rawat	Marathon	2:15:48 (2016)	Army; Bhandari, later K. C. Ramu	Mumbai Indian course record

Pattern 1: the soldier-athlete

Ten of the twelve men in the table served in the Army; Hari Chand served in the CRPF. Only Ajay Kumar Saroj's service affiliation could not be confirmed. The Army provides more than a salary. Sable's postings took him from Siachen to the Rajasthan desert; Gulveer did patrolling duties at altitude in Arunachal Pradesh. These are incidental environmental exposures, not planned training, but they show how physically demanding the pathway is before any coach gets involved.

Pattern 2: the late discovery

Many of these men did not arrive as polished juniors. Sable joined the Army after Class 12 and took up running through Army cross country in 2015; an Army coach, Amrish

Kumar, moved him into the steeplechase around 2017. Within two years he was in a World Championships final. Sawan started as a 1500 m runner at 14 and ran his first marathon at 28. Gopi ran his best marathon at 37. For coaches, this is an encouraging pattern: in Indian men's distance running, the late developer is common, and aerobic development can continue well into the thirties.

KEY IDEA

Indian male distance runners often reach elite level late, after enlisting or after years on the track. Talent identification that closes at 17 would have missed several of the country's record holders.

Pattern 3: the Colorado and Ooty effect

Two altitude hubs dominate the recent men's story. Domestically, the Nilgiris hosted the Rio 2016 marathon group under Surendra Singh Bhandari, Gopi before his 2025 Valencia run, and the winter base of Sawan's Project 2:09 group. Internationally, Colorado Springs under the American coach Scott Simmons became the distance hub from 2019, used by Jinson, Sable from 2022 and Gulveer from 2024. Sable also trained at St Moritz before the 2023 World Championships in Budapest.

The timing is striking. The steeplechase record, 1981 to 2018; the 5000 m record, 1992 to 2022; the 10,000 m record, 2008 to 2024; the marathon record, 1978 to 2026. All fell in the era of foreign coaching, planned altitude blocks and racing abroad. Correlation is not proof, and the same era also brought better shoes, better pacing and TOPS funding. But the pattern is consistent across events.

Pattern 4: records set abroad

Almost every men's national record since 2022 was set at an overseas meet with fast pacing: The TEN and Sound Running in California, Portland, Niigata, Hachioji, Boston indoors, Budapest, the Paris Diamond League and Rotterdam. Indian domestic races rarely offer record pace, deep fields or cool evenings. This matters for selection: an athlete who never races abroad may never discover how fast they can run.

COACH'S CORNER

For a developing senior athlete, plan at least one race a season in conditions built for fast times: cool weather, a deep field and pacing. Treat it as a measurement, not just a competition. A single honest time in good conditions is worth more for planning than several wins in the heat of a domestic meet.

Pattern 5: fragile bodies at the top

The best men have also paid a physical price. Jinson Johnson, still the 1500 m record holder, described a severe Achilles injury from 2019 and retired in January 2026. Sable

injured the ACL and meniscus of his right knee at the Monaco Diamond League in July 2025, missed the Tokyo World Championships and the 2026 Commonwealth Games, and was fourth at the 2026 Asian Games on his comeback. He later reflected that he may have rushed his return. A small elite group means each injury costs the nation a large share of its capacity.

The marathon depth problem

The men's marathon has more depth than it first appears. In 2025–26, Sawan (2:11:37), Gopi (2:12:23), Man Singh (2:13:25) and A. B. Belliappa (2:14:07) all ran under 2:14:10, and Gopi and Man Singh won Asian Marathon Championships titles in 2017 and 2024. Yet no Indian has ever run under 2:10. In 2019 alone, 293 sub-2:10 performances were recorded worldwide. The gap is not at the top of the Indian list; it is between the top of the Indian list and the world.

MYTH VS EVIDENCE

Myth: Indians are track runners, not marathoners.

Evidence: Shivnath Singh's 2:12:00 in 1978 was a strong mark for its era, and four Indians ran under 2:14:10 in 2025–26. The limiting factor looks more like marathon-specific preparation, racing opportunities and late conversion than any fixed trait.

CHECKPOINT

Look at the men's table. Which athlete's pathway most resembles that of a runner you coach? What single element of that pathway, whether altitude, overseas racing or a specialist coach, could you realistically reproduce this year?

CHAPTER 92

92. The Women: Steeple to Marathon

PART XVIII — INDIA'S ELITE RUNNERS: AN AEI ANALYSIS

At the Rio Olympics in August 2016, a farmer's daughter from Mohi village in Satara district lined up in the final of the 3000 m steeplechase. As a child, Lalita Babar was reported to have run 4 km each way to school and fetched water for a 17-person household in a drought-prone area. At 15 she won Rs 10,000 in prize money in a 10 km road race. A Belarusian coach, Nikolai Snesev, introduced her to the steeplechase. In the heats in Rio she set a national record of 9:19.76, and in the final she finished tenth, the first Indian to reach an Olympic track final since P. T. Usha in 1984. That record

lasted seven years, until a Railways ticket examiner from a sugarcane village near Meerut broke it.

This chapter profiles India's leading women from 1500 m to the marathon. Their story is different from the men's in almost every structural way: different employers, different coaches, different clusters, and records that have either leapt forward or stood still for more than a decade.

The profiles

Athlete	Event	PB (year)	Base / coach	Notes
Parul Chaudhary	Steeple / 5000 m	9:08.67 (2026); 15:04.26 (2026)	Railways, UP Police; Simmons	NRs 3000 m, 5000 m, steeple
Lalita Babar	Steeplechase	9:19.76 (2016)	Railways; Snesarev	Rio 2016 finalist, 10th
Sudha Singh	Steeple / marathon	9:26.55 (2016); 2:34:55 (2019)	Central Railway; Snesarev	Asian Games gold 2010
Preeja Sreedharan	10,000 m	31:50.47 (2010)	Southern Railway; Snesarev	NR stands since 2010
Kavita Raut	10,000 m / half	31:51.44 (2010)	Nashik, Vijender Singh; Snesarev	CWG 2010 10,000 m bronze
O. P. Jaisha	Marathon	2:34:43 (2015)	Snesarev	NR stands since 2015
L. Suriya	Half marathon	1:10:31 (2017)	Kavinadu club (reported)	NR stands since 2017
Seema	10,000 m	32:02.43 (2026)	SAI hostel; SAI Bengaluru	Asian Games bronze 2026
Ankita Dhyani	Steeplechase	9:19.24 (2026)	Bhopal NCOE; EDRP	2000 m steeple NR
K. M. Deeksha	1500 m	4:04.78 (2024)	MP academy; S. K. Prasad	NR
Harmilan Bains	1500 m	4:05.39 (2021)	Punjab; J. S. Bhatia, S. K. Saini	Asian Games double silver
Sunita Rani	1500 m / 5000 m	4:06.03 (2002)	Punjab Police	1500 m NR until 2021

Pattern 1: the job as contract

Where the men have the Army, the women have the Railways and the police. Four of the twelve women in the table held Railways jobs. Parul Chaudhary joined Western Railway in 2015 and became a Deputy Superintendent of Police in Uttar Pradesh in 2024. Sunita Rani joined Punjab Police as a DSP in 2002. The public record contains a

stark counter-example in Jyoti Gawate, who reached a 2:45:48 marathon in 2019 while living on prize money, training in Parbhani with her long-time coach and without national-camp access. The job is not a perk. It is what makes full-time training possible.

KEY IDEA

For Indian women, a secure job has functioned as the professional contract that allows elite training. The absence of one is a measurable disadvantage, not just a personal hardship.

Pattern 2: the Snesev generation

Between 2005 and 2016, one coach shaped an entire generation. Nikolai Snesev arrived in 2005 to coach Preeja Sreedharan and Kavita Raut; at the 2010 Asian Games in Guangzhou they finished first and second in the 10,000 m, and Sudha Singh won the first women's steeplechase ever held at the Asian Games. He later guided Lalita Babar to the Rio final. Olympics.com has described him as an advocate of natural methods. He left the national set-up in 2019 and died in March 2021 at NIS Patiala, shortly after being re-appointed.

The strength of that generation is also a warning. Three of the seven women's distance records, 10,000 m (2010), marathon (2015) and half marathon (2017), were set by this cohort or its contemporaries and still stand. When a single coach-led group lifts the ceiling and then disperses, the ceiling can stay where it was for a long time.

Pattern 3: Parul and the new ceiling

The recent women's story is, to a remarkable degree, one athlete. Parul Chaudhary started running barefoot in a school 800 m race in 2011, with no practice, at her father's suggestion. She trained on paths between sugarcane fields. From 2022 to 2026 she set national records at 3000 m (8:57.19, the first Indian woman under nine minutes), 5000 m (15:10.35 and then 15:04.26) and the steeplechase (9:15.31, 9:13.39, 9:12.46 and 9:08.67). She won Asian Games 5000 m gold and steeplechase silver in Hangzhou, and two bronze medals in Nagoya in 2026. She trains with Scott Simmons in Colorado Springs.

Her rise tells us that the ceiling is not fixed. It also tells us how thin the depth is behind her: at the 2026 Asian Games, Ankita Dhyani ran 9:19.24 for fourth, a large personal best, but still more than ten seconds behind.

COACH'S CORNER

If you coach a promising woman at 800 m or 1500 m, do not assume that is her event. Parul moved from 800 m to 5000 m and the steeplechase; Lalita moved from road races to the steeplechase; Jaisha moved from 1500 m to the marathon. Test the athlete across distances every season and keep the door to longer events open.

Pattern 4: event switching and hills

Event switching is common and often successful among Indian women, and the new generation adds a geographical twist. Ankita Dhyani grew up at about 1,400 m in Pauri Garhwal and, as a girl, ran hill roads alongside young men training for Army fitness tests. Her coach at Bhopal has credited that upbringing with giving her an altitude background. Seema, from Reta village in Chamba, grew up running barefoot while herding cattle, came through a SAI hostel and won 10,000 m bronze at the 2026 Asian Games, only the third Indian woman to medal in the event and the first in 16 years.

Pattern 5: the stalled long events

The 10,000 m record has stood since 2010 and the marathon since 2015. The closest recent attempts, Parul's 32:02.08 and Seema's 32:02.43 on the track and Seema's 1:11:23 in the half marathon, are close but not through. At the 2026 national marathon championship in New Delhi the winning time was 2:41:15, and no Indian woman met the Asian Games qualifying mark of 2:31:52. The women's marathon is where the gap to the world is largest, as Chapter 94 will show.

CHECKPOINT

Of the women you coach, how many have a realistic route to a secure job through sport? What would change in your planning if you treated that route as seriously as their next race?

CHAPTER 93

93. Records Over Time

PART XVIII — INDIA'S ELITE RUNNERS: AN AEI ANALYSIS

Picture a noticeboard in a stadium office somewhere in India, the national records typed on a sheet and pinned up years ago. For some events, the sheet has been retyped so often that the paper is new. For others, the ink has faded in the sun. The men's steeplechase line changed ten times in six years; before that it had not changed in thirty-seven. The women's 10,000 m line has not changed since Preeja Sreedharan crossed the line in Guangzhou in November 2010. A record progression is a history of a nation's system, written in seconds. The flat stretches show when nothing was working; the steps show when something finally did.

This chapter sets out the verified progressions of India's main distance records and asks what their shapes reveal. Dates and times come from Olympics.com, World Athletics and major news reports. Earlier marks that could not be verified are left out rather than guessed.

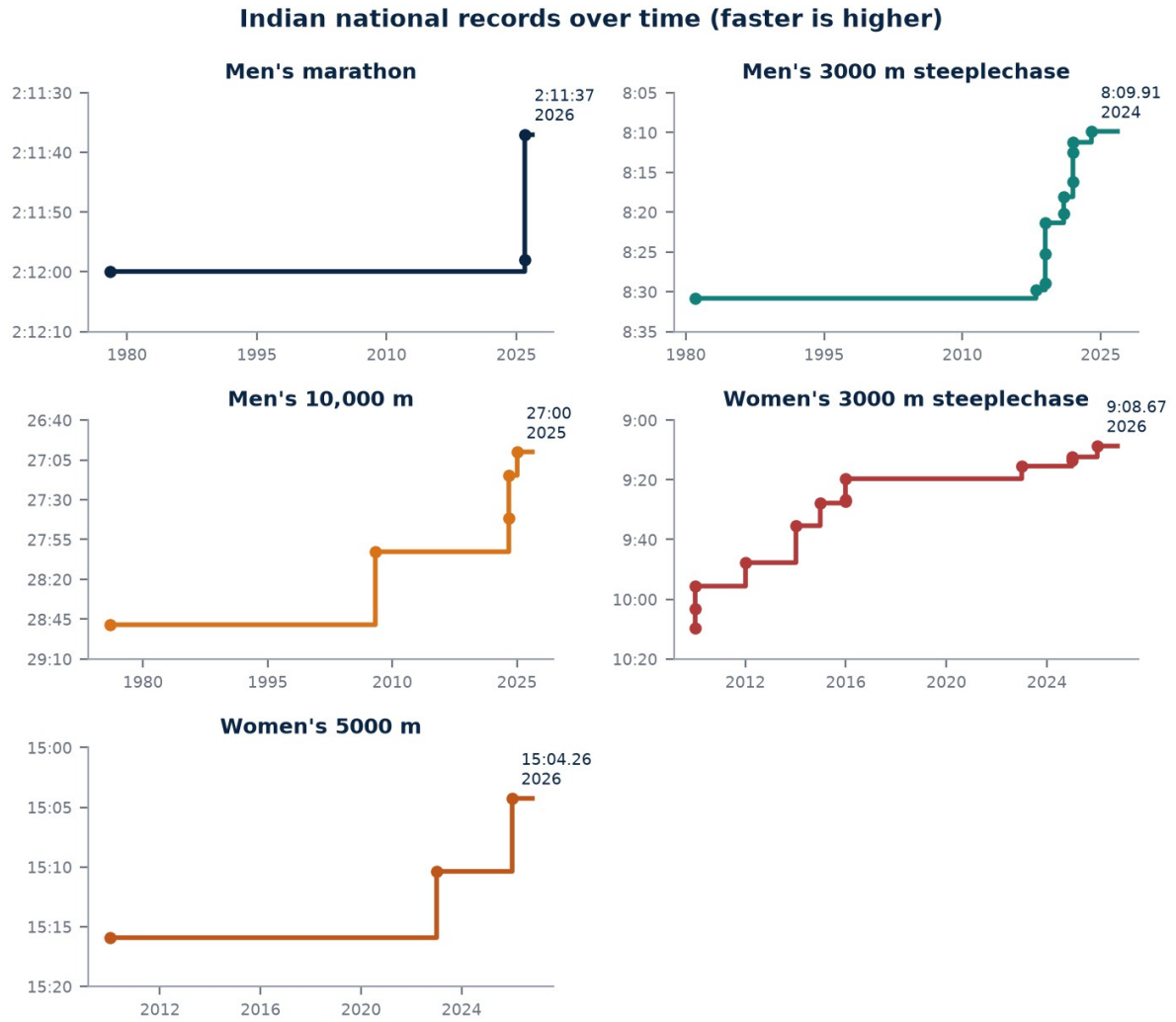


Figure 35. Verified Indian national-record progressions in five events: men's marathon, men's 3000 m steeplechase, men's 10,000 m, women's 3000 m steeplechase and women's 5000 m. Long flat periods are followed by rapid clusters of new records.

Men: long walls, then sudden falls

Event	Record	Athlete	Year
Marathon	2:12:00	Shivnath Singh	1978
Marathon	2:11:58	Sawan Barwal	2026 (Apr)
Marathon	2:11:37	Sawan Barwal	2026 (Sep)
10,000 m	28:48.72	Hari Chand	1976
10,000 m	28:02.89	Surendra Singh	2008
10,000 m	27:41.81	Gulveer Singh	2024
10,000 m	27:14.88	Gulveer Singh	2024
10,000 m	27:00.22	Gulveer Singh	2025
5000 m	13:29.70	Bahadur Prasad	1992
5000 m	13:25.65	Avinash Sable	2022
5000 m	13:19.30	Avinash Sable	2023
5000 m	13:18.92	Gulveer Singh	2024
5000 m	13:11.82	Gulveer Singh	2024
5000 m	12:59.77 (short track)	Gulveer Singh	2025

The men's marathon record stood for 48 years, the 10,000 m record for 32 years from 1976, and the 5000 m for about 30 years. Then the 10,000 m record fell by more than a minute in about a year, with three new records between March 2024 and March 2025.

The steeplechase staircase

Event	Record	Athlete	Year
Steeplechase (M)	8:30.88	Gopal Saini	1981
Steeplechase (M)	8:29.80	Avinash Sable	2018
Steeplechase (M)	8:21.37	Avinash Sable	2019
Steeplechase (M)	8:18.12	Avinash Sable	2021
Steeplechase (M)	8:12.48	Avinash Sable	2022
Steeplechase (M)	8:11.20	Avinash Sable	2022
Steeplechase (M)	8:09.91	Avinash Sable	2024

The table shows selected steps; Sable broke the record ten times in all, from 8:29.80 in 2018 to 8:09.91 at the Paris Diamond League in 2024. Several of those records were set in championship heats and finals, at the 2019 World Championships, the Tokyo

Olympics and the 2022 Commonwealth Games. That is unusual: most athletes run their fastest times in paced meets, not championships.

KEY IDEA

Indian distance records tend to move in bursts. A single athlete with the right coach, environment and racing programme can rewrite a record five or ten times, after decades of stagnation.

Women: the two generations

Event	Record	Athlete	Year
Steeplechase (W)	10:09.56	Sudha Singh	2010
Steeplechase (W)	10:03.05	O. P. Jaisha	2010
Steeplechase (W)	9:55.67	Sudha Singh	2010
Steeplechase (W)	9:47.70	Sudha Singh	2012
Steeplechase (W)	9:35.37	Lalita Babar	2014
Steeplechase (W)	9:27.86	Lalita Babar	2015
Steeplechase (W)	9:19.76	Lalita Babar	2016
Steeplechase (W)	9:15.31	Parul Chaudhary	2023
Steeplechase (W)	9:12.46	Parul Chaudhary	2025
Steeplechase (W)	9:08.67	Parul Chaudhary	2026
5000 m (W)	15:15.89	Preeja Sreedharan	2010
5000 m (W)	15:10.35	Parul Chaudhary	2023
5000 m (W)	15:04.26	Parul Chaudhary	2026
10,000 m (W)	31:50.47	Preeja Sreedharan	2010
Half marathon (W)	1:10:31	L. Suriya	2017
Marathon (W)	2:39:10	Vally Satyabhama	1995
Marathon (W)	2:37:29	O. P. Jaisha	2015
Marathon (W)	2:34:43	O. P. Jaisha	2015

The table shows selected steps. The women's steeplechase shows two distinct bursts. The first, from 2010 to 2016, came from the Snegarev group: Sudha Singh, Jaisha and Lalita Babar brought the record down by almost 50 seconds, partly because the event was new to Indian women and the early marks were soft. After a seven-year gap, Parul Chaudhary started the second burst in 2023. The women's 5000 m record has improved only twice since 2010, both times by Parul Chaudhary.

The 1500 m: slow steps

Event	Record	Athlete	Year
1500 m (W)	4:06.03	Sunita Rani	2002
1500 m (W)	4:05.39	Harmilan Bains	2021
1500 m (W)	4:04.78	K. M. Deeksha	2024
1500 m (M)	3:35.24	Jinson Johnson	2019

The 1500 m records move differently from the longer events. Sunita Rani's 4:06.03 from the 2002 Asian Games stood for 19 years, and the two records since have improved it by little more than a second in total. On the men's side, Jinson Johnson has described breaking a 23-year-old 1500 m record at the 2018 Commonwealth Games before setting the current 3:35.24 in Berlin in 2019, which still stands. The middle distances have not yet had their Sable or Gulveer.

What stagnation tells us

Long flat periods are rarely about a lack of talent. The same villages, the same Army recruitment and the same Railways quotas existed throughout. What the flat periods share is the absence of three things: a coach with a clear long-term method, sustained altitude or specialist training blocks, and access to fast races. When all three were present, for the Snesev group from 2005 and for the Colorado Springs and Ooty groups from about 2019, records fell. When they disappeared, as after the Snesev group dispersed, records stood.

There is a second, less comfortable lesson. Some flat periods may reflect the fact that the old records were genuinely strong. Shivnath Singh's 2:12:00 and Preeja Sreedharan's 31:50.47 were not easy targets, and the athletes who approached them later did so with far better equipment and support.

COACH'S CORNER

Look at your own club or state records the same way. List them with their dates. Records that have not moved in more than ten years point to a gap in your system: a missing coach, a missing training environment or a missing race opportunity. Records that are moving fast tell you which parts of your system are working.

What breakthroughs tell us

Breakthroughs cluster around specific people and environments rather than spreading evenly. Sable and Gulveer account for almost all recent men's track records; Parul for almost all recent women's records. This is a fragile structure. A single injury, as Sable's 2025 knee injury showed, can remove a nation's main record-breaker for a season or

more. A healthy system should produce several athletes pressing the same record at once.

CHECKPOINT

Pick one national record that has not moved in more than ten years. What would need to be true, in coaching, environment and racing, for an Indian athlete to break it within four years?

CHAPTER 94

94. How Far from the World?

PART XVIII — INDIA'S ELITE RUNNERS: AN AEI ANALYSIS

At the Tata Mumbai Marathon in January 2026, the first Indian man home, Kartik Karkera, crossed the line in 2:19:55. The overall winner, Tadu Abate of Ethiopia, had already finished in 2:09:55, ten minutes earlier. At the Delhi Half Marathon in 2025, the women's race was won by Kenya's Lilian Rengeruk in 1:07:20; the first Indian woman, Seema, ran 1:11:23, a strong run just outside the national record. These races are held in Indian conditions, on Indian roads, in front of Indian crowds, and the gap is visible to anyone standing at the finish. The question for this chapter is how big that gap really is at the very top, event by event, when India's best-ever marks are set against the best in the world.

Here we compare India's current national records with world records and with Asian podium levels. The gap is calculated as a percentage: $\text{gap \%} = (\text{Indian record time} - \text{world record time}) \div \text{world record time} \times 100$. A gap of 3% means the Indian record is 3% slower than the world record.

The comparison table

World records are those standing at the end of 2025 unless stated; any records broken during 2026 are not included except the men's marathon, which is noted. Treat all world-record values as approximate for 2026 and check the current World Athletics list before quoting them.

Event	Indian NR	World record	Gap %
Men 1500 m	3:35.24	3:26.00	4.5
Men 3000 m	7:34.49	7:17.55	3.9
Men 5000 m (outdoor)	13:03.93	12:35.36	3.8
Men 10,000 m	27:00.22	26:11.00	3.1
Men steeplechase	8:09.91	7:52.11	3.8
Men marathon	2:11:37	2:00:35	9.1
Women 1500 m	4:04.78	3:49.04	6.9
Women 3000 m	8:57.19	8:06.11	10.5
Women 5000 m	15:04.26	13:58.06	7.9
Women 10,000 m	31:50.47	28:54.14	10.2
Women steeplechase	9:08.67	8:44.32	4.6
Women marathon	2:34:43	2:09:56	19.1

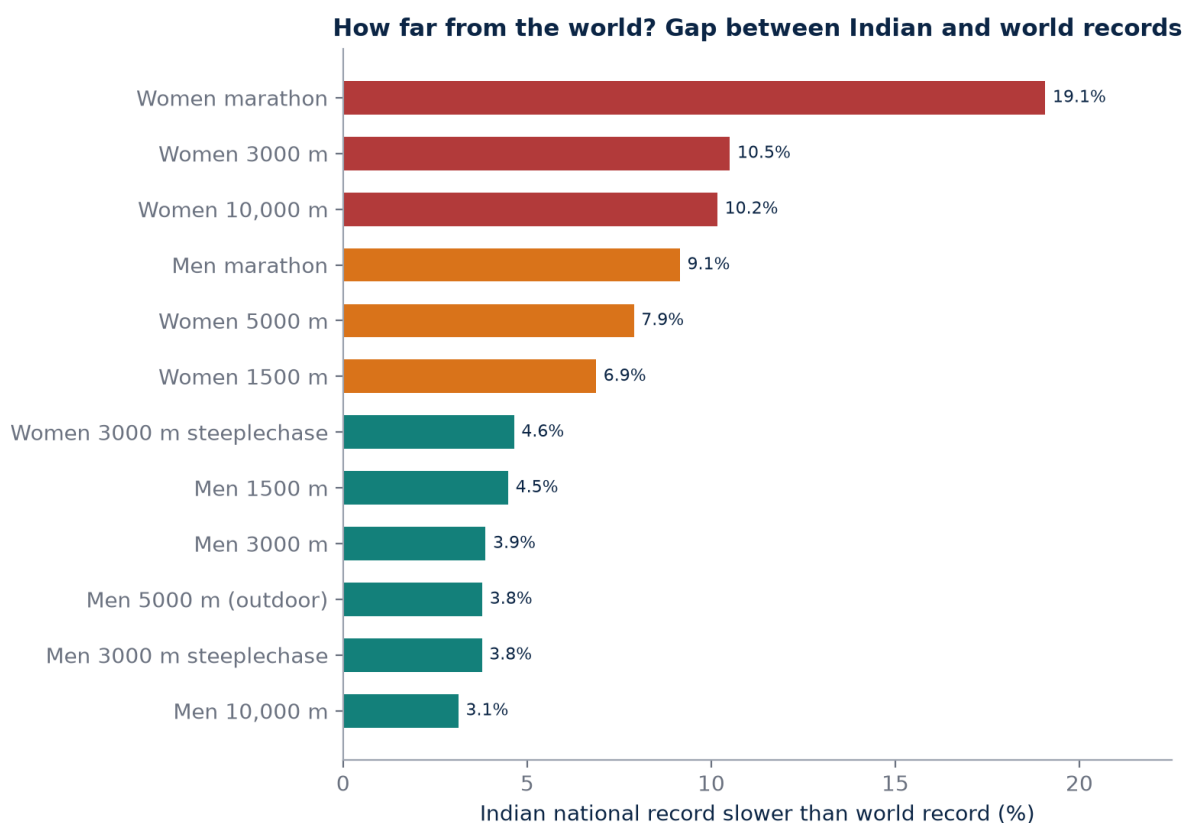


Figure 36. Percentage gap between Indian national records and world records, by event and sex. Men's track records sit within about 3–5% of world level; the gaps widen sharply in the women's long events and in both marathons.

Notes: the men's 5000 m uses Gulveer Singh's outdoor best of 13:03.93; his short-track 12:59.77, ratified as the absolute record, would give a gap of about 3.2%. For the men's marathon, the table uses Kelvin Kiptum's 2:00:35 (2023). If Sabastian Sawe's reported 1:59:30 from London in April 2026 is ratified, the gap rises to about 10.1%. The women's marathon record used is Ruth Chepngetich's 2:09:56 (2024), which stands because it predates her later doping ban.

Reading the gaps

Three patterns jump out.

First, men's track events are closest. Gulveer Singh's 27:00.22 is about 3.1% from the world 10,000 m record, and Sable's 8:09.91 is about 3.8% from the steeplechase record. In practical terms, 3% in a 10,000 m is about 49 seconds; at world-record pace that is about three-quarters of a lap. It is a big gap in world terms, but it is one that Indian athletes have been closing quickly.

Second, the gaps widen with distance, especially for women. The women's steeplechase (4.6%) is close to the men's track events, thanks almost entirely to Parul Chaudhary. But the women's 10,000 m (10.2%), 3000 m (10.5%) and marathon (19.1%) are far behind. A 19% gap in the marathon is about 25 minutes.

Third, the marathon is India's widest gap for both sexes. The men's marathon gap (9.1%) is more than twice the men's track gaps.

KEY IDEA

India's distance gap to the world is not one number. It is smallest in men's track events and the women's steeplechase, about 3–5%, and largest in the marathons and the women's long track events, about 9–19%.

Against Asian podiums

World records are the far horizon. Asian podiums are the nearer target, and the 2026 Asian Games give a direct reading. Parul Chaudhary's 9:08.67 won bronze behind the winner Yavi's 9:04.36, a gap of only about 0.8%. Sable won Asian Games steeplechase gold in 2023 and Asian Championships gold in 2025. Gulveer won 10,000 m silver, 1500 m silver and 5000 m bronze at the 2026 Asian Games, and Sawan won marathon silver. Seema took the women's 10,000 m bronze.

In other words, India is already at Asian podium level in several distance events. The step from Asian podium to Olympic final is the steep one. Sable reached the Paris 2024 final and finished 11th in 8:14.18, about 0.9% slower than his own national record, which shows that championship finals are raced differently from paced record attempts.

COACH'S CORNER

Use percentage gaps to set realistic goals. An athlete 10% off the national record needs a multi-year development plan; an athlete 2% off can target it within a season or two. The same arithmetic works for any club runner: express the goal as a percentage improvement, and check that it is plausible over the time available.

Why the marathon gap is so large

The marathon gap cannot be explained by talent alone, because the same Indian men who are 3–4% from world level on the track are 9% from it over 42.2 km. Several explanations are plausible, and all are hypotheses. Indian runners have tended to move to the marathon late. Marathon-specific volume and long-run durability may be under-developed. Domestic marathons are often run in heat and humidity. And very few Indians race in fast, deep, cool-weather marathons. Sawan's move to about 200 km a week and alternate-week 40–42 km long runs, and his debut in Rotterdam rather than a domestic race, can be read as an attempt to address exactly these factors.

Limits of the comparison

These gaps compare single best-ever marks, not the depth of a nation's field. They also compare a small number of Indian athletes, some of whom set records once in ideal conditions, with world records set by the best athletes in history. They are a useful map of relative strengths, not a verdict on potential.

CHECKPOINT

Which event in the table has the smallest gap relative to the investment India currently makes in it? Which has the largest gap relative to the number of athletes competing? Where would you put the next rupee?

CHAPTER 95**95. Through the AEI Lens**

PART XVIII — INDIA'S ELITE RUNNERS: AN AEI ANALYSIS

Imagine a national coach in Bengaluru opening a folder for India's ten best distance runners, hoping to compare them across the ten AEI domains introduced earlier in this book. The Performance page is full: race results, dates, venues, splits. The Altitude page has some entries: Ooty, Colorado Springs, St Moritz, dates of camps. Then the pages go quiet. Aerobic capacity: blank. Threshold: blank. Economy: blank. Haemoglobin mass: blank. Heat tolerance: blank. That is not necessarily because these tests are

never done inside Indian camps; it is because almost nothing about them is public, and a coach cannot learn from numbers that nobody can see. For a book built on measurement, that silence is itself a finding.

This chapter applies the AEI domains to what is publicly known about Indian elites. It must be read with a clear warning: public data on the physiology of Indian distance runners are very limited. Everything below that goes beyond race results is a hypothesis, stated so that it can be tested, not a conclusion.

Domain by domain

Domain	What is public	Hypothesis	Data to collect
Performance	Full race histories	Men's track near world level; marathons lag	Standardised race conditions log
Aerobic	One 1991 study of Bengali girls (VO ₂ max 43.0)	Modern elites far higher; no current data	VO ₂ max and HR–pace markers each season
Threshold	None	Threshold development drives recent 10,000 m gains	Lactate or critical-speed tests
Economy	None	May limit marathon more than track	Running economy at marathon pace
Oxygen	Anaemia widespread in Indian women	Iron deficiency a key risk for women	Ferritin, Hb, Hb mass
Cardiovascular	Screening guidance for marathons	Unknown; screening matters	ECG and echo screening records
Recovery	Injury reports (Sable, Jinson, Harmilan)	Small elite, high injury cost	HRV, sleep, load and injury logs
Heat	No deliberate protocol documented	Heat exposure incidental, not planned	Heat acclimation and sweat testing
Altitude	Camp locations and durations	Altitude blocks helped record-breakers	Hb mass before and after camps
Durability	Marathon results only	Largest single gap for marathoners	Late-race pace decay in long runs

Performance and durability

The Performance domain is the only one with rich public data, and Chapter 94 has already read it: men's track gaps of about 3–5%, marathon gaps of 9% for men and 19% for women. The simplest hypothesis consistent with this pattern is a Durability deficit. An athlete whose 10,000 m is within 3% of world level but whose marathon is 9% away is, in AEI terms, losing more over the final third of the race than their speed would predict. Sawan Barwal's alternate-week long runs of 40–42 km, and his stated focus in Nagoya on maintaining the pace to the finish, point to the kind of work that addresses durability.

KEY IDEA

For Indian marathoners, the Durability domain is the most likely single limiter. It is also one of the cheapest to measure: the pace and heart-rate decay over the last 10 km of long runs and races.

Aerobic, threshold and economy

The only peer-reviewed aerobic capacity figure we found for Indian distance runners is from 1991: Bengali female long-distance runners averaged a relative VO_2max of 43.0 ml/kg/min on a cycle ergometer. It is useful as history, not as a description of today's elites. By comparison, one widely cited study found Kenyan runners averaging about 79.9 ml/kg/min at sea level. No public threshold or economy data exist for current Indian elites. The hypothesis worth testing is that recent track gains owe much to threshold development under structured coaching, and that economy at marathon pace may be a hidden limiter.

Oxygen transport and the women's question

The Oxygen domain raises the most urgent public-health signal. A venous-blood survey across eight Indian states found anaemia in 44% of adolescent girls and 41% of adult women, although iron deficiency explained less than a third of it. A 2025 study of 104 Tamil Nadu female athletes, not runners, found about 53.8% were anaemic. If even part of this burden reaches young female distance runners, it could help explain why the women's long events lag further behind world level than the men's. This is a hypothesis, but an easily testable one.

COACH'S CORNER

For every female distance runner you coach, ask a doctor to arrange haemoglobin and ferritin testing at least twice a year and before and after any altitude block, and let the clinician interpret the results. It is cheap, it is available in almost every town, and an untreated iron deficiency can undo months of good training.

Heat and altitude

The Altitude domain has the best circumstantial evidence. Ooty (2,240 m) is roughly Iten-equivalent in elevation. Gopi credited it after his Valencia personal best. Sable's coach in Colorado Springs described his adaptation on return to altitude as almost instant. But no peer-reviewed study of Indian athletes' haematology at Ooty was found, so the dose, response and individual variation remain unknown.

The Heat domain is the most surprising gap. India is one of the hottest places in the world to train, and domestic races such as the 2026 TCS World 10K in Bengaluru were run in hot, humid conditions. Yet we found no documented, deliberate heat-acclimation protocol for Indian elites. Heat exposure appears incidental rather than planned.

Indian runners may be well heat-habituated but not systematically heat-adapted for specific races.

Recovery and cardiovascular

In a small elite group, injuries are expensive. Sable's knee, Jinson's Achilles, Harmilan Bains' knee surgery, Priti Lamba's ankle fracture at a water jump and Monika Athare's cartilage injury all removed key athletes from competition. A Recovery domain built on load, sleep, heart-rate variability and injury history is not a luxury here. For the Cardiovascular domain, the Indian Heart Journal has published guidance on preventing sudden cardiac death in Indian marathoners, which argues for structured pre-participation screening.

What India should start collecting

The single most useful step would be a standard, confidential national athlete data set that follows every senior and junior distance runner. At minimum it would include standardised race conditions, twice-yearly aerobic and threshold tests, iron status, haemoglobin mass around altitude camps, heat-acclimation sessions, long-run pace decay and injury history. The National Centre of Sports Science and Research in New Delhi, inaugurated in April 2025, and its partner university departments could anchor this work.

CHECKPOINT

For one athlete you coach, fill in the ten AEI domains using only what you actually have on file. How many are blank? Which two blanks, if filled, would most change your coaching decisions?

CHAPTER 96

96. A Roadmap for the Next Indian Elite

PART XVIII — INDIA'S ELITE RUNNERS: AN AEI ANALYSIS

Somewhere in India today, a 14-year-old is running to school along a road that has never been measured. It might be a hill road in Chamba, a canal path near Meerut, a dusty track in Beed. The runner does not know yet that the national 10,000 m record for women has stood since before they were born, or that no Indian man has ever broken 2:10 for the marathon.

Whether that runner ever reaches a start line in a World Championships final depends less on their lungs than on a chain of decisions made by other people: a school coach who notices, a hostel that has a place, a job that pays, a camp at altitude, a race abroad, a test that catches an iron

deficiency, a coach who knows when to hold them back. This chapter is about building that chain.

The recommendations below come directly from the evidence in this Part. Each is linked to a pattern observed in Indian elites and to principles set out earlier in this book. They are offered for discussion, not as official policy.

Six priorities

Priority	Evidence in this Part	Action	Success looks like
Talent pathways	Army and Railways carry most elites; late developers common	Link schools, SAI hostels and services trials; keep doors open to age 25+	More athletes entering from more states
Heat and altitude	Altitude used; heat unplanned	Plan altitude blocks and heat acclimation by season	Records set at home and abroad
Coaching education	Records fell under strong coach-led groups	Mentor Indian coaches alongside foreign experts	Several coach-led groups producing records
Data systems	Almost no public physiology	National AEI-style data set	Every elite with ten-domain profile
Anti-doping culture	India in AIU Category A since 2026	Education, testing, transparent support	Fewer violations; Category improves
Women's pathway	Long women's events lag most	Jobs, iron screening, marathon development	Women's 10,000 m and marathon records move

1. Talent pathways

The Army and the Railways already act as India's distance academies. The priority is to connect them to the places where talent begins. SAI hostels, state academies such as Madhya Pradesh's in Bhopal, and long-serving local coaches such as Vijender Singh in Nashik should have clear routes to services and Railways trials. Talent identification should stay open into the mid-twenties, because many of India's record holders, including Sable, Sawan and Gopi, reached their peaks late. Khelo India support, set at ₹6.28 lakh per athlete per year, can bridge the gap between a hostel place and a job.

KEY IDEA

India does not need a new pathway so much as better links between the ones it already has: school, hostel, services or Railways job, national camp.

2. Use heat and altitude deliberately

India has a rare combination: some of the world's hottest training conditions and genuine high-altitude venues at Ooty (2,240 m) and Shilaroo (about 2,450 m), with

Shillong planned. The roadmap should treat both as planned tools. Altitude blocks should be timed to competitions and monitored with haemoglobin and, where possible, haemoglobin mass. Heat acclimation should be scheduled before hot championships, using the protocols set out in Part III and Chapter 74, rather than left to chance.

COACH'S CORNER

Write the season plan with two extra lines: where the athlete will sleep at altitude, for how long and when; and when the heat-acclimation block will start before a warm-weather race. If neither is planned, the environment is coaching the athlete instead of you.

3. Coaching education

Records fell fastest when a clear method met committed athletes: Snesev's group from 2005, Amrith Kumar with Sable, Simmons in Colorado Springs, Markose's Project 2:09. The lesson is not to import more foreign coaches but to build Indian coaching depth alongside them. A ministry official has estimated that India is short of roughly 750 coaches. Pairing Indian coaches with every foreign expert, and requiring a written, shared training philosophy for national groups, would help the method outlast any single coach.

4. Data systems

Chapter 95 showed how little is publicly known about Indian elites beyond their race results. The roadmap should create a confidential national athlete data set using the ten AEI domains, anchored at the National Centre of Sports Science and Research and its partner universities, with clear consent and data-protection rules. Even simple, standardised field data, collected consistently, would transform the evidence base within five years.

5. Anti-doping culture

This must be stated plainly and neutrally. In April 2026 the Athletics Integrity Unit moved India's athletics federation into Category A, its highest doping-risk group, alongside Russia, Belarus, Ethiopia, Kenya, Nigeria and Ukraine. WADA's 2024 figures show 260 adverse analytical findings from 7,113 Indian tests across all sports. Among the distance runners discussed in this Part, two sustained cases are documented: Sanjivani Jadhav received a two-year ban in 2019 for the masking agent probenecid, and Kartik Kumar accepted a three-year ban from USADA in 2025 for anabolic steroids. A credible system protects clean athletes by combining education, frequent out-of-competition testing, transparent medical support and supplement guidance. Category A rules already require at least three no-notice tests before World Championships and Olympics, including blood-passport and EPO tests for events from 800 m upward.

6. The women's pathway

The women's long events show the largest gaps. The roadmap should guarantee secure jobs for leading women, screen iron status routinely, and build a deliberate pathway from the 5000 m and steeplechase into the 10,000 m, half marathon and marathon, with access to fast international races.

A four-year roadmap

1. Year 1: agree a national AEI-style data set; screen iron status for all national-camp women; schedule altitude and heat blocks in every national group's plan.
2. Year 2: link SAI hostels and state academies to services and Railways trials; begin paired Indian–foreign coaching for each event group.
3. Year 3: send a group of 5000 m and 10,000 m runners into marathon-specific development, with one fast international marathon each year.
4. Year 4: review the data set; publish anonymised findings; set new targets for the 2030s, including an Indian man under 2:10 and new women's 10,000 m and marathon records.

CHECKPOINT

Of the six priorities, which one could your own club, school or state association start on this year without any new funding? What is the first step, and who will take it?

PART XVIII AT A GLANCE

The key idea from each chapter, for quick revision.

90

The Indian Distance Map

For Indian men, the Army is not one pathway among many; it is the main pathway. Any national talent strategy that ignores the services, or duplicates what they already do, will waste scarce resources.

91

The Men: From 1500 m to the Marathon

Indian male distance runners often reach elite level late, after enlisting or after years on the track. Talent identification that closes at 17 would have missed several of the country's record holders.

92

The Women: Steeple to Marathon

For Indian women, a secure job has functioned as the professional contract that allows elite training. The absence of one is a measurable disadvantage, not just a personal hardship.

93

Records Over Time

Indian distance records tend to move in bursts. A single athlete with the right coach, environment and racing programme can rewrite a record five or ten times, after decades of stagnation.

94**How Far from the World?**

India's distance gap to the world is not one number. It is smallest in men's track events and the women's steeplechase, about 3–5%, and largest in the marathons and the women's long track events, about 9–19%.

95**Through the AEI Lens**

For Indian marathoners, the Durability domain is the most likely single limiter. It is also one of the cheapest to measure: the pace and heart-rate decay over the last 10 km of long runs and races.

96**A Roadmap for the Next Indian Elite**

India does not need a new pathway so much as better links between the ones it already has: school, hostel, services or Railways job, national camp.

Appendix A — AEI v0.1 Research Specification

AEI is proposed here as an original AKURA research framework. It is not presented as a validated replacement for VDOT, VO₂max testing, medical assessment or established performance models. The domains, fields and formula below are a research specification for prototype use only and require independent validation before any high-stakes application.

Domain	Primary examples	Interpretation	Confidence
AEI-P Performance	5K/10K/HM/Marathon results	Current race expression	High when recent and standardised
AEI-A Aerobic	VO ₂ -related tests, submax HR-pace	Aerobic capacity	Protocol dependent
AEI-T Threshold	Lactate/threshold pace	Sustainable high intensity	Protocol dependent
AEI-E Economy	O ₂ cost at fixed speed	Movement efficiency	Needs repeatability
AEI-O Oxygen	Hb mass, Hb concentration	O ₂ transport context	Specialist testing for Hb mass
AEI-C Cardiovascular	HR, stroke-volume-related measures	Circulatory response	Context dependent
AEI-R Recovery	Sleep, HRV trend, RHR, fatigue	Today's expression capacity	Baseline dependent
AEI-H Heat	Heat exposure, thermal response	Heat adaptation	Environment must be logged
AEI-X Altitude	Sleeping altitude, Hb mass response	Hypoxic adaptation	Dose dependent
AEI-D Durability	Long-run degradation	Marathon-specific resilience	Needs standardised protocol

Proposed data fields

Category	Fields
Identification	Athlete ID, Date/time, Device, Protocol version
Session and performance	Race/test type, Distance, Time, Pace
Physiological response	HR, RPE, HRV, RHR, Body mass
Environment	Temperature, Humidity/WBGT, Elevation
Recovery and wellness	Sleep, Fatigue, Soreness
Load and inputs	Training load, Fuel/hydration
Intervention	Intervention, Pre/post status
Quality and notes	Clinician/technician note, Data quality, Confidence

Calculation philosophy

- Do not assign arbitrary weights simply to produce a 0–100 score.
- Keep raw measurements permanently available.
- Normalise within athlete and, where appropriate, against a reference population.
- Use recent performance as the primary expression anchor.
- Separate Fitness from Readiness.
- Attach confidence and missing-data penalties.
- Validate every derived score prospectively before high-stakes use.

Example research formula (prototype only)

A future AEI score could be modelled as a calibrated function rather than a hand-built average:

$AEI = f(\text{Performance, Economy, Threshold, Oxygen Transport, Cardiovascular Response, Durability, Recovery, Environment})$

Here f is learned and validated from longitudinal athlete data, not chosen by hand. Until that validation exists, the components should be reported separately, each with its own raw data, date, method and confidence level.

Appendix B — Athlete Research Worksheet

Use one worksheet per athlete per intervention. Record the baseline and post-intervention values under the same protocol, note the change, and add a confidence rating and short coach note in the final column.

Field	Baseline	Post	Change	Confidence / Note
Race performance				
Easy pace at fixed HR				
Threshold marker				
VO ₂ -related measure				
Running economy				
Hb / Hb mass				
Resting HR				
HRV trend				
Sleep				
Body mass				
Heat exposure				
Altitude exposure				
Long-run durability				

Appendix C — Heat Session Log Template

Complete one log for every heat session. Stop the session immediately if any heat-illness warning sign appears, begin cooling and seek emergency medical care.

Field	Entry
Date and start time	
Location and surface	
Temperature / humidity / WBGT	
Session type (run, heated room, passive)	
Duration and target pace or intensity	
Average and peak HR	
RPE and thermal comfort (end)	
Body mass before / after	
Fluid intake	
Core or skin temperature (if measured)	
Symptoms or warning signs	
Supervisor and coach note	

Appendix D — Altitude Camp Log Template

Complete daily during any camp at Ooty, Kodaikanal or elsewhere. Report worsening headache, vomiting, breathlessness at rest or confusion to the medical lead immediately.

Field	Entry
Date and camp day number	
Sleeping altitude	
Training altitude	
Hours at altitude	
Morning SpO ₂ and resting HR	
Sleep duration and quality	
Session and training load	
RPE and HR at fixed pace	
Wellness and appetite	
Iron supplementation (as prescribed)	
Symptoms or warning signs	
Coach and clinician note	

Appendix E — Daily Readiness Check-in

A short morning check-in supports the AISRi readiness layer. It is a conversation starter, not a diagnosis; any concerning symptom should be discussed with the coach and, where needed, a clinician.

Item	Scale	Notes
Sleep duration	Hours	
Sleep quality	1 (very poor) – 5 (excellent)	
Resting HR	beats per minute	
HRV (if available)	Device value; compare to trend	
Fatigue	1 (fresh) – 5 (exhausted)	
Muscle soreness	1 (none) – 5 (severe)	
Mood / stress	1 (very low) – 5 (very good)	
Illness symptoms	Yes / No	
Menstrual-cycle note (optional)	Free text	
Unusual context (travel, work, exams)	Free text	

Glossary

AEI: AKURA Endurance Index; a proposed research framework for organising endurance performance data into domains. Not yet validated.

AISri: The AKURA daily readiness and training decision layer in the proposed coaching ecosystem.

Cardiac output: The volume of blood pumped by the heart per minute; the product of heart rate and stroke volume.

Confidence interval: A range of values, calculated from data, that expresses the uncertainty around an estimate such as a mean change.

Decoupling: The drift of heart rate relative to pace (or power) during prolonged steady exercise; often used as an informal marker of aerobic durability.

Durability: The ability to maintain performance and physiological characteristics as exercise duration and fatigue increase.

EPO: Erythropoietin; a hormone, produced mainly by the kidneys, involved in red-cell production.

Erythropoiesis: The production of new red blood cells, which can be stimulated by hypoxia.

Ferritin: A protein that stores iron; blood ferritin is commonly used as an indicator of iron stores and should be interpreted by a clinician.

Fractional utilisation: The percentage of VO_2max an athlete can sustain over a given duration or race distance.

Glycogen: The stored form of carbohydrate in muscle and liver; an important fuel in prolonged and high-intensity exercise.

Haemoglobin mass: The total mass of haemoglobin in the circulation; distinct from haemoglobin concentration.

Heat acclimation vs acclimatisation: Acclimation refers to adaptation produced by artificial or controlled heat exposure (for example a heated room); acclimatisation refers to adaptation produced by natural environmental exposure.

HRV: Heart-rate variability; a context-dependent autonomic and recovery signal best interpreted as a trend.

Hypoxia: A reduced availability of oxygen to the body or tissues, as occurs at altitude.

Lactate threshold: A workload region where lactate dynamics change markedly; its exact definition depends on the protocol used.

Normalisation: Expressing a value relative to a reference, such as the athlete's own baseline or a population range, so that different measures can be compared.

Partial pressure: The pressure exerted by one gas within a mixture. At altitude the oxygen fraction of air is unchanged, but the partial pressure of oxygen falls because total pressure falls.

Plasma volume: The volume of the liquid component of blood.

RPE: Rating of perceived exertion; the athlete's subjective rating of how hard an effort feels, on a defined scale.

Running economy: The oxygen (or energy) cost of running at a given submaximal speed.

Smallest worthwhile change: The smallest change in a measure that is considered practically meaningful for performance or health.

Stroke volume: The volume of blood ejected by the heart with each beat.

Thermoregulation: The body's processes for controlling core temperature, including sweating and changes in skin blood flow.

Typical error: The typical variation in a measurement when it is repeated under the same conditions; used to judge whether a change is real.

VDOT: An established running-performance and training framework based on current running ability.

VO₂max: Maximum oxygen uptake during severe or maximal exercise.

WBGT: Wet-bulb globe temperature; an environmental heat-stress index that combines temperature, humidity, radiant heat and air movement.

Final Research Principles

1. Never confuse concentration with total mass.
2. Never confuse oxygen fraction with oxygen partial pressure.
3. Never confuse VO_2max with complete running ability.
4. Never confuse heat adaptation with altitude adaptation.
5. Never treat a single wearable metric as a diagnosis.
6. Never assign score weights without validation.
7. Always document the protocol.
8. Always separate current fitness from today's readiness.
9. Always allow the data to disagree with the hypothesis.
10. Always keep the athlete at the centre of the system.

These ten principles are the shortest possible summary of this book. They will outlast any particular version of AEI. Equations will be revised, devices will change and new research will overturn some of today's assumptions. What should remain is the habit of careful measurement, honest interpretation and respect for the individual athlete. If this edition helps even a few coaches ask better questions and test their beliefs more carefully, it will have done its work.

References and Further Reading

The sources below were used as starting points for the evidence discussed in this book. Readers should consult the original papers; inclusion here does not imply that the authors endorse the AKURA Endurance Index.

- Al Jazeera. India in "extremely high" doping risk bracket: Athletics Integrity Unit. 2026.
<https://www.aljazeera.com/sports/2026/4/20/india-in-extremely-high-doping-risk-bracket-athletics-integrity-unit>
- Armstrong LE, Casa DJ, Millard-Stafford M, Moran DS, Pyne SW, Roberts WO. American College of Sports Medicine position stand: exertional heat illness during training and competition. *Medicine & Science in Sports & Exercise*. 2007;39(3):556–572.
- Army Sports Institute, Pune. About ASI. <https://armysportsinstitute.com/>
- Atkinson G, Nevill AM. Statistical methods for assessing measurement error (reliability) in variables relevant to sports medicine. *Sports Medicine*. 1998;26(4):217–238.
- Balsalobre-Fernández C, Santos-Concejero J, Grivas GV. Effects of strength training on running economy in highly trained runners: a systematic review with meta-analysis of controlled trials. *Journal of Strength and Conditioning Research*. 2016;30(8):2361–2368.
- Bassett DR Jr, Howley ET. 2000. Limiting factors for maximum oxygen uptake and determinants of endurance performance. *Medicine and Science in Sports and Exercise*.
- Beattie K, Kenny IC, Lyons M, Carson BP. The effect of strength training on performance in endurance athletes. *Sports Medicine*. 2014;44(6):845–865.
- Bouchard C, An P, Rice T, Skinner JS, Wilmore JH, Gagnon J, Pérusse L, Leon AS, Rao DC. 1999. Familial aggregation of VO_2max response to exercise training: results from the HERITAGE Family Study. *Journal of Applied Physiology*.
- Bourdon PC, Cardinale M, Murray A, et al. Monitoring athlete training loads: consensus statement. *International Journal of Sports Physiology and Performance*. 2017;12(Suppl 2):S2161–S2170.
- Brooks GA. 2018. The science and translation of lactate shuttle theory. *Cell Metabolism*.
- Brown HA, et al. Quantifying Exercise Heat Acclimatisation in Athletes and Military Personnel: A Systematic Review and Meta-analysis. *Sports Medicine*. 2024;54(3):727–741. PMID 38051495.
- Burke LM, Hawley JA, Wong SHS, Jeukendrup AE. 2011. Carbohydrates for training and competition. *Journal of Sports Sciences*.
- Casa DJ, DeMartini JK, Bergeron MF, et al. National Athletic Trainers' Association position statement: exertional heat illnesses. *Journal of Athletic Training*. 2015;50(9):986–1000.
- Chapman RF, Stray-Gundersen J, Levine BD. Individual variation in response to altitude training. *Journal of Applied Physiology*. 1998;85(4):1448–1456.
- Chatterjee S, Saha SK, Saha D, Nag SK. Maximal aerobic capacity of Bengali girl athletes of different sports activities. *Japanese Journal of Physiology*. 1991. PMID 1960887.
- Collins GS, Reitsma JB, Altman DG, Moons KGM. Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): the TRIPOD statement. *BMJ*. 2015;350:g7594.
- Coyle EF, González-Alonso J. Cardiovascular drift during prolonged exercise: new perspectives. *Exercise and Sport Sciences Reviews*. 2001;29(2):88–92.

- Daniels J. Daniels' Running Formula. 3rd ed. Champaign, IL: Human Kinetics; 2014.
- Daniels J. Daniels' Running Formula. 4th ed. Champaign, IL: Human Kinetics; 2022 (VDOT framework and training paces). VDOT® is a registered trademark of The Run SMART Project, LLC.
- Dill DB, Costill DL. 1974. Calculation of percentage changes in volumes of blood, plasma, and red cells in dehydration. *Journal of Applied Physiology*.
- ESPN India. No turning back: Ankita Dhyani. 2021.
https://www.espn.in/athletics/story/_/id/30859627/no-turning-back-ankita-dhyani-mastery-improbable-chases-makes-unbeatable
- ESPN. Thonakal Gopi exclusive interview. 2016.
https://www.espn.com/athletics/story/_/id/17475379/india-marathon-runner-olympics-thonackal-gopi-talks-espn-exclusive-interview
- Faude O, Kindermann W, Meyer T. Lactate threshold concepts: how valid are they? *Sports Medicine*. 2009;39(6):469–490.
- Foster C, Florhaug JA, Franklin J, et al. A new approach to monitoring exercise training. *Journal of Strength and Conditioning Research*. 2001;15(1):109–115.
- Gabbett TJ. The training–injury prevention paradox: should athletes be training smarter and harder? *British Journal of Sports Medicine*. 2016;50(5):273–280.
- Ghosh S, Laxmaiah A, Chandak GR, et al. Anaemia and iron deficiency in India: a venous blood-based survey of adolescents, adults, and the elderly in eight states. *European Journal of Clinical Nutrition*. 2025;79(5):443–451. PMID 39779946.
- Gore CJ, Sharpe K, Garvican-Lewis LA, et al. Altitude training and haemoglobin mass from the optimised carbon monoxide rebreathing method determined by a meta-analysis. *British Journal of Sports Medicine*. 2013;47(Suppl 1):i31–i39. PMID 24282204.
- Halsen SL. Monitoring training load to understand fatigue in athletes. *Sports Medicine*. 2014;44(Suppl 2):S139–S147.
- Hecksteden A, Kraushaar J, Scharhag-Rosenberger F, et al. Individual response to exercise training – a statistical perspective. *Journal of Applied Physiology*. 2015;118(12):1450–1459.
- Hindustan Times. High-altitude training centre to come up in Shillong. 2026.
<https://www.hindustantimes.com/sports/others/high-altitude-training-centre-to-come-up-in-shillong-101780588041378.html>
- Holloszy JO. 1967. Biochemical adaptations in muscle: effects of exercise on mitochondrial oxygen uptake and respiratory enzyme activity in skeletal muscle. *Journal of Biological Chemistry*.
- Hopkins WG, Marshall SW, Batterham AM, Hanin J. Progressive statistics for studies in sports medicine and exercise science. *Medicine and Science in Sports and Exercise*. 2009;41(1):3–13.
- Hopkins WG. Measures of reliability in sports medicine and science. *Sports Medicine*. 2000;30(1):1–15.
- Indian Express. Running through sugarcane fields: Parul Chaudhary. 2023.
<https://indianexpress.com/article/sports/asian-games/running-sugarcane-field-parul-chaudhary-asian-games-2023-hangzhou-8943068/>
- Jegatheesan SL, Kumbamoorthy SS, Sanjaykumar S, Rajkumar NCJ. Unlocking the connection: hematological status, dietary iron intake and endurance in Indian female athletes. *Physical Activity and Nutrition*. 2025;29(1):31–37. PMID 40443248.

- Jeukendrup A. A step towards personalized sports nutrition: carbohydrate intake during exercise. *Sports Medicine*. 2014;44(Suppl 1):S25–S33.
- Jones AM, Kirby BS, Clark IE, et al. Physiological demands of running at 2-hour marathon race pace. *Journal of Applied Physiology*. 2021;130(2):369–379.
- Jones AM. The physiology of the world record holder for the women's marathon. *International Journal of Sports Science & Coaching*. 2006;1(2):101–116.
- Joyner MJ, Coyle EF. Endurance exercise performance: the physiology of champions. *Journal of Physiology*. 2008;586(1):35–44.
- Levine BD, Stray-Gundersen J. "Living high-training low": effect of moderate-altitude acclimatization with low-altitude training on performance. *Journal of Applied Physiology*. 1997;83(1):102–112.
- Little RJA, Rubin DB. *Statistical Analysis with Missing Data*. 2nd ed. Hoboken, NJ: Wiley; 2002.
- Lorenzo S, Halliwill JR, Sawka MN, Minson CT. Heat acclimation improves exercise performance. *Journal of Applied Physiology*. 2010;109(4):1140–1147.
- Maunder E, Seiler S, Mildenhall MJ, Kilding AE, Plews DJ. The importance of 'durability' in the physiological profiling of endurance athletes. *Sports Medicine*. 2021;51(8):1619–1628.
- Meeusen R, Duclos M, Foster C, et al. Prevention, diagnosis, and treatment of the overtraining syndrome: joint consensus statement of the European College of Sport Science and the American College of Sports Medicine. *Medicine & Science in Sports & Exercise*. 2013;45(1):186–205.
- Midgley AW, McNaughton LR, Polman R, Marchant D. Criteria for determination of maximal oxygen uptake: a brief critique and recommendations for future research. *Sports Medicine*. 2007;37(12):1019–1028.
- Ministry of Youth Affairs and Sports. National Centre of Sports Sciences and Research. <https://yas.gov.in/en/sports/national-centre-sports-sciences-and-research-ncssr-0>
- Olympics.com. Gulveer Singh 10,000 m national record at The TEN. 2025. <https://www.olympics.com/en/news/gulveer-singh-india-10000m-national-record-the-ten-2025-athletics-california>
- Olympics.com. Nikolai Snegarev: the late Belarusian coach's contribution to Indian athletics. 2021. <https://www.olympics.com/en/news/nikolai-snegarev-the-late-belarusian-coach-s-contribution-to-indian-athletics>
- Olympics.com. Parul Chaudhary national record 5000 m, Meeting Nikaia. 2026. <https://www.olympics.com/en/news/parul-chaudhary-national-record-5000m-meeting-nikaia-2026-athletics>
- Olympics.com. Who is Avinash Sable. 2024. <https://www.olympics.com/en/news/who-is-avinash-sable-india-athlete-steeplechase>
- Outlook India. India highest doping positivity rate in WADA 2024 testing report. 2025. <https://www.outlookindia.com/sports/others/india-highest-doping-positivity-rate-wada-2024-testing-report>
- Outlook India. Seema Kumari: Asian Games 2026 bronze medal. 2026. <https://www.outlookindia.com/sports/others/seema-kumari-asian-games-2026-bronze-medal-mother-health-athletics-india>

- Paavolainen L, Häkkinen K, Härmäläinen I, Nummela A, Rusko H. Explosive-strength training improves 5-km running time by improving running economy and muscle power. *Journal of Applied Physiology*. 1999;86(5):1527–1533.
- Peeling P, Dawson B, Goodman C, Landers G, Trinder D. Athletic induced iron deficiency: new insights into the role of inflammation, cytokines and hormones. *European Journal of Applied Physiology*. 2008;103(4):381–391.
- Pelliccia A, Sharma S, Gati S, et al. 2020 ESC Guidelines on sports cardiology and exercise in patients with cardiovascular disease. *European Heart Journal*. 2021;42(1):17–96.
- Périard JD, Racinais S, Sawka MN. Adaptations and mechanisms of human heat acclimation: applications for competitive athletes and sports. *Scandinavian Journal of Medicine & Science in Sports*. 2015;25(Suppl 1):20–38.
- Plews DJ, Laursen PB, Stanley J, Kilding AE, Buchheit M. Training adaptation and heart rate variability in elite endurance athletes: opening the door to effective monitoring. *Sports Medicine*. 2013;43(9):773–781.
- Press Information Bureau. Khelo India athlete support. 2024.
<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2004017>
- Racinais S, Alonso JM, Coutts AJ, et al. Consensus recommendations on training and competing in the heat. *British Journal of Sports Medicine*. 2015;49(18):1164–1173.
- Rahimi GRM, Albanaqi AL, Van der Touw T, et al. Physiological Responses to Heat Acclimation: A Systematic Review and Meta-Analysis of Randomized Controlled Trials. *Journal of Sports Science and Medicine*. 2019;18(2):316–326. PMID 31191102.
- Rediff. Asian Games: after double bronze haul, Parul now craves family time. 2026.
<https://www.rediff.com/sports/report/asian-games-after-double-bronze-haul-parul-now-craves-family-time/20260928.htm>
- Rønnestad BR, Mujika I. Optimizing strength training for running and cycling endurance performance: a review. *Scandinavian Journal of Medicine & Science in Sports*. 2014;24(4):603–612.
- Rubin DB. Inference and missing data. *Biometrika*. 1976;63(3):581–592.
- Saltin B, Larsen H, Terrados N, Bangsbo J, Bak T, Kim CK, Svedenhag J, Rolf CJ. 1995. Aerobic exercise capacity at sea level and at altitude in Kenyan boys, junior and senior runners compared with Scandinavian runners. *Scandinavian Journal of Medicine and Science in Sports*.
- Saunders PU, Pyne DB, Telford RD, Hawley JA. Factors affecting running economy in trained distance runners. *Sports Medicine*. 2004;34(7):465–485.
- Saw AE, Main LC, Gastin PB. Monitoring the athlete training response: subjective self-reported measures trump commonly used objective measures: a systematic review. *British Journal of Sports Medicine*. 2016;50(5):281–291.
- Sawka MN, Burke LM, Eichner ER, et al. American College of Sports Medicine position stand: Exercise and fluid replacement. *Medicine and Science in Sports and Exercise*. 2007;39(2):377–390.
- Sawka MN, Leon LR, Montain SJ, Sanna LA. Integrated physiological mechanisms of exercise performance, adaptation, and maladaptation to heat stress. *Comprehensive Physiology*. 2011;1(4):1883–1928.
- Schmidt W, Prommer N. The optimised CO-rebreathing method: a new tool to determine total haemoglobin mass routinely. *European Journal of Applied Physiology*. 2005;95(5–6):486–495.

- Scroll.in. How a former SAI coach is making Nashik the hub for long-distance running in India. 2018. <https://scroll.in/field/865878/how-a-former-sai-coach-is-making-nashik-the-hub-for-long-distance-running-in-india>
- Scroll.in. Sanjivani Jadhav handed two-year suspension for doping violation. 2019. <https://scroll.in/field/931143/athletics-long-distance-runner-sanjivani-jadhav-handed-two-year-suspension-for-doping-violation>
- Seiler KS, Kjerland GØ. Quantifying training intensity distribution in elite endurance athletes: is there evidence for an "optimal" distribution? *Scandinavian Journal of Medicine & Science in Sports*. 2006;16(1):49–56.
- Sharma S, Drezner JA, Baggish A, et al. International recommendations for electrocardiographic interpretation in athletes. *Journal of the American College of Cardiology*. 2017;69(8):1057–1075.
- Shyam Gopan. Gopi 23 seconds short of national record at Valencia Marathon. 2025. <https://shyamgopan.com/2025/12/11/gopi-23-secs-short-of-national-record-sets-new-pb-at-valencia-marathon/>
- Solomon TPJ, Laye MJ. The effect of post-exercise heat exposure (passive heat acclimation) on endurance exercise performance: a systematic review and meta-analysis. *BMC Sports Science, Medicine and Rehabilitation*. 2025. PMID 39762944.
- Sterne JAC, White IR, Carlin JB, et al. Multiple imputation for missing data in epidemiological and clinical research: potential and pitfalls. *BMJ*. 2009;338:b2393.
- Stray-Gundersen J, Chapman RF, Levine BD. "Living high-training low" altitude training improves sea level performance in male and female elite runners. *Journal of Applied Physiology*. 2001;91(3):1113–1120.
- The Hindu. Udhagamandalam boasts south India's only high-altitude sports training centre. 2024. <https://www.thehindu.com/news/national/tamil-nadu/udhagamandalam-boasts-south-indias-only-high-altitude-sports-training-centre/article68333440.ece>
- The Quint. From running in fields to the Rio tracks: Lalita Babar's story. 2016. <https://www.thequint.com/sports/sports-buzz/from-running-in-fields-to-the-rio-tracks-lalita-babars-story>
- The Tribune. Who is Sawan Barwal, Himachal runner who challenged marathon myth to win Asian Games silver. 2026. <https://www.tribuneindia.com/news/sports/who-is-sawan-barwal-himachal-runner-who-challenged-marathon-myth-to-win-asian-games-silver/>
- Thomas DT, Erdman KA, Burke LM. American College of Sports Medicine Joint Position Statement: Nutrition and Athletic Performance. *Medicine and Science in Sports and Exercise*. 2016;48(3):543–568.
- Tibshirani R. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B*. 1996;58(1):267–288.
- USADA. Kartik Kumar accepts doping sanction. 2025. <https://www.usada.org/sanction/kartik-kumar-accepts-doping-sanction/>
- Vora A, Burkule N, Contractor A, Bhargava K. Prevention of sudden cardiac death in athletes, sportspersons and marathoners in India. *Indian Heart Journal*. 2018. <https://pmc.ncbi.nlm.nih.gov/articles/PMC5903013/>
- Wilber RL, Pitsiladis YP. 2012. Kenyan and Ethiopian distance runners: what makes them so good? *International Journal of Sports Physiology and Performance*.
- Wilber RL. *Altitude Training and Athletic Performance*. Champaign, IL: Human Kinetics; 2004.

World Athletics. Athlete profiles (Gulveer Singh, Avinash Sable, Sawan Barwal, Parul Chaudhary, Preeja Sreedharan). <https://worldathletics.org/athletes>

Yaglou CP, Minard D. Control of heat casualties at military training centers. *AMA Archives of Industrial Health*. 1957;16(4):302–316.

About the Author

Kura is the founder of the AKURA Endurance Research System and the AISRI coaching ecosystem, based in Chennai, India. The work grows from years of practical involvement with endurance athletes training in South Indian heat and humidity, and from a conviction that Indian coaching can build its own evidence base rather than borrowing assumptions from elsewhere. This Prototype Edition 0.1 is offered as a starting framework for discussion, collaboration and validation with coaches, athletes, scientists and clinicians. Feedback, criticism and research partnerships are welcome.